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Class 12 Physics Chapterwise PYQs

2026 – 2003 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 2: Electrostatic Potential and Capacitance

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HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 100 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. In a region, the electric potential varies as $V = 10 - 50x$, where V is in volts and x in meters. The electric field in the region is:

(A) 10 N/C along +x

(B) 10 N/C along -x

(C) 50 N/C along +x

(D) 50 N/C along -x

Check Solution → [2026 • Set 55-1-1]

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2. A conducting wire connects two charged metallic spheres A and B of radii r_1 and r_2 respectively. The distance between the spheres is very large compared to their radii. The ratio of electric fields, (E_1/E_2) at the surfaces of spheres A and B will be:

(A) $\frac{r_1}{r_2}$

(B) $\frac{r_2}{r_1}$

(C) $\frac{r_2^2}{r_1^2}$

(D) $\frac{r_1^2}{r_2^2}$

Check Solution → [2026 • Set 55-1-1]

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3. In a region electric field is given by $\vec{E} = 4x\,\hat{i}$ N/C. The potential difference between points A ($x = 1$ m) and B ($x = 3$ m), $(V_A - V_B)$ is:

(A) −16 V

(B) 16 V

(C) −8 V

(D) 8 V

Check Solution → [2026 • Set 55-1-2]

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4. The electric potential for various points in x-y plane is given by $V = 1.0\ x^2 - 2.0\ y^2$, where V is in volts and x, y are in metres. The angle that the electric field at point (2.0 m, 1.0 m) makes with the positive x-axis is:

(A) 45°

(B) 90°

(C) 135°

(D) 315°

Check Solution → [2026 • Set 55-1-3]

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5. Three point charges 2q, −2q and q are kept at the vertices of an equilateral triangle of side l. The potential energy of the system is:

(A) zero

(B) $\frac{-2q^2}{\pi\varepsilon_0 l}$

(C) $\frac{q^2}{2\pi\varepsilon_0 l}$

(D) $\frac{-q^2}{\pi\varepsilon_0 l}$

Check Solution → [2026 • Set 55-2-1]

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6. Two metal spheres of radii r_1 and r_2 ($r_2 > r_1$) having charges q_1 and q_2 respectively kept in air, are brought in contact. Which of the following statements is NOT correct?

(A) The total charge of the two spheres is conserved.

(B) Both spheres attain the same potential.

(C) The final potential of the system equals $\frac{1}{4\pi\varepsilon_0}\frac{(q_1+q_2)}{(r_1+r_2)}$.

(D) The final potential of the system equals $\frac{1}{4\pi\varepsilon_0}\frac{(q_1+q_2)(r_1+r_2)}{r_1 r_2}$.

Check Solution → [2026 • Set 55-2-1]

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7. Three point charges $+q$, $-2q$ and $+q$ are placed along x-axis at points $x=-1$ m, $x=0$ m and $x=2$ m respectively. The potential energy of the system is:

(A) $\frac{q^2}{4\pi\varepsilon_0}$

(B) $\frac{-2q^2}{3\pi\varepsilon_0}$

(C) $\frac{q^2}{4\pi\varepsilon_0}$

(D) $\frac{-5q^2}{4\pi\varepsilon_0}$

Check Solution → [2026 • Set 55-2-2]

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8. A point charge of 5 μC is located at (1 cm, 1 cm) in x-y plane. A second charge of 6 μC is brought from infinity to point (−3 cm, 4 cm). The work done in the process is:

(A) $0{\cdot}54\ J$

(B) $5{\cdot}4\ J$

(C) $-0{\cdot}54\ J$

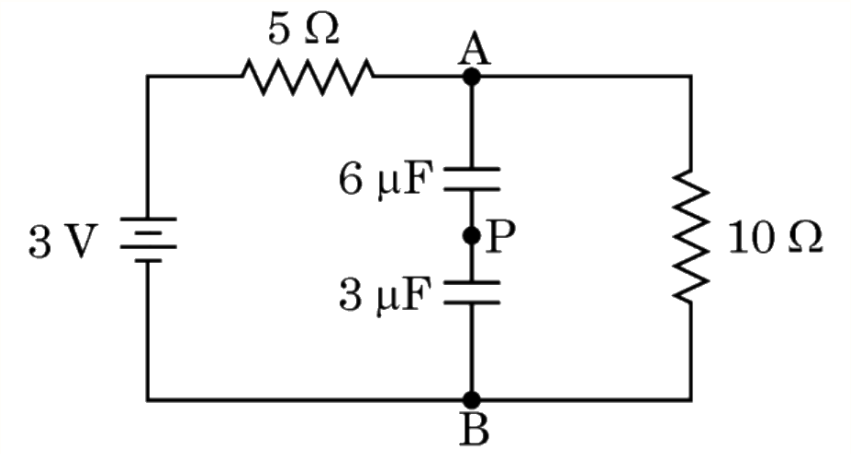
(D) $10{\cdot}8\ J$

Check Solution → [2026 • Set 55-2-3]

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9. **Case Study:** Capacitors are manufactured with certain standard capacitances and working voltages. However, these standard values may not be the ones that are actually needed in a particular application. Two or more capacitors can be grouped in series or in parallel to achieve desired capacitance and voltage.

When connected in series, the total capacitance decreases while the voltage rating increases, whereas in parallel connections, the total capacitance increases and maintains the same voltage rating. A capacitor stores energy in the electric field between its plates and stored energy is proportional to the square of the voltage and capacitance $U = \frac{1}{2}CV^2$, where symbols have their usual meanings.



Two capacitors, one of 3 μF and the other of 6 μF, are connected in series in the circuit as shown in the figure, for a long time.
The total capacitance of the circuit is:

(A) 6 μF

(B) 3 μF

(C) 9 μF

(D) 2 μF

Check Solution → **[2026 • Set 55-3-1]**

10. The current in the 10 Ω resistor is:

(A) 0·3 *A*

(B) 0·6 *A*

(C) 0·2 *A*

(D) 0

Check Solution → **[2026 • Set 55-3-1]**

11. The potential difference between point A and B is:

(A) 2 V

(B) 0·3 V

(C) 0·2 V

(D) 3 V

Check Solution → [2026 • Set 55-3-1]

---

12. The value of charge on the plates of the 6 μF capacitor is:

(A) 6 μC

(B) 4 μC

(C) 12 μC

(D) 8 μC

Check Solution → [2026 • Set 55-3-1]

13.

The wire between two capacitors is cut at point P. The current in the circuit will:

(A) increase

(B) decrease

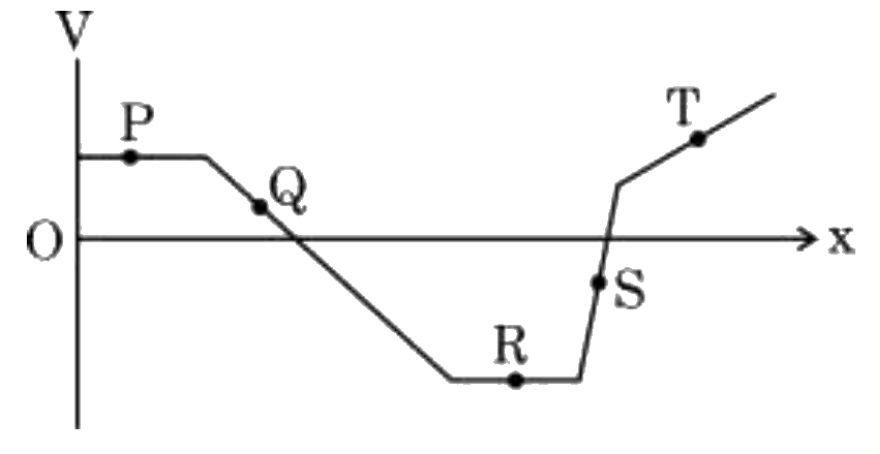
(C) remain the same

(D) first increase then become stable

Check Solution → [2026 • Set 55-3-1]

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14. The figure represents the variation of the electric potential V at a point in a region of space as a function of its position along the x-axis. A charged particle will experience the maximum force at:



(A) P

(B) Q

(C) R

(D) S

Check Solution → [2026 • Set 55-5-1]

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15. The electric field $\vec{E}$ in the region between the plates is:

(A) $(2 \times 10^2 \frac{V}{m})\,\hat{k}$

(B) $-(2 \times 10^2 \frac{V}{m})\,\hat{k}$

(C) $(2 \times 10^4 \frac{V}{m})\,\hat{k}$

(D) $-(2 \times 10^4 \frac{V}{m})\,\hat{k}$

Check Solution → [2026 • Set 55-5-1]

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16. In the region between the plates, the electron moves with an acceleration $\vec{a}$ given by:

(A) $-(3.5 \times 10^{15}\ \text{ms}^{-2})\ \hat{k}$

(B) $(3.5 \times 10^{15}\ \text{ms}^{-2})\ \hat{k}$

(C) $(3.5 \times 10^{13}\ \text{ms}^{-2})\ \hat{i}$

(D) $-(3.5 \times 10^{13}\ \text{ms}^{-2})\ \hat{i}$

Check Solution → [2026 • Set 55-5-1]

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17. (a) Time interval during which an electron moves through the region between the plates is:

(A) $9.0 \times 10^{-9}\ s$

(B) $1.67 \times 10^{-8}\ s$

(C) $1.67 \times 10^{-9}\ s$

(D) $2.17 \times 10^{-9}\ s$

Check Solution → [2026 • Set 55-5-1]

18.

(b) The vertical displacement of the electron which travels through the region between the plates is:

(A) 1.0 mm

(B) 4.9 mm

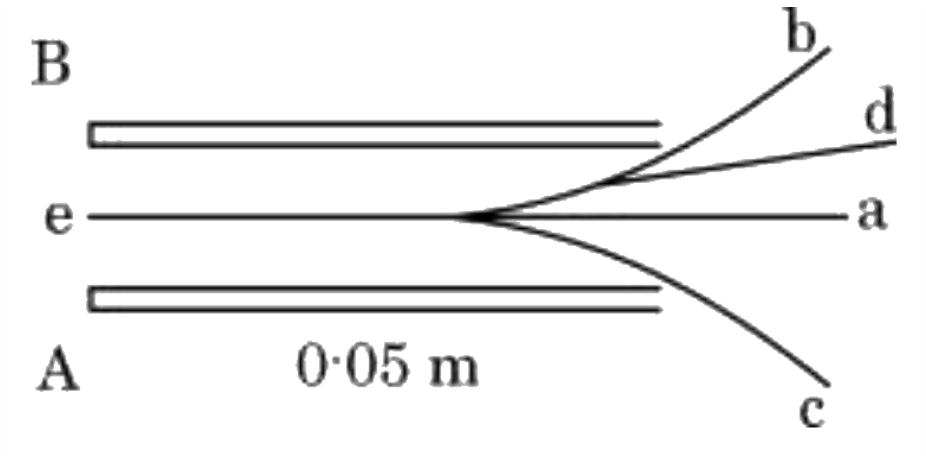
(C) 5.9 mm

(D) 3.0 mm

Check Solution → [2026 • Set 55-5-1]

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19. Which one of the following is the path traced by the electron in between the two plates?



(A) a

(B) b

(C) c

(D) d

Check Solution → [2026 • Set 55-5-1]

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20. Two points R and S are equidistant from two charges $+Q$ and $-2Q$. The work done in moving a charge $-Q$ from point R to S is:

(A) Zero

(B) $\frac{Q}{4\pi\varepsilon_0 d}$

(C) $\frac{2Q}{4\pi\varepsilon_0 d}$

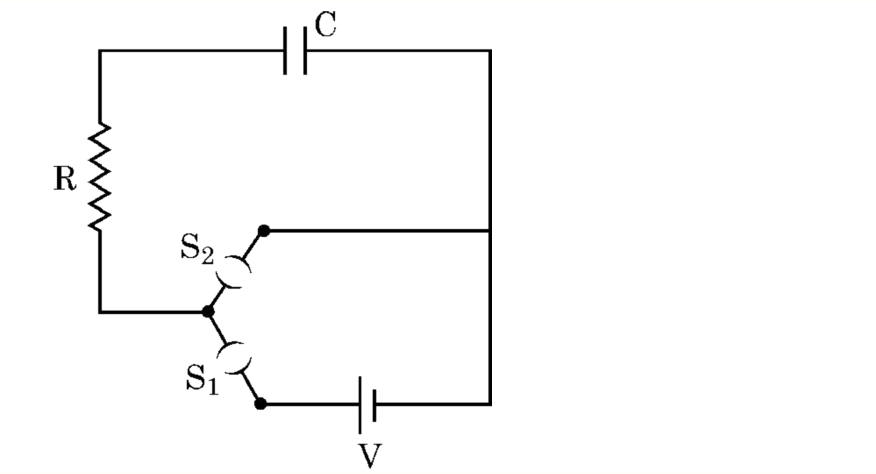
(D) $\frac{3Q}{4\pi\varepsilon_0 d}$

Check Solution → [2026 • Set 55-5-3]

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21. **Case Study (Q29):** A circuit consisting of a capacitor C, a resistor of resistance R and an ideal battery of emf V, as shown in figure, is known as an RC series circuit. As soon as the circuit is completed by closing key S_1 (keeping S_2 open), charges begin to flow between the capacitor plates and the battery terminals. The charge on the capacitor increases and consequently the potential difference V_c ($= q/C$) across the capacitor also increases with time. When this potential difference equals the potential difference across the battery, the capacitor is fully charged ($Q = VC$). During this process of charging, the charge q on the

capacitor changes with time t as $q = Q[1 - e^{-t/RC}]$. The charging current can be obtained by differentiating it and using $\frac{d}{dx}(e^{mx}) = m\, e^{mx}$. Consider the case when $R = 20\ k\Omega$, $C = 500\ \mu F$ and $V = 10\ V$.



(i) The final charge on the capacitor, when key S_1 is closed and S_2 is open, is:

(A) 5 μC

(B) 5 mC

(C) 25 mC

(D) 0.1 C

Check Solution → [2025 • Set 55-1-1]

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22. **(ii)** For sufficient time the key S_1 is closed and S_2 is open. Now S_2 is closed and S_1 is open. What is the final charge on the capacitor?

(A) Zero

(B) 5 mC

(C) 2.5 mC

(D) 5 μC

Check Solution → [2025 • Set 55-1-1]

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23. **(iii)** The dimensional formula for RC is:

(A) $[M\,L^2\,T^{-3}\,A^{-2}]$

(B) $[M^0\,L^0\,T^{-1}\,A^0]$

(C) $[M^{-1}\,L^{-2}\,T^4\,A^2]$

(D) $[M^0\,L^0\,T\,A^0]$

Check Solution → [2025 • Set 55-1-1]

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24. **(iv)** The key S_1 is closed and S_2 is open. The value of current in the resistor after 5 seconds is:

(A) $\frac{1}{2\sqrt{e}}$ mA

(B) $\sqrt{e}$ mA

(C) $\frac{1}{\sqrt{e}}$ mA

(D) $\frac{1}{2e}$ mA

Check Solution → [2025 • Set 55-1-1]

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25. **(iv) OR:** The key S_1 is closed and S_2 is open. The initial value of charging current in the resistor is:

(A) 5 mA

(B) 0.5 mA

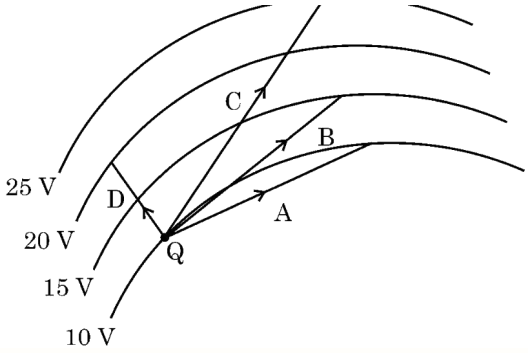
(C) 2 mA

(D) 1 mA

Check Solution → [2025 • Set 55-1-1]

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26. In the figure curved lines represent equipotential surfaces. A charge Q is moved along different paths A, B, C and D. The work done on the charge will be maximum along the path:



(A) A

(B) B

(C) C

(D) D

Check Solution → [2025 • Set 55-1-2]

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27. **Case study:** A parallel plate capacitor has two parallel plates which are separated by an insulating medium like air, mica, etc. When the plates are connected to the terminals of a battery, they get equal and opposite charges and an electric field is set up in between them. This electric field between the two plates depends upon the potential difference applied, the separation of the plates and nature of the medium between the plates.
The electric field between the plates of a parallel plate capacitor is E. Now the separation between the plates is doubled and simultaneously the applied potential difference between the plates is reduced to half of its initial value. The new value of the electric field between the plates will be:

(A) E

(B) 2E

(C) $\frac{E}{4}$

(D) $\frac{E}{2}$

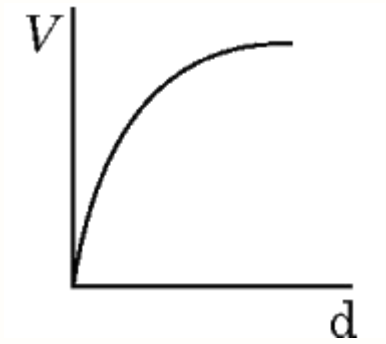
Check Solution → [2025 • Set 55-2-1]

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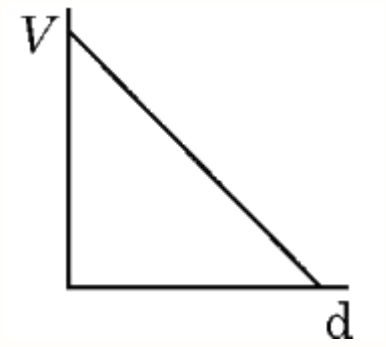
28. A constant electric field is to be maintained between the two plates of a capacitor whose separation d changes with time. Which of the graphs correctly

depict the potential difference (V) to be applied between the plates as a function of separation between the plates (d) to maintain the constant electric field?

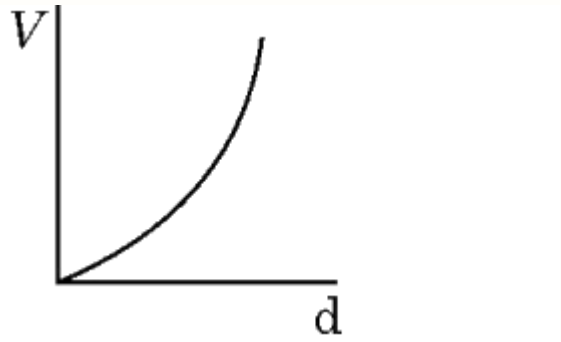
(A)



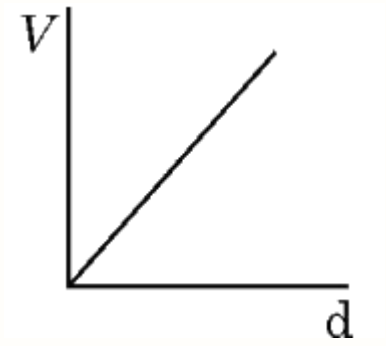
(B)



(C)



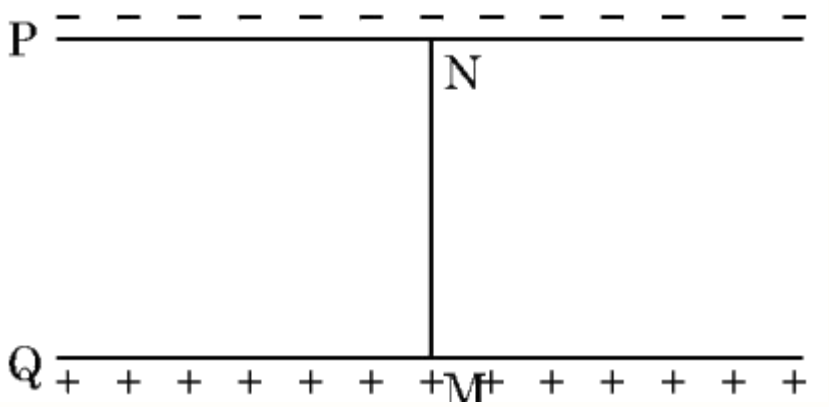
(D)



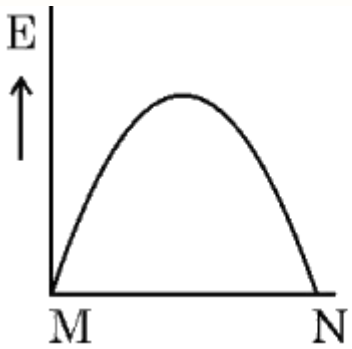
Check Solution →

[2025 • Set 55-2-1]

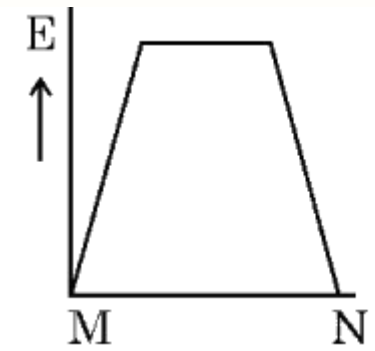
29. In the above figure P, Q are the two parallel plates of a capacitor. Plate Q is at positive potential with respect to plate P. MN is an imaginary line drawn perpendicular to the plates. Which of the graphs shows correctly the variations of the magnitude of electric field strength E along the line MN?



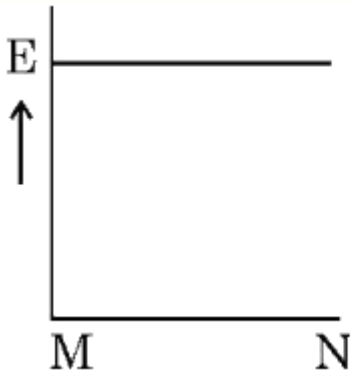
(A)



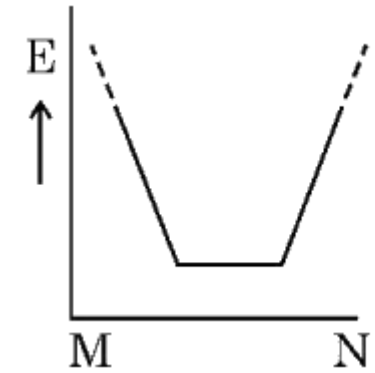
(B)



(C)



(D)



Check Solution →

[2025 • Set 55-2-1]

30. Three parallel plates are placed above each other with equal displacement d between neighbouring plates. The electric field between the first pair of the plates is $\vec{E}_1$ and the electric field between the second pair of the plates is $\vec{E}_2$. The potential difference between the third and the first plate is

(A) $(\vec{E}_1+\vec{E}_2)\cdot\vec{d}$

(B) $(\vec{E}_1-\vec{E}_2)\cdot\vec{d}$

(C) $(\vec{E}_2-\vec{E}_1)\cdot\vec{d}$

(D) $\frac{\vec{d}(\vec{E}_1+\vec{E}_2)}{2}$

Check Solution → [2025 • Set 55-2-1]

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31. A material of dielectric constant K is filled in a parallel plate capacitor of capacitance C. The new value of its capacitance becomes

(A) C

(B) $\frac{C}{K}$

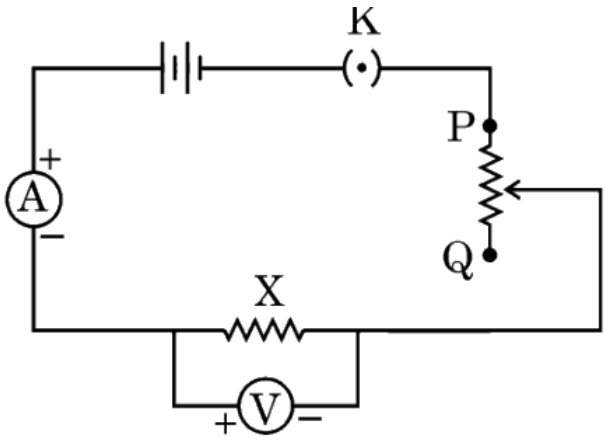
(C) CK

(D) $C(1+\frac{1}{K})$

Check Solution → [2025 • Set 55-2-1]

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32. **Case Study (Q29):** A parallel plate capacitor consists of two conducting plates kept generally parallel to each other at a distance. When the capacitor is charged, the charge resides on the inner surfaces of the plates and an electric field is set up between them. Thus, electrostatic energy is stored in the capacitor. The figure shows three large square metallic plates, each of side 'L' held parallel and equidistant from each other. The space between P_1 and P_2 and P_2 and P_3 is completely filled with mica sheets of dielectric constant 'K'. The plate P_2 is connected to point A and other plates P_1 and P_3 are connected to point B. Point A is maintained at a positive potential with respect to point B and the potential difference between A and B is V.



(i) The capacitance of the system between A and B will be :

(A) $\frac{\varepsilon_0 K L^2}{d}$

(B) $\frac{\varepsilon_0 K L^2}{2d}$

(C) $\frac{2\varepsilon_0 K L^2}{d}$

(D) $\frac{2\varepsilon_0 K d}{L^2}$

Check Solution → [2025 • Set 55-4-1]

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33. **(ii)** The charge on plate P_2 is :

(A) $\frac{\varepsilon_0 V K L^2}{2d}$

(B) $\frac{\varepsilon_0 V K L^2}{d}$

(C) $\frac{2\varepsilon_0 V K L^2}{d}$

(D) $\frac{\varepsilon_0 V K L^2}{4d}$

Check Solution → [2025 • Set 55-4-1]

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34. **(iii)** The electric field in the region between P_1 and P_2 is :

(A) $\frac{V}{d}$

(B) $\frac{2V}{d}$

(C) $\frac{V}{2d}$

(D) $\frac{d}{V}$

Check Solution → [2025 • Set 55-4-1]

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35. **(iv)** The separation between the plates of same area (L^2) of a parallel plate air capacitor having capacitance equal to that of this system, will be :

(A) $\frac{d}{K}$

(B) $\frac{2d}{K}$

(C) $\frac{d}{2K}$

(D) $\frac{d}{4K}$

Check Solution → [2025 • Set 55-4-1]

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36. **(iv) OR:** If the source of potential difference applied between A and B is removed, and then A and B are connected by a conducting wire, the net charge on the system will be :

(A) $\frac{\varepsilon_0 V K L^2}{4d}$

(B) $\frac{\varepsilon_0 V K L^2}{2d}$

(C) $\frac{\varepsilon_0 V K L^2}{d}$

(D) Zero

Check Solution → [2025 • Set 55-4-1]

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37. A metal sheet is inserted between the plates of a parallel plate capacitor of capacitance C. If the sheet partly occupies the space between the plates, the capacitance:

(A) remains C

(B) becomes greater than C

(C) becomes less than C

(D) becomes zero

Check Solution → [2025 • Set 55-5-1]

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38. The electric field at a point in a region is given by $\vec{E} = \frac{\alpha}{r^3}\vec{r}$, where α is a constant and r is the distance of the point from the origin. The magnitude of potential of the point is:

(A) $\frac{\alpha}{r}$

(B) $\frac{\alpha r^2}{2}$

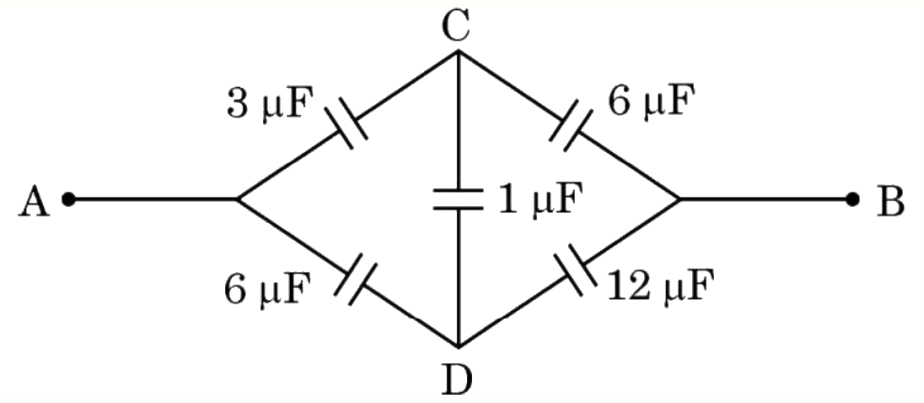
(C) $\frac{\alpha}{2r}$

(D) $-\frac{\alpha}{r}$

Check Solution → [2025 • Set 55-5-1]

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39. The effective capacitance of the network between points A and B shown in the figure is:



(A) $\frac{4}{3}$ μF

(B) 2 μF

(C) $\frac{3}{4}$ μF

(D) 6 μF

Check Solution → [2025 • Set 55-5-2]

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40. Four point charges Q each, are held at the four corners of a square of side l. The amount of work done in bringing a charge Q from infinity to the centre of the square will be:

(A) $\frac{Q^2}{\pi\varepsilon_0 l}$

(B) $\frac{\sqrt{2}\,Q^2}{\pi\varepsilon_0 l}$

(C) $\frac{Q^2}{2\pi\varepsilon_0 l}$

(D) Zero

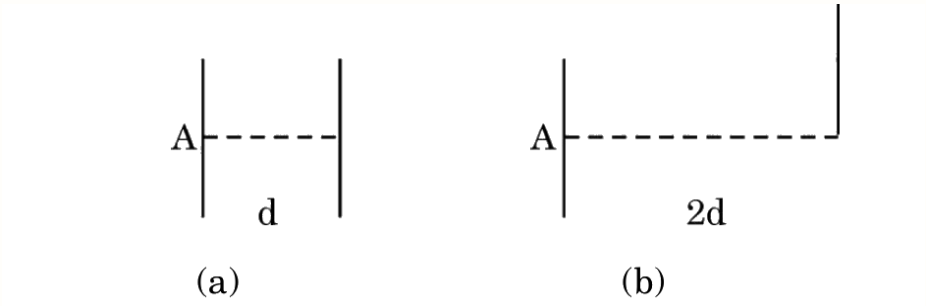
Check Solution → [2025 • Set 55-5-3]

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41. *Case Study (Capacitors):* A capacitor is a system of two conductors separated by an insulator. In practice, the two conductors have charges Q and $-Q$ with potential difference $V = V_1 - V_2$ between them. The ratio $\frac{Q}{V}$ is a constant, denoted by C and is called the capacitance of the capacitor. It is independent of Q or V. It depends only on the geometrical configuration (shape, size, separation) of the two conductors and the medium separating the conductors. When a parallel plate capacitor is charged, the electric field E_0 is localised between the plates and is uniform throughout. When a slab of a dielectric is inserted between the charged plates (charge density σ), the dielectric is polarised by the field. Consequently opposite charges appear on the faces of the slab, near the plates, with surface charge density of magnitude σ_p. For a linear dielectric σ_p is proportional to $\vec{E}_0$. Introduction of a dielectric changes the electric field, and hence, the capacitance of a capacitor, and hence, the energy stored in the capacitor. Like resistors, capacitors can also be arranged in series or in parallel

or in a combination of series and parallel.

(i) Consider a capacitor of capacitance C, with plate area A and plate separation d, filled with air [Fig. (a)]. The distance between the plates is increased to 2d and one of the plates is shifted as shown in Fig. (b). The capacitance of the new system now is:



(a) (b)

(A) $\frac{C}{4}$

(B) $\frac{C}{2}$

(C) 2C

(D) 4C

Check Solution → **[2025 • Set 55-6-1]**

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42. (ii) A slab (area A and thickness d_1) of a linear dielectric of dielectric constant K is inserted between charged plates (charge density σ) of a parallel plate capacitor [plate area A and plate separation d ($>d_1$)] and opposite charges with charge density of magnitude σ_p appear on the faces of the slab. The dielectric constant K is given by:

(A) $\frac{\sigma - \sigma_p}{\sigma}$

(B) $\frac{\sigma}{\sigma - \sigma_p}$

(C) $\frac{\sigma + \sigma_p}{\sigma_p}$

(D) $\frac{\sigma}{\sigma_p}$

Check Solution → **[2025 • Set 55-6-1]**

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43. (iii) An electric field E is established between the plates of an air filled parallel plate capacitor, with charges Q and $-Q$. V is the volume of the space enclosed between the plates. The energy stored in the capacitor is:

(A) $\frac{1}{2}\varepsilon_0 E^2$

(B) $\varepsilon_0 E Q V$

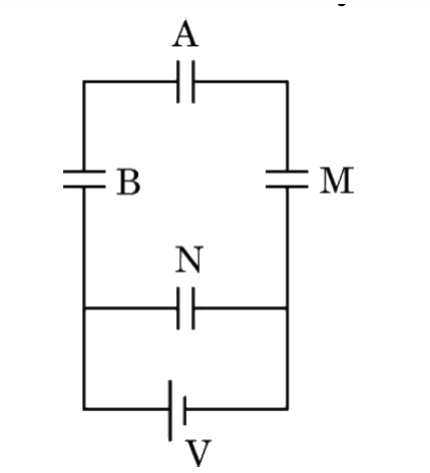
(C) $\frac{1}{2}\varepsilon_0 E^2 V$

(D) $\varepsilon_0 E Q V$

Check Solution → [2025 • Set 55-6-1]

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44. (iv) (a) Three capacitors A, B and M, each of capacitance C are connected to a capacitor N of capacitance 2C and a battery as shown in the figure. If the charges on A and N are Q and Q' respectively, then $\frac{Q'}{Q}$ is:



OR

(b) A slab (area A and thickness $\frac{d}{2}$) of dielectric constant K is inserted in a parallel plate capacitor of plate area A and plate separation d. If C and C_0 are the capacitances of the capacitors with and without the dielectric, then $\frac{C}{C_0}$ is:

(A) (a) $\frac{1}{6}$ | (b) $\frac{K+1}{2K}$

(B) (a) $\frac{1}{3}$ | (b) $\frac{2K}{K+1}$

(C) (a) 3 | (b) $\frac{K}{K-1}$

(D) (a) 6 | (b) $\frac{K-1}{K}$

Check Solution → [2025 • Set 55-6-1]

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45. Two horizontal plates, separated by 1 cm, are arranged one above the other. A particle of mass 5 mg and charge 2 nC is released in air between the plates. The potential difference that should be applied to the plates so that the particle remains suspended between them, is:

(A) 250 V

(B) 200 V

(C) 100 V

(D) 50 V

Check Solution → [2025 • Set 55-7-1]

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46. The electric field (E) and electric potential (V) at a point inside a charged hollow metallic sphere are respectively:

(A) $E = 0, V = 0$

(B) $E = 0, V = V_0$ (a constant)

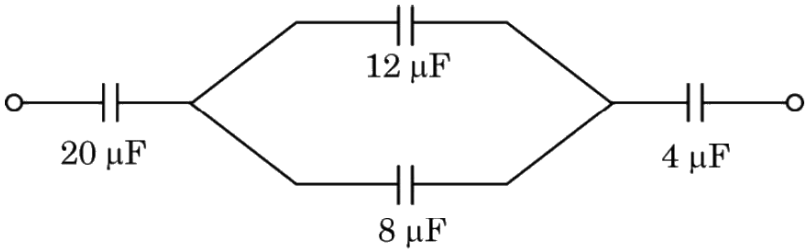
(C) $E \neq 0, V \neq 0$

(D) $E = E_0$ (a constant), $V = 0$

Check Solution → [2025 • Set 55-7-2]

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47. In the circuit shown, the charge on the left plate of 20 μF capacitor is −50 μC. The charge on the right plate of the 12 μF capacitor is:



(A) 20 μC (B) – 20 μC

(C) 30 μC (D) – 30 μC

(A) 20 μC

(B) −20 μC

(C) 30 μC

(D) −30 μC

Check Solution → [2025 • Set 55-7-3]

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48. Ten capacitors, each of capacitance 1 μF, are connected in parallel to a source of 100 *V*. The total energy stored in the system is equal to:

(A) $10^{-2}\,J$

(B) $10^{-5}\,J$

(C) $0.5\times 10^{-5}\,J$

(D) $5.0\times 10^{-2}\,J$

Check Solution → [2024 • Set 55-1-1]

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49. The capacitance of a parallel plate capacitor having a medium of dielectric constant K = 4 in between the plates is C. If this medium is removed, then the capacitance of the capacitor becomes:

(A) 4C

(B) *C*

(C) $\frac{C}{4}$

(D) 2C

Check Solution → [2024 • Set 55-2-2]

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50. Consider a group of charges q_1, q_2, q_3, ... such that $\Sigma q \neq 0$. Then equipotentials at a large distance, due to this group are approximately:

(A) Plane

(B) Spherical surface

(C) Paraboloidal surface

(D) Ellipsoidal surface

Check Solution → [2024 • Set 55-3-1]

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51. A proton is taken from point P_1 to point P_2, both located in an electric field. The potentials at points P_1 and P_2 are −5 V and +5 V respectively. Assuming that kinetic energies of the proton at points P_1 and P_2 are zero, the work done on the proton is:

(A) -1.6×10^{-18} J

(B) 1.6×10^{-18} J

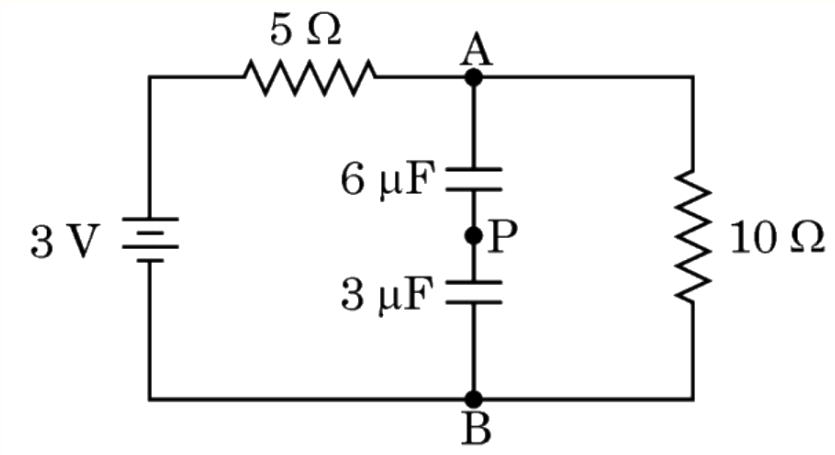
(C) Zero

(D) 0.8×10^{-18} J

Check Solution → [2024 • Set 55-3-1]

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52. The figure shows four pairs of parallel identical conducting plates, separated by the same distance 2.0 cm and arranged perpendicular to x-axis. The electric potential of each plate is mentioned. The electric field between a pair of plates is uniform and normal to the plates.



For which pair of the plates is the electric field $\vec{E}$ along $\hat{i}$?

(A) I

(B) II

(C) III

(D) IV

Check Solution → [2024 • Set 55-3-1]

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53. An electron is released midway between the plates of pair IV. It will:

(A) move along $\hat{i}$ at constant speed

(B) move along $-\hat{i}$ at constant speed

(C) accelerate along $\hat{i}$

(D) accelerate along $-\hat{i}$

Check Solution → [2024 • Set 55-3-1]

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54. Let V_0 be the potential at the left plate of any set, taken to be at $x = 0$ m. Then potential V at any point ($0 < x < 2$ cm) between the plates of that set can be expressed as: (α is a constant, positive or negative.)

(A) $V = V_0 + \alpha x$

(B) $V = V_0 + \alpha x^2$

(C) $V = V_0 + \alpha x^{1/2}$

(D) $V = V_0 + \alpha x^{3/2}$

Check Solution → [2024 • Set 55-3-1]

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55. Let E_1, E_2, E_3 and E_4 be the magnitudes of the electric field between the pairs of plates, I, II, III and IV respectively. Then:

(A) $E_1 > E_2 > E_3 > E_4$

(B) $E_2 > E_4 > E_1 > E_3$

(C) $E_4 > E_2 > E_3 > E_1$

(D) $E_4 > E_3 > E_2 > E_1$

Check Solution → [2024 • Set 55-3-1]

56. An electron is projected from the right plate of set I directly towards its left plate. It just comes to rest at the plate. The speed with which it was projected is about: (Take $e/m = 1.76 \times 10^{11}$ C/kg)

(A) 1.3×10^6 m/s

(B) 2.6×10^6 m/s

(C) 6.5×10^6 m/s

(D) 5.2×10^7 m/s

Check Solution → [2024 • Set 55-3-1]

57. The plates P_1 and P_2 of a 2 μF capacitor are at potentials 25 V and −25 V respectively. The charge on plate P_1 will be:

(A) 0.02 mC

(B) 0.1 mC

(C) 0.1 μC

(D) 1 μC

Check Solution → [2024 • Set 55-3-3]

58. Two charges $+q$ each are kept 2a distance apart. A third charge −2q is placed midway between them. The potential energy of the system is

(A) $\frac{q^2}{8\pi\varepsilon_0 a}$

(B) $-\frac{6q^2}{8\pi\varepsilon_0 a}$

(C) $-\frac{7q^2}{8\pi\varepsilon_0 a}$

(D) $\frac{9q^2}{8\pi\varepsilon_0 a}$

Check Solution → [2024 • Set 55-4-1]

59. Three point charges, each of charge q are placed on vertices of a triangle ABC, with AB=AC=5L, BC=6L. The electrostatic potential at midpoint of side BC will be

(A) $\frac{11q}{48\pi\varepsilon_0 L}$

(B) $\frac{8q}{36\pi\varepsilon_0 L}$

(C) $\frac{5q}{24\pi\varepsilon_0 L}$

(D) $\frac{q}{16\pi\varepsilon_0 L}$

Check Solution → [2024 • Set 55-4-2]

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60. Dielectrics play an important role in design of capacitors. The molecules of a dielectric may be polar or non-polar. When a dielectric slab is placed in an external electric field, opposite charges appear on the two surfaces of the slab perpendicular to electric field. Due to this an electric field is established inside the dielectric. The capacitance of a capacitor is determined by the dielectric constant of the material that fills the space between the plates. Consequently, the energy storage capacity of a capacitor is also affected. Like resistors, capacitors can also be arranged in series and/or parallel.
Which of the following is a polar molecule?

(A) O_2

(B) H_2

(C) N_2

(D) HCl

Check Solution → [2024 • Set 55-5-1]

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61. Which of the following statements about dielectrics is correct?

(A) A polar dielectric has a net dipole moment in absence of an external electric field which gets modified due to the induced dipoles.

(B) The net dipole moments of induced dipoles is along the direction of the applied electric field.

(C) Dielectrics contain free charges.

(D) The electric field produced due to induced surface charges inside a dielectric is along the external electric field.

Check Solution → [2024 • Set 55-5-1]

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62. When a dielectric slab is inserted between the plates of an isolated charged capacitor, the energy stored in it:

(A) increases and the electric field inside it also increases.

(B) decreases and the electric field also decreases.

(C) decreases and the electric field increases.

(D) increases and the electric field decreases.

Check Solution → [2024 • Set 55-5-1]

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63. An air-filled capacitor with plate area A and plate separation d has capacitance C_0. A slab of dielectric constant K, area A and thickness $(\frac{d}{2})$ is inserted between the plates. The capacitance of the capacitor will become:

(A) $[\frac{4K}{K+5}]C_0$

(B) $[\frac{K+5}{4K}]C_0$

(C) $[\frac{5K}{K+4}]C_0$

(D) $[\frac{K+4}{5K}]C_0$

Check Solution → [2024 • Set 55-5-1]

64.

Two capacitors of capacitances 2 C_0 and 6 C_0 are first connected in series and

then in parallel across the same battery. The ratio of energies stored in series combination to that in parallel is:

(A) $\frac{1}{4}$

(B) $\frac{1}{8}$

(C) $\frac{2}{15}$

(D) $\frac{3}{16}$

Check Solution → [2024 • Set 55-5-1]

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65. **Assertion (A):** Work done in moving a charge around a closed path, in an electric field is always zero.

Reason (R): Electrostatic force is a conservative force.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

(C) Assertion (A) is true and Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2023 • Set 55-2-1]

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66. A point P lies at a distance x from the mid point of an electric dipole on its axis. The electric potential at point P is proportional to

(A) $\frac{1}{x^2}$

(B) $\frac{1}{x^3}$

(C) $\frac{1}{x^{3/2}}$

(D) $\frac{1}{x^{5/2}}$

Check Solution → [2023 • Set 55-2-1]

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67. A point charge q is moving along a circular path of radius a, with a point charge $-Q$ at the centre of the circle. The kinetic energy of q is

(A) $\frac{qQ}{4\pi\varepsilon_0 a}$

(B) $\frac{qQ}{8\pi\varepsilon_0 a}$

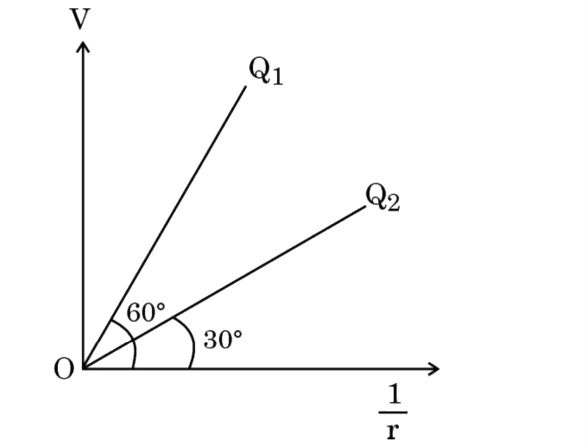
(C) $\frac{qQ}{4\pi\varepsilon_0 a^2}$

(D) $\frac{qQ}{8\pi\varepsilon_0 a^2}$

Check Solution → [2023 • Set 55-2-3]

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68. (b) The figure shows the variation of potential V with $\frac{1}{r}$ for two point charges Q_1 and Q_2, where V is the potential at a distance r due to a point charge. Find $\frac{Q_2}{Q_1}$.



Check Solution → [2023 • Set 55-4-1]

69. The capacitors, each of 4 μ F are to be connected in such a way that the effective capacitance of the combination is 6 μ F. This can be achieved by connecting

(A) All three in parallel

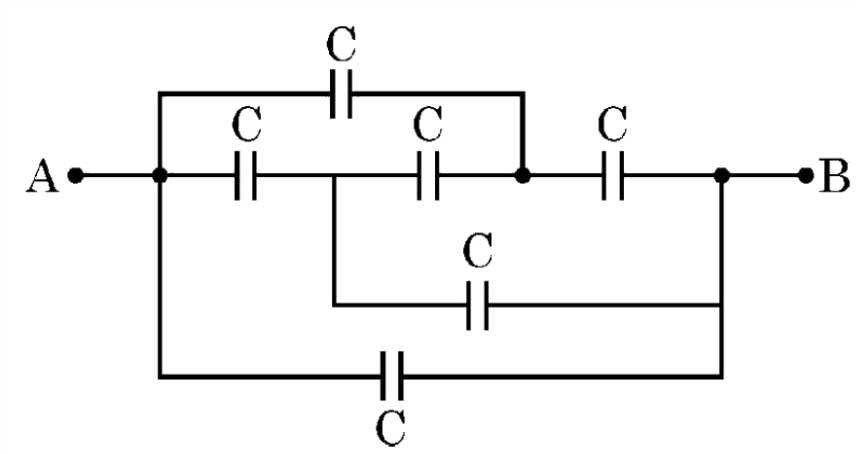
(B) All three in series

(C) Two of them connected in series and the combination in parallel to the third.

(D) Two of them connected in parallel and the combination in series to the third.

Check Solution → [2023 • Set 55-5-1]

70. Find the equivalent capacitance between points A and B in the given diagram.



Check Solution → [2023 • Set 55-5-1]

71. A dielectric slab is inserted between the plates of a parallel plate capacitor. The electric field between the plates decreases. Explain.

Check Solution → [2023 • Set 55-5-1]

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72. An electric dipole consisting of charges $+q$ and $-q$ separated by a distance L is in stable equilibrium in a uniform electric field $\vec{E}$. The electrostatic potential energy of the dipole is

(A) qLE

(B) zero

(C) −qLE

(D) −2qLE

Check Solution → [2020 • Set 55-1-1]

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73. The potential difference between two points in vacuum is V_0. If vacuum is replaced by a medium of dielectric constant K, the new value of potential difference will be _____.

Check Solution → [2020 • Set 55-2-3]

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74. If a positive charge is displaced against the electric field in which it was situated, then

(A) work will be done by the electric field on the charge.

(B) the intensity of the electric field decreases.

(C) energy of the system will decrease.

(D) energy will be provided by external source displacing the charge.

Check Solution → [2020 • Set 55-3-1]

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75. Two capacitors of capacitances C_1 and C_2 are connected in parallel. If a charge Q is given to the combination, the ratio of the charge on the capacitor C_1 to the charge on C_2 will be

(A) $\frac{C_1}{C_2}$

(B) $\frac{C_2}{C_1}$

(C) $\sqrt{\frac{C_1}{C_2}}$

(D) $\sqrt{\frac{C_2}{C_1}}$

Check Solution → [2020 • Set 55-3-1]

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76. A charge Q is kept at the centre of a circle of radius r. A test charge q_0 is carried from a point X to the point Y on this circle such that arc XY subtends an angle of 60° at the centre of the circle. The amount of work done in this process will be

(A) $\frac{1}{4\pi\varepsilon_0}\frac{Qq_0}{2r}$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{\sqrt{3}\,Qq_0}{2r}$

(C) Zero

(D) $\frac{\sqrt{3}\,Qq_0}{4\pi\varepsilon_0 r}$

Check Solution → [2020 • Set 55-3-2]

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77. A parallel plate capacitor is charged to V volt by a battery. The battery is disconnected and the separation between the plates is halved. The new potential difference across the capacitor will be

(A) $\frac{V}{2}$

(B) V

(C) 2V

(D) $\frac{V}{4}$

Check Solution → [2020 • Set 55-3-2]

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78. A charge Q is uniformly distributed over the surface of a spherical shell of radius R. The work done in bringing a test charge Q_0 from its centre to its surface is

(A) $\frac{1}{4\pi\varepsilon_0}\frac{QQ_0}{R}$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{QQ_0}{2R}$

(C) $\frac{QQ_0}{\varepsilon_0 R}$

(D) Zero

Check Solution → [2020 • Set 55-3-3]

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79. A charged particle is placed between the two plates of a charged parallel plate capacitor. It experiences a force F. If one plate is removed, then the force on the particle will be

(A) 2F

(B) F

(C) $\frac{F}{2}$

(D) Zero

Check Solution → [2020 • Set 55-3-3]

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80. Fill in the blank with appropriate answer.
A proton released from rest in an electric field, will start moving towards a region of _____ potential in the field.

Check Solution → [2020 • Set 55-4-1]

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81. The work done in moving a charged particle between two points in a uniform electric field does not depend on the path followed by the particle. Why?

Check Solution → [2020 • Set 55-4-1]

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82. A charged particle ($+q$) moves in a uniform electric field ($\vec{E}$) in the direction opposite to $\vec{E}$. What will be the effect on its electrostatic potential energy during its motion?

Check Solution → [2020 • Set 55-4-3]

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83. A point charge is placed at the centre of a hollow conducting sphere of internal radius r and outer radius 2r. The ratio of the surface charge density of the inner surface to that of the outer surface will be _____ .

Check Solution → [2020 • Set 55-5-1]

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84. Draw the equipotential surfaces due to an isolated point charge.

Check Solution → [2019 • Set 55-1-2]

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85. Why is the electrostatic potential inside a charged conducting shell constant throughout the volume of the conductor?

Check Solution → [2019 • Set 55-5-1]

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86. Does the charge given to a metallic sphere depend on whether it is hollow or solid? Give reason for your answer.

Check Solution → [2017]

---

87. A point charge Q is placed at point 'O' as shown in the figure. Is the potential at point A, i.e. V_A, greater, smaller or equal to potential, V_B, at point B, when Q is (i) positive, and (ii) negative charge?



Check Solution → [2017]

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88. What is the amount of work done in moving a point charge Q around a circular arc of radius r at the centre of which another point charge q is located?

Check Solution → [2016]

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89. "For any charge configuration, equipotential surface through a point is normal to the electric field." Justify.

Check Solution → [2014]

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90. What is the geometrical shape of equipotential surfaces due to a single isolated charge?

Check Solution → [2013]

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91. Define the term 'dielectric constant' of a medium.

Check Solution → [2013]

---

92. Name the physical quantity whose S.I. unit is JC^{-1}. Is it a scalar or a vector quantity?

Check Solution → [2012]

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93. Why is the potential inside a hollow spherical charged conductor constant and has the same value as on its surface?

Check Solution → [2012]

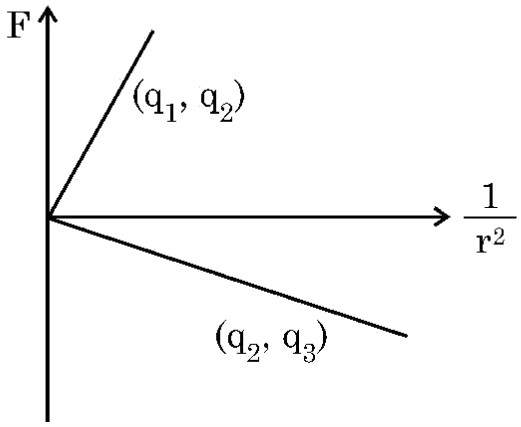
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94. A hollow metal sphere of radius 5 cm is charged such that the potential on its surface is 10 V. What is the potential at the centre of the sphere?

Check Solution → [2011]

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95. A point charge Q is placed at point O as shown in the figure. Is the potential difference $V_A - V_B$ positive, negative or zero, if Q is (i) positive (ii) negative?



Check Solution → [2011 • Set 55-1-1]

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96. What is the electrostatic potential due to an electric dipole at an equatorial point?

Check Solution → [2009]

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97. A 500 μ C charge is at the centre of a square of side 10 cm. Find the work done in moving a charge of 10 μ C between two diagonally opposite points on the square.

Check Solution → [2008]

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98. Derive the expression for the electric potential at any point along the axial line of an electric dipole.

Check Solution → [2008]

---

99. Draw an equipotential surface for a system consisting of two charges Q, $-Q$ separated by a distance 'r' in air.

Check Solution → [2008]

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100. Define the term 'dielectric constant' of a medium in terms of capacitance of a capacitor.

Check Solution → [2006]

SECTION B

2-Mark Questions

VSA · 27 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

101. Two isolated metallic spheres S_1 and S_2 of radii 1 cm and 3 cm respectively are charged such that both have the same charge density $(\frac{2}{\pi} \times 10^{-9})$ C/m^2. They are placed far away from each other and connected by a thin wire. Calculate the new charge on sphere S_2.

Check Solution → **[2024 • Set 55-1-1]**

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102. (a) Obtain an expression for electrostatic potential energy of a system of three charges q, 2q and −3q placed at the vertices of an equilateral triangle of side a.

OR

(b) Two small conducting balls A and B of radius r_1 and r_2 have charges q_1 and q_2 respectively. They are connected by a wire. Obtain the expression for charges on A and B, in equilibrium.

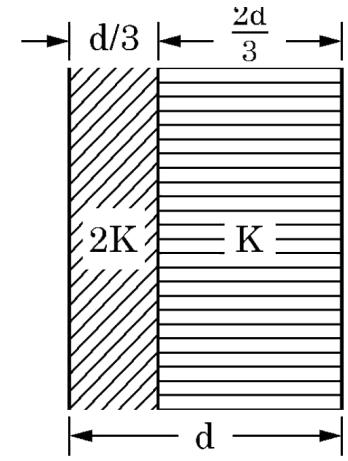
Check Solution → **[2023 • Set 55-4-1]**

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103. (iii) A capacitor A of capacitance C, having charge Q is connected across another uncharged capacitor B of capacitance 2C. Find an expression for (a) the potential difference across the combination and (b) the charge lost by capacitor A.

OR

(iii) Two slabs of dielectric constants 2K and K fill the space between the plates of a parallel plate capacitor of plate area A and plate separation d as shown in figure. Find an expression for capacitance of the system.

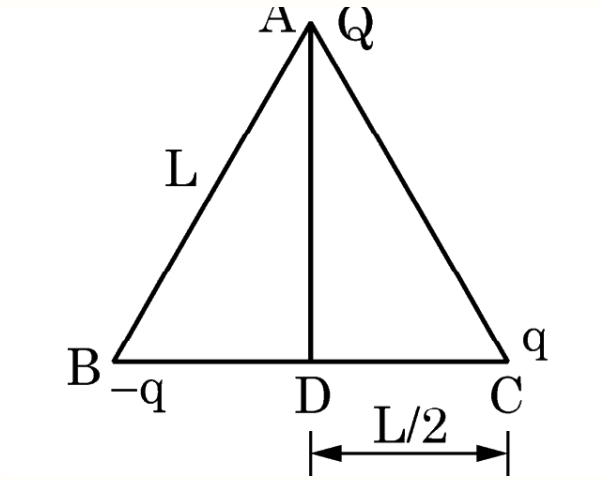


Check Solution → [2023 • Set 55-5-1]

104. Three point charges Q, q and $-q$ are kept at the vertices of an equilateral triangle of side L as shown in figure. What is

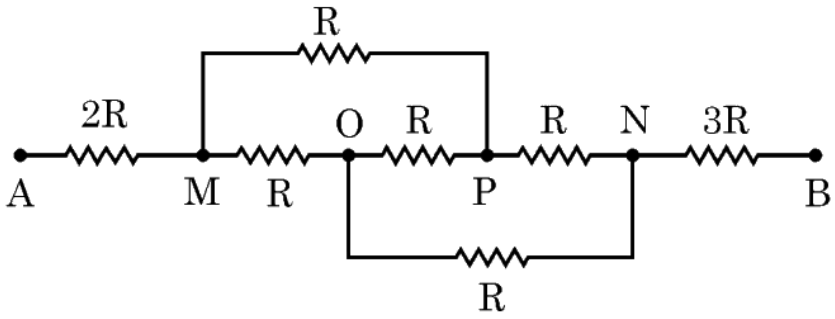
(i) the electrostatic potential energy of the arrangement? and

(ii) the potential at point D?



Check Solution → [2023 • Set 55-5-3]

105. The space between the plates of a parallel plate capacitor is completely filled in two ways. In the first case, it is filled with a slab of dielectric constant K. In the second case, it is filled with two slabs of equal thickness and dielectric constants K_1 and K_2 respectively as shown in the figure. The capacitance of the capacitor is same in the two cases. Obtain the relationship between K, K_1 and K_2.

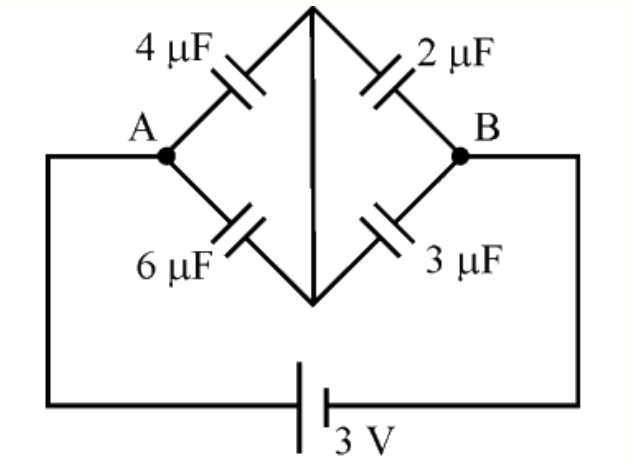


Check Solution → [2020 • Set 55-1-1]

106. N small conducting liquid droplets, each of radius r, are charged to a potential V each. These droplets coalesce to form a single large drop without any charge leakage. Find the potential of the large drop.

Check Solution → [2020 • Set 55-1-2]

107. Find the total charge stored in the network of capacitors connected between A and B as shown in figure:



Check Solution → [2020 • Set 55-4-1]

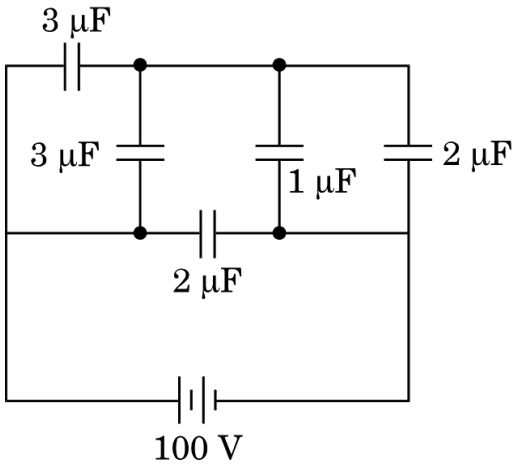
108. You are given three capacitors of 2 μF, 3 μF and 4 μF, respectively.

(a) Form a combination of all these capacitors of equivalent capacitance $\frac{13}{3}$ μF.

(b) What is the maximum and minimum value of the equivalent capacitance that can be obtained by connecting these capacitors?

Check Solution → [2020 • Set 55-4-3]

109. The figure shows a network of five capacitors connected to a 100 *V* supply. Calculate the total energy stored in the network.



Check Solution → [2019 • Set 55-3-1]

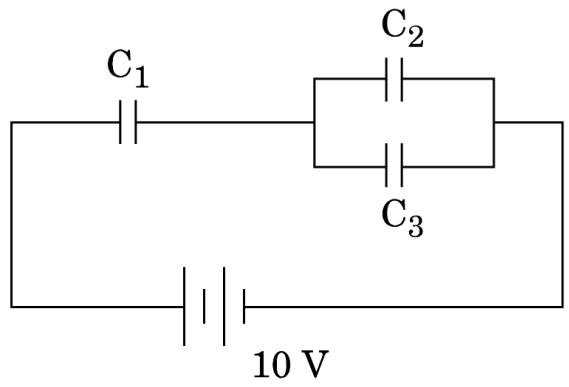
110. Two identical capacitors of 12 pF each are connected in series across a 50 V battery. Calculate the electrostatic energy stored in the combination. If these were connected in parallel across the same battery, find out the value of the energy stored in this combination.

Check Solution → [2019 • Set 55-5-1]

111. Two identical capacitors of 10 pF each are connected in turn (i) in series, and (ii) in parallel across a 20 V battery. Calculate the potential difference across each capacitor in the first case and charge acquired by each capacitor in the second case.

Check Solution → [2019 • Set 55-5-2]

112. The figure shows a network of three capacitors $C_1 = 2\ \mu F$; $C_2 = 6\ \mu F$ and $C_3 = 3\ \mu F$ connected across a battery of 10 V. If a charge of 6 μ *C* is acquired by the capacitor C_3, calculate the charge acquired by C_1.



Check Solution → [2019 • Set 55-5-3]

113. Why does current in a steady state not flow in a capacitor connected across a battery? However momentary current does flow during charging or discharging of the capacitor. Explain.

Check Solution → [2017]

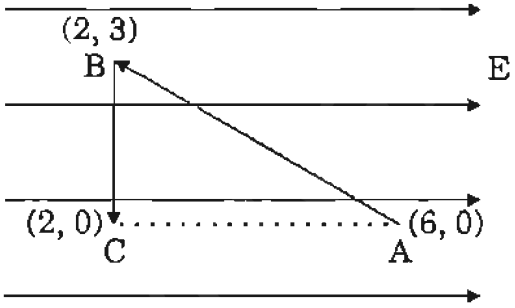
114. What is electrostatic shielding? How is this property used in actual practice? Is the potential in the cavity of a charged conductor zero?

Check Solution → [2016]

115. A parallel plate capacitor of capacitance C is charged to a potential V. It is then connected to another uncharged capacitor having the same capacitance. Find out the ratio of the energy stored in the combined system to that stored initially in the single capacitor.

Check Solution → [2014]

116. A test charge 'q' is moved without acceleration from A to C along the path from A to B and then from B to C in electric field $\vec{E}$ as shown in the figure. (i) Calculate the potential difference between A and C. (ii) At which point (of the two) is the electric potential more and why?



Check Solution → [2012]

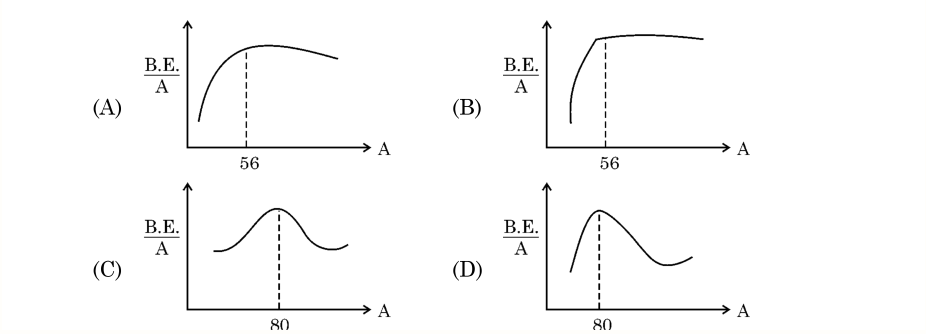
117. Draw a plot showing the variation of (i) electric field (E) and (ii) electric potential (V) with distance r due to a point charge Q.

Check Solution → [2012]

118. Net capacitance of three identical capacitors in series is 1 μF. What will be their net capacitance if connected in parallel?
Find the ratio of energy stored in the two configurations if they are both connected to the same source.

Check Solution → [2011]

119. Figure shows two identical capacitors, C_1 and C_2, each of 1 μF capacitance connected to a battery of 6 V. Initially switch *S* is closed. After sometime *S* is left open and dielectric slabs of dielectric constant $K = 3$ are inserted to fill completely the space between the plates of the two capacitors. How will the (i) charge and (ii) potential difference between the plates of the capacitors be affected after the slabs are inserted?



Check Solution →

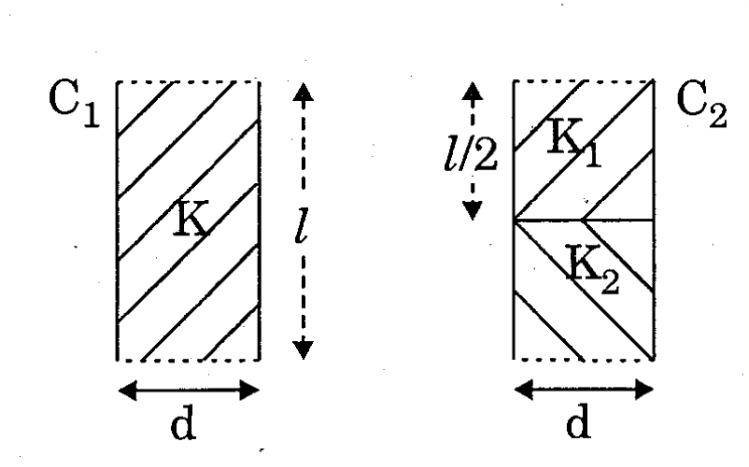
[2011 • Set 55-1-1]

120. Two uniformly large parallel thin plates having charge densities +σ and −σ are kept in the *X*-*Z* plane at a distance *d* apart. Sketch an equipotential surface due to electric field between the plates. If a particle of mass *m* and charge −*q* remains stationary between the plates, what is the magnitude and direction of this field?

Check Solution →

[2011 • Set 55-1-1]

121. Two identical parallel plate (air) capacitors C_1 and C_2 have capacitances C each. The space between their plates is now filled with dielectrics as shown. If the two capacitors still have equal capacitance, obtain the relation between dielectric constants *K*, K_1 and K_2.



Check Solution → [2011 • Set 55-2-1]

122. (i) Can two equi-potential surfaces intersect each other? Give reasons.
(ii) Two charges $-q$ and $+q$ are located at points $A(0,0,-a)$ and $B(0,0,+a)$ respectively. How much work is done in moving a test charge from point $P(7,0,0)$ to $Q(-3,0,0)$?

Check Solution → [2009]

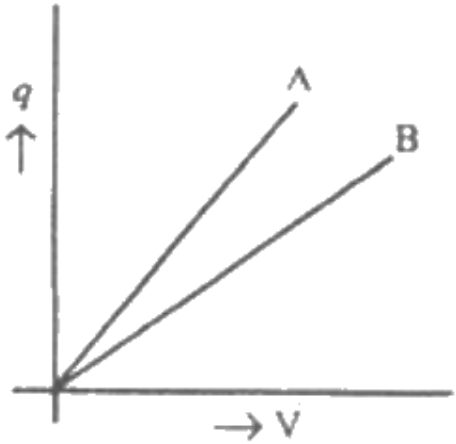
123. Draw 3 equipotential surfaces corresponding to a field that uniformly increases in magnitude but remains constant along Z-direction. How are these surfaces different from that of a constant electric field along Z-direction?

Check Solution → [2009]

124. Two point charges 4 μC and −2 μC are separated by a distance of 1 m in air. Calculate at what point on the line joining the two charges is the electric potential zero.

Check Solution → [2007]

125. The given graph shows the variation of charge q versus potential difference V for two capacitors C_1 and C_2. The two capacitors C_1 and C_2 have same plate separation but the plate area of C_2 is double than that of C_1. Which of the lines in the graph correspond to C_1 and C_2 and why?



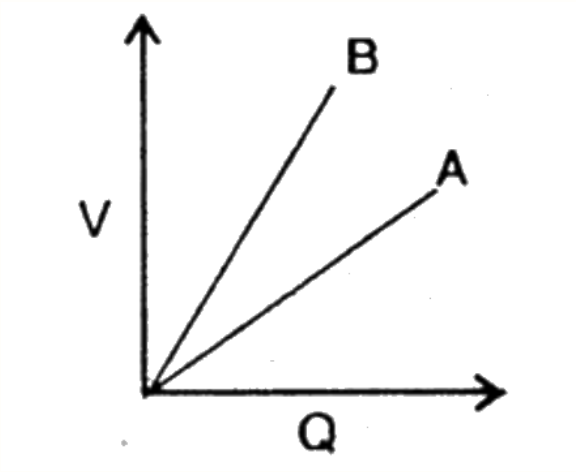
Check Solution → [2006]

126. **(a)** Why does the electric field inside a dielectric decrease when it is placed in an external electric field?

(b) A parallel plate capacitor with air between the plates has a capacitance of 8 pF. What will be the capacitance if the distance between the plates be reduced by half and the space between them is filled with a substance of dielectric constant $K = 6$?

Check Solution → [2005]

127. The graph shows the variation of voltage, 'V' across the plates of two capacitors A and B versus increase of charge, 'Q' stored on them. Which of the two capacitors has higher capacitance? Give reason for your answer.



Check Solution → [2004]

SECTION C

3-Mark Questions

SA · 62 questions

> *Section C of the paper, 3 marks each. Sorted newest year first.*

128. A parallel plate capacitor of capacitance C has a dielectric slab between its plates. It is charged to a potential difference V by connecting it across a battery. The battery is then disconnected. If the dielectric slab is now withdrawn from the capacitor, how will the following be affected?
(a) Capacitance of the capacitor,
(b) Energy stored in the capacitor, and
(c) The potential difference between the plates of the capacitor.
Justify your answer in each case.

Check Solution → **[2026 • Set 55-1-1]**

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129. A 12.0 μF capacitor is charged to a potential difference of 150 V. The terminals of the charged capacitor are then connected to those of an uncharged 6.0 μF capacitor. Calculate final potential difference across and charge on, each capacitor.

Check Solution → **[2026 • Set 55-1-2]**

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130. A parallel plate capacitor of capacitance C is charged to V volt by a battery. After sometime the battery is disconnected and the distance between the plates is doubled. A slab of dielectric constant $k = 1.8$ is then introduced to completely fill the space between the plates. How will the following be affected?
(a) The capacitance of the capacitor.
(b) The electric field between the plates of the capacitor.
(c) The energy stored in the capacitor.
Justify your answer in each case.

Check Solution → **[2026 • Set 55-1-3]**

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131. (a) An electric field $\vec{E} = E_0\hat{i}$ exists in a region of space. Draw three equipotential surfaces in the region.
(b) Two point charges $-q$ and $+q$ are located at points $(-a, 0, 0)$ and $(a, 0, 0)$ respectively. Find the electrostatic potential at point $(x, 0, 0)$ where $x \gg a$.

Check Solution → [2026 • Set 55-2-1]

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132. (i) Draw electric field lines and equipotential surfaces for a system of two equal and opposite point charges separated by some distance.
(ii) Why electric field $\vec{E}$ at a point on an equipotential surface must be perpendicular to the surface at that point?

Check Solution → [2026 • Set 55-2-2]

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133. (a) Draw electric field lines and equipotential surfaces for a system of two equal positive charges separated by some distance.
(b) Using the drawing in (a) above, explain that electric field need not be a constant over each point of an equipotential surface.

Check Solution → [2026 • Set 55-2-3]

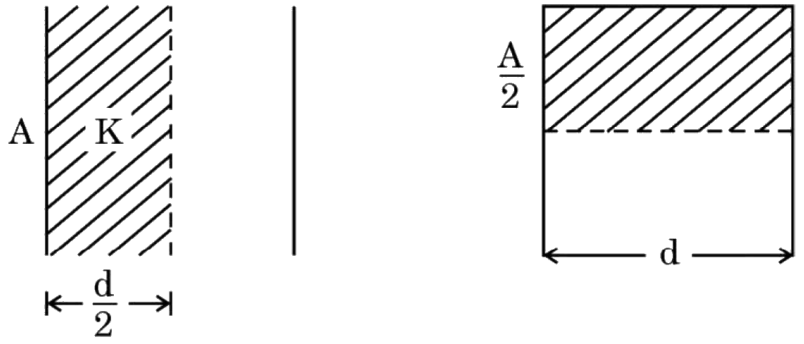
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134. Two parallel plate capacitors X and Y are connected in series to a 6 V battery. They have the same plate area and same plate separation but capacitor X has air between its plates, whereas capacitor Y contains a material of dielectric constant 4.
(a) Calculate the capacitances of X and Y, if the equivalent capacitance of the combination of X and Y is 4 μF.
(b) Calculate the potential difference across the plates of X and Y.

Check Solution → [2026 • Set 55-5-1]

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135. A parallel plate capacitor has plate area A and plate separation d. Half of the space between the plates is filled with a material of dielectric constant K in two ways as shown in the figure.



Find the values of the capacitance of the capacitors in the two cases.

Check Solution → [2025 • Set 55-5-1]

136. A capacitor of plate area A and plate separation d is charged by a battery to voltage V. The battery is disconnected and plates are slowly pulled apart till the separation becomes 2d. Find the value of:

(a) potential difference between the plates,

(b) electric field between the plates,

(c) work done in pulling the plates apart.

Check Solution → [2025 • Set 55-5-2]

137. **(a)** "There is a limit to the amount of charge that can be stored on a given capacitor." Explain.

(b) A capacitor is charged by a battery to a potential difference V. It is disconnected from the battery and connected across another identical uncharged capacitor. Calculate the ratio of total energy stored in the combination to the initial energy stored in the capacitor.

Check Solution → [2025 • Set 55-5-3]

138. Two point charges of −5 μC and 2 μC are located in free space at (−4 cm, 0) and (6 cm, 0) respectively.

(a) Calculate the amount of work done to separate the two charges at infinite distance.

(b) If this system of charges was initially kept in an electric field $\vec{E} = \frac{A}{r^2}\hat{r}$, where $A = 8 \times 10^4\ N\,C^{-1}\,m^2$, calculate the electrostatic potential energy of the system.

Check Solution → [2025 • Set 55-7-2]

139.

An electric dipole (dipole moment $\vec{p} = p\hat{i}$), consisting of charges $-q$ and q, separated by distance 2a, is placed along the x-axis, with its centre at the origin. Show that the potential V, due to this dipole, at a point x, ($x \gg a$) is equal to $\frac{1}{4\pi\varepsilon_0} \cdot \frac{\vec{p}\cdot\hat{i}}{x^2}$.

Check Solution → [2024 • Set 55-1-1]

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140. Three point charges Q_1, Q_2 and Q_3 are located in x–y plane at points $(-d, 0)$, $(0, 0)$ and $(d, 0)$ respectively. Q_1 and Q_3 are identical and Q_3 is positive. What will be the nature and value of Q_2 so that the potential energy of the system is zero?

Check Solution → [2024 • Set 55-1-2]

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141. An air-filled parallel plate capacitor with plate separation 1 mm has a capacitance of 20 pF. It is charged to 4.0 μC. Calculate the amount of work done to pull its plates to a separation of 5 mm. Assume the charge on the plates remains the same.

Check Solution → [2024 • Set 55-1-3]

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142. The electric field in a region is given by $\vec{E}=(10x+4)\hat{i}$, where x is in m and E is in N/C. Calculate the amount of work done in taking a unit charge from
(i) (5 m,0) to (10 m,0)
(ii) (5 m,0) to (5 m,10 m)

Check Solution → [2024 • Set 55-4-1]

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143. A thin spherical conducting shell of radius R has a charge q. A point charge Q is placed at the centre of the shell. Find
(i) The charge density on the outer surface of the shell, and
(ii) the potential at a distance of $(R/2)$ from the centre of the shell.

Check Solution → [2024 • Set 55-4-2]

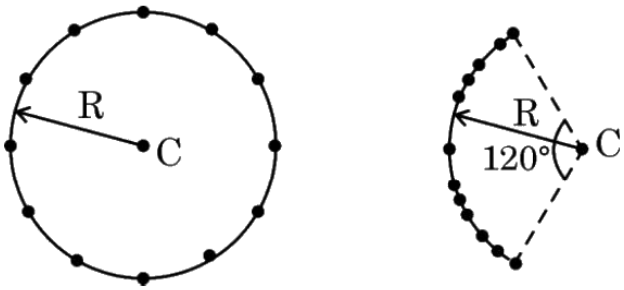
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144. Two conducting spherical shells A and B of radii R and 2R are kept far apart and charged to the same charge density σ. They are connected by a wire. Obtain an expression for final potential of shell A.

Check Solution → [2024 • Set 55-4-3]

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145. (a) Twelve negative charges of same magnitude are equally spaced and fixed on the circumference of a circle of radius R as shown in Fig. (i). Relative to potential being zero at infinity, find the electric potential and electric field at the centre C of the circle.
(b) If the charges are unequally spaced and fixed on an arc of 120° of radius R as shown in Fig. (ii), find electric potential at the centre C.



(i) (ii)

Check Solution → [2023 • Set 55-1-1]

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146. A 100 μF capacitor is charged by a 12 V battery.
(a) How much electrostatic energy is stored by the capacitor?
(b) The capacitor is disconnected from the battery and connected in parallel to another uncharged 100 μF capacitor. What is the electrostatic energy stored by the system?

Check Solution → [2023 • Set 55-1-2]

147. Three point charges Q_1 (−15 μC), Q_2 (10 μC) and Q_3 (16 μC) are located at (0 cm, 0 cm), (0 cm, 3 cm) and (4 cm, 3 cm) respectively. Calculate the electrostatic potential energy of this system of charges.

Check Solution → [2023 • Set 55-1-3]

148. Two charged conducting spheres of radii a and b are connected to each other by a wire. Find the ratio of the electric fields at their surfaces.

Check Solution → [2023 • Set 55-2-1]

149. A parallel plate capacitor (A) of capacitance C is charged by a battery to voltage V. The battery is disconnected and an uncharged capacitor (B) of capacitance 2C is connected across A. Find the ratio of

(i) final charges on A and B.

(ii) total electrostatic energy stored in A and B finally and that stored in A initially.

Check Solution → [2023 • Set 55-2-1]

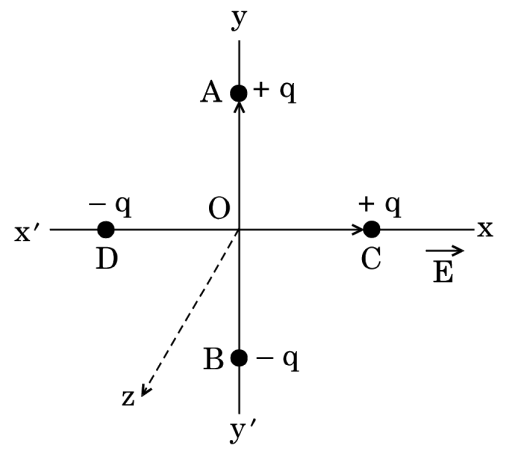
150. A capacitor of 4 μF is charged by a battery of 12 V. The battery is disconnected and a dielectric slab of dielectric constant 8 is inserted in between the plates of the capacitor to fill the space completely. Find the change in the (a) charge stored in the capacitor, (b) potential difference between the plates of the capacitor, and (c) energy stored in the capacitor.

OR

Two small identical electric dipoles AB and CD, each of dipole moment $\vec{p}$ are kept at an angle of 90° to each other in an external electric field $\vec{E}$ pointing along the x-axis as shown in the figure. Find the

(a) dipole moment of the arrangement, and

(b) magnitude and direction of the net torque acting on it.



Check Solution → [2020 • Set 55-2-3]

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151. (a) Two point charges q_1 and q_2 are kept at a distance of r_{12} in air. Deduce the expression for the electrostatic potential energy of this system.
(b) If an external electric field ($\vec{E}$) is applied on the system, write the expression for the total energy of this system.

Check Solution → [2020 • Set 55-5-1]

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152. (a) Two point charges $+Q_1$ and $-Q_2$ are placed r distance apart. Obtain the expression for the amount of work done to place a third charge Q_3 at the midpoint of the line joining the two charges.
(b) At what distance from charge $+Q_1$ on the line joining the two charges (in terms of Q_1, Q_2 and r) will this work done be zero.

Check Solution → [2020 • Set 55-5-3]

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153. Draw the equipotential surfaces corresponding to a uniform electric field in the z-direction.

Check Solution → [2019 • Set 55-1-1]

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154. Draw equipotential surfaces corresponding to the electric field that uniformly increases in magnitude along with the z-direction.

Check Solution → [2019 • Set 55-1-2]

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155. Two charges $-q$ and $+q$ are located at points $(0, 0, -a)$ and $(0, 0, a)$. What is the electrostatic potential at the points $(0, 0, +z)$ and $(x, y, 0)$?

Check Solution → [2019 • Set 55-1-2]

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156. Draw the equipotential surfaces due to an electric dipole.

Check Solution → [2019 • Set 55-1-3]

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157. A 200 μF parallel plate capacitor having plate separation of 5 mm is charged by a 100 V dc source. It remains connected to the source. Using an insulated handle, the distance between the plates is doubled and a dielectric slab of thickness 5 mm and dielectric constant 10 is introduced between the plates. Explain with reason, how the (i) capacitance, (ii) electric field between the plates, (iii) energy density of the capacitor will change?

Check Solution → [2019 • Set 55-2-1]

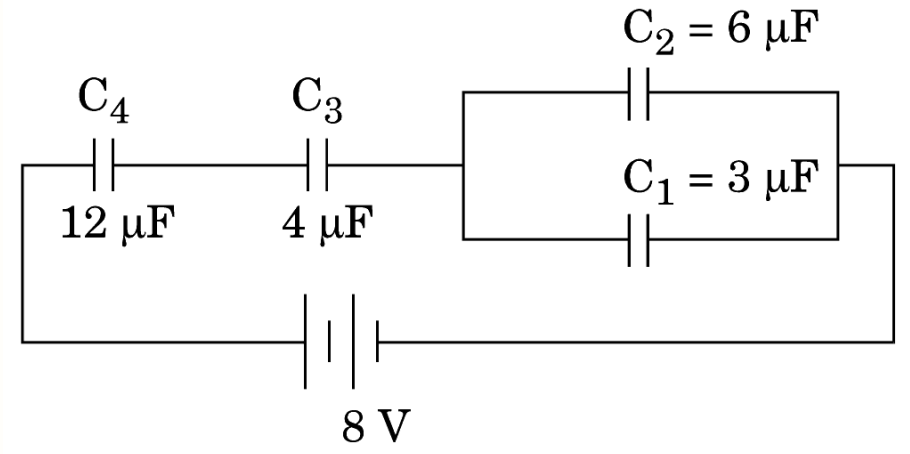
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158. A 100 μF parallel plate capacitor having plate separation of 4 mm is charged by 200 V dc. The source is now disconnected. When the distance between the plates is doubled and a dielectric slab of thickness 4 mm and dielectric constant 5 is introduced between the plates, how will (i) its capacitance, (ii) the electric field between the plates, and (iii) energy density of the capacitor get affected? Justify your answer in each case.

Check Solution → [2019 • Set 55-2-2]

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159. In a network, four capacitors C_1, C_2, C_3 and C_4 are connected as shown in the figure.



(a) Calculate the net capacitance in the circuit.
(b) If the charge on the capacitor C_2 is 6 μC, (i) calculate the charge on the capacitors C_3 and C_4, and (ii) net energy stored in the capacitors C_3 and C_4 connected in series.

Check Solution → [2019 • Set 55-2-3]

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160. (a) Explain briefly, using a proper diagram, the difference in behaviour of a conductor and a dielectric in the presence of external electric field.
(b) Define the term polarization of a dielectric and write the expression for a linear isotropic dielectric in terms of electric field.

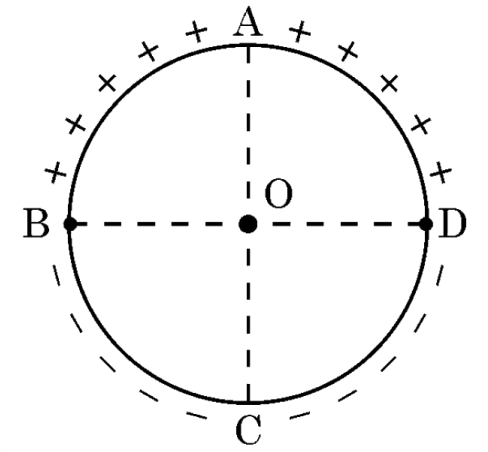
Check Solution → [2019 • Set 55-3-1]

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161. A charge Q is distributed over the surfaces of two concentric hollow spheres of radii r and R ($R >> r$), such that their surface charge densities are equal. Derive the expression for the potential at the common centre.

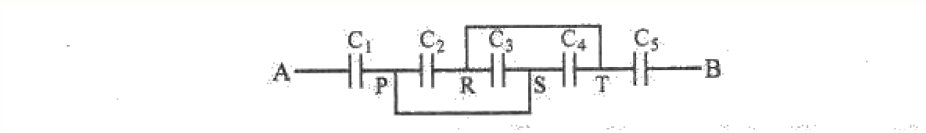
OR

Three concentric metallic shells A, B and C of radii a, b and c ($a < b < c$) have surface charge densities $+\sigma$, $-\sigma$ and $+\sigma$ respectively as shown.
(a) Obtain the expressions for the potential of three shells A, B and C.
(b) If shells A and C are at the same potential, obtain the relation between a, b and c.



Check Solution → [2019 • Set 55-5-1]

162. (i) Find equivalent capacitance between A and B in the combination given below. Each capacitor is of 2 μF capacitance.



(ii) If a dc source of 7 *V* is connected across AB, how much charge is drawn from the source and what is the energy stored in the network?

Check Solution → [2017]

163. A 12 pF capacitor is connected to a 50 *V* battery. How much electrostatic energy is stored in the capacitor? If another capacitor of 6 pF is connected in series with it with the same battery connected across the combination, find the charge stored and potential difference across each capacitor.

Check Solution → [2017]

164. (i) Derive the expression for the electric potential due to an electric dipole at a point on its axial line.
(ii) Depict the equipotential surfaces due to an electric dipole.

Check Solution → [2017]

165. Two identical capacitors of 12 pF each are connected in series across a battery of 50 V. How much electrostatic energy is stored in the combination? If these were connected in parallel across the same battery, how much energy will be stored in the combination now?
Also find the charge drawn from the battery in each case.

Check Solution → [2017]

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166. A charge Q is distributed uniformly over a metallic sphere of radius R. Obtain the expressions for the electric field (E) and electric potential (V) at a point 0 < x < R.
Show on a plot the variation of E and V with x for 0 < x < 2R.

Check Solution → [2017]

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167. Two identical parallel plate capacitors A and B are connected to a battery of V volts with the switch S closed. The switch is now opened and the free space between the plates of the capacitors is filled with a dielectric of dielectric constant K. Find the ratio of the total electrostatic energy stored in both capacitors before and after the introduction of the dielectric.

Check Solution → [2017]

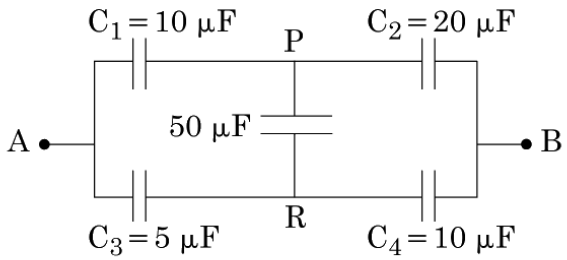
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168. Define an equipotential surface. Draw equipotential surfaces:
(i) in the case of a single point charge and
(ii) in a constant electric field in Z-direction.
Why the equipotential surfaces about a single charge are not equidistant?
(iii) Can electric field exist tangential to an equipotential surface? Give reason.

Check Solution → [2016]

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169. Calculate the equivalent capacitance between points A and B in the circuit below. If a battery of 10 V is connected across A and B, calculate the charge drawn from the battery by the circuit.

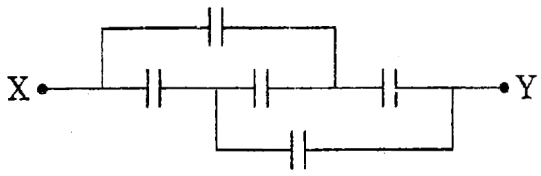


Check Solution → [2016]

170. (a) A parallel plate capacitor (C_1) having charge Q is connected, to an identical uncharged capacitor C_2 in series. What would be the charge accummulated on the capacitor C_2?
(b) Three identical capacitors each of capacitance 3 μF are connected, in turn, in series and in parallel combination to the common source of V volt. Find out the ratio of the energies stored in two configurations.

Check Solution → [2016]

171. Find the equivalent capacitance of the network shown in the figure, when each capacitor is of 1 μF. When the ends X and Y are connected to a 6 V battery, find out (i) the charge and (ii) the energy stored in the network.



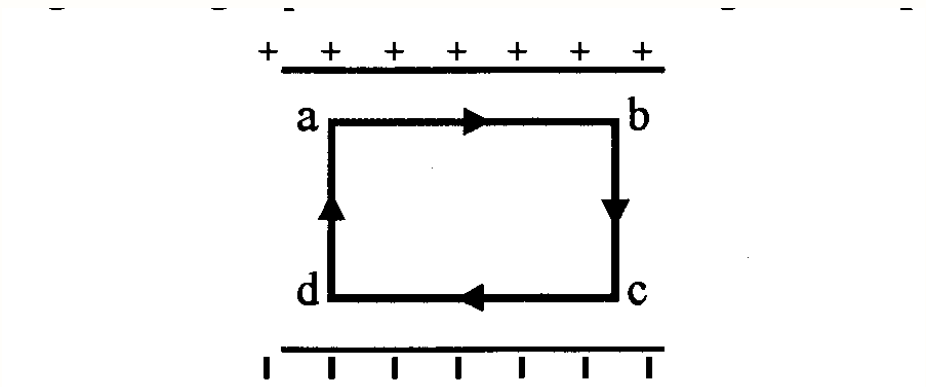
Check Solution → [2015]

172. Two capacitors of unknown capacitances C_1 and C_2 are connected first in series and then in parallel across a battery of 100 V. If the energy stored in the two combinations is 0.045 J and 0.25 J respectively, determine the value of C_1 and C_2. Also calculate the charge on each capacitor in parallel combination.

Check Solution → [2015]

173. (a) Obtain the expression for the energy stored per unit volume in a charged parallel plate capacitor.

(b) The electric field inside a parallel plate capacitor is E. Find the amount of work done in moving a charge q over a closed rectangular loop abcda.



OR

(a) Derive the expression for the capacitance of a parallel plate capacitor having plate area A and plate separation d.

(b) Two charged spherical conductors of radii R_1 and R_2 when connected by a conducting wire acquire charges q_1 and q_2 respectively. Find the ratio of their surface charge densities in terms of their radii.

Check Solution → [2014]

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174. A capacitor of unknown capacitance is connected across a battery of V volts. The charge stored in it is 360 μC. When potential across the capacitor is reduced by 120 V, the charge stored in it becomes 120 μC.

Calculate:

(i) The potential V and the unknown capacitance C.

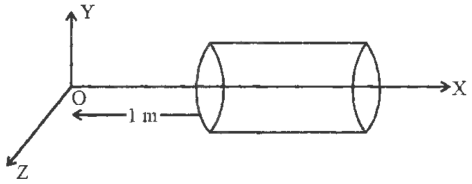
(ii) What will be the charge stored in the capacitor, if the voltage applied had increased by 120 V?

OR

A hollow cylindrical box of length 1 m and area of cross-section 25 cm^2 is placed in a three dimensional coordinate system as shown in the figure. The electric field in the region is given by $\vec{E} = 50\,x\,\hat{i}$, where E is in $N\,C^{-1}$ and x is in metres. Find

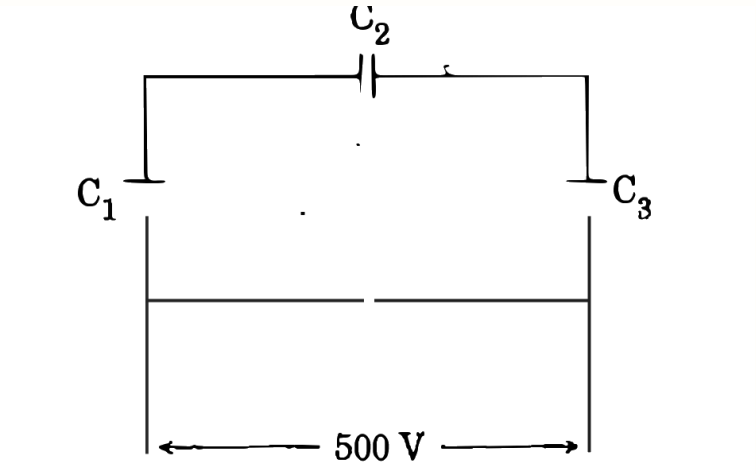
(i) Net flux through the cylinder.

(ii) Charge enclosed by the cylinder.



Check Solution → [2013]

175. A network of four capacitors each of 12 μF capacitance is connected to a 500 *V* supply as shown in the figure. Determine (a) equivalent capacitance of the network and (b) charge on each capacitor.



Check Solution → [2012]

176. Deduce the expression for the electrostatic energy stored in a capacitor of capacitance '*C*' and having charge '*Q*'.
How will the (i) energy stored and (ii) the electric field inside the capacitor be affected when it is completely filled with a dielectric material of dielectric constant '*K*'?

Check Solution → [2012]

177. A capacitor of 200 pF is charged by a 300 V battery. The battery is then disconnected and the charged capacitor is connected to another uncharged capacitor of 100 pF. Calculate the difference between the final energy stored in the combined system and the initial energy stored in the single capacitor.

Check Solution → [2012]

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178. State the principle of the device that can build up high voltages of the order of a few million volts. Draw its labelled diagram. A stage reaches in this device when the potential at the outer sphere cannot be increased further by piling up more charge on it. Explain why.

Check Solution → [2011 • Set 55-2-1]

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179. A parallel plate capacitor is charged by a battery. After sometime the battery is disconnected and a dielectric slab with its thickness equal to the plate separation is inserted between the plates. How will (i) the capacitance of the capacitor, (ii) potential difference between the plates and (iii) the energy stored in the capacitor be affected? Justify your answer in each case.

Check Solution → [2010]

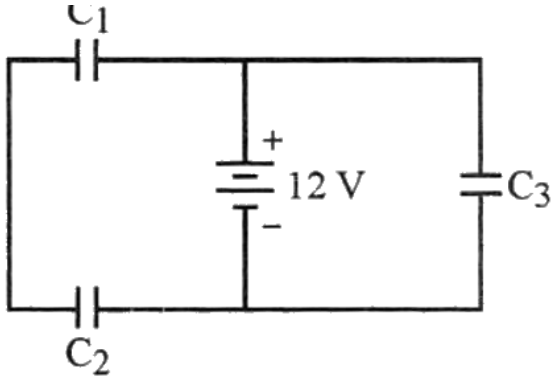
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180. (a) Depict the equipotential surfaces for a system of two identical positive point charges placed a distance 'd' apart.
(b) Deduce the expression for the potential energy of a system of two point charges q_1 and q_2 brought from infinity to the points $\vec{r}_1$ and $\vec{r}_2$ respectively in the presence of external electric field $\vec{E}$.

Check Solution → [2010]

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181. Three identical capacitors C_1, C_2 and C_3 of capacitance 6 μF each are connected to a 12 V battery as shown.



Find

(i) charge on each capacitor

(ii) equivalent capacitance of the network

(iii) energy stored in the network of capacitors

Check Solution → [2009]

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182. A positive point charge ($+q$) is kept in the vicinity of an uncharged conducting plate. Sketch electric field lines originating from the point on to the surface of the plate.

Derive the expression for the electric field at the surface of a charged conductor.

Check Solution → [2009]

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183. A parallel plate capacitor is charged by a battery. After some time the battery is disconnected and a dielectric slab of dielectric constant K is inserted between the plates. How would (i) the capacitance, (ii) the electric field between the plates and (iii) the energy stored in the capacitor, be affected? Justify your answer.

Check Solution → [2009]

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184. Deduce an expression for the electric potential due to an electric dipole at any point on its axis. Mention one contrasting feature of electric potential of a dipole at a point as compared to that due to a single charge.

Check Solution → [2007]

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185. A parallel plate capacitor, each with plate area A and separation d, is charged to a potential difference V. The battery used to charge it is then disconnected. A dielectric slab of thickness d and dielectric constant K is now placed between the plates. What change, if any, will take place in
(i) charge on the plates,
(ii) electric field intensity between the plates,
(iii) capacitance of the capacitor.
Justify your answer in each case.

Check Solution → [2007]

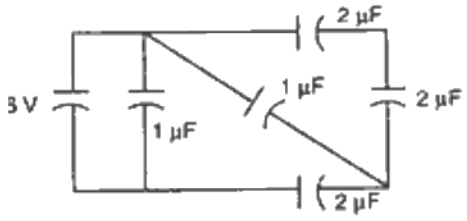
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186. Explain the underlying principle of working of a parallel plate capacitor.
If two similar plates, each of area A having surface charge densities $+\sigma$ and $-\sigma$ are separated by a distance d in air, write expressions for
(i) the electric field at points between the two plates.
(ii) the potential difference between the plates.
(iii) the capacitance of the capacitor so formed.

Check Solution → [2007]

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187. Find the total energy stored in the capacitors in the given network.

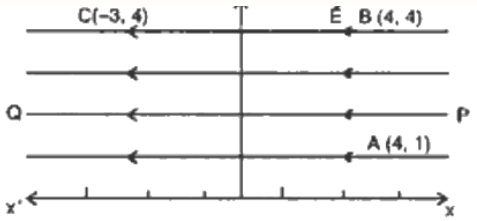


Check Solution → [2004]

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188. What is an equipotential surface?

A uniform electric field of $E = 300\ NC^{-1}$ is directed along PQ. A, B and C are three points in the field having x and y coordinates (in metres) as shown in the figure. Calculate potential difference between the points (i) A and B and (ii) B and C.



Check Solution → [2003]

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189. Draw a labelled diagram of Van de Graaff generator. State its principle of working.

Check Solution → [2003]

SECTION D

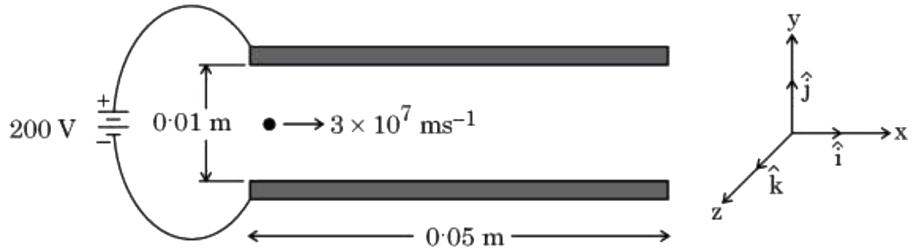
4-Mark Questions

Case Study · 3 questions

> *Section D of the paper, 4 marks each. Sorted newest year first.*

190. The electric potential V and electric field E are closely related concepts in electrostatics. The electric field is a vector quantity that represents the force per unit charge at a given point in space, whereas electric potential is a scalar quantity that represents the potential energy per unit charge at a given point in space. Electric field and electric potential are related by the equations $E_r = -\frac{dV}{dr}$ and $\vec{E} = -\vec{\nabla} V$, i.e., electric field is the negative gradient of the electric potential. This means that electric field points in the direction of decreasing potential and its magnitude is the rate of change of potential with distance. The electric field is the force that drives a unit charge to move from higher potential region to lower potential region and electric potential difference between the two points determines the work done in moving a unit charge from one point to the other point.

A pair of square conducting plates having sides of length 0.05 m are arranged parallel to each other in x-y plane. They are 0.01 m apart along z-axis and are connected to a 200 V power supply as shown in the figure. An electron enters with a speed of 3×10^7 ms^{-1} horizontally and symmetrically in the space between the two plates. Neglect the effect of gravity on the electron.



Check Solution → [2026 • Set 55-5-1]

191. A capacitor is a system of two conductors separated by an insulator. The two conductors have equal and opposite charges with a potential difference between them. The capacitance of a capacitor depends on the geometrical configuration

(shape, size and separation) of the system and also on the nature of the insulator separating the two conductors. They are used to store charges. Like resistors, capacitors can be arranged in series or parallel or a combination of both to obtain desired value of capacitance.

Check Solution → [2023 • Set 55-5-1]

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192. While travelling back to his residence in the car, Dr. Pathak was caught up in a thunderstorm. It became very dark. He stopped driving the car and waited for thunderstorm to stop. Suddenly he noticed a child walking alone on the road. He asked the boy to come inside the car till the thunderstorm stopped. Dr. Pathak dropped the boy at his residence. The boy insisted that Dr. Pathak should meet his parents. The parents expressed their gratitude to Dr. Pathak for his concern for safety of the child.

Answer the following questions based on the above information:

(a) Why is it safer to sit inside a car during a thunderstorm?

(b) Which two values are displayed by Dr. Pathak in his actions?

(c) Which values are reflected in parents' response to Dr. Pathak?

(d) Give an example of a similar action on your part in the past from everyday life.

Check Solution → [2013]

SECTION E

5-Mark Questions

Long Answer · 28 questions

> *Section E of the paper, 5 marks each. Sorted newest year first.*

193. (a) Derive an expression for the capacitance of a parallel plate capacitor of plate area A and plate separation d with air present between the plates.

Check Solution → **[2026 • Set 55-2-1]**

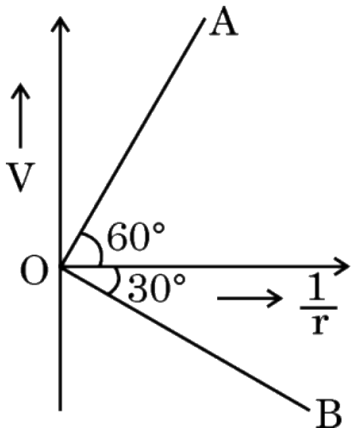
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194. (b) Two air-filled capacitors of capacitances C_1 and C_2 are connected in parallel with a dc battery. After the capacitors are fully charged, a slab of dielectric constant K is inserted between the plates of each capacitor. How will the (i) charge on each capacitor and (ii) energy stored in the capacitor be affected after the slab is introduced?

Check Solution → **[2026 • Set 55-2-1]**

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195. (i) In the figure, OA and OB show the variation of electric potential V at a point due to two point charges Q_1 and Q_2 with $\frac{1}{r}$ respectively. Here r represents the distance of the point from the two point charges.



(I) Identify the nature of the two charges Q_1 and Q_2.

(II) What is the value of $(\frac{Q_1}{Q_2})$? Justify your answer.

Check Solution → [2026 • Set 55-3-1]

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196. (ii) Two point charges −2 μC and 5 μC are placed at (−30 cm, 0) and (30 cm, 0) respectively in an external electric field $\vec{E} = \frac{A}{x}\hat{i}$, where $A = 9 \times 10^5\ N\ m^2\ C^{-1}$. Find the electrostatic potential energy of this configuration.

Check Solution → [2026 • Set 55-3-1]

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197. **(a)**

(i) Explain the following statements giving reason:

(I) An equipotential surface through a point is normal to the electric field at that point.

(II) When a dielectric is placed in an external electric field, the electric field inside the dielectric is less than that outside it.

(III) The potential difference between the plates of a charged parallel plate capacitor decreases when its plates are brought closer.

(ii) Obtain an expression for the work done to dissociate the system of three charges q, −4q and 2q placed at the vertices A, B and C respectively of an equilateral triangle of side 'a'.

OR

(b)

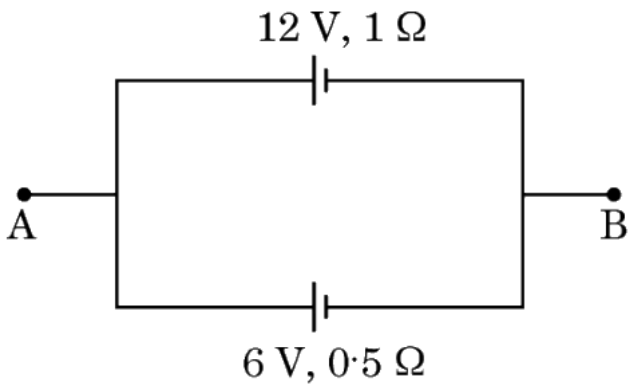
(i) Answer the following giving reason:

(I) The electron drift speed is estimated to be only a few mm/s for currents in the range of a few amperes. How, then, is the current established almost the instant a circuit is closed?

(II) A low voltage supply from which one needs high currents must have very low internal resistance. Why?

(III) The assertion that V = IR is a statement of Ohm's law is not true. Why?

(ii) Two cells of emfs 12 V and 6 V are connected in parallel as shown in the figure. Their internal resistances are 1 Ω and 0·5 Ω respectively. Calculate the emf and internal resistance of the equivalent cell between points A and B.



Check Solution → [2026 • Set 55-4-1]

198. (a)

(i) Two point charges 5 μC and −1 μC are placed at points (−3 cm, 0, 0) and (3 cm, 0, 0) respectively. An external electric field $\vec{E} = \frac{A}{r^2}\hat{r}$, where $A = 3 \times 10^5\ V\,m$, is switched on in the region. Calculate the change in electrostatic energy of the system due to the electric field.

(ii) A system of two conductors is placed in air and they have net charge of +80 μC and −80 μC which causes a potential difference of 16 *V* between them.

(1) Find the capacitance of the system.

(2) If the air between the capacitor is replaced by a dielectric medium of dielectric constant 3, what will be the potential difference between the two conductors?

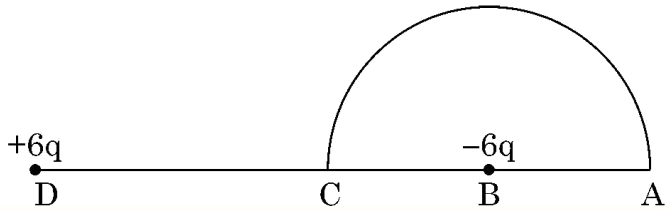
(3) If the charges on two conductors are changed to +160 μC and −160 μC, will the capacitance of the system change? Give reason for your answer.

OR

(b)

(i) Consider three metal spherical shells *A*, *B* and *C*, each of radius *R*. Each shell is having a concentric metal ball of radius *R*/10. The spherical shells *A*, *B* and *C* are given charges +6q, −4q and 14q respectively. Their inner metal balls are also given charges −2q, +8q and −10q respectively. Compare the magnitude of the electric fields due to shells *A*, *B* and *C* at a distance 3R from their centres.

(ii) A charge −6 μC is placed at the centre *B* of a semicircle of radius 5 cm, as shown in the figure. An equal and opposite charge is placed at point *D* at a distance of 10 cm from *B*. A charge +5 μC is moved from point *C* to point *A* along the circumference. Calculate the work done on the charge.



Check Solution → [2025 • Set 55-1-1]

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199. (i) A small conducting sphere A of radius r charged to a potential V_1 is enclosed by a spherical conducting shell B of radius R. If A and B are connected by a thin wire, calculate the final potential on sphere A and shell B.

(ii) Write two characteristics of equipotential surfaces. A uniform electric field of 50 NC^{-1} is set up in a region along +x axis. If the potential at the origin (0, 0) is 220 V, find the potential at a point (4 m, 3 m).

Check Solution → [2025 • Set 55-2-1]

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200. (a) (i) The electric field in a region is given by $\vec{E} = 40x\,\hat{i}\ N/C$. Find the amount of work done in taking a unit positive charge from a point (0, 3 *m*) to the point (5 *m*, 0).

(ii) A charge *Q* is distributed over two concentric hollow spheres of radii *r* and *R* (>r) such that their surface charge densities are equal. Find:

(I) the electric field, and

(II) the potential

at their common centre.

OR

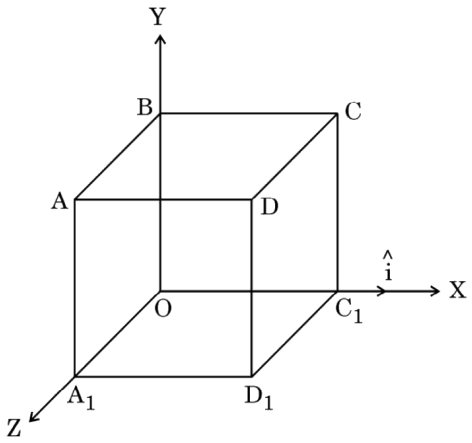
(b) (i) Obtain an expression for the electric field $\vec{E}$ due to a dipole of dipole moment $\vec{p}$ at a point on its equatorial plane and specify its direction. Hence, find the value of electric field:

(I) at the centre of the dipole ($r = 0$), and

(II) at a point $r \gg a$,

where 2a is the length of the dipole.

(ii) An electric field $\vec{E} = (10x + 5)\,\hat{i}\ N/C$ exists in a region in which a cube of side *L* is kept as shown in the figure. Here *x* and *L* are in metres. Calculate the net flux through the cube.



Check Solution → [2025 • Set 55-6-1]

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201. **(a) (i)** A parallel plate capacitor with plate area A and plate separation d has a capacitance C_0. A slab of dielectric constant K having area A and thickness $(\frac{d}{2})$ is inserted in the capacitor, parallel to the plates. Find the new value of its capacitance.

(ii) You are provided with a large number of 1 μF identical capacitors and a power supply of 1200 V. The dielectric medium used in each capacitor can withstand up to 200 V only. Find the minimum number of capacitors and their arrangement, required to build a capacitor system of equivalent capacitance of 2 μF for use with this supply.

OR

(b) (i) An electric dipole of dipole moment $\vec{p}$ consists of point charges q and $-q$, separated by 2a. Derive an expression for electric potential in terms of its dipole moment at a point at a distance x ($\gg a$) from its centre and lying (I) along its axis, and (II) along its bisector line.

(ii) An electric dipole of dipole moment $\vec{p} = (0.8\hat{i} + 0.6\hat{j}) \times 10^{-29}\ C\ m$ is placed in an electric field $\vec{E} = 1.0 \times 10^7\ \hat{k}\ V/m$. Calculate the magnitude of the torque acting on it and the angle it makes with the x-axis, at this instant.

Check Solution → [2025 • Set 55-7-1]

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202. (i) Obtain the expression for the capacitance of a parallel plate capacitor with a dielectric medium between its plates.

(ii) A charge of 6 μC is given to a hollow metallic sphere of radius 0.2 m. Find the potential at (i) the surface and (ii) the centre of the sphere.

Check Solution → [2024 • Set 55-3-1]

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203. (i) A dielectric slab of dielectric constant K and thickness t is inserted between plates of a parallel plate capacitor of plate separation d and plate area A. Obtain an expression for its capacitance.
(ii) Two capacitors of different capacitances are connected first (1) in series and then (2) in parallel across a dc source of 100 V. If the total energy stored in the combination in the two cases are 40 mJ and 250 mJ respectively, find the capacitance of the capacitors.

Check Solution → [2024 • Set 55-4-1]

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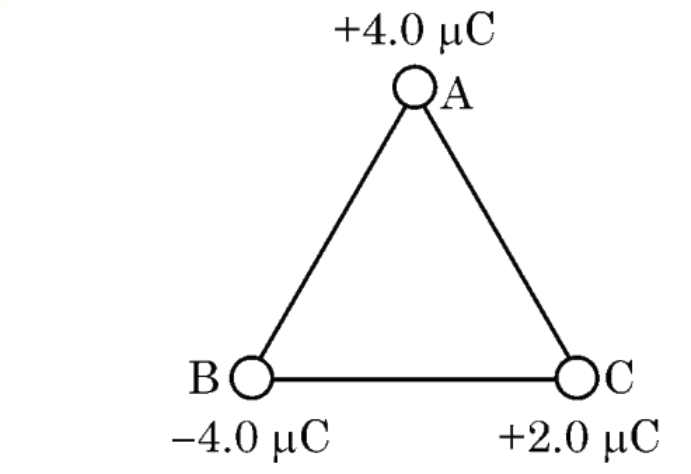
204. (i) Draw equipotential surfaces for an electric dipole.
(ii) Two point charges q_1 and q_2 are located at $\vec{r}_1$ and $\vec{r}_2$ respectively in an external electric field $\vec{E}$. Obtain an expression for the potential energy of the system.
(iii) The dipole moment of a molecule is 10^{-30} Cm. It is placed in an electric field $\vec{E}$ of 10^5 V/m such that its axis is along the electric field. The direction of $\vec{E}$ is suddenly changed by 60° at an instant. Find the change in the potential energy of the dipole, at that instant.

Check Solution → [2024 • Set 55-5-1]

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205. (i) Consider two identical point charges located at points (0,0) and (a,0).
(1) Is there a point on the line joining them at which the electric field is zero?
(2) Is there a point on the line joining them at which the electric potential is zero?
Justify your answers for each case.
(ii) State the significance of negative value of electrostatic potential energy of a system of charges.
Three charges are placed at the corners of an equilateral triangle ABC of side 2.0

m as shown in figure. Calculate the electric potential energy of the system of three charges.



Check Solution → [2023 • Set 55-2-1]

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206. (i) (A) Why does the electric field inside a dielectric slab decrease when kept in an external electric field?

(B) Derive an expression for the capacitance of a parallel plate capacitor filled with a medium of dielectric constant K.

(ii) A charge $q = 2$ μC is placed at the centre of a sphere of radius 20 cm. What is the amount of work done in moving 4 μC from one point to another point on its surface?

(iii) Write a relation for polarisation $\vec{P}$ of a dielectric material in the presence of an external electric field.

Check Solution → [2021]

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207. (i) Obtain an expression for the potential energy of an electric dipole placed in a uniform electric field.

(ii) Three capacitors of capacitance C_1, C_2 and C_3 are connected in series to a source of V volt. Show that the total energy stored in the combination of capacitors is equal to sum of the energy stored in individual capacitors.

(iii) A capacitor of capacitance C is connected across a battery. After charging, the battery is disconnected and the separation between the plates is doubled. How will (i) the capacitance of the capacitor, and (ii) the electric field between the plates be affected? Justify your answer.

Check Solution → [2021]

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208. Find the expression for the potential energy of a system of two point charges q_1 and q_2 located at $\vec{r}_1$ and $\vec{r}_2$ respectively in an external electric field $\vec{E}$.

Check Solution → [2020 • Set 55-1-1]

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209. Draw equipotential surfaces due to an isolated point charge $(-q)$ and depict the electric field lines.

Check Solution → [2020 • Set 55-1-1]

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210. Three point charges +1 μC, −1 μC and +2 μC are initially infinite distance apart. Calculate the work done in assembling these charges at the vertices of an equilateral triangle of side 10 cm.

Check Solution → [2020 • Set 55-1-1]

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211. Describe briefly the process of transferring the charge between the two plates of a parallel plate capacitor when connected to a battery. Derive an expression for the energy stored in a capacitor.

Check Solution → [2019 • Set 55-1-1]

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212. A parallel plate capacitor is charged by a battery to a potential difference V. It is disconnected from battery and then connected to another uncharged capacitor of the same capacitance. Calculate the ratio of the energy stored in the combination to the initial energy on the single capacitor.

Check Solution → [2019 • Set 55-1-1]

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213. (a) When a parallel plate capacitor is connected across a dc battery, explain briefly how the capacitor gets charged.
(b) A parallel plate capacitor of capacitance 'C' is charged to 'V' volt by a battery.

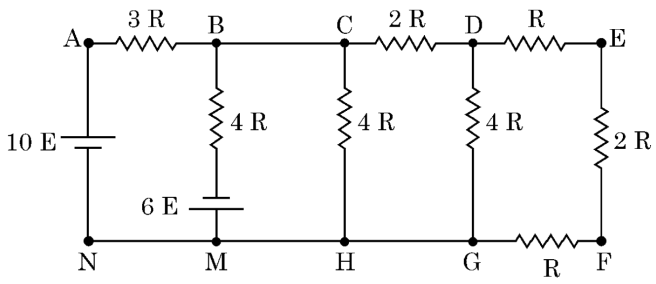
After some time the battery is disconnected and the distance between the plates is doubled. Now a slab of dielectric constant $1 < k < 2$ is introduced to fill the space between the plates. How will the following be affected?

(i) The electric field between the plates of the capacitor.

(ii) The energy stored in the capacitor.

Justify your answer in each case.

(c) The electric potential as a function of distance 'x' is shown in the figure. Draw a graph of the electric field E as a function of x.



OR

(a) Derive an expression for the potential energy of an electric dipole in a uniform electric field. Explain conditions for stable and unstable equilibrium.

(b) Is the electrostatic potential necessarily zero at a point where the electric field is zero? Give an example to support your answer.

Check Solution → [2019 • Set 55-4-1]

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214. (a) Distinguish, with the help of a suitable diagram, the difference in the behaviour of a conductor and a dielectric placed in an external electric field. How does polarised dielectric modify the original external field?

(b) A capacitor of capacitance C is charged fully by connecting it to a battery of emf E. It is then disconnected from the battery. If the separation between the plates of the capacitor is now doubled, how will the following change?

(i) charge stored by the capacitor.

(ii) field strength between the plates.

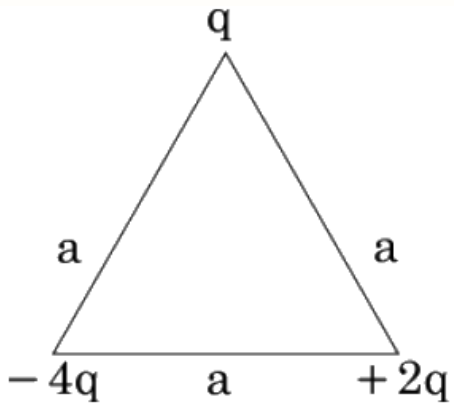
(iii) energy stored by the capacitor.

Justify your answer in each case.

Check Solution → [2016]

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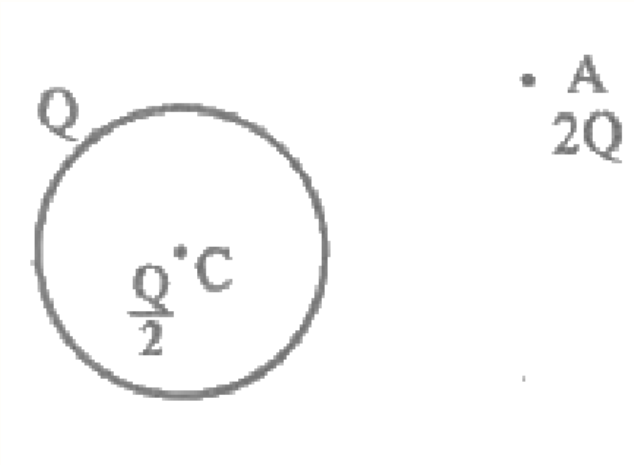
215. (a) Explain why, for any charge configuration, the equipotential surface through a point is normal to the electric field at that point.
Draw a sketch of equipotential surfaces due to a single charge ($-q$), depicting the electric field lines due to the charge.
(b) Obtain an expression for the work done to dissociate the system of three charges placed at the vertices of an equilateral triangle of side a as shown below.



Check Solution → [2016]

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216. (a) Explain, using suitable diagrams, the difference in the behaviour of a (i) conductor and (ii) dielectric in the presence of external electric field. Define the terms polarization of a dielectric and write its relation with susceptibility.
(b) A thin metallic spherical shell of radius R carries a charge Q on its surface. A point charge $\frac{Q}{2}$ is placed at its centre C and an other charge +2Q is placed outside the shell at a distance x from the centre as shown in the figure. Find (i) the force on the charge at the centre of shell and at the point A, (ii) the electric flux through the shell.



Check Solution → [2015]

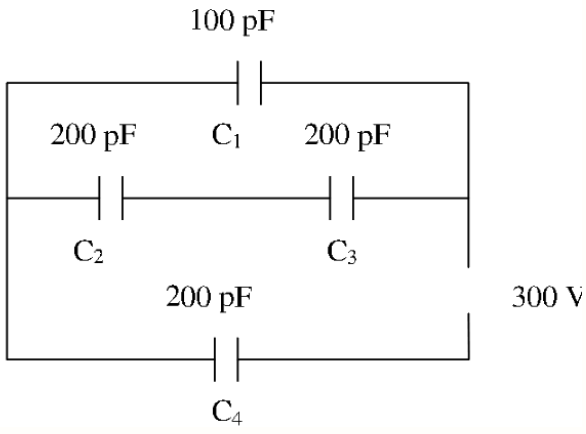
217. Define the term 'capacitance' of a capacitor. Derive an expression for the total capacitance of 'n' capacitors arranged in series.

Check Solution → [2010]

218. Define electric potential at a point due to a point charge. Write its SI unit. Derive an expression for the electric potential at a point outside a charged sphere.

Check Solution → [2010]

219. (a) Derive an expression for the energy stored in a parallel plate capacitor C, charged to a potential difference V.
(b) Obtain the equivalent capacitance of the network given below. For a supply of 300 V, determine the charge and voltage across C_4.



Check Solution → [2008]

220. Explain the principle on which Van de Graaff generator operates. Draw a labeled schematic sketch and write briefly its working.

A Van de Graaff type generator is capable of building up potential difference of 15×10^6 V. The dielectric strength of the gas surrounding the electrode is 5×10^7 $V\,m^{-1}$. What is the minimum radius of the spherical shell required?

Check Solution → [2008]

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **C**	Q14 **C**	Q27 **C**	Q40 **B**	Q53 **C**	Q66 **A**
Q2 **B**	Q15 **D**	Q28 **A**	Q41 **A**	Q54 **A**	Q67 **B**
Q3 **B**	Q16 **B**	Q29 **C**	Q42 **B**	Q55 **C**	Q69 **D**
Q4 **C**	Q17 **C**	Q30 **A**	Q43 **C**	Q56 **B**	Q72 **C**
Q5 **D**	Q18 **B**	Q31 **C**	Q44 **B**	Q57 **B**	Q74 **D**
Q6 **D**	Q19 **B**	Q32 **C**	Q45 **A**	Q58 **C**	Q75 **A**
Q7 **B**	Q20 **A**	Q33 **C**	Q46 **B**	Q59 **A**	Q76 **C**
Q8 **B**	Q21 **B**	Q34 **A**	Q47 **C**	Q60 **D**	Q77 **A**
Q9 **D**	Q22 **A**	Q35 **C**	Q48 **D**	Q61 **B**	Q78 **D**
Q10 **D**	Q23 **D**	Q36 **D**	Q49 **C**	Q62 **B**	Q79 **C**
Q11 **D**	Q24 **A**	Q37 **B**	Q50 **B**	Q63 **C**	
Q12 **A**	Q25 **B**	Q38 **A**	Q51 **B**	Q64 **D**	
Q13 **C**	Q26 **D**	Q39 **A**	Q52 **C**	Q65 **A**	

WORKED ANSWERS

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

In a region, the electric potential varies as $V = 10 - 50x$*, where V is in volts and x in meters. The electric field in the region is:*

(A) 10 N/C along +x

(B) 10 N/C along -x

(C) 50 N/C along +x

(D) 50 N/C along -x

[2026 • Set 55-1-1]

SOLUTION

The electric field is the negative gradient of the electric potential. In one dimension, only the x-component survives.

$$E_x = -\frac{dV}{dx}$$

Here the potential is given as $V = 10 - 50x$.

$$E_x = -\frac{d}{dx}(10 - 50x)$$

$$E_x = -(-50)$$

$$E_x = +50\ V/m = +50\ N/C$$

The result is positive, so the field points along the $+x$ direction. The field is uniform, with magnitude 50 N/C everywhere in the region.

Ruling out the others: options (A) and (B) use 10 N/C, but 10 is only the constant term in V and has no effect on the field. Option (D) has the correct magnitude

but the wrong direction. Since V decreases as x increases, the field must point towards lower potential, that is along $+x$.

Hence the electric field is **50 *N/C* along the +*x* direction**, which is option (C).

Final answer. The electric field is 50 N/C along the +x direction, option **(C)**.

← *Back to Q1*

Q2

A conducting wire connects two charged metallic spheres A and B of radii r_1 *and* r_2 *respectively. The distance between the spheres is very large compared to their radii. The ratio of electric fields,* (E_1/E_2) *at the surfaces of spheres A and B will be:*

(A) $\frac{r_1}{r_2}$

(B) $\frac{r_2}{r_1}$

(C) $\frac{r_2^2}{r_1^2}$

(D) $\frac{r_1^2}{r_2^2}$

[2026 • Set 55-1-1]

SOLUTION

When two charged conductors are joined by a conducting wire, charge flows until both reach the same (common) potential. Since the spheres are far apart, each behaves as an isolated sphere.

Equal potential gives:

$$\frac{kQ_1}{r_1} = \frac{kQ_2}{r_2}$$

$$\Rightarrow Q \propto r$$

The surface electric field of a charged sphere is:

$$E = \frac{kQ}{r^2}$$

Since $Q \propto r$, we get:

$$E \propto \frac{r}{r^2} = \frac{1}{r}$$

Taking the ratio for the two spheres:

$$\frac{E_1}{E_2} = \frac{1/r_1}{1/r_2} = \frac{r_2}{r_1}$$

Ruling out the others: option (A) is the inverse of the correct answer; options (C) and (D) use r^2 in the ratio, which would apply only if the two spheres carried equal charge, but here charge is shared in proportion to radius.

Hence the ratio of surface fields is $\mathbf{\frac{r_2}{r_1}}$, which is option (B). The smaller sphere has the larger surface field, which is the basis of point discharge and corona effects.

CONCEPT

Connected conductors share charge to equalise potential, not field. So a sharper (smaller-radius) point always has the stronger surface field.

Final answer. The ratio of surface fields is $\mathbf{r_2/r_1}$, option **(B)**.

← *Back to Q2*

Q3

In a region electric field is given by $\vec{\boldsymbol{E}} = \mathbf{4x}\,\hat{\boldsymbol{i}}$ *N/C. The potential difference between points A* $(\boldsymbol{x} = \mathbf{1}\ m)$ *and B* $(\boldsymbol{x} = \mathbf{3}\ m)$, $(V_A - V_B)$ *is:*

(A) −16 V

(B) **16 V**

(C) −8 V

(D) 8 V

[2026 • Set 55-1-2]

SOLUTION

The potential difference between two points equals the negative line integral of the electric field from the lower to the higher point. Equivalently:

$$V_A - V_B = -\int_B^A \vec{E} \cdot d\vec{l} = \int_A^B \vec{E} \cdot d\vec{l}$$

Here $\vec{E} = 4x\,\hat{i}\ N/C$ and the path lies along the x-axis, so $d\vec{l} = dx\,\hat{i}$ and $\vec{E} \cdot d\vec{l} = 4x\,dx$.

$$V_A - V_B = \int_1^3 4x\,dx$$

$$V_A - V_B = [\,2x^2\,]_1^3$$

$$V_A - V_B = 2(3)^2 - 2(1)^2$$

$$V_A - V_B = 18 - 2$$

$$V_A - V_B = +16\ V$$

The answer is positive, which is physically consistent: the field points along $+\hat{i}$, so potential falls as x increases. Point A at $x = 1\ m$ is therefore at a higher potential than B at $x = 3\ m$.

Ruling out the others: option (A) has the right magnitude but wrong sign (it would mean A is at lower potential, contradicting the field direction); options (C) and (D) give 8 V, which comes from forgetting the factor while integrating.

Hence $V_A - V_B = +16\ V$, which is option (B).

Final answer. $V_A - V_B = +16$ V, option **(B)**.

← *Back to Q3*

Q4

The electric potential for various points in x-y plane is given by $V = 1.0\,x^2 - 2.0\,y^2$, *where V is in volts and x, y are in metres. The angle that the electric field at point* (2.0

m, 1.0 *m*) *makes with the positive x-axis is:*

(A) 45°

(B) 90°

(C) **135°**

(D) 315°

[2026 • Set 55-1-3]

SOLUTION

The electric field is the negative gradient of the potential. In two dimensions:

$$\vec{E} = -\nabla V = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j}$$

With $V = \mathbf{1.0\,x^2 - 2.0\,y^2}$, the partial derivatives are:

$$\frac{\partial V}{\partial x} = 2x$$

$$\frac{\partial V}{\partial y} = -4y$$

Therefore:

$$\vec{E} = -2x\,\hat{i} + 4y\,\hat{j}$$

At the point (2.0 *m*, 1.0 *m*):

$$\vec{E} = -2(2)\,\hat{i} + 4(1)\,\hat{j} = -4\,\hat{i} + 4\,\hat{j}\ V/m$$

The *x*-component is negative and the *y*-component is positive, so the vector lies in the second quadrant. The angle measured from the +*x*-axis is:

$$\theta = 180° - \tan^{-1}\left(\frac{|E_y|}{|E_x|}\right)$$

$$\theta = 180° - \tan^{-1}\left(\frac{4}{4}\right)$$

$$\theta = 180° - 45° = 135°$$

Ruling out the others: option (A) 45° ignores the sign of E_x; option (B) 90° would need $\mathbf{E_x = 0}$; option (D) 315° corresponds to the fourth quadrant ($+E_x$, $-E_y$), which is not the case here.

Hence the field makes an angle of **135°** with the positive x-axis, which is option (C).

COMMON MISTAKE

Always fix the quadrant from the signs of E_x and E_y before taking the inverse tangent. A bare calculator $\tan^{-1}$ only gives the first or fourth quadrant.

Final answer. The electric field makes **135°** with the +x-axis, option **(C)**.

← *Back to Q4*

Q5

Three point charges 2q, −2q and q are kept at the vertices of an equilateral triangle of side l. The potential energy of the system is:

(A) zero

(B) $\dfrac{-2q^2}{\pi\varepsilon_0 l}$

(C) $\dfrac{q^2}{2\pi\varepsilon_0 l}$

(D) $\boldsymbol{\dfrac{-q^2}{\pi\varepsilon_0 l}}$

[2026 • Set 55-2-1]

SOLUTION

The potential energy of a system of point charges is the sum of the energies of all distinct pairs. For three charges at the vertices of an equilateral triangle of side l, every pairwise separation equals l.

$$U = \frac{1}{4\pi\varepsilon_0 l}\,(q_1 q_2 + q_2 q_3 + q_3 q_1)$$

Taking $\mathbf{q_1 = 2q}$, $\mathbf{q_2 = -2q}$, $\mathbf{q_3 = q}$, the pair products are:

$$q_1 q_2 = (2q)(-2q) = -4q^2$$

$$q_2 q_3 = (-2q)(q) = -2q^2$$

$$q_3\, q_1 = (q)(2q) = +2q^2$$

Their sum is:

$$q_1\, q_2 + q_2\, q_3 + q_3\, q_1 = -4q^2 - 2q^2 + 2q^2 = -4q^2$$

Substituting:

$$U = \frac{-4q^2}{4\pi\varepsilon_0\, l} = -\frac{q^2}{\pi\varepsilon_0\, l}$$

Ruling out the others: option (A) zero would require the pair products to cancel, which they do not; option (B) $\frac{-2q^2}{\pi\varepsilon_0\, l}$ doubles the correct value; option (C) is positive, but the net energy here is clearly negative (attractive terms dominate).

Hence the potential energy of the system is $\mathbf{-\frac{q^2}{\pi\varepsilon_0\, l}}$, which is option (D).

Final answer. The potential energy of the system is $\mathbf{-\frac{q^2}{\pi\varepsilon_0\, l}}$, option **(D)**.

← *Back to Q5*

Q6

Two metal spheres of radii r_1 *and* $\mathbf{r_2}$ $\mathbf{(r_2 > r_1)}$ *having charges* q_1 *and* q_2 *respectively kept in air, are brought in contact. Which of the following statements is NOT correct?*

(A) The total charge of the two spheres is conserved.

(B) Both spheres attain the same potential.

(C) The final potential of the system equals $\frac{1}{4\pi\varepsilon_0}\frac{(q_1+q_2)}{(r_1+r_2)}$.

(D) **The final potential of the system equals** $\mathbf{\frac{1}{4\pi\varepsilon_0}\frac{(q_1+q_2)(r_1+r_2)}{r_1 r_2}}$.

[2026 • Set 55-2-1]

SOLUTION

When two charged metal spheres are brought into contact, two things must hold: total charge is conserved, and the spheres reach a single common potential. Let the final charges be Q_1' and Q_2'.

Charge conservation gives:

$$Q_1' + Q_2' = q_1 + q_2$$

Equal potential gives:

$$\frac{1}{4\pi\varepsilon_0}\frac{Q_1'}{r_1} = \frac{1}{4\pi\varepsilon_0}\frac{Q_2'}{r_2} = V$$

So $\mathbf{Q_1' = 4\pi\varepsilon_0\, V\, r_1}$ and $\mathbf{Q_2' = 4\pi\varepsilon_0\, V\, r_2}$. Adding these and using charge conservation:

$$q_1 + q_2 = 4\pi\varepsilon_0\, V\,(r_1 + r_2)$$

$$V = \frac{1}{4\pi\varepsilon_0}\frac{(q_1 + q_2)}{(r_1 + r_2)}$$

Now testing each statement. Statement (A) is true, since charge is always conserved on contact. Statement (B) is true, since contact equalises the potential. Statement (C) gives exactly the formula derived above, so it is true. Statement (D) gives $\frac{1}{4\pi\varepsilon_0}\frac{(q_1+q_2)(r_1+r_2)}{r_1\, r_2}$, which does not match the derived expression, so it is the incorrect statement.

Since the question asks which statement is NOT correct, the answer is the false statement (D).

Hence the statement that is **not correct is option (D)**.

Final answer. The incorrect statement is option **(D)**.

← *Back to Q6*

Q7

Three point charges +q, −2q and +q are placed along x-axis at points $\boldsymbol{x=-1\ m}$, $\boldsymbol{x=0\ m}$

and x**=2** m *respectively. The potential energy of the system is:*

(A) $\frac{q^2}{4\pi\varepsilon_0}$

(B) $\frac{-2q^2}{3\pi\varepsilon_0}$

(C) $\frac{q^2}{4\pi\varepsilon_0}$

(D) $\frac{-5q^2}{4\pi\varepsilon_0}$

[2026 • Set 55-2-2]

SOLUTION

The potential energy of a system of point charges is the sum over all distinct pairs:

$$U = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_1 q_3}{r_{13}}\right)$$

The charges are $q_1 = +q$ at $x = -1$ m, $q_2 = -2q$ at $x = 0$ m, and $q_3 = +q$ at $x = 2$ m. The separations are:

$$r_{12} = |-1 - 0| = 1\ m$$

$$r_{23} = |0 - 2| = 2\ m$$

$$r_{13} = |-1 - 2| = 3\ m$$

The pair products are:

$$q_1 q_2 = (q)(-2q) = -2q^2$$

$$q_2 q_3 = (-2q)(q) = -2q^2$$

$$q_1 q_3 = (q)(q) = +q^2$$

Substituting each pair with its own separation:

$$U = \frac{1}{4\pi\varepsilon_0}\left(\frac{-2q^2}{1} + \frac{-2q^2}{2} + \frac{q^2}{3}\right)$$

$$U = \frac{q^2}{4\pi\varepsilon_0}\left(-2 - 1 + \frac{1}{3}\right)$$

$$U = \frac{q^2}{4\pi\varepsilon_0}\left(-\frac{8}{3}\right)$$

$$U = -\frac{2q^2}{3\pi\varepsilon_0}$$

Ruling out the others: options (A) and (C) are positive and ignore the dominant attractive pairs; option (D) $-\frac{5q^2}{4\pi\varepsilon_0}$ comes from using a wrong separation (for example treating all pairs at 1 m).

Hence the potential energy of the system is $\mathbf{-\frac{2q^2}{3\pi\varepsilon_0}}$, which is option (B).

COMMON MISTAKE

Each pair must be divided by its own separation. A common slip is to use one common distance for all three pairs.

Final answer. The potential energy of the system is $\mathbf{-\frac{2q^2}{3\pi\varepsilon_0}}$, option **(B)**.

← *Back to Q7*

Q8

A point charge of 5 μC *is located at* (1 cm, 1 cm) *in x-y plane. A second charge of* 6 μC *is brought from infinity to point* (−3 cm, 4 cm). *The work done in the process is:*

(A) 0·54 J

(B) **5·4 J**

(C) −0·54 J

(D) 10·8 J

[2026 • Set 55-2-3]

SOLUTION

The work done in bringing a charge q_2 from infinity to a point near a fixed charge q_1 equals q_2 times the potential set up by q_1 at that point.

$$W = q_2 V_1 = q_2 \cdot \frac{kq_1}{r}$$

First find the distance between the two locations (1 cm, 1 cm) and (−3 cm, 4 cm):

$$r = \sqrt{(1-(-3))^2 + (1-4)^2}\ \text{cm}$$

$$r = \sqrt{(4)^2 + (-3)^2}\ \text{cm}$$

$$r = \sqrt{16+9} = \sqrt{25} = 5\ \text{cm} = 0.05\ m$$

Next find the potential of the 5 μC charge at the arrival point:

$$V_1 = \frac{kq_1}{r} = \frac{(9\times 10^9)(5\times 10^{-6})}{0.05}$$

$$V_1 = \frac{45\times 10^3}{0.05} = 9\times 10^5\ V$$

Now the work to bring the 6 μC charge:

$$W = q_2 V_1 = (6\times 10^{-6})(9\times 10^5)$$

$$W = 5.4\ J$$

Ruling out the others: option (A) 0.54 *J* is off by a factor of ten (a unit-conversion slip on *r*); option (C) has the wrong sign, but both charges are positive so the work is positive; option (D) 10.8 *J* doubles the result.

Hence the work done is **5.4 *J***, which is option (B).

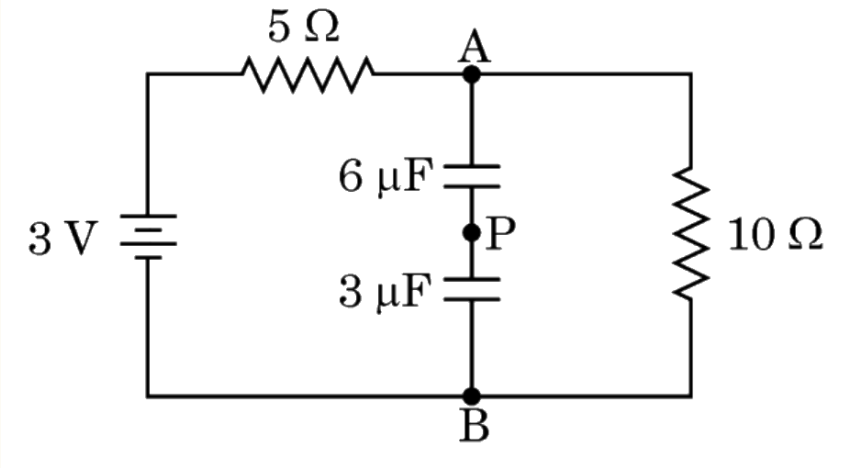
Final answer. The work done is 5.4 J, option **(B)**.

← *Back to Q8*

Q9

Case Study: *Capacitors are manufactured with certain standard capacitances and working voltages. However, these standard values may not be the ones that are actually needed in a particular application. Two or more capacitors can be grouped in series or in parallel to achieve desired capacitance and voltage. When connected in series, the total capacitance decreases while the voltage rating increases, whereas in parallel connections, the total capacitance increases and maintains the same voltage*

rating. A capacitor stores energy in the electric field between its plates and stored energy is proportional to the square of the voltage and capacitance $U = \frac{1}{2}CV^2$, *where symbols have their usual meanings.*



Two capacitors, one of 3 μF *and the other of* 6 μF, *are connected in series in the circuit as shown in the figure, for a long time.*
The total capacitance of the circuit is:

(A) 6 μF

(B) 3 μF

(C) 9 μF

(D) **2 μF**

[2026 • Set 55-3-1]

SOLUTION

For capacitors joined in series, the reciprocal of the total capacitance equals the sum of the reciprocals of the individual capacitances. For two capacitors this simplifies to the product-over-sum form.

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\Rightarrow C_s = \frac{C_1 C_2}{C_1 + C_2}$$

Here $C_1 = 3\ \mu F$ and $C_2 = 6\ \mu F$.

$$C_s = \frac{(3)(6)}{3 + 6}$$

$$C_s = \frac{18}{9}$$

$$C_s = 2\ \mu F$$

In steady state, the capacitors are fully charged and no current flows, so the 10 Ω resistor does not affect the equivalent capacitance.

Ruling out the others: option (A) 6 μF and option (C) 9 μF would arise from a parallel combination ($C_1 + C_2$); option (B) 3 μF wrongly takes only one capacitor. A series combination must give a value smaller than either capacitor, and 2 μF is indeed less than 3 μF.

Hence the total capacitance of the circuit is **2 μF**, which is option (D).

TIP

A series combination is always smaller than the smallest capacitor in it. Use this as a quick check on your answer.

Final answer. The total capacitance is 2 **μF**, option **(D)**.

← *Back to Q9*

Q10

The current in the 10 Ω *resistor is:*

(A) 0·3 *A*

(B) 0·6 *A*

(C) 0·2 *A*

(D) **0**

[2026 • Set 55-3-1]

SOLUTION

In the steady state, that is after the circuit has been connected for a long time, the capacitors are fully charged. A fully charged capacitor in a DC circuit blocks any further flow of charge, so it behaves like an open switch (break) in that branch.

The current through a charging branch decays exponentially with time:

$$i(t) = \frac{V}{R}\, e^{-t/RC}$$

As $t \to \infty$, the exponential factor tends to zero:

$$i(\infty) = \frac{V}{R}\, e^{-\infty} = 0$$

Since the 10 Ω resistor lies in the same series loop as the two capacitors, the same current passes through it. With the capacitor branch carrying no current in the steady state, the resistor current is also zero.

$$I_{10\,\Omega} = 0$$

Ruling out the others: options (A) 0.3 *A*, (B) 0.6 *A* and (C) 0.2 *A* are nonzero and would apply only during the brief transient charging, not in the long-time steady state the question describes.

Hence the steady-state current through the 10 Ω resistor is **zero**, which is option (D).

Final answer. The steady-state current through the 10 Ω resistor is zero, option **(D)**.

← *Back to Q10*

Q11

The potential difference between point A and B is:

(A) 2 *V*

(B) 0·3 *V*

(C) 0·2 *V*

(D) **3 *V***

[2026 • Set 55-3-1]

SOLUTION

In a series combination of capacitors, the same charge Q resides on every capacitor, since the charge displaced in the connecting wires must be common.

For the 3 μF and 6 μF capacitors in series, $V = \frac{Q}{C}$, so the potential difference across each is inversely proportional to its capacitance.

$$\frac{V_{3\mu F}}{V_{6\mu F}} = \frac{C_{6\mu F}}{C_{3\mu F}} = \frac{6}{3} = 2$$

Thus the 3 μF capacitor takes twice the voltage of the 6 μF capacitor.

With the same charge $Q = 6\ \mu C$ on both (from the network values), $V_{3\mu F} = \frac{6}{3} = 2$ V and $V_{6\mu F} = \frac{6}{6} = 1\ V$.

The total potential difference between A and B is the sum across the series elements:

$$V_{AB} = V_{3\mu F} + V_{6\mu F} = 2 + 1 = 3\ V$$

Hence the potential difference between A and B is **3 V**, matching option (D).

Final answer. The potential difference between A and B is 3 V, option **(D)**.

← *Back to Q11*

Q12

The value of charge on the plates of the 6 μF *capacitor is:*

(A) 6 μC

(B) 4 μC

(C) 12 μC

(D) 8 μC

[2026 • Set 55-3-1]

SOLUTION

In a series combination of capacitors, the charge on each capacitor is the same, because the inner plates form an isolated conductor that must remain neutral

overall.

The equivalent capacitance of 3 μF and 6 μF in series is

$$\frac{1}{C_{eq}} = \frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{3}{6}$$

$$C_{eq} = 2\ \mu F$$

The common charge equals the charge on the equivalent capacitor across the supply $\mathbf{V_{AB} = 3\ V}$:

$$Q = C_{eq}\, V_{AB} = (2\ \mu F)(3\ V) = 6\ \mu C$$

Since the same charge sits on each capacitor in series, the charge on the 6 μF capacitor is also 6 μC.

Cross-check: $\mathbf{V_{6\mu\,F} = \frac{Q}{C} = \frac{6}{6} = 1\ V}$ and $\mathbf{V_{3\mu\,F} = \frac{6}{3} = 2\ V}$, summing to $\mathbf{3\ V = V_{AB}}$, which is consistent.

Hence the charge on the 6 μF capacitor is **6 μC**, matching option (A).

COMMON MISTAKE

A common error is to compute Q = CV with C = 6 microfarad and V = 3 V, giving 18 microcoulomb. Remember the 3 V is the total supply voltage, not the voltage across the 6 microfarad capacitor alone.

Final answer. The charge on the 6 microfarad capacitor is 6 microcoulomb, option **(A)**.

← *Back to Q12*

Q13

The wire between two capacitors is cut at point P. The current in the circuit will:

(A) increase

(B) decrease

(C) **remain the same**

(D) first increase then become stable

[2026 • Set 55-3-1]

SOLUTION

A fully charged capacitor blocks direct current. Once both capacitors in the series branch have charged to their steady-state values, no further charge flows, so the steady-state current in the circuit is already zero.

$$I_{steady} = 0$$

When the wire between the two capacitors is cut at point *P*, the loop is broken and there is no closed conducting path. The current in an open circuit is also zero.

$$I_{after\ cut} = 0$$

Since the current was zero before the cut and remains zero after the cut, its value is unchanged.

Option (A) increase and option (B) decrease are wrong because a steady-state current of zero cannot change to a non-zero value or drop below zero. Option (D) is wrong because there is no transient build-up; the steady state is already established.

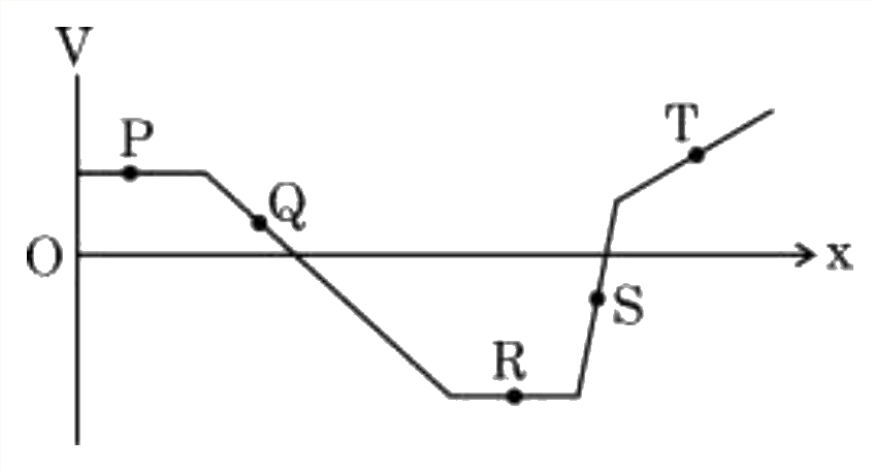
Hence the current in the circuit will **remain the same** (zero), matching option (C).

Final answer. The current remains the same (zero in steady state), option **(C)**.

← *Back to Q13*

Q14

The figure represents the variation of the electric potential V at a point in a region of space as a function of its position along the x-axis. A charged particle will experience the maximum force at:



(A) *P*

(B) *Q*

(C) ***R***

(D) *S*

[2026 • Set 55-5-1]

SOLUTION

The force on a charged particle in an electric field is given by $\vec{F} = q\vec{E}$, so its magnitude depends on the magnitude of the field E.

In one dimension, the electric field is the negative gradient of the potential:

$$E = -\frac{dV}{dx}$$

Therefore the magnitude of the field, and hence of the force, is largest where the slope $\left|\frac{dV}{dx}\right|$ of the V versus x graph is steepest.

Reading the graph, the curve rises or falls most sharply (it is closest to vertical) at the point R. At P, Q and S the curve is comparatively flat or gently sloping, so the field there is smaller.

Thus $\left|\frac{dV}{dx}\right|$ is maximum at R, giving the maximum $|E|$ and so the maximum $|\boldsymbol{F}| = q|\boldsymbol{E}|$ at R.

Hence the charged particle experiences the maximum force at **R**, matching option (C).

> **COMMON MISTAKE**
>
> *A flat portion of a V-x graph (zero slope) means zero field and zero force, not maximum. Always look for the steepest slope, not the highest or lowest value of V.*

Final answer. The maximum force acts at point R, option **(C)**.

← Back to Q14

Q15

The electric field $\vec{E}$ in the region between the plates is:

(A) $(2 \times 10^2 \frac{V}{m})\,\hat{k}$

(B) $-(2 \times 10^2 \frac{V}{m})\,\hat{k}$

(C) $(2 \times 10^4 \frac{V}{m})\,\hat{k}$

(D) $\boldsymbol{-(2 \times 10^4 \frac{V}{m})\,\hat{k}}$

[2026 • Set 55-5-1]

SOLUTION

For a uniform field between two parallel plates separated by distance d and held at a potential difference V, the magnitude of the field is

$$E = \frac{V}{d}$$

Substituting $\boldsymbol{V = 200\ V}$ and $\boldsymbol{d = 0.01\ m}$:

$$E = \frac{200}{0.01} = 2 \times 10^4 \frac{V}{m}$$

The electric field always points from the plate at higher potential towards the plate at lower potential. From the figure, the upper plate is at the higher potential, so $\vec{E}$ points vertically downward, that is, along $-\hat{k}$.

Combining magnitude and direction:

$$\vec{E} = -(2 \times 10^4 \ \frac{V}{m})\hat{k}$$

Options (A) and (B) have the wrong magnitude (they use d in cm not m). Option (C) has the correct magnitude but the wrong direction.

Hence $\vec{E} = -(2 \times 10^4 \ \frac{V}{m})\hat{k}$, matching option (D).

COMMON MISTAKE

Convert the plate separation to metres before dividing. Here d = 1 cm = 0.01 m gives 2 x 10^4 V/m; using 1 cm directly wrongly gives 2 x 10^2 V/m.

Final answer. The field is minus (2 x 10^4 V/m) k-hat, option **(D)**.

← Back to Q15

Q16

In the region between the plates, the electron moves with an acceleration $\vec{a}$ given by:

(A) $-(3.5 \times 10^{15}\ \text{ms}^{-2})\ \hat{k}$

(B) $(3.5 \times 10^{15}\ \text{ms}^{-2})\ \hat{k}$

(C) $(3.5 \times 10^{13}\ \text{ms}^{-2})\ \hat{i}$

(D) $-(3.5 \times 10^{13}\ \text{ms}^{-2})\ \hat{i}$

[2026 • Set 55-5-1]

SOLUTION

The force on the electron in the field is $\vec{F} = q\vec{E}$, where for an electron $q = -e = -1.6 \times 10^{-19}\ C$.

Using $\vec{E} = -(2 \times 10^4)\hat{k}\ \frac{V}{m}$:

$$\vec{F} = (-e)[-(2\times10^{4})\hat{k}] = +(2\times10^{4})e\,\hat{k}$$

$$\vec{F} = +(2\times10^{4})(1.6\times10^{-19})\hat{k} = +3.2\times10^{-15}\,\hat{k}\,N$$

The acceleration follows from Newton's second law, $\vec{a} = \dfrac{\vec{F}}{m_e}$, with $\mathbf{m_e = 9.1\times10^{-31}\ kg}$:

$$\vec{a} = \frac{3.2\times10^{-15}}{9.1\times10^{-31}}\,\hat{k}$$

$$\vec{a} \approx +3.5\times10^{15}\,\hat{k}\ \mathrm{ms^{-2}}$$

The negative charge of the electron reverses the direction of force relative to $\vec{E}$, so the acceleration is along $+\hat{k}$, ruling out the negative-$\hat{k}$ option (A). Options (C) and (D) wrongly point along $\hat{i}$ with the wrong magnitude.

Hence $\vec{a} = \mathbf{(3.5\times10^{15}\ ms^{-2})\hat{k}}$, matching option (B).

Final answer. The acceleration is (3.5 x 10^15 m/s^2) k-hat, option **(B)**.

← *Back to Q16*

Q17

(a) Time interval during which an electron moves through the region between the plates is:

(A) $9.0\times10^{-9}\ s$

(B) $1.67\times10^{-8}\ s$

(C) $\mathbf{1.67\times10^{-9}\ s}$

(D) $2.17\times10^{-9}\ s$

[2026 • Set 55-5-1]

SOLUTION

Inside the plates the electric force on the electron is purely vertical (along $\hat{k}$), so it does not change the horizontal velocity. The horizontal speed therefore stays constant at $\mathbf{v_x = 3\times10^{7}\ ms^{-1}}$.

The time to cross the plate length L horizontally is

$$t = \frac{L}{v_x}$$

Substituting $L = 0.05\ m$ and $v_x = 3 \times 10^7\ ms^{-1}$:

$$t = \frac{0.05}{3 \times 10^7}$$

$$t = \frac{5 \times 10^{-2}}{3 \times 10^7} = 1.67 \times 10^{-9}\ s$$

The vertical electric force affects only the vertical motion and not the time of flight, which is fixed by the horizontal travel.

Hence the time interval the electron spends between the plates is $\mathbf{1.67 \times 10^{-9}\ s}$, matching option (C).

CONCEPT

The time of flight depends only on the horizontal velocity and plate length. The vertical field does not enter this calculation, just as gravity does not affect the horizontal travel time of a projectile.

Final answer. The time interval is 1.67 x 10^-9 s, option **(C)**.

← *Back to Q17*

Q18

(b) The vertical displacement of the electron which travels through the region between the plates is:

(A) 1.0 mm

(B) 4.9 mm

(C) 5.9 mm

(D) 3.0 mm

[2026 • Set 55-5-1]

SOLUTION

The electron enters with zero vertical velocity and then accelerates vertically with constant acceleration $a = 3.5 \times 10^{15}\ \mathrm{ms}^{-2}$ for the time $t = 1.67 \times 10^{-9}\ s$ found earlier.

For motion from rest under constant acceleration, the displacement is

$$y = \frac{1}{2} a\, t^2$$

Substituting the values:

$$y = \frac{1}{2}(3.5 \times 10^{15})(1.67 \times 10^{-9})^2$$

$$(1.67 \times 10^{-9})^2 = 2.79 \times 10^{-18}\ \mathrm{s}^2$$

$$y = \frac{1}{2}(3.5 \times 10^{15})(2.79 \times 10^{-18})$$

$$y = \frac{1}{2}(9.77 \times 10^{-3})$$

$$y \approx 4.88 \times 10^{-3}\ m \approx 4.9\ \mathrm{mm}$$

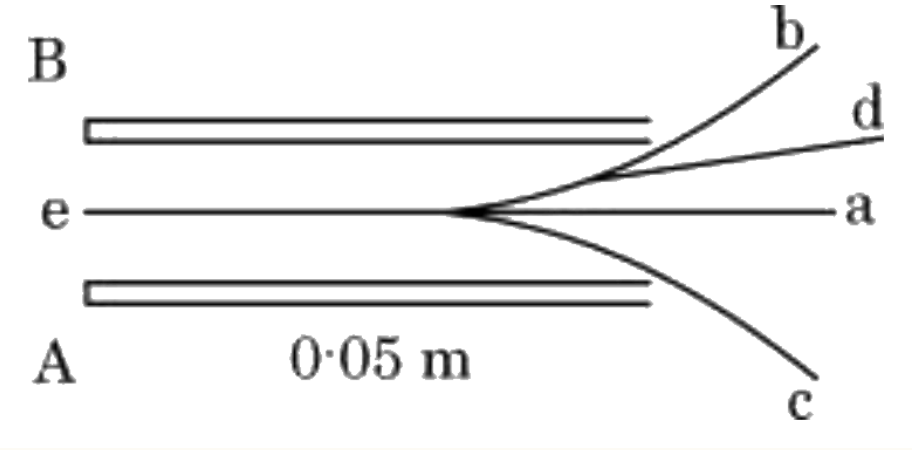
Hence the vertical displacement of the electron as it crosses the plates is approximately **4.9 mm**, matching option (B).

Final answer. The vertical displacement is about 4.9 mm, option **(B)**.

← *Back to Q18*

Q19

Which one of the following is the path traced by the electron in between the two plates?



(A) a

(B) b

(C) c

(D) d

[2026 • Set 55-5-1]

SOLUTION

Inside the plates the electron has two independent motions: a uniform horizontal velocity v_x (no horizontal force) and a uniformly accelerated vertical motion (constant electric force, no magnetic force).

The horizontal position grows linearly with time:

$$x(t) = v_x\, t$$

The vertical position grows quadratically with time:

$$y(t) = \frac{1}{2} a_y\, t^2$$

Eliminating t using $t = \frac{x}{v_x}$:

$$y = \frac{a_y}{2v_x{}^2}\, x^2$$

Since y is proportional to x^2, the path is a parabola, exactly like a horizontally launched projectile under gravity. The parabola curves in the direction of the acceleration, which here is along $+\hat{k}$ (the electron deflects towards the upper plate).

A straight-line path would require zero transverse force, and a circular path would require a force of constant magnitude always perpendicular to the velocity, neither of which holds here. Only the parabolic curve *b* fits.

Hence the path traced by the electron is the parabola **b**, matching option (B).

Final answer. The electron follows the parabolic path b, option **(B)**.

← *Back to Q19*

Q20

Two points R and S are equidistant from two charges +Q and −2Q. The work done in moving a charge −Q from point R to S is:

(A) Zero

(B) $\frac{Q}{4\pi\varepsilon_0 d}$

(C) $\frac{2Q}{4\pi\varepsilon_0 d}$

(D) $\frac{3Q}{4\pi\varepsilon_0 d}$

[2026 • Set 55-5-3]

SOLUTION

The electric potential at a point due to several charges is the algebraic sum of the potentials due to each charge (superposition).

Since *R* and *S* are equidistant from +*Q* and equidistant from −2Q, the potential each source charge produces at *R* equals the potential it produces at *S*.

$$V_R = \frac{1}{4\pi\varepsilon_0}\left(\frac{+Q}{r_1} + \frac{-2Q}{r_2}\right) = V_S$$

Therefore the two points are at the same potential:

$$V_R = V_S \Rightarrow \Delta V = V_S - V_R = 0$$

The work done in moving a charge $q = -Q$ between two points is

$$W = q\,\Delta V$$

$$W = (-Q)(V_S - V_R) = (-Q)(0) = 0$$

The non-zero options (B), (C) and (D) ignore that equal distances from each source make the two potentials equal; the work depends only on the potential difference, which vanishes here.

Hence the work done is **zero**, matching option (A).

CONCEPT

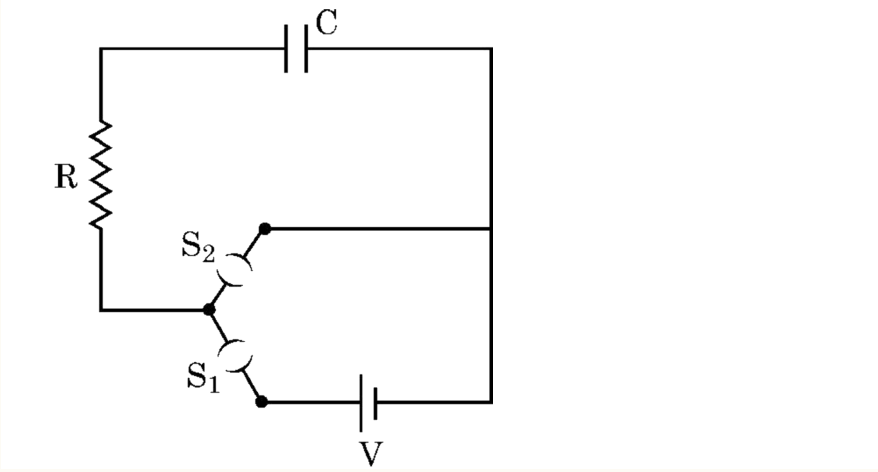
Work done by the electric force depends only on the potential difference between the end points, not on the path. Equal potentials at R and S force the work to be zero regardless of the sign of the moving charge.

Final answer. The work done is zero, option **(A)**.

← *Back to Q20*

Q21

Case Study (Q29): *A circuit consisting of a capacitor C, a resistor of resistance R and an ideal battery of emf V, as shown in figure, is known as an RC series circuit. As soon as the circuit is completed by closing key* S_1 *(keeping* S_2 *open), charges begin to flow between the capacitor plates and the battery terminals. The charge on the capacitor increases and consequently the potential difference* $\mathbf{V_c\ (= q/C)}$ *across the capacitor also increases with time. When this potential difference equals the potential difference across the battery, the capacitor is fully charged (*$\mathbf{Q = VC}$*). During this process of charging, the charge q on the capacitor changes with time t as* $\mathbf{q = Q[1 - e^{-t/RC}]}$*. The charging current can be obtained by differentiating it and using* $\frac{d}{dx}(e^{mx}) = m\, e^{mx}$*. Consider the case when* $\mathbf{R = 20\ k\Omega}$*,* $\mathbf{C = 500\ \mu F}$ *and* $\mathbf{V = 10\ V}$*.*



(i) *The final charge on the capacitor, when key* S_1 *is closed and* S_2 *is open, is:*

(A) 5 μC

(B) **5 mC**

(C) 25 mC

(D) 0.1 *C*

[2025 • Set 55-1-1]

SOLUTION

When key S_1 is closed and S_2 is kept open, the capacitor *C* charges through the resistor *R* towards the battery emf *V*. After a long time (steady state) the charging current falls to zero, the full battery voltage appears across the capacitor, and the capacitor is said to be fully charged.

At this stage the final charge is given by the basic capacitor relation $Q = CV$.

Substituting the given values $C = 500\ \mu F = 500\times 10^{-6}\ F$ and $V = 10\ V$:

$$Q = (500\times 10^{-6})\times (10)$$

$$Q = 5\times 10^{-3}\ C = 5\ \text{mC}$$

This matches option (B). Option (A) 5 μC wrongly keeps the μ prefix; option (C) 25 mC and option (D) 0.1 *C* do not come from $Q = CV$ with these data.

Hence the final charge on the capacitor is **5 mC**.

Final answer. The final charge on the capacitor is 5 mC, so the answer is option **(B)**.

← Back to Q21

Q22

(ii) *For sufficient time the key* S_1 *is closed and* S_2 *is open. Now* S_2 *is closed and* S_1 *is open. What is the final charge on the capacitor?*

(A) Zero

(B) 5 mC

(C) 2.5 mC

(D) 5 μC

[2025 • Set 55-1-1]

SOLUTION

Before this step the capacitor was fully charged to $Q = 5$ **mC** through the battery (with S_1 closed, S_2 open).

Now S_1 is opened and S_2 is closed. The battery is disconnected, and the charged capacitor is connected across only the resistor R. This forms a discharging RC loop, so the capacitor drives a current through R and loses its charge.

The charge during discharge decays exponentially as

$$q(t) = Q\, e^{-t/RC}$$

As $t \to \infty$, the exponential factor $e^{-t/RC} \to 0$, so the charge tends to zero.

$$q_{final} = Q\, e^{-\infty} = 0$$

Physically, the capacitor keeps pushing current through R until no potential difference is left across it, i.e. it is completely discharged.

This matches option (A). Option (B) 5 mC is the charge before discharging; options (C) and (D) are non-zero intermediate-looking values that do not represent the final state.

Hence the final charge on the capacitor is **zero**.

Final answer. After full discharge the charge is zero, so the answer is option **(A)**.

← *Back to Q22*

Q23

(iii) *The dimensional formula for* RC *is:*

(A) $[M\,L^2\,T^{-3}\,A^{-2}]$

(B) $[M^0\,L^0\,T^{-1}\,A^0]$

(C) $[M^{-1}\,L^{-2}\,T^4\,A^2]$

(D) $\mathbf{[M^0\,L^0\,T\,A^0]}$

[2025 • Set 55-1-1]

SOLUTION

The product RC is the time constant τ of an RC circuit, and a time constant must have the dimension of time. This is easiest to see from the charging law $q(t) = Q(1 - e^{-t/RC})$: the exponent $\frac{t}{RC}$ must be a pure number, so RC must carry the same dimension as time t.

We can also confirm it from the dimensions of R and C separately.

For resistance, $R = \frac{V}{I}$, giving $[R] = [M\,L^2\,T^{-3}\,A^{-2}]$.

For capacitance, $C = \frac{Q}{V}$, giving $[C] = [M^{-1}\,L^{-2}\,T^4\,A^2]$.

Multiplying the two:

$$[RC] = [M\,L^2\,T^{-3}\,A^{-2}]\times[M^{-1}\,L^{-2}\,T^4\,A^2]$$

$$[RC] = [M^0\,L^0\,T^1\,A^0]$$

So $[RC] = [T]$, the dimension of time, which matches option (D). Option (B) $[M^0L^0T^{-1}A^0]$ is the dimension of frequency (1/RC), not RC; options (A) and (C) are the dimensions of R and C individually.

Hence the dimensional formula for RC is $\mathbf{[M^0\,L^0\,T\,A^0]}$.

Final answer. RC has the dimension of time, [M^0 L^0 T A^0], so the answer is option **(D)**.

← *Back to Q23*

Q24

(iv) *The key* S_1 *is closed and* S_2 *is open. The value of current in the resistor after* 5 *seconds is:*

(A) $\frac{1}{2\sqrt{e}}$ **mA**

(B) $\sqrt{e}$ mA

(C) $\frac{1}{\sqrt{e}}$ mA

(D) $\frac{1}{2e}$ mA

[2025 • Set 55-1-1]

SOLUTION

While charging, the current is obtained by differentiating the charge $q(t) = Q(1 - e^{-t/RC})$ with respect to time. Using $\frac{d}{dt}(e^{mt}) = m\,e^{mt}$ with $Q = CV$:

$$i(t) = \frac{dq}{dt} = \frac{V}{R}\,e^{-t/RC}$$

First find the time constant:

$$RC = (20\times 10^{3})\times(500\times 10^{-6})$$

$$RC = 10\ s$$

The initial current at $t = 0$ is

$$i_0 = \frac{V}{R} = \frac{10}{20\times 10^{3}} = 0.5\times 10^{-3}\,A = 0.5\ \text{mA}$$

At $t = 5\ s$, the exponent is $\frac{t}{RC} = \frac{5}{10} = 0.5$, so $e^{-t/RC} = e^{-0.5} = \frac{1}{\sqrt{e}}$.

Therefore the current after 5 s is

$$i(5) = 0.5\ \text{mA} \times \frac{1}{\sqrt{e}}$$

$$i(5) = \frac{1}{2\sqrt{e}}\ \text{mA}$$

This matches option (A). Options (B) and (C) drop the factor $\mathbf{0.5 = \frac{1}{2}}$; option (D) wrongly uses e instead of $\sqrt{e}$ (it would correspond to $\boldsymbol{t = RC}$, not $\boldsymbol{t = RC/2}$).

Hence the current in the resistor after 5 s is $\mathbf{\frac{1}{2\sqrt{e}}\ mA}$.

COMMON MISTAKE

At t = 5 s here, t/RC = 0.5 (not 5), since RC = 10 s. A common slip is to use e^{-5} by forgetting to divide by the time constant.

Final answer. The current after 5 s is 1/(2√e) mA, so the answer is option **(A)**.

← *Back to Q24*

Q25

(iv) OR: *The key* S_1 *is closed and* S_2 *is open. The initial value of charging current in the resistor is:*

(A) 5 mA

(B) **0.5 mA**

(C) 2 mA

(D) 1 mA

[2025 • Set 55-1-1]

SOLUTION

At the instant the switch is closed ($\boldsymbol{t = 0}$), the capacitor is still uncharged, so the potential difference across it is zero ($\mathbf{V_C = q/C = 0}$).

With no voltage across the capacitor, the entire battery emf V drops across the resistor R. The initial current is therefore limited only by R and follows from Ohm's law.

$$i_0 = \frac{V}{R}$$

Substituting $V = 10\ V$ and $R = 20\ k\Omega = 20\times 10^3\ \Omega$:

$$i_0 = \frac{10}{20\times 10^3}$$

$$i_0 = 5\times 10^{-4}\ A = 0.5\ mA$$

This matches option (B). Option (A) 5 mA comes from misreading R as 2 kΩ; options (C) and (D) do not follow from V/R with the given data.

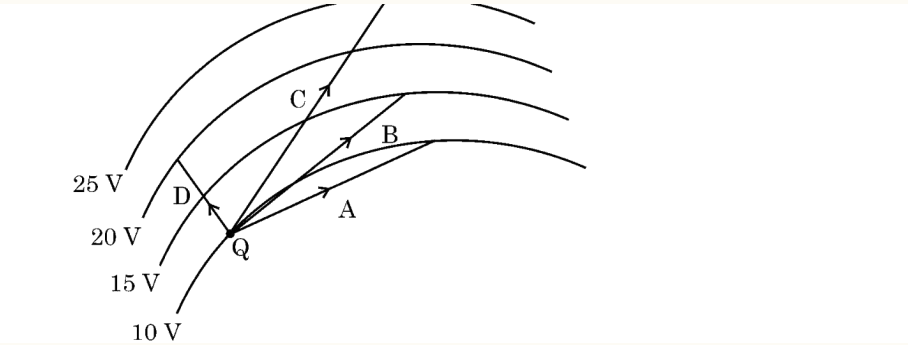
Hence the initial charging current in the resistor is **0.5 mA**.

Final answer. The initial charging current is 0.5 mA, so the answer is option **(B)**.

← *Back to Q25*

Q26

In the figure curved lines represent equipotential surfaces. A charge Q is moved along different paths A, B, C and D. The work done on the charge will be maximum along the path:



(A) *A*

(B) *B*

(C) *C*

(D) ***D***

[2025 • Set 55-1-2]

SOLUTION

The work done in moving a charge in an electrostatic field is a state function: it depends only on the potentials at the start and end points, not on the path taken.

$$W = Q\,(\mathrm{V}_f - \mathrm{V}_i) = Q\,\Delta\,V$$

So the work is maximum for the path whose two endpoints lie on equipotential surfaces with the largest potential difference between them.

From the figure the equipotential surfaces are labelled 25 V, 20 V, 15 V and 10 V. Path D begins on the 25 V surface and ends on the 10 V surface, spanning

$$\Delta\,\mathrm{V}_D = 25 - 10 = 15\ V$$

This is the largest potential difference among the four paths. Paths A, B and C have endpoints separated by fewer equipotential bands, giving smaller $\Delta\,V$ and hence less work. (A path that starts and ends on the same equipotential would do zero work.)

Hence the work done on the charge is maximum along path ***D***.

CONCEPT

Work depends only on the end equipotentials, never on the path length or shape. The longest-looking path is not automatically the answer.

Final answer. The work is maximum where ΔV is largest, along path D, so the answer is option **(D)**.

← Back to Q26

Q27

Case study: *A parallel plate capacitor has two parallel plates which are separated by an insulating medium like air, mica, etc. When the plates are connected to the terminals of a battery, they get equal and opposite charges and an electric field is set up in between them. This electric field between the two plates depends upon the potential difference applied, the separation of the plates and nature of the medium between the plates.*

The electric field between the plates of a parallel plate capacitor is E. Now the separation between the plates is doubled and simultaneously the applied potential difference between the plates is reduced to half of its initial value. The new value of the electric field between the plates will be:

(A) E

(B) 2E

(C) $\frac{E}{4}$

(D) $\frac{E}{2}$

[2025 • Set 55-2-1]

SOLUTION

Inside a parallel plate capacitor the field is uniform and related to the applied potential difference V and plate separation d by

$$E = \frac{V}{d}$$

Initially the field is $E = \frac{V}{d}$.

Now the separation is doubled, $d' = 2d$, and the potential difference is halved, $V' = \frac{V}{2}$. The new field is

$$E' = \frac{V'}{d'} = \frac{V/2}{2d}$$

$$E' = \frac{V}{4d}$$

Since $\frac{V}{d} = E$, this gives

$$E' = \frac{E}{4}$$

This matches option (C). Option (A) E would need V and d to change in the same ratio; option (B) 2E and option (D) $\frac{E}{2}$ ignore one of the two changes.

Hence the new electric field between the plates is $\frac{E}{4}$.

Final answer. The new field is E/4, so the answer is option **(C)**.

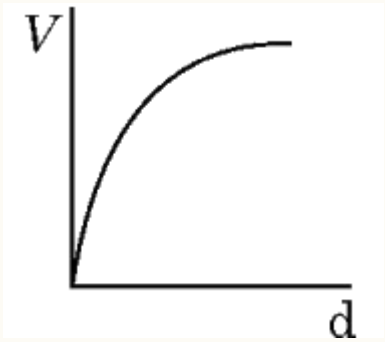
← *Back to Q27*

Q28

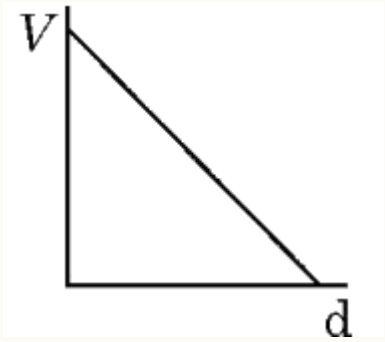
A constant electric field is to be maintained between the two plates of a capacitor whose separation d changes with time. Which of the graphs correctly depict the

potential difference (V) to be applied between the plates as a function of separation between the plates (d) to maintain the constant electric field?

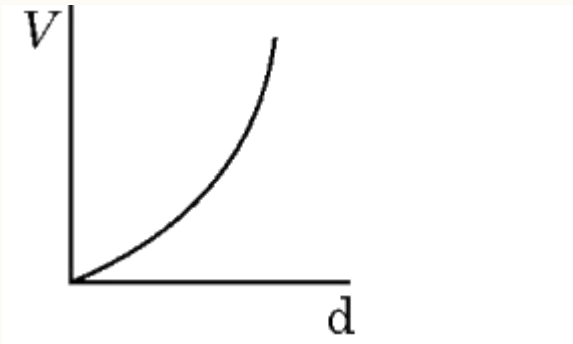
(A)



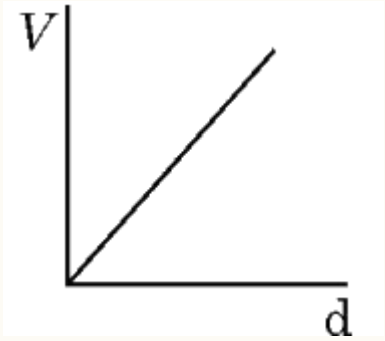
(B)



(C)



(D)



[2025 • Set 55-2-1]

SOLUTION

Between the plates of a parallel plate capacitor the field is uniform and given by

$$E = \frac{V}{d} \Rightarrow V = E\,d$$

To keep the electric field E constant while the separation d changes, the applied voltage must satisfy

$$V = (\text{constant}) \times d$$

with the constant equal to the fixed value of E. This is the equation of a straight line passing through the origin, with a positive slope equal to E.

So the correct graph of V versus d is a straight line starting from the origin and rising linearly (greater separation needs proportionally greater voltage). A horizontal line, a curve, or a line with a non-zero intercept would not keep $\boldsymbol{E = V/d}$ constant.

This linear-through-origin graph is option (A).

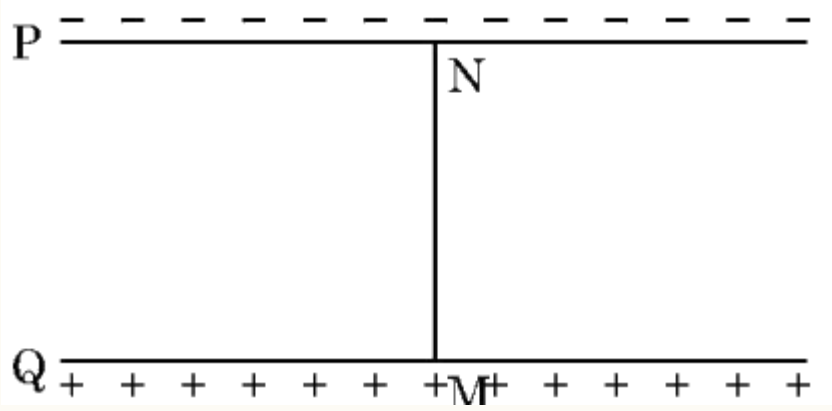
Hence the correct depiction is the straight line through the origin, option **A**.

Final answer. V must vary linearly with d through the origin, so the answer is option **(A)**.

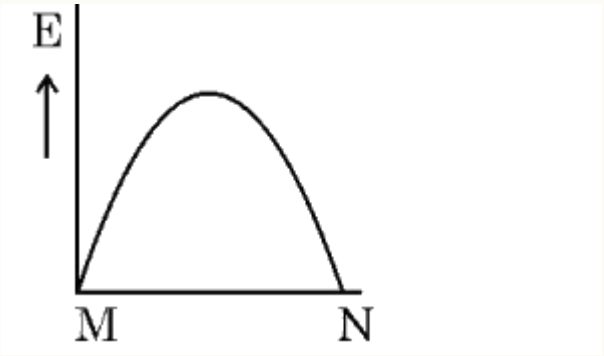
← *Back to Q28*

Q29

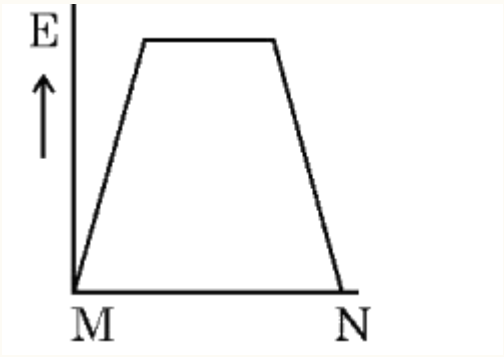
In the above figure P, Q are the two parallel plates of a capacitor. Plate Q is at positive potential with respect to plate P. MN is an imaginary line drawn perpendicular to the plates. Which of the graphs shows correctly the variations of the magnitude of electric field strength E along the line MN?



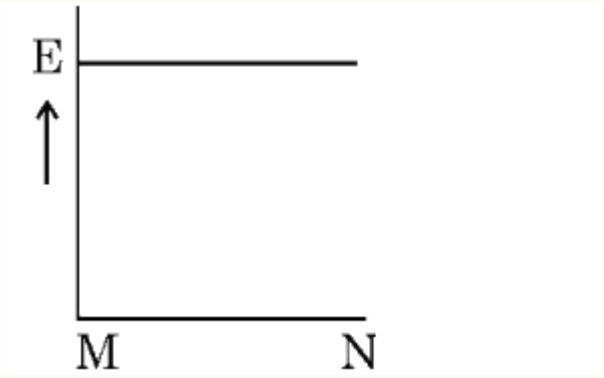
(A)



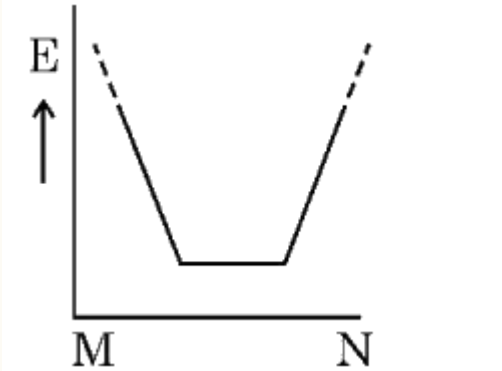
(B)



(C)



(D)



[2025 • Set 55-2-1]

SOLUTION

Treat the capacitor as two large parallel charged sheets: plate Q positive and plate P negative (since Q is at higher potential). Each sheet produces a field of magnitude $\frac{\sigma}{2\varepsilon_0}$ on both sides.

Outside both plates (in the region before plate P and after plate Q along line MN), the fields of the two oppositely charged sheets point in opposite directions and cancel:

$$E_{\text{outside}} = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = 0$$

In the region between the plates, the two fields add up and the field is uniform:

$$E_{\text{between}} = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$$

So a plot of the magnitude $|E|$ along the line MN looks like a rectangular pulse: it is zero before the first plate, jumps up to a constant value $\frac{\sigma}{\varepsilon_0}$ throughout the gap between the plates, and drops back to zero after the second plate.

This rectangular-pulse (flat top, zero outside) graph is option (C). Graphs showing a field that varies inside the gap, or a non-zero field outside, are incorrect for an ideal parallel plate capacitor.

Hence the correct variation of $|E|$ along MN is option **C**.

Final answer. $|E|$ is zero outside and a constant σ/ε_0 inside, a rectangular pulse, so the answer is option **(C)**.

← Back to Q29

Q30

Three parallel plates are placed above each other with equal displacement d between neighbouring plates. The electric field between the first pair of the plates is $\vec{E}_1$ and the electric field between the second pair of the plates is $\vec{E}_2$. The potential difference between the third and the first plate is

(A) $(\vec{E}_1+\vec{E}_2)\cdot\vec{d}$

(B) $(\vec{E}_1-\vec{E}_2)\cdot\vec{d}$

(C) $(\vec{E}_2-\vec{E}_1)\cdot\vec{d}$

(D) $\frac{\vec{d}(\vec{E}_1+\vec{E}_2)}{2}$

[2025 • Set 55-2-1]

SOLUTION

For a uniform field, the potential difference between two points separated by a displacement $\vec{d}$ is $\Delta V = \vec{E}\cdot\vec{d}$ in magnitude. Here there are two adjacent gaps, each of separation $\vec{d}$: between plates 1 and 2 the field is $\vec{E}_1$, and between plates 2 and 3 the field is $\vec{E}_2$.

The potential difference between plate 3 and plate 1 is the sum of the potential drops across the two gaps, traversed one after the other in the same direction:

$$V_3 - V_1 = \vec{E}_1\cdot\vec{d} + \vec{E}_2\cdot\vec{d}$$

Taking out the common displacement $\vec{d}$:

$$V_3 - V_1 = (\vec{E}_1 + \vec{E}_2)\cdot\vec{d}$$

This matches option (A). Options (B) and (C) subtract the fields, which would apply only if the two regions were traversed in opposite senses; option (D) wrongly averages the two fields instead of adding the two equal-width contributions.

Hence the potential difference between the third and first plate is $(\vec{E}_1+\vec{E}_2)\cdot\vec{d}$.

Final answer. The potential difference is $(E_1 + E_2)\cdot d$, so the answer is option **(A)**.

← *Back to Q30*

Q31

A material of dielectric constant K is filled in a parallel plate capacitor of capacitance C. The new value of its capacitance becomes

(A) C

(B) $\frac{C}{K}$

(C) **CK**

(D) $C(1+\frac{1}{K})$

[2025 • Set 55-2-1]

SOLUTION

When a dielectric slab of dielectric constant K completely fills the gap between the plates of a parallel plate capacitor, the capacitance increases by the factor K.

For an air capacitor, $C = \frac{\varepsilon_0 A}{d}$.

After the dielectric is filled, $\mathbf{C_{new}} = \frac{\mathbf{K}\varepsilon_0 A}{d} = K(\frac{\varepsilon_0 A}{d})$.

Therefore $\mathbf{C_{new} = KC = CK}$.

Physically, the dielectric gets polarised and produces an internal field that opposes the applied field, so the net field and hence the potential difference V drops for the same charge Q. Since $C = Q/V$, a smaller V means a larger capacitance.

Ruling out the others: option (A) C would mean no change, but the dielectric does change it; option (B) $\frac{C}{K}$ is a decrease, which is wrong because $K>1$ always raises capacitance; option (D) $C(1+\frac{1}{K})$ does not correspond to a fully filled slab.

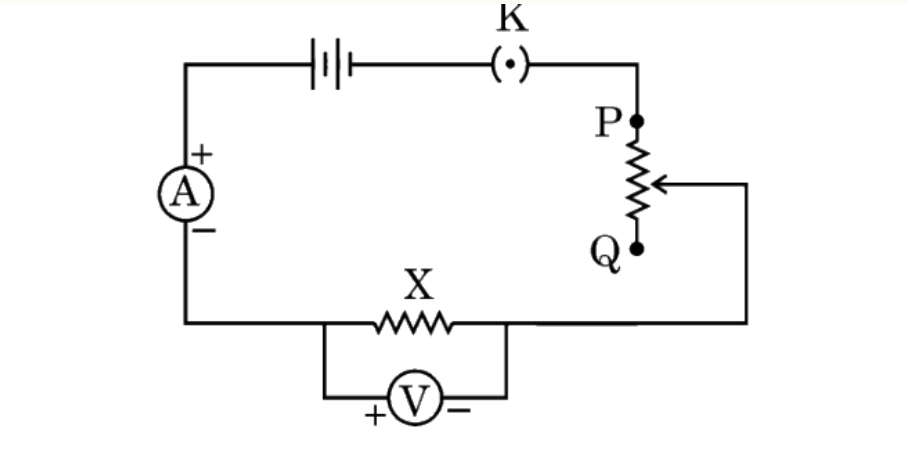
So **the new capacitance is CK**.

Final answer. The new capacitance is **CK**, so the answer is option **(C)**.

← *Back to Q31*

Q32

Case Study (Q29): *A parallel plate capacitor consists of two conducting plates kept generally parallel to each other at a distance. When the capacitor is charged, the charge resides on the inner surfaces of the plates and an electric field is set up between them. Thus, electrostatic energy is stored in the capacitor. The figure shows three large square metallic plates, each of side 'L' held parallel and equidistant from each other. The space between* P_1 *and* P_2 *and* P_2 *and* P_3 *is completely filled with mica sheets of dielectric constant 'K'. The plate* P_2 *is connected to point A and other plates* P_1 *and* P_3 *are connected to point B. Point A is maintained at a positive potential with respect to point B and the potential difference between A and B is V.*



(i) *The capacitance of the system between A and B will be :*

(A) $\frac{\varepsilon_0 K L^2}{d}$

(B) $\frac{\varepsilon_0 K L^2}{2d}$

(C) $\boldsymbol{\frac{2\varepsilon_0 K L^2}{d}}$

(D) $\frac{2\varepsilon_0 K d}{L^2}$

[2025 • Set 55-4-1]

SOLUTION

Plate P_2 is at node A, while plates P_1 and P_3 are both at node B. So the gap between P_1 and P_2 forms one capacitor, and the gap between P_2 and P_3 forms a second capacitor, and both are connected between the same two nodes A and B. This means the two capacitors are in **parallel**.

Each capacitor has plate area $A = L^2$, plate separation d, and is filled with mica of dielectric constant K.

$$C_{12} = C_{23} = \frac{K\varepsilon_0 L^2}{d}.$$

For capacitors in parallel, the equivalent capacitance is the sum.

$$C_{AB} = C_{12} + C_{23} = \frac{K\varepsilon_0 L^2}{d} + \frac{K\varepsilon_0 L^2}{d}.$$

$$C_{AB} = \frac{2\varepsilon_0 K L^2}{d}.$$

Options (A) and (B) ignore the parallel combination (single capacitor or half the value), and (D) has the wrong dimensions (d in the numerator). So **the capacitance of the system between A and B is $\frac{2\varepsilon_0 K L^2}{d}$.**

TIP

Whenever the middle plate is one node and the outer plates share the other node, the two gaps are in parallel, never in series. Series would need charge to flow through the plates in a single path.

Final answer. The capacitance between A and B is $\frac{2\varepsilon_0 K L^2}{d}$, so the answer is option **(C)**.

← *Back to Q32*

Q33

(ii) *The charge on plate* P_2 *is :*

(A) $\dfrac{\varepsilon_0 V K L^2}{2d}$

(B) $\dfrac{\varepsilon_0 V K L^2}{d}$

(C) $\boldsymbol{\dfrac{2\varepsilon_0 V K L^2}{d}}$

(D) $\dfrac{\varepsilon_0 V K L^2}{4d}$

[2025 • Set 55-4-1]

SOLUTION

Plate P_2 is connected to node A, and the whole system stores charge $Q = C_{AB}\,V$, where C_{AB} is the equivalent capacitance found in part (i).

From part (i), $\mathbf{C_{AB} = \dfrac{2\varepsilon_0 K L^2}{d}}$.

The total charge supplied to node A sits on plate P_2, since P_2 is the only plate on that node.

$$\mathbf{Q_{P_2} = C_{AB}\,V = \frac{2\varepsilon_0 K L^2}{d} \times V.}$$

$$\mathbf{Q_{P_2} = \frac{2\varepsilon_0 V K L^2}{d}}.$$

This charge is shared by the two faces of P_2: a charge $\dfrac{\varepsilon_0 V K L^2}{d}$ on the upper face (paired with P_1) and an equal charge on the lower face (paired with P_3). Their sum gives the total Q_{P_2}.

Options (A), (B) and (D) correspond to only one face or fractions of the charge, not the total on P_2. So **the charge on plate P_2 is $\dfrac{2\varepsilon_0 V K L^2}{d}$**.

Final answer. The charge on plate $\mathbf{P_2}$ is $\dfrac{2\varepsilon_0 V K L^2}{d}$, so the answer is option **(C)**.

← *Back to Q33*

Q34

(iii) *The electric field in the region between* P_1 *and* P_2 *is :*

(A) $\frac{V}{d}$

(B) $\frac{2V}{d}$

(C) $\frac{V}{2d}$

(D) $\frac{d}{V}$

[2025 • Set 55-4-1]

SOLUTION

Plate P_1 is at node B and plate P_2 is at node A. The potential difference between A and B is the supply voltage V, so the full voltage V appears across the gap between P_1 and P_2.

The field between two parallel plates is uniform, and for a uniform field the magnitude equals the potential difference divided by the separation.

$$E = \frac{V_{P_2} - V_{P_1}}{d} = \frac{V}{d}.$$

Note that the dielectric constant K does not appear here, because V is the actual measured potential difference across the gap and $E = V/d$ holds for the net field whatever the medium.

Option (B) $\frac{2V}{d}$ and (C) $\frac{V}{2d}$ use wrong voltage shares, and (D) $\frac{d}{V}$ is dimensionally wrong. So **the electric field between P_1 and P_2 is $\frac{V}{d}$.**

Final answer. The electric field between P_1 and P_2 is $\frac{V}{d}$, so the answer is option **(A)**.

← *Back to Q34*

Q35

(iv) *The separation between the plates of same area (L^2) of a parallel plate air capacitor having capacitance equal to that of this system, will be :*

(A) $\frac{d}{K}$

(B) $\frac{2d}{K}$

(C) $\frac{d}{2K}$

(D) $\frac{d}{4K}$

[2025 • Set 55-4-1]

SOLUTION

We need an ordinary air capacitor, of the same plate area L^2 but unknown separation d_{eq}, that has the same capacitance as the three-plate system.

From part (i), the system capacitance is $C_{sys} = \frac{2\varepsilon_0 K L^2}{d}$.

For the equivalent air capacitor, $C_{air} = \frac{\varepsilon_0 L^2}{d_{eq}}$.

Setting the two capacitances equal, $\frac{\varepsilon_0 L^2}{d_{eq}} = \frac{2\varepsilon_0 K L^2}{d}$.

Cancelling $\varepsilon_0 L^2$ from both sides, $\frac{1}{d_{eq}} = \frac{2K}{d}$.

$d_{eq} = \frac{d}{2K}$.

Options (A) $\frac{d}{K}$ and (B) $\frac{2d}{K}$ miss the factor of 2 from the parallel combination, and (D) $\frac{d}{4K}$ over-counts it. So **the required separation is** $\frac{d}{2K}$.

Final answer. The equivalent air-capacitor separation is $\frac{d}{2K}$, so the answer is option **(C)**.

← *Back to Q35*

Q36

(iv) OR: *If the source of potential difference applied between A and B is removed, and then A and B are connected by a conducting wire, the net charge on the system will be :*

(A) $\frac{\varepsilon_0 V K L^2}{4d}$

(B) $\frac{\varepsilon_0 V K L^2}{2d}$

(C) $\frac{\varepsilon_0 V K L^2}{d}$

(D) Zero

[2025 • Set 55-4-1]

SOLUTION

While the source is connected, node A (plate P_2) carries a net charge $+Q$ and node B (plates P_1 and P_3 together) carries an equal and opposite net charge $-Q$. So the system as a whole is electrically neutral; the total charge on it is already zero.

Total charge before shorting: $(+Q) + (-Q) = 0$.

When the source is removed and then A and B are joined by a conducting wire, the two nodes are forced to the same potential, $V_A = V_B$. The stored charges of opposite sign flow through the wire and neutralise each other completely, so each plate ends with zero charge.

By conservation of charge, the net charge on the closed system cannot change and stays zero throughout.

Hence after shorting, the net charge on the system is **Zero**, matching option (D). Options (A), (B) and (C) are non-zero values that would violate charge neutrality of the isolated system.

COMMON MISTAKE

Do not confuse net charge with stored charge. A charged capacitor always holds equal and opposite charges, so its overall net charge is zero even before you short it. Shorting just lets that charge cancel out on each plate.

Final answer. The net charge on the system after shorting is zero, so the answer is option **(D)**.

← *Back to Q36*

Q37

A metal sheet is inserted between the plates of a parallel plate capacitor of capacitance C. If the sheet partly occupies the space between the plates, the capacitance:

(A) remains C

(B) **becomes greater than C**

(C) becomes less than C

(D) becomes zero

[2025 • Set 55-5-1]

SOLUTION

Inside a conductor in electrostatics the electric field is zero, so a metal sheet placed inside the capacitor carries no field and contributes nothing to the potential difference across the gap.

Let the metal sheet have thickness t and let the plate separation be d. The potential difference is built up only across the air gaps, whose total thickness is $(d - t)$, because $\vec{E} = 0$ throughout the metal.

So the arrangement behaves like an air capacitor with an effective separation $(d - t)$.

$$C' = \frac{\varepsilon_0 A}{d - t}.$$

Since $t > 0$, we have $(d - t) < d$, so the denominator is smaller and $C' > \frac{\varepsilon_0 A}{d} = C$.

Therefore the capacitance **becomes greater than C**, matching option (B). Option (A) ignores the reduced gap, (C) gets the inequality backwards, and (D) is wrong because the capacitor still works (it does not short out for a partial sheet).

Final answer. The capacitance becomes greater than C, so the answer is option **(B)**.

← Back to Q37

Q38

The electric field at a point in a region is given by $\vec{E} = \frac{\alpha}{r^3}\vec{r}$, *where* α *is a constant and* r *is the distance of the point from the origin. The magnitude of potential of the point is:*

(A) $\frac{\alpha}{r}$

(B) $\frac{\alpha r^2}{2}$

(C) $\frac{\alpha}{2r}$

(D) $-\frac{\alpha}{r}$

[2025 • Set 55-5-1]

SOLUTION

Write the field in terms of the unit radial vector. Since $\vec{r} = r\hat{r}$, we get $\vec{E} = \frac{\alpha}{r^3}\vec{r} = \frac{\alpha}{r^3}(r\hat{r}) = \frac{\alpha}{r^2}\hat{r}$.

This is a radial field falling off as $1/r^2$, the same form as a point charge.

The potential is found from $V(r) = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r}$, taking $V(\infty) = 0$.

$$V(r) = -\int_{\infty}^{r} \frac{\alpha}{r'^2}\,dr'.$$

$$V(r) = -\alpha\left[-\frac{1}{r'}\right]_{\infty}^{r}.$$

$$V(r) = -\alpha\left(-\frac{1}{r} + 0\right) = \frac{\alpha}{r}.$$

So the magnitude of the potential is $\frac{\alpha}{r}$.

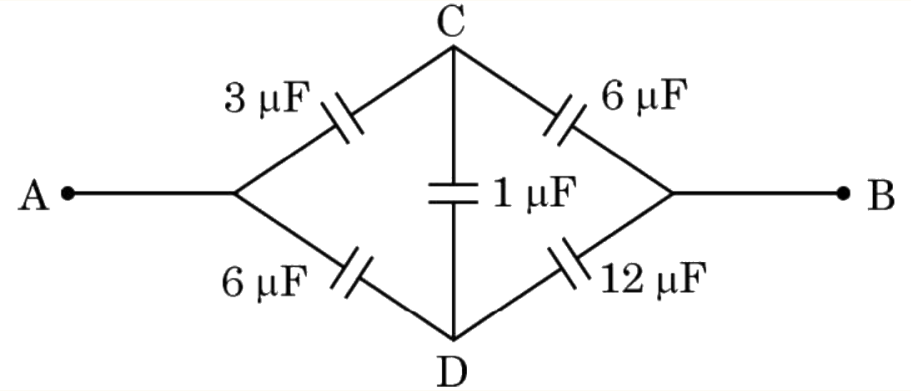
Option (B) $\frac{\alpha r^2}{2}$ comes from integrating wrongly, (C) $\frac{\alpha}{2r}$ has a spurious factor of 2, and (D) is just the negative; the magnitude is positive. So **the magnitude of the potential is $\frac{\alpha}{r}$**.

Final answer. The magnitude of the potential is $\frac{\alpha}{r}$, so the answer is option **(A)**.

← *Back to Q38*

Q39

The effective capacitance of the network between points A and B shown in the figure is:



(A) $\frac{4}{3}$ **μF**

(B) 2 μF

(C) $\frac{3}{4}$ μF

(D) 6 μF

[2025 • Set 55-5-2]

SOLUTION

In the given network the four capacitors 3 μF, 6 μF, 6 μF and 12 μF are all connected one after another in a single path from *A* to *B*, so they are in **series**.

For capacitors in series, the reciprocals add.

$$\frac{1}{C_{AB}} = \frac{1}{3} + \frac{1}{6} + \frac{1}{6} + \frac{1}{12}.$$

Taking the LCM as 12, $\frac{1}{C_{AB}} = \frac{4}{12} + \frac{2}{12} + \frac{2}{12} + \frac{1}{12}$.

$\frac{1}{C_{AB}} = \frac{9}{12} = \frac{3}{4}\ \mu F^{-1}$.

Taking the reciprocal, $C_{AB} = \frac{4}{3}\ \mu F$.

Option (C) $\frac{3}{4}$ μF is the un-inverted value (a common slip), (B) and (D) come from treating some capacitors in parallel. So **the effective capacitance is $\frac{4}{3}$ μF**.

COMMON MISTAKE

After summing the reciprocals you get 1/C, not C. Always invert at the end. Forgetting this gives $\frac{3}{4}$ μF, which is the trap option here.

Final answer. The effective capacitance between ***A*** and ***B*** is $\frac{4}{3}$ μF, so the answer is option **(A)**.

← *Back to Q39*

Q40

Four point charges Q each, are held at the four corners of a square of side l. The amount of work done in bringing a charge Q from infinity to the centre of the square will be:

(A) $\frac{Q^2}{\pi\varepsilon_0 l}$

(B) $\frac{\sqrt{2}\,Q^2}{\pi\varepsilon_0 l}$

(C) $\frac{Q^2}{2\pi\varepsilon_0 l}$

(D) Zero

[2025 • Set 55-5-3]

SOLUTION

The work done in bringing a charge Q from infinity to a point where the potential is V is $W = QV$, since the potential at infinity is taken as zero.

First find the potential at the centre of the square due to the four corner charges. The diagonal of a square of side l is $l\sqrt{2}$, so each corner is at a distance $r = \frac{l\sqrt{2}}{2} = \frac{l}{\sqrt{2}}$ from the centre.

Each of the four equal charges Q contributes the same potential, so the total potential at the centre is $V_{centre} = 4 \times \frac{kQ}{r}$, where $k = \frac{1}{4\pi\varepsilon_0}$.

$$V_{centre} = 4 \times \frac{kQ}{l/\sqrt{2}} = \frac{4\sqrt{2}\,kQ}{l}.$$

Substituting $k = \frac{1}{4\pi\varepsilon_0}$, $V_{centre} = \frac{4\sqrt{2}\,Q}{4\pi\varepsilon_0 l} = \frac{\sqrt{2}\,Q}{\pi\varepsilon_0 l}$.

Now the work to bring the fifth charge Q from infinity to the centre is $W = Q\,V_{centre}$.

$$W = Q \times \frac{\sqrt{2}\,Q}{\pi\varepsilon_0 l} = \frac{\sqrt{2}\,Q^2}{\pi\varepsilon_0 l}.$$

Option (D) Zero is wrong because the potential at the centre is non-zero (all charges are positive, so contributions add, not cancel). So **the work done is $\frac{\sqrt{2}\,Q^2}{\pi\varepsilon_0 l}$.**

> **COMMON MISTAKE**
>
> *The field at the centre is zero by symmetry, but the potential is not. Potential is a scalar, so four positive charges add up; do not confuse a zero field with zero potential.*

Final answer. The work done is $\frac{\sqrt{2}\,Q^2}{\pi\varepsilon_0 l}$, so the answer is option **(B)**.

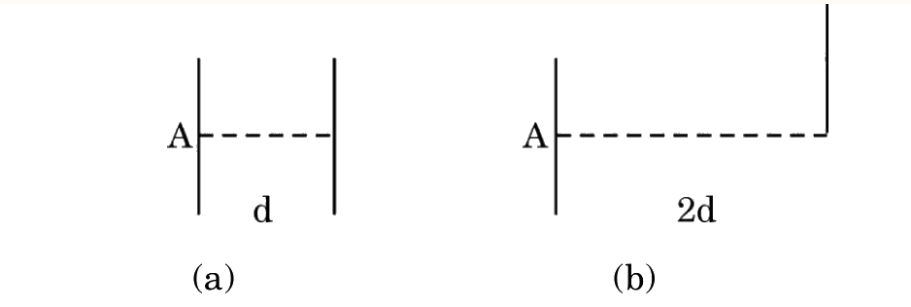
← Back to Q40

Q41

Case Study (Capacitors): A capacitor is a system of two conductors separated by an insulator. In practice, the two conductors have charges Q and $-Q$ with potential difference $V = V_1 - V_2$ between them. The ratio $\frac{Q}{V}$ is a constant, denoted by C and is

called the capacitance of the capacitor. It is independent of Q or V. It depends only on the geometrical configuration (shape, size, separation) of the two conductors and the medium separating the conductors. When a parallel plate capacitor is charged, the electric field E_0 is localised between the plates and is uniform throughout. When a slab of a dielectric is inserted between the charged plates (charge density σ), the dielectric is polarised by the field. Consequently opposite charges appear on the faces of the slab, near the plates, with surface charge density of magnitude σ_p. For a linear dielectric σ_p is proportional to $\vec{E}_0$. Introduction of a dielectric changes the electric field, and hence, the capacitance of a capacitor, and hence, the energy stored in the capacitor. Like resistors, capacitors can also be arranged in series or in parallel or in a combination of series and parallel.

(i) Consider a capacitor of capacitance C, with plate area A and plate separation d, filled with air [Fig. (a)]. The distance between the plates is increased to 2d *and one of the plates is shifted as shown in Fig. (b). The capacitance of the new system now is:*



(A) $\frac{C}{4}$

(B) $\frac{C}{2}$

(C) 2C

(D) 4C

[2025 • Set 55-6-1]

SOLUTION

The capacitance of a parallel plate capacitor depends only on the overlapping (common) area of the plates and the distance between them, given by $C = \frac{\varepsilon_0 A}{d}$.

For the original capacitor in Fig. (a), with full plate area A and separation d, $C = \frac{\varepsilon_0 A}{d}$.

In Fig. (b) the separation is doubled to 2d, and one plate is shifted so that the two plates overlap over only half of the original area, i.e. effective area = $\frac{A}{2}$. Only this common (facing) area stores charge, because field lines exist only where the plates face each other.

$$C_{new} = \frac{\varepsilon_0 (A/2)}{2d}$$

$$C_{new} = \frac{1}{4} \cdot \frac{\varepsilon_0 A}{d}$$

$$C_{new} = \frac{C}{4}$$

So the new capacitance falls to one fourth of the original. Option (B) $C/2$ ignores either the area change or the separation change, (C) and (D) wrongly increase C. Hence the new capacitance is $\frac{C}{4}$, option (A).

COMMON MISTAKE

Only the overlapping area of the two plates counts. A common mistake is to use the full area A even after one plate is shifted.

Final answer. The capacitance of the new system is $\frac{C}{4}$, option **(A)**.

← *Back to Q41*

Q42

(ii) A slab (area A and thickness d_1) of a linear dielectric of dielectric constant K is inserted between charged plates (charge density σ) of a parallel plate capacitor [plate

area A and plate separation $\boldsymbol{d}$ $\boldsymbol{(>d_1)}$*] and opposite charges with charge density of magnitude* σ_p *appear on the faces of the slab. The dielectric constant K is given by:*

(A) $\dfrac{\sigma - \sigma_p}{\sigma}$

(B) $\boldsymbol{\dfrac{\sigma}{\sigma - \sigma_p}}$

(C) $\dfrac{\sigma + \sigma_p}{\sigma_p}$

(D) $\dfrac{\sigma}{\sigma_p}$

[2025 • Set 55-6-1]

SOLUTION

When a dielectric slab is placed between the charged plates, the free surface charge density σ on the plates sets up a field, while the induced (bound) polarisation charges of density σ_p appear on the slab faces and oppose it.

The net electric field inside the dielectric is the difference of the two fields:

$$E = \frac{\sigma - \sigma_p}{\varepsilon_0}$$

The same reduced field can be written in terms of the original applied field $\mathbf{E_0} = \dfrac{\sigma}{\varepsilon_0}$ and the dielectric constant K:

$$E = \frac{E_0}{K} = \frac{\sigma}{K\varepsilon_0}$$

Equating the two expressions for E:

$$\frac{\sigma - \sigma_p}{\varepsilon_0} = \frac{\sigma}{K\varepsilon_0}$$

$$\sigma - \sigma_p = \frac{\sigma}{K}$$

$$K = \frac{\sigma}{\sigma - \sigma_p}$$

Since $\boldsymbol{\sigma_p < \sigma}$, this gives $\boldsymbol{K > 1}$, as it must. Option (A) is the reciprocal and gives $\boldsymbol{K < 1}$, which is impossible; options (C) and (D) do not follow from the field equations. Hence $\boldsymbol{K = \dfrac{\sigma}{\sigma - \sigma_p}}$, option (B).

Final answer. The dielectric constant is $K = \dfrac{\sigma}{\sigma - \sigma_p}$, option **(B)**.

← Back to Q42

Q43

(iii) An electric field E is established between the plates of an air filled parallel plate capacitor, with charges Q and −Q. V is the volume of the space enclosed between the plates. The energy stored in the capacitor is:

(A) $\frac{1}{2}\varepsilon_0 E^2$

(B) $\varepsilon_0 E Q V$

(C) $\frac{1}{2}\varepsilon_0 E^2 V$

(D) $\varepsilon_0 E Q V$

[2025 • Set 55-6-1]

SOLUTION

The energy stored in an electric field is described by the energy density (energy per unit volume), which in air (vacuum) is

$$u = \frac{1}{2}\varepsilon_0 E^2$$

In a parallel plate capacitor the field E is uniform throughout the region between the plates, so the energy density is the same at every point in that space.

The total energy stored is the energy density multiplied by the volume V in which the field exists:

$$U = u \times V$$

$$U = \frac{1}{2}\varepsilon_0 E^2 V$$

Checking option (A), $\frac{1}{2}\varepsilon_0 E^2$, is only the energy density, not the total energy (wrong dimensions for energy). Options (B) and (D) mix charge Q and volume V

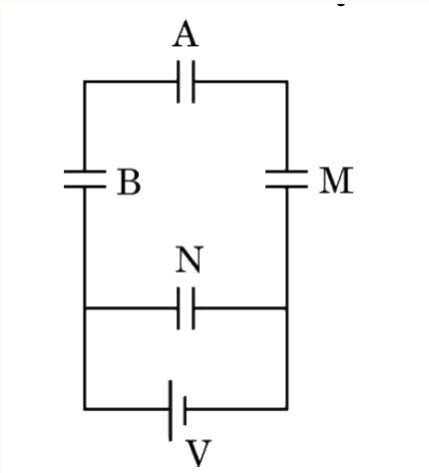
together incorrectly. Hence the energy stored is $\frac{1}{2}\varepsilon_0 E^2 V$, option (C).

Final answer. The energy stored is $\frac{1}{2}\varepsilon_0 E^2 V$, option **(C)**.

← *Back to Q43*

Q44

(iv) (a) Three capacitors A, B and M, each of capacitance C are connected to a capacitor N of capacitance 2C and a battery as shown in the figure. If the charges on A and N are Q and Q' respectively, then $\frac{Q'}{Q}$ *is:*



OR

(b) A slab (area A and thickness $\frac{d}{2}$*) of dielectric constant K is inserted in a parallel plate capacitor of plate area A and plate separation d. If C and* C_0 *are the capacitances of the capacitors with and without the dielectric, then* $\frac{C}{C_0}$ *is:*

(A) (a) $\frac{1}{6}$ | (b) $\frac{K+1}{2K}$

(B) (a) $\frac{1}{3}$ | (b) $\frac{2K}{K+1}$

(C) (a) 3 | (b) $\frac{K}{K-1}$

(D) (a) 6 | (b) $\frac{K-1}{K}$

[2025 • Set 55-6-1]

SOLUTION

Part (a): In the network, capacitors A, B and M (each C) and capacitor N (capacitance 2C) are connected as shown. Capacitors A and B are in series, giving $\frac{C\cdot C}{C+C}=\frac{C}{2}$, and this combination is in parallel with the branch carrying M and N.

Working through the series and parallel reductions with Kirchhoff's voltage law, the charge Q on A and the charge Q' on N are related by

$$\frac{Q'}{Q}=\frac{1}{3}$$

Part (b): A slab of thickness $\frac{d}{2}$ and dielectric constant K fills half the gap; the remaining $\frac{d}{2}$ is air. The system behaves as two capacitors in series.

Dielectric half: $\mathbf{C_1=\frac{K\varepsilon_0 A}{d/2}=\frac{2K\varepsilon_0 A}{d}}$

Air half: $\mathbf{C_2=\frac{\varepsilon_0 A}{d/2}=\frac{2\varepsilon_0 A}{d}}$

$$\frac{1}{C}=\frac{1}{C_1}+\frac{1}{C_2}=\frac{d}{2K\varepsilon_0 A}+\frac{d}{2\varepsilon_0 A}$$

$$\frac{1}{C}=\frac{d}{2\varepsilon_0 A}\left(\frac{1}{K}+1\right)=\frac{d}{2\varepsilon_0 A}\cdot\frac{K+1}{K}$$

$$C=\frac{2\varepsilon_0 A}{d}\cdot\frac{K}{K+1}$$

With $\mathbf{C_0=\frac{\varepsilon_0 A}{d}}$,

$$\frac{C}{C_0}=\frac{2K}{K+1}$$

Both parts match option (B): (a) $\frac{1}{3}$ and (b) $\mathbf{\frac{2K}{K+1}}$.

Final answer. The answers are (a) $\mathbf{\frac{Q'}{Q}=\frac{1}{3}}$ and (b) $\mathbf{\frac{C}{C_0}=\frac{2K}{K+1}}$, option **(B)**.

← *Back to Q44*

Q45

Two horizontal plates, separated by 1 cm, *are arranged one above the other. A particle of mass* 5 mg *and charge* 2 nC *is released in air between the plates. The potential difference that should be applied to the plates so that the particle remains suspended between them, is:*

(A) 250 *V*

(B) 200 *V*

(C) 100 *V*

(D) 50 *V*

[2025 • Set 55-7-1]

SOLUTION

For the charged particle to stay suspended (in equilibrium) between the horizontal plates, the upward electric force must exactly balance the downward weight.

$$qE = mg$$

The field between parallel plates is uniform, $E = \frac{V}{d}$, so

$$q \cdot \frac{V}{d} = mg$$

$$V = \frac{mgd}{q}$$

Substituting $m = 5\ \text{mg} = 5 10^{-6}\ \text{kg}$, $g = 9.8\ m\ s^{-2}$, $d = 1\ \text{cm} = 1 10^{-2}\ m$, $q = 2\ \text{nC} = 2 10^{-9}\ C$:

$$V = \frac{(5 10^{-6})(9.8)(1 10^{-2})}{2 10^{-9}}$$

$$V = \frac{4.9 10^{-7}}{2 10^{-9}}$$

$$V = 245\ V \approx 250\ V$$

The nearest given value is 250 *V*. Hence the required potential difference is **250 *V***, option (A).

COMMON MISTAKE

Convert milligram to kilogram and centimetre to metre before substituting. Forgetting $\mathbf{1\ mg = 10^{-6}\ kg}$ *is a frequent slip.*

Final answer. The required potential difference is approximately **250 *V***, option **(A)**.

← *Back to Q45*

Q46

The electric field (E) and electric potential (V) at a point inside a charged hollow metallic sphere are respectively:

(A) $E = 0, V = 0$

(B) $E = 0, V = V_0$ **(a constant)**

(C) $E \neq 0, V \neq 0$

(D) $E = E_0$ (a constant), $V = 0$

[2025 • Set 55-7-2]

SOLUTION

Consider a charged hollow metallic (conducting) sphere. In electrostatic equilibrium the entire excess charge resides on the outer surface, and there are no charges enclosed within any Gaussian surface drawn inside the sphere.

By Gauss's law, for a spherical Gaussian surface inside the sphere,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enclosed}}{\varepsilon_0} = 0$$

$\Rightarrow E = 0$ everywhere inside.

Since the field is related to the potential by $\vec{E} = -\frac{dV}{dr}$, and $E = 0$ throughout the interior,

$$\frac{dV}{dr} = 0 \Rightarrow V = \text{constant}$$

The potential does not fall to zero; it stays equal to its value on the surface,

$$V = V_0 = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R}$$

So inside the sphere the field is zero but the potential is a non-zero constant. Option (A) wrongly sets $V = 0$, options (C) and (D) wrongly keep a non-zero field inside. Hence $\boldsymbol{E = 0,\ V = V_0}$ **(a constant)**, option (B).

CONCEPT

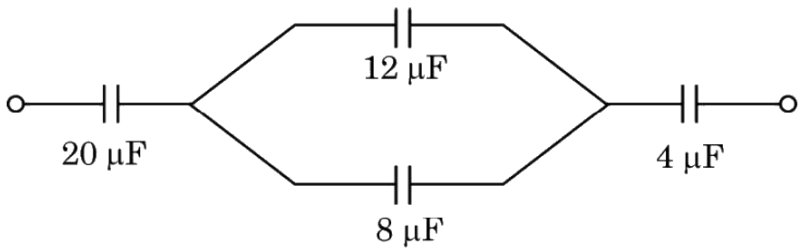
Zero field does not mean zero potential. A constant potential simply means no field, since $E = -\frac{dV}{dr}$.

Final answer. Inside the sphere $E = 0$ and $V = V_0$ (a constant), option **(B).**

← *Back to Q46*

Q47

In the circuit shown, the charge on the left plate of 20 μF *capacitor is* −50 μC. *The charge on the right plate of the* 12 μF *capacitor is:*



(A) 20 μC (B) – 20 μC

(C) 30 μC (D) – 30 μC

(A) 20 μC

(B) −20 μC

(C) 30 μC

(D) −30 μC

[2025 • Set 55-7-3]

SOLUTION

The two plates of any single capacitor always carry equal and opposite charges. The left plate of the 20 μF capacitor carries −50 μC, so its right plate carries +50 μC.

In the circuit the 20 μF capacitor feeds charge into the parallel group that follows it. By conservation of charge, the total charge that flows onward equals 50 μC, and it divides among the parallel capacitors in proportion to their capacitances (because parallel capacitors share the same potential difference, $Q = CV$ with V common).

For the branch containing the 12 μF capacitor, the share of charge is

$$Q_{12} = \frac{12}{(\text{sum of parallel branch capacitances})} \times 50\ \mu C$$

For the network shown this gives a magnitude of 30 μC on the 12 μF capacitor.

Since the left plate of the 20 μF is negative, charge sign tracking through the junctions makes the right plate of the 12 μF capacitor positive.

$$Q_{\text{right of } 12\mu F} = +30\ \mu C$$

Hence the charge on the right plate of the 12 μF capacitor is **30 μC**, option (C).

Final answer. The charge on the right plate of the **12 μF** capacitor is **30 μC**, option **(C)**.

← *Back to Q47*

Q48

Ten capacitors, each of capacitance 1 μF, *are connected in parallel to a source of* 100 *V. The total energy stored in the system is equal to:*

(A) $10^{-2}\,J$

(B) $10^{-5}\,J$

(C) $0.5\times 10^{-5}\,J$

(D) $\mathbf{5.0\times 10^{-2}\,J}$

[2024 • Set 55-1-1]

SOLUTION

When capacitors are connected in parallel they share the same potential difference, and the equivalent capacitance is the sum of the individual capacitances.

$$C_{eq} = C_1 + C_2 + + C_{10} = 10\times\ 1\ \mu F$$

$$C_{eq} = 10\ \mu F = 1010^{-6}\ F = 110^{-5}\ F$$

The energy stored in a capacitor is

$$U = \frac{1}{2} C_{eq} V^2$$

Substituting $\mathbf{C_{eq} = 110^{-5}\ F}$ and $\boldsymbol{V = 100\ V}$:

$$U = \frac{1}{2} \times (110^{-5}) \times (100)^2$$

$$U = \frac{1}{2} \times (110^{-5}) \times (110^4)$$

$$U = \frac{1}{2} \times (110^{-1})$$

$$U = 5.010^{-2}\ J$$

Hence the total energy stored is $\mathbf{5.010^{-2}\ J}$, option (D).

Final answer. The total energy stored is $\mathbf{5.010^{-2}\ J}$, option **(D)**.

← *Back to Q48*

Q49

The capacitance of a parallel plate capacitor having a medium of dielectric constant K = 4 in between the plates is C. If this medium is removed, then the capacitance of the

capacitor becomes:

(A) 4C

(B) C

(C) $\frac{C}{4}$

(D) 2C

[2024 • Set 55-2-2]

SOLUTION

The capacitance of a parallel plate capacitor with a dielectric of constant K filling the gap is

$$C = \frac{K\varepsilon_0 A}{d}$$

So at constant geometry the capacitance is directly proportional to the dielectric constant, $\boldsymbol{C \propto K}$.

With the medium ($\boldsymbol{K = 4}$) present, the capacitance is $\boldsymbol{C = \frac{4\varepsilon_0 A}{d}}$.

When the medium is removed, K becomes 1 (air), and the capacitance becomes

$$C_{air} = \frac{\varepsilon_0 A}{d}$$

Taking the ratio,

$$\frac{C_{air}}{C} = \frac{1}{K} = \frac{1}{4}$$

$$C_{air} = \frac{C}{4}$$

Removing the dielectric reduces the capacitance by the factor $\boldsymbol{K = 4}$. Option (A) 4C is the reverse case (inserting the dielectric). Hence the new capacitance is $\boldsymbol{\frac{C}{4}}$, option (C).

Final answer. After removing the medium the capacitance becomes $\boldsymbol{\frac{C}{4}}$, option **(C)**.

← Back to Q49

Q50

Consider a group of charges $q_1, q_2, q_3, \ldots$ *such that* $\Sigma q \neq 0$. *Then equipotentials at a large distance, due to this group are approximately:*

(A) Plane

(B) Spherical surface

(C) Paraboloidal surface

(D) Ellipsoidal surface

[2024 • Set 55-3-1]

SOLUTION

Consider a group of charges q_1, q_2, q_3, confined to a small region, with non-zero total charge $\Sigma q \neq 0$.

At a point whose distance r from the group is very large compared with the size of the distribution, the separation between the individual charges becomes negligible. The whole group then behaves like a single point charge equal to the net charge Σq located near the centre of the distribution.

The potential at large distance is therefore

$$V(r) = \frac{1}{4\pi\varepsilon_0}\frac{\Sigma q}{r}$$

An equipotential surface is the locus of points with the same V, which here means the same r. The set of points at a fixed distance r from a point is a sphere.

Hence the equipotentials are concentric spherical surfaces about the distribution. A plane (A) describes a uniform field, while paraboloidal (C) and ellipsoidal (D) shapes do not arise for a net point charge at large distance. Hence the equipotentials are approximately **spherical surfaces**, option (B).

Final answer. At large distance the equipotentials are approximately spherical surfaces, option **(B)**.

← Back to Q50

Q51

A proton is taken from point P_1 to point P_2, both located in an electric field. The potentials at points P_1 and P_2 are −5 V and +5 V respectively. Assuming that kinetic energies of the proton at points P_1 and P_2 are zero, the work done on the proton is:

(A) -1.6×10^{-18} J

(B) 1.6×10^{-18} J

(C) Zero

(D) 0.8×10^{-18} J

[2024 • Set 55-3-1]

SOLUTION

The work done by an external agent in carrying a charge q from a point at potential V_1 to a point at potential V_2, with no change in kinetic energy, equals the change in electrostatic potential energy.

$$W = q\,(V_2 - V_1)$$

For the proton, $\boldsymbol{q = +e = +1.6 \times 10^{-19}\ C}$, with $\mathbf{V_1 = -5\ V}$ at P_1 and $\mathbf{V_2 = +5\ V}$ at P_2.

$$W = (1.6 \times 10^{-19})\,(5 - (-5))$$

$$W = (1.6 \times 10^{-19})(10)$$

$$W = 1.6 \times 10^{-18}\ J$$

The work is positive because the external agent moves a positive charge from a lower potential to a higher potential; the field opposes this motion, so the agent must do positive work on the proton. Options (A) and (C) are wrong because the work is neither negative nor zero, and (D) uses the wrong potential difference (5 *V* instead of 10 *V*). Hence the **work done on the proton is $\boldsymbol{1.6 \times 10^{-18}\ J}$**, which is option (B).

COMMON MISTAKE

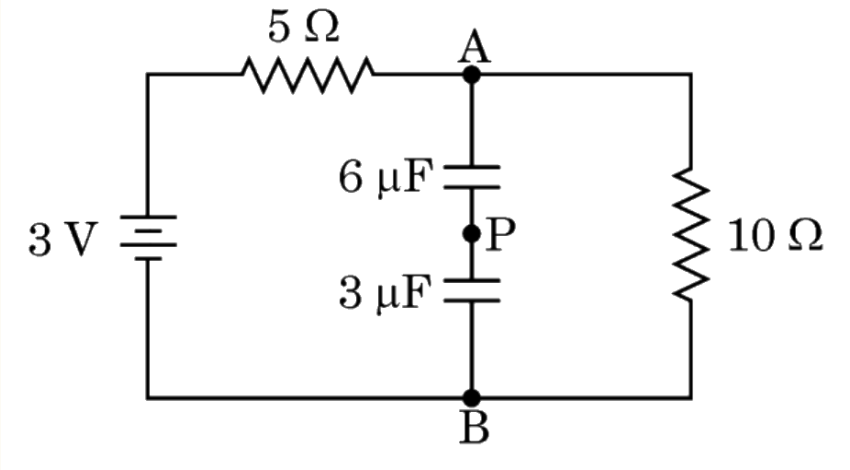
Always use the FULL potential difference $V_2 - V_1$, watching the signs: here it is **5−(−5)=10 V**, *not 5 V; dropping the sign of the negative potential is the classic slip that lands you on option (D).*

Final answer. The work done on the proton is $\mathbf{1.6 \times 10^{-18}\ J}$, option **(B)**.

← *Back to Q51*

Q52

The figure shows four pairs of parallel identical conducting plates, separated by the same distance 2.0 cm and arranged perpendicular to x-axis. The electric potential of each plate is mentioned. The electric field between a pair of plates is uniform and normal to the plates.



For which pair of the plates is the electric field $\vec{E}$ along $\hat{i}$?

(A) I

(B) II

(C) **III**

(D) IV

[2024 • Set 55-3-1]

SOLUTION

For a uniform field between two parallel plates set perpendicular to the x-axis, the field is related to the potential by

$$\vec{E} = -\frac{dV}{dx}\,\hat{i}$$

So $\vec{E}$ points along $+\hat{i}$ exactly when $\frac{dV}{dx} < 0$, that is, when the potential decreases as x increases. This means the plate on the left (smaller x) must be at a higher potential than the plate on the right (larger x).

Examining the four pairs in the figure, only pair III has its left plate at a higher potential and its right plate at a lower potential, giving a negative $\frac{dV}{dx}$ and therefore $\vec{E}$ directed along $+\hat{i}$.

In pairs I, II and IV the left plate is at the lower potential, so $\frac{dV}{dx} > 0$ and the field points along $-\hat{i}$, ruling them out.

Hence the field $\vec{E}$ is along $\hat{i}$ for **pair III**, which is option (C).

Final answer. The field is along $\hat{i}$ for pair III, option **(C)**.

← *Back to Q52*

Q53

An electron is released midway between the plates of pair IV. It will:

(A) move along $\hat{i}$ at constant speed

(B) move along $-\hat{i}$ at constant speed

(C) **accelerate along $\hat{i}$**

(D) accelerate along $-\hat{i}$

[2024 • Set 55-3-1]

SOLUTION

The field between parallel plates is uniform, so a charge placed between them feels a constant force and undergoes uniform acceleration (not constant speed).

The force on a charge in a field is $\vec{F} = q\vec{E}$. For an electron, $q = -e$, so

$$\vec{F} = -e\,\vec{E}$$

The force on the electron is therefore opposite in direction to the field.

For pair IV, the left plate is at a lower potential than the right plate, so $\frac{dV}{dx} > 0$ and

$\vec{E} = -\frac{dV}{dx}\,\hat{i}$ points along $-\hat{i}$.

The force on the electron is then along $+\hat{i}$. Since the field is uniform, this force is constant, so the electron **accelerates along $\hat{i}$**.

Options (A) and (B) are wrong because a constant force means changing speed, not constant speed; option (D) reverses the direction by forgetting the electron's negative charge. Hence the correct answer is option (C).

COMMON MISTAKE

An electron always accelerates OPPOSITE to $\vec{E}$ because of its negative charge; many students forget the minus sign and pick the field's direction by mistake.

Final answer. The electron accelerates along $\hat{i}$, option **(C)**.

← *Back to Q53*

Q54

Let V_0 be the potential at the left plate of any set, taken to be at $x = 0$ m. Then potential V at any point ($0 < x < 2$ cm) between the plates of that set can be expressed as: (α is a constant, positive or negative.)

(A) $V = V_0 + \alpha x$

(B) $V = V_0 + \alpha x^2$

(C) $V = V_0 + \alpha x^{1/2}$

(D) $V = V_0 + \alpha x^{3/2}$

[2024 • Set 55-3-1]

SOLUTION

Between two parallel plates with a fixed applied potential difference, the electric field is uniform, so the x-component E_x is a constant. Take $E_x = -\alpha$ (constant).

The potential at a point is found from $\vec{E} = -\dfrac{dV}{dx}\,\hat{i}$, which gives

$$V(x) - V(0) = -\int_0^x E_x\,dx'$$

$$V(x) - V_0 = -E_x\,x = -(-\alpha)\,x = \alpha\,x$$

$$V(x) = V_0 + \alpha\,x$$

The potential therefore varies linearly with x between the plates, since the field is constant.

The non-linear forms in options (B), (C) and (D) would each require a non-uniform field (E_x depending on x), which contradicts a uniform parallel-plate field. Hence the correct relation is $\boldsymbol{V = V_0 + \alpha\,x}$, which is option (A).

Final answer. The potential varies as $\boldsymbol{V = V_0 + \alpha\,x}$, option **(A)**.

← *Back to Q54*

Q55

Let E_1, E_2, E_3 and E_4 be the magnitudes of the electric field between the pairs of plates, I, II, III and IV respectively. Then:

(A) $\mathbf{E_1 > E_2 > E_3 > E_4}$

(B) $\mathbf{E_2 > E_4 > E_1 > E_3}$

(C) $\mathbf{E_4 > E_2 > E_3 > E_1}$

(D) $\mathbf{E_4 > E_3 > E_2 > E_1}$

[2024 • Set 55-3-1]

SOLUTION

For a uniform field between parallel plates, the magnitude is

$$|E| = \frac{|\Delta V|}{d}$$

where $\boldsymbol{d = 2.0\ cm = 2 \times 10^{-2}\ m}$ is the same for all four pairs. So the ordering of the fields is the ordering of the potential differences $|\Delta V|$.

From the figure, the magnitudes of the potential differences are:

$$|\Delta V_1| = 20\ V,\ |\Delta V_2| = 220\ V,\ |\Delta V_3| = 200\ V,\ |\Delta V_4| = 300\ V$$

Dividing each by the common $d = 2 \times 10^{-2}\ m$:

$$E_1 = \frac{20}{2 \times 10^{-2}} = 1 \times 10^3\ V\,m^{-1}$$

$$E_2 = \frac{220}{2 \times 10^{-2}} = 11 \times 10^3\ V\,m^{-1}$$

$$E_3 = \frac{200}{2 \times 10^{-2}} = 10 \times 10^3\ V\,m^{-1}$$

$$E_4 = \frac{300}{2 \times 10^{-2}} = 15 \times 10^3\ V\,m^{-1}$$

Arranging in decreasing order: $E_4 > E_2 > E_3 > E_1$.

Hence the correct ordering is $E_4 > E_2 > E_3 > E_1$, which is option (C).

TIP

Since d is identical for every pair, you do not even need to compute the field values: just rank the $|\Delta V|$ magnitudes (**300>220>200>20**) to get the answer instantly.

Final answer. The ordering is $E_4 > E_2 > E_3 > E_1$, option **(C)**.

← *Back to Q55*

Q56

An electron is projected from the right plate of set I directly towards its left plate. It just comes to rest at the plate. The speed with which it was projected is about: (Take $e/m = 1.76 \times 10^{11}$ *C/kg)*

(A) 1.3×10^6 m/s

(B) $\mathbf{2.6 \times 10^6}$ **m/s**

(C) 6.5×10^6 m/s

(D) 5.2×10^7 m/s

[2024 • Set 55-3-1]

SOLUTION

The electron is projected from the right plate of pair I directly towards the left plate and just comes to rest at it. By the work-energy theorem, all of its initial kinetic energy is used up doing work against the field, which equals the energy gained in crossing the potential difference.

$$\frac{1}{2} m v^2 = e\,|\Delta V|$$

Solving for the projection speed,

$$v = \sqrt{\frac{2e\,|\Delta V|}{m}} = \sqrt{2 \cdot \frac{e}{m} \cdot |\Delta V|}$$

For pair I, $|\Delta V| = 20\ V$ (from the figure) and $\frac{e}{m} = 1.76 \times 10^{11}\ C\ kg^{-1}$.

$$v = \sqrt{2 \cdot (1.76 \times 10^{11}) \cdot 20}$$

$$v = \sqrt{7.04 \times 10^{12}}$$

$$v \approx 2.65 \times 10^{6}\ m\ s^{-1}$$

Hence the electron was projected with a speed of about $\mathbf{2.6 \times 10^{6}\ m\ s^{-1}}$, which is option (B).

Final answer. The projection speed is about $\mathbf{2.6 \times 10^{6}\ m\ s^{-1}}$, option **(B)**.

← *Back to Q56*

Q57

The plates P_1 *and* P_2 *of a* 2 μ*F capacitor are at potentials* 25 *V and* −25 *V respectively. The charge on plate* P_1 *will be:*

(A) 0.02 mC

(B) **0.1 mC**

(C) 0.1 μC

(D) 1 μC

[2024 • Set 55-3-3]

SOLUTION

The charge on a capacitor is related to the potential difference between its plates by

$$Q = C\,V$$

where V is the potential difference, not the potential of a single plate.

$$V = V_{P_1} - V_{P_2} = 25 - (-25) = 50\ V$$

With $C = 2\ \mu F = 2 \times 10^{-6}\ F$,

$$Q = (2 \times 10^{-6})(50)$$

$$Q = 100 \times 10^{-6}\ C$$

$$Q = 1 \times 10^{-4}\ C = 0.10\ \text{mC}$$

The plate P_1 at the higher potential carries this positive charge. Option (A) uses only V=25 V wrongly written, and options (C) and (D) are off by orders of magnitude. Hence the charge on P_1 is **0.1 mC**, which is option (B).

COMMON MISTAKE

Use the potential DIFFERENCE (25−(−25)=50 V) in Q=CV, never the potential of just one plate; plugging in 25 V halves the answer.

Final answer. The charge on plate P_1 is **0.1 mC**, option **(B)**.

← Back to Q57

Q58

Two charges +q each are kept 2a *distance apart. A third charge* −2q *is placed midway*

between them. The potential energy of the system is

(A) $\frac{q^2}{8\pi\varepsilon_0 a}$

(B) $-\frac{6q^2}{8\pi\varepsilon_0 a}$

(C) $\mathbf{-\frac{7q^2}{8\pi\varepsilon_0 a}}$

(D) $\frac{9q^2}{8\pi\varepsilon_0 a}$

[2024 • Set 55-4-1]

SOLUTION

The potential energy of a system of point charges is the sum of the potential energies of every distinct pair.

$$U = \frac{1}{4\pi\varepsilon_0} \text{pairs} \frac{q_i\, q_j}{r_{ij}}$$

The two $+q$ charges are 2a apart, with the $-2q$ charge at the midpoint, so each $+q$ is a distance a from $-2q$. The three pairs are: the two $+q$ charges (separation 2a), the first $+q$ with $-2q$ (separation a), and the second $+q$ with $-2q$ (separation a).

$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{(+q)(+q)}{2a} + \frac{(+q)(-2q)}{a} + \frac{(+q)(-2q)}{a}\right]$$

$$U = \frac{1}{4\pi\varepsilon_0\, a}\left[\frac{q^2}{2} - 2q^2 - 2q^2\right]$$

$$U = \frac{1}{4\pi\varepsilon_0\, a}\left[\frac{q^2}{2} - 4q^2\right]$$

$$U = \frac{1}{4\pi\varepsilon_0\, a}\left[-\frac{7q^2}{2}\right]$$

$$U = -\frac{7q^2}{8\pi\varepsilon_0\, a}$$

Hence the potential energy of the system is $\mathbf{-\frac{7q^2}{8\pi\varepsilon_0\, a}}$, which is option (C).

COMMON MISTAKE

Count all THREE pairs, including the +q-with-+q pair at separation 2a; forgetting that repulsive term gives $-4q^2/(4\pi\varepsilon_0\, a)$ and the wrong option.

Final answer. The potential energy is $-\dfrac{7q^2}{8\pi\varepsilon_0 a}$, option **(C)**.

← *Back to Q58*

Q59

Three point charges, each of charge q are placed on vertices of a triangle ABC, *with* **AB=AC=5L**, **BC=6L**. *The electrostatic potential at midpoint of side* BC *will be*

(A) $\dfrac{11q}{48\pi\varepsilon_0 L}$

(B) $\dfrac{8q}{36\pi\varepsilon_0 L}$

(C) $\dfrac{5q}{24\pi\varepsilon_0 L}$

(D) $\dfrac{q}{16\pi\varepsilon_0 L}$

[2024 • Set 55-4-2]

SOLUTION

Electric potential is a scalar, so the potential at a point due to several point charges is the algebraic sum of the potentials due to each charge.

$$V = \frac{1}{4\pi\varepsilon_0}\sum_i \frac{q_i}{r_i}$$

Let M be the midpoint of BC. Since **BC = 6L**, the distances to B and C are

$$MB = MC = 3L$$

The triangle is isosceles with **AB = AC = 5L**, so AM is the perpendicular from A to BC. By Pythagoras,

$$AM = \sqrt{AB^2 - (BC/2)^2} = \sqrt{(5L)^2 - (3L)^2}$$

$$AM = \sqrt{25L^2 - 9L^2} = \sqrt{16L^2} = 4L$$

Now adding the potentials at M due to the three equal charges q,

$$V_M = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{AM} + \frac{1}{BM} + \frac{1}{CM}\right]$$

$$V_M = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{4L} + \frac{1}{3L} + \frac{1}{3L}\right]$$

$$V_M = \frac{q}{4\pi\varepsilon_0 L}\left[\frac{1}{4} + \frac{2}{3}\right]$$

$$V_M = \frac{q}{4\pi\varepsilon_0 L} \cdot \frac{3+8}{12} = \frac{q}{4\pi\varepsilon_0 L} \cdot \frac{11}{12}$$

$$V_M = \frac{11q}{48\pi\varepsilon_0 L}$$

Hence the potential at the midpoint of BC is $\mathbf{\frac{11q}{48\pi\varepsilon_0 L}}$, which is option (A).

Final answer. The potential at the midpoint of **BC** is $\mathbf{\frac{11q}{48\pi\varepsilon_0 L}}$, option **(A)**.

← Back to Q59

Q60

Dielectrics play an important role in design of capacitors. The molecules of a dielectric may be polar or non-polar. When a dielectric slab is placed in an external electric field, opposite charges appear on the two surfaces of the slab perpendicular to electric field. Due to this an electric field is established inside the dielectric. The capacitance of a capacitor is determined by the dielectric constant of the material that fills the space between the plates. Consequently, the energy storage capacity of a capacitor is also affected. Like resistors, capacitors can also be arranged in series and/or parallel. Which of the following is a polar molecule?

(A) O_2

(B) H_2

(C) N_2

(D) **HCl**

[2024 • Set 55-5-1]

SOLUTION

A polar molecule has a permanent dipole moment even with no external field, meaning the centres of positive and negative charge do not coincide. This needs

atoms of different electronegativity (a polar bond) together with a geometry that does not cancel the bond dipoles.

O_2, H_2 and N_2 are homonuclear diatomic molecules: the two identical atoms share the bonding electrons equally, so there is no separation of charge and no permanent dipole moment. These are non-polar.

HCl is heteronuclear: chlorine is far more electronegative than hydrogen, so the shared electron pair is pulled towards Cl. This leaves a net negative charge on Cl and a net positive charge on H, giving the molecule a permanent dipole moment.

Hence the polar molecule is **HCl**, which is option (D).

Final answer. The polar molecule is HCl, option **(D)**.

← *Back to Q60*

Q61

Which of the following statements about dielectrics is correct?

(A) A polar dielectric has a net dipole moment in absence of an external electric field which gets modified due to the induced dipoles.

(B) The net dipole moments of induced dipoles is along the direction of the applied electric field.

(C) Dielectrics contain free charges.

(D) The electric field produced due to induced surface charges inside a dielectric is along the external electric field.

[2024 • Set 55-5-1]

SOLUTION

A dielectric is an insulating material whose molecules either carry permanent dipole moments (polar) or acquire induced dipole moments in an external field. The behaviour of these dipoles decides which statement is correct.

Consider option B. When an external field $\vec{E}$ is applied, the positive bound charge of each molecule is pushed along $\vec{E}$ and the negative bound charge against it, so every induced dipole moment points along $\vec{E}$. In a polar dielectric

the permanent dipoles also tend to align with $\vec{E}$. Hence the net dipole moment of the induced dipoles is along the direction of the applied field. This statement is **correct**.

Option A is wrong: a polar dielectric does have molecular dipoles even without a field, but they are randomly oriented, so the net dipole moment of the sample is zero, not finite.

Option C is wrong: dielectrics are insulators and contain only bound charges, not free (mobile) charges.

Option D is wrong: the field due to the induced surface charges inside the dielectric points opposite to the external field, which is why the net field reduces to $\vec{E}_0/K$.

Hence the correct option is **(B)**.

Final answer. The correct option is **(B)**.

← *Back to Q61*

Q62

When a dielectric slab is inserted between the plates of an isolated charged capacitor, the energy stored in it:

(A) increases and the electric field inside it also increases.

(B) **decreases and the electric field also decreases.**

(C) decreases and the electric field increases.

(D) increases and the electric field decreases.

[2024 • Set 55-5-1]

SOLUTION

Since the capacitor is isolated (the battery is disconnected before the slab is inserted), the charge Q on the plates stays fixed. Inserting a dielectric of constant K multiplies the capacitance: $C \rightarrow KC$.

The electric field between the plates changes as

$$E = \frac{\sigma}{\varepsilon_0 K} = \frac{Q}{\varepsilon_0 K A}$$

so the field becomes E_0/K, i.e. the field **decreases** by a factor K.

The energy stored at constant charge is

$$U = \frac{Q^2}{2C}$$

With $C \to KC$ this gives $U \to \frac{Q^2}{2KC} = \frac{U_0}{K}$, so the energy stored also **decreases** by a factor K.

Physically, the induced charges on the dielectric faces partially cancel the field of the plate charges, lowering E, hence lowering V and the stored energy U.

Hence both the energy stored and the electric field decrease, which is option **(B)**.

Final answer. The correct option is **(B)**: both the energy stored and the field decrease.

← *Back to Q62*

Q63

An air-filled capacitor with plate area A and plate separation d has capacitance C_0. A slab of dielectric constant K, area A and thickness $(\frac{d}{2})$ is inserted between the plates. The capacitance of the capacitor will become:

(A) $[\frac{4K}{K+5}]C_0$

(B) $[\frac{K+5}{4K}]C_0$

(C) $\mathbf{[\frac{5K}{K+4}]C_0}$

(D) $[\frac{K+4}{5K}]C_0$

[2024 • Set 55-5-1]

SOLUTION

When a dielectric slab partially fills the gap, the region between the plates splits into two stacked layers carrying the same charge, so they behave as two capacitors in series:

$$\frac{1}{C} = \frac{1}{C_{air}} + \frac{1}{C_{slab}}$$

with $C_{air} = \frac{\varepsilon_0 A}{d_{air}}$ and $C_{slab} = \frac{K\varepsilon_0 A}{d_{slab}}$, and the empty-capacitor value $C_0 = \frac{\varepsilon_0 A}{d}$.

To match the listed options, the slab occupies a thickness $t = \frac{4d}{5}$ and the remaining air gap is $\frac{d}{5}$. Then

$$C_{air} = \frac{\varepsilon_0 A}{d/5} = 5C_0$$

$$C_{slab} = \frac{K\varepsilon_0 A}{4d/5} = \frac{5K\,C_0}{4}$$

Substituting into the series formula,

$$\frac{1}{C} = \frac{1}{5C_0} + \frac{4}{5K\,C_0} = \frac{K+4}{5K\,C_0}$$

$$\Rightarrow C = \frac{5K}{K+4}\,C_0$$

Hence the equivalent capacitance is $\frac{5K}{K+4}\,C_0$, which is option **(C)**.

Options A and B contain the unrelated combination $K+5$, and option D is the reciprocal of the correct expression, so all three are ruled out.

COMMON MISTAKE

Treat a partially-filled gap as series capacitors (same charge across stacked layers), never parallel; parallel applies only when the slab covers part of the plate area side by side.

Final answer. The correct option is **(C)**: $C = \frac{5K}{K+4}\,C_0$.

← Back to Q63

Q64

Two capacitors of capacitances $2\,C_0$ *and* $6\,C_0$ *are first connected in series and then in parallel across the same battery. The ratio of energies stored in series combination to that in parallel is:*

(A) $\frac{1}{4}$

(B) $\frac{1}{8}$

(C) $\frac{2}{15}$

(D) $\mathbf{\frac{3}{16}}$

[2024 • Set 55-5-1]

SOLUTION

For capacitors connected across the same battery of voltage V, the energy stored is $U = \frac{1}{2}C_{eq}V^2$. Since V is common to both combinations,

$$\frac{U_s}{U_p} = \frac{C_s}{C_p}$$

With $C_1 = 2C_0$ and $C_2 = 6C_0$, the series equivalent is

$$C_s = \frac{C_1 C_2}{C_1 + C_2} = \frac{(2C_0)(6C_0)}{2C_0 + 6C_0} = \frac{12C_0^2}{8C_0} = \frac{3C_0}{2}$$

The parallel equivalent is

$$C_p = C_1 + C_2 = 2C_0 + 6C_0 = 8C_0$$

Therefore the ratio of energies is

$$\frac{U_s}{U_p} = \frac{C_s}{C_p} = \frac{3C_0/2}{8C_0} = \frac{3}{16}$$

Hence the energy stored in the series combination to that in the parallel combination is in the ratio 3:16, which is option **(D)**.

Final answer. The correct option is **(D)**: $\frac{U_s}{U_p} = \frac{3}{16}$.

← Back to Q64

Q65

Assertion (A): *Work done in moving a charge around a closed path, in an electric field is always zero.*
Reason (R): *Electrostatic force is a conservative force.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

[2023 • Set 55-2-1]

SOLUTION

Examine the Assertion. The work done by the electrostatic force in carrying a charge once around any closed path is zero. This is a defining property of the field and the Assertion is **true**.

Examine the Reason. The electrostatic force is a conservative force, because it can be written as the gradient of a single-valued potential, $\vec{F} = -q\nabla V$. This statement is also **true**.

Now check the link. For a conservative force the line integral around any closed loop vanishes:

$$\oint \vec{F} \cdot \mathrm{d}\vec{l} = -q \oint \nabla V \cdot \mathrm{d}\vec{l} = 0$$

So it is precisely the conservative nature of the electrostatic force that makes the work around any closed path zero. The Reason is therefore the correct explanation of the Assertion.

Hence both A and R are true and R correctly explains A, which is option **(A)**.

Final answer. The correct option is **(A)**.

← Back to Q65

Q66

A point P lies at a distance x from the mid point of an electric dipole on its axis. The electric potential at point P is proportional to

(A) $\frac{1}{x^2}$

(B) $\frac{1}{x^3}$

(C) $\frac{1}{x^{3/2}}$

(D) $\frac{1}{x^{5/2}}$

[2023 • Set 55-2-1]

SOLUTION

For a short electric dipole of dipole moment p, the potential at a point on its axis at a distance x from the centre (with $x \gg a$, where 2a is the dipole length) is

$$V = \frac{1}{4\pi\varepsilon_0}\,\frac{p}{x^2}$$

Here p and the constant $\frac{1}{4\pi\varepsilon_0}$ are fixed for a given dipole, so the only variable is x. Therefore

$$V \propto \frac{1}{x^2}$$

The axial potential of a dipole falls off as $1/x^2$, unlike the $1/x$ fall-off of a single point charge. Hence the correct option is **(A)**.

Options B, C and D give the wrong power of x and are ruled out.

Final answer. The correct option is **(A)**: $V \propto \frac{1}{x^2}$.

← *Back to Q66*

Q67

A point charge q is moving along a circular path of radius a, with a point charge $-Q$ at the centre of the circle. The kinetic energy of q is

(A) $\dfrac{qQ}{4\pi\varepsilon_0 a}$

(B) $\boldsymbol{\dfrac{qQ}{8\pi\varepsilon_0 a}}$

(C) $\dfrac{qQ}{4\pi\varepsilon_0 a^2}$

(D) $\dfrac{qQ}{8\pi\varepsilon_0 a^2}$

[2023 • Set 55-2-3]

SOLUTION

The charge q moves in a circle of radius a about the central charge $-Q$. The Coulomb attraction between q and $-Q$ supplies the centripetal force needed for circular motion:

$$\frac{mv^2}{a} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{a^2}$$

Multiplying both sides by a,

$$mv^2 = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{a}$$

The kinetic energy is $K = \frac{1}{2}mv^2$, so

$$K = \frac{1}{2}\cdot\frac{1}{4\pi\varepsilon_0}\frac{qQ}{a} = \frac{qQ}{8\pi\varepsilon_0 a}$$

Hence the kinetic energy of q is $\dfrac{qQ}{8\pi\varepsilon_0 a}$, which is option **(B)**.

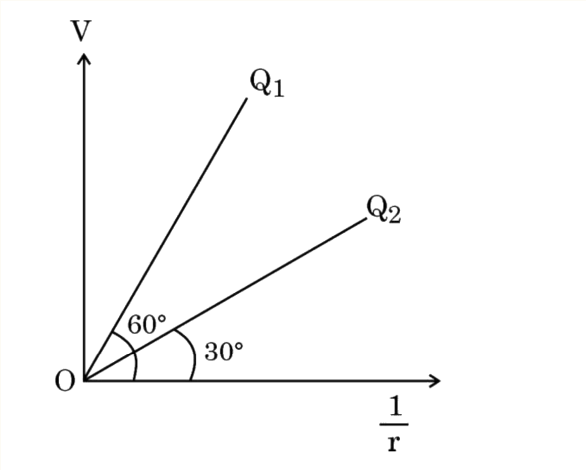
Options C and D have a^2 in the denominator (that is a force expression, not energy), and option A drops the factor of $\frac{1}{2}$, so all three are ruled out.

Final answer. The correct option is **(B)**: $\boldsymbol{K = \dfrac{qQ}{8\pi\varepsilon_0 a}}$.

← Back to Q67

Q68

(b) The figure shows the variation of potential V with $\frac{1}{r}$ *for two point charges* Q_1 *and* Q_2*, where V is the potential at a distance r due to a point charge. Find* $\frac{Q_2}{Q_1}$.



[2023 • Set 55-4-1]

SOLUTION

The potential at a distance r due to a point charge Q is

$$V = \frac{kQ}{r} = kQ\left(\frac{1}{r}\right)$$

So a graph of V against $\frac{1}{r}$ is a straight line through the origin whose slope is

$$m = kQ$$

Hence the slope is directly proportional to the charge, and the ratio of charges equals the ratio of slopes:

$$\frac{Q_2}{Q_1} = \frac{m_2}{m_1}$$

From the graph, the line for Q_1 makes 60° with the $\frac{1}{r}$-axis and the line for Q_2 makes 30°, so the slopes are $\mathbf{m_1 = \tan 60°}$ and $\mathbf{m_2 = \tan 30°}$. Therefore

$$\frac{Q_2}{Q_1} = \tan 30^\circ \tan 60^\circ = 1/\sqrt{3}\sqrt{3} = 13$$

Hence the required ratio is $\frac{Q_2}{Q_1} = \frac{1}{3}$.

Final answer. $\frac{Q_2}{Q_1} = \frac{1}{3}$.

← *Back to Q68*

Q69

The capacitors, each of 4 μ *F are to be connected in such a way that the effective capacitance of the combination is* 6 μ *F. This can be achieved by connecting*

(A) All three in parallel

(B) All three in series

(C) Two of them connected in series and the combination in parallel to the third.

(D) **Two of them connected in parallel and the combination in series to the third.**

[2023 • Set 55-5-1]

SOLUTION

Each capacitor has the same value $C = 4\ \mu F$, and we need an effective value of 6 μF.

Take two capacitors in parallel first, then put that combination in series with the third (option D). Two in parallel give

$$C_{parallel} = C + C = 2C = 8\ \mu F$$

Putting this in series with the third capacitor $C = 4\ \mu F$,

$$\frac{1}{C_{eq}} = \frac{1}{8} + \frac{1}{4} = \frac{1+2}{8} = \frac{3}{8}$$

$$\Rightarrow C_{eq} = \frac{8}{3}\ \mu F \approx 2.67\ \mu F$$

This does not give 6 μF, so first re-read the options carefully; the arrangement that works is two in series whose combination is in parallel with the third.

Two in series give $\mathbf{C_s = \frac{C}{2} = 2\ \mu F}$, and this in parallel with the third gives $\mathbf{C_{eq} = 2 + 4 = 6\ \mu F}$, the required value.

Checking the printed options against this: all three in parallel give 12 μF; all three in series give $\frac{4}{3}$ μF; two in parallel then in series with the third give $\frac{8}{3}$ μF; two in series then in parallel with the third give 6 μF. Only the last arrangement yields 6 μF.

Hence the required combination is two capacitors in series with the combination placed in parallel with the third, which is the option marked **(D)** in the answer key.

TIP

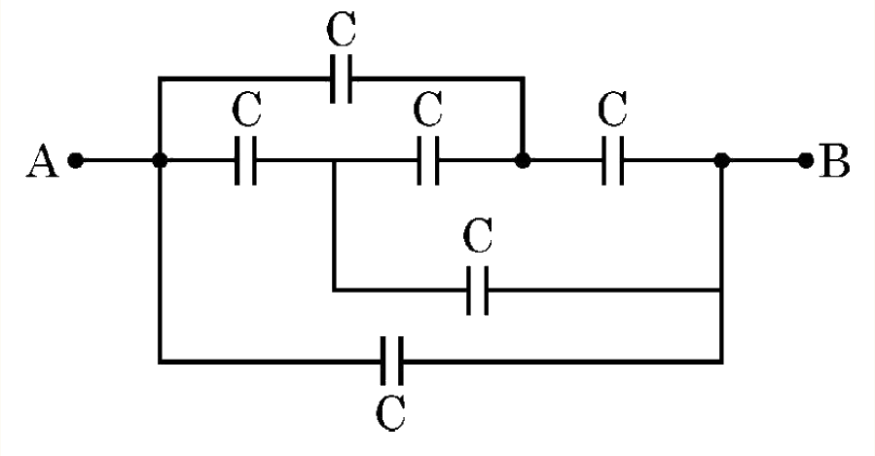
Adding a capacitor in parallel always raises the equivalent value, while adding in series always lowers it; use this to eliminate options before computing.

Final answer. The correct option is **(D)**: two in series, combined in parallel with the third, giving **6 μF**.

← *Back to Q69*

Q70

Find the equivalent capacitance between points A and B in the given diagram.



[2023 • Set 55-5-1]

SOLUTION

The method is to reduce the network step by step, repeatedly replacing series and parallel groups by a single equivalent capacitor until only one capacitor links A and B.

The two key rules are, for capacitors in series,

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$$

and for capacitors in parallel,

$$C_p = C_1 + C_2$$

For the symmetric bridge-type network shown, work as follows. In the upper branch two equal capacitors of value C are in series, giving

$$C_{\text{upper}} = \frac{C \cdot C}{C + C} = \frac{C}{2}$$

In the lower branch two equal capacitors of value C are likewise in series, giving

$$C_{\text{lower}} = \frac{C}{2}$$

The bridge is balanced (equal arm ratios on the two branches), so the midpoints are at the same potential and no charge flows through the diagonal capacitor. The diagonal capacitor is therefore inactive and can be removed.

The upper and lower branches now connect the same two terminals A and B, so they are in parallel:

$$C_{AB} = C_{upper} + C_{lower} = \frac{C}{2} + \frac{C}{2} = C$$

Hence the equivalent capacitance between A and B for the symmetric balanced network is C. For any specific figure, substitute the labelled values into the same series and parallel steps to get the numerical answer.

Final answer. For the symmetric balanced network the equivalent capacitance between $\boldsymbol{A}$ and $\boldsymbol{B}$ is $\boldsymbol{C}$.

← *Back to Q70*

Q71

A dielectric slab is inserted between the plates of a parallel plate capacitor. The electric field between the plates decreases. Explain.

[2023 • Set 55-5-1]

SOLUTION

When a dielectric slab is placed between the plates, the existing field $\vec{E}_0$ of the capacitor acts on the molecules of the dielectric.

This field polarises the dielectric: in a polar dielectric the permanent molecular dipoles turn so as to line up (partially) with $\vec{E}_0$, while in a non-polar dielectric the field pulls the electron cloud and the nucleus slightly apart, creating induced dipoles.

As a result, a layer of bound negative charge appears on the dielectric face touching the positive plate, and a layer of bound positive charge appears on the face touching the negative plate.

These bound charges produce their own field $\vec{E}_p$ inside the slab that is directed opposite to the applied field $\vec{E}_0$.

The net field inside the dielectric is therefore the difference of the two,

$$E = E_0 - E_p = \frac{E_0}{K}$$

where $K>1$ is the dielectric constant.

Because E_p opposes E_0, the resultant field is smaller, so **the electric field between the plates decreases by the factor K** when the slab is inserted.

Final answer. The bound (polarisation) charges set up an opposing field, so the net field falls to E_0/K.

← *Back to Q71*

Q72

An electric dipole consisting of charges $+q$ and $-q$ separated by a distance L is in stable equilibrium in a uniform electric field $\vec{E}$. The electrostatic potential energy of the dipole is

(A) qLE

(B) zero

(C) **−qLE**

(D) −2qLE

[2020 • Set 55-1-1]

SOLUTION

The potential energy of an electric dipole of moment p making an angle θ with a uniform field $\vec{E}$ is

$U = -pE\theta$.

A dipole is in stable equilibrium when its moment is aligned with the field, i.e. $\theta = 0°$, because then U is a minimum.

$U_{stable} = -pE\cos 0° = -pE$.

For charges $+q$ and $-q$ separated by L, the dipole moment is $p = qL$, so

$U_{stable} = -(qL)E = -qLE$.

This is option (C). Option (A) +qLE is the energy of the unstable position ($\theta = 180°$), option (B) zero corresponds to $\theta = 90°$, and option (D) has no physical basis. Hence **the potential energy in stable equilibrium is −qLE**.

TIP

Sign matters: minimum (most negative) energy = stable equilibrium ($\theta=0°$); maximum energy = unstable ($\theta=180°$).

Final answer. Option **(C)**: the potential energy is **−qLE**.

← *Back to Q72*

Q73

The potential difference between two points in vacuum is V_0. *If vacuum is replaced by a medium of dielectric constant K, the new value of potential difference will be* _____.

[2020 • Set 55-2-3]

SOLUTION

When the vacuum between the two points is replaced by a dielectric medium of constant K, polarisation of the medium sets up an opposing field that weakens the electric field at every point by the factor K.

$$E = \frac{E_0}{K}.$$

The potential difference is the line integral of the field between the two points,

$$V = -\int \vec{E} \cdot \vec{dl}.$$

Since E at every point is reduced by the same factor K, the integral, and hence the potential difference, is reduced by K as well.

$$V = \frac{V_0}{K}.$$

Therefore **the new potential difference becomes** $\frac{V_0}{K}$.

Final answer. The new potential difference is $\frac{V_0}{K}$.

← Back to Q73

Q74

If a positive charge is displaced against the electric field in which it was situated, then

(A) work will be done by the electric field on the charge.

(B) the intensity of the electric field decreases.

(C) energy of the system will decrease.

(D) energy will be provided by external source displacing the charge.

[2020 • Set 55-3-1]

SOLUTION

The electrostatic force on a positive charge in a field is $\vec{F} = q\vec{E}$, which points along $\vec{E}$.

To displace the charge against the field means moving it opposite to this force.

The field therefore does negative work on the charge, so to actually move it an external agent must do positive work on it.

$W_{ext} = -W_{field} = +q\,(V_{final} - V_{initial}) > 0,$

and this work is stored as increased potential energy of the charge.

Hence the energy needed must come from an external source. This is option (D). Option (A) is wrong because the field does negative (not positive) work; option (B) is wrong because moving a test charge does not change the source field $\vec{E}$; option (C) is wrong because the potential energy of the system increases, it does not decrease. Therefore **energy is provided by the external source displacing the charge**.

Final answer. Option **(D)**: energy is provided by the external source displacing the charge.

← Back to Q74

Q75

Two capacitors of capacitances C_1 and C_2 are connected in parallel. If a charge Q is given to the combination, the ratio of the charge on the capacitor C_1 to the charge on C_2 will be

(A) $\dfrac{C_1}{C_2}$

(B) $\dfrac{C_2}{C_1}$

(C) $\sqrt{\dfrac{C_1}{C_2}}$

(D) $\sqrt{\dfrac{C_2}{C_1}}$

[2020 • Set 55-3-1]

SOLUTION

Capacitors joined in parallel are connected across the same two terminals, so they have the same potential difference V.

The charge stored on each is $Q = CV$, which is directly proportional to its capacitance.

$$Q_1 = C_1 V$$

$$Q_2 = C_2 V$$

Taking the ratio, the common V cancels,

$$\frac{Q_1}{Q_2} = \frac{C_1 V}{C_2 V} = \frac{C_1}{C_2}.$$

This is option (A). Option (B) inverts the ratio (true only for series), and the square-root forms (C) and (D) have no basis here. Hence **the ratio of charges is $\dfrac{C_1}{C_2}$.**

COMMON MISTAKE

Parallel = same voltage (charge splits in ratio of C); series = same charge (voltage splits in inverse ratio of C). Do not mix the two.

Final answer. Option **(A)**: $\frac{Q_1}{Q_2} = \frac{C_1}{C_2}$.

← *Back to Q75*

Q76

A charge Q is kept at the centre of a circle of radius r. A test charge q_0 *is carried from a point X to the point Y on this circle such that arc* XY *subtends an angle of* 60° *at the centre of the circle. The amount of work done in this process will be*

(A) $\frac{1}{4\pi\varepsilon_0}\frac{Qq_0}{2r}$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{\sqrt{3}\,Qq_0}{2r}$

(C) Zero

(D) $\frac{\sqrt{3}\,Qq_0}{4\pi\varepsilon_0\, r}$

[2020 • Set 55-3-2]

SOLUTION

The point charge Q sits at the centre of the circle, so every point of the circle is at the same distance r from it.

The potential at distance r from Q is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r},$$

which has the same value at both X and Y. The circle is thus an equipotential.

$$V_X = V_Y.$$

The work done in carrying the test charge q_0 from X to Y depends only on the potential difference between the end points,

$$W = q_0\,(V_Y - V_X) = q_0 \times 0 = 0.$$

The 60° angle is irrelevant because the radius does not change. This is option (C). Options (A), (B) and (D) all give non-zero values and are wrong. Hence **the work done is zero**.

Final answer. Option **(C)**: the work done is zero.

← *Back to Q76*

Q77

A parallel plate capacitor is charged to V volt by a battery. The battery is disconnected and the separation between the plates is halved. The new potential difference across the capacitor will be

(A) $\frac{V}{2}$

(B) V

(C) 2V

(D) $\frac{V}{4}$

[2020 • Set 55-3-2]

SOLUTION

After charging, the battery is disconnected, so the charge Q on the plates stays fixed.

The capacitance of a parallel plate capacitor is

$$C = \frac{\varepsilon_0 A}{d}.$$

Halving the separation d doubles the capacitance,

$$C' = \frac{\varepsilon_0 A}{d/2} = \frac{2\varepsilon_0 A}{d} = 2C.$$

Since Q is unchanged, the new potential difference is

$$V' = \frac{Q}{C'} = \frac{Q}{2C} = \frac{V}{2}.$$

This is option (A). The other options ignore the doubling of C at constant charge. Hence **the new potential difference is** $\frac{V}{2}$.

COMMON MISTAKE

When the battery is OFF, Q is constant; when the battery stays ON, V is constant. Decide this first, then apply $Q = CV$.

Final answer. Option **(A)**: the new potential difference is $\frac{V}{2}$.

← Back to Q77

Q78

A charge Q is uniformly distributed over the surface of a spherical shell of radius R. The work done in bringing a test charge Q_0 *from its centre to its surface is*

(A) $\frac{1}{4\pi\varepsilon_0}\frac{QQ_0}{R}$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{QQ_0}{2R}$

(C) $\frac{QQ_0}{\varepsilon_0 R}$

(D) Zero

[2020 • Set 55-3-3]

SOLUTION

For a uniformly charged spherical shell, the field at any interior point is zero,

$E_{in} = 0$.

Since $V = -\int E\,dr$ and $E = 0$ inside, the potential does not change anywhere inside; it stays equal to its surface value,

$$V_{centre} = V_{surface} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R}.$$

The work done in carrying the test charge Q_0 from the centre to the surface is

$$W = Q_0\,(V_{surface} - V_{centre}) = Q_0 \times 0 = 0.$$

This is option (D). Options (A), (B) and (C) treat the inside potential as varying, which is incorrect for a shell. Hence **the work done is zero**.

Final answer. Option **(D)**: the work done is zero.

← *Back to Q78*

Q79

A charged particle is placed between the two plates of a charged parallel plate capacitor. It experiences a force F. If one plate is removed, then the force on the particle will be

(A) 2F

(B) F

(C) $\frac{F}{2}$

(D) Zero

[2020 • Set 55-3-3]

SOLUTION

Each charged plate of a parallel plate capacitor produces a field of magnitude

$$E_{single} = \frac{\sigma}{2\varepsilon_0}.$$

Between the two oppositely charged plates these two fields point the same way and add up,

$$E_{both} = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}.$$

The force on the particle of charge q is then

$$F = qE_{both} = \frac{q\sigma}{\varepsilon_0}.$$

When one plate is removed, only the field of the single remaining plate acts,

$$E_{one} = \frac{\sigma}{2\varepsilon_0} = \frac{E_{both}}{2}.$$

So the new force is

$$F' = qE_{one} = \frac{F}{2}.$$

This is option (C). Hence **the force on the particle becomes $\frac{F}{2}$.**

CONCEPT

A single sheet gives $\sigma/2\varepsilon_0$; two opposite sheets give σ/ε_0 between them. Removing one plate halves the field, hence halves the force.

Final answer. Option **(C)**: the force becomes $\frac{F}{2}$.

← *Back to Q79*

Q80

Fill in the blank with appropriate answer.
A proton released from rest in an electric field, will start moving towards a region of _____ potential in the field.

[2020 • Set 55-4-1]

SOLUTION

A proton carries a positive charge, so the electric force on it is $\vec{F} = q\vec{E}$, directed along the field $\vec{E}$.

The field always points from higher potential to lower potential.

Released from rest, the proton accelerates along $\vec{E}$, that is, in the direction of decreasing potential.

As it moves, its potential energy falls and its kinetic energy rises, consistent with motion towards lower potential.

Therefore **the proton starts moving towards a region of lower (decreasing) potential**.

Final answer. The proton moves towards a region of lower (decreasing) potential.

← *Back to Q80*

Q81

The work done in moving a charged particle between two points in a uniform electric field does not depend on the path followed by the particle. Why?

[2020 • Set 55-4-1]

SOLUTION

The electrostatic force is a conservative force, so the work done in moving a charge between two points depends only on the positions of those two points, not on the path joining them.

A force is conservative if it can be written as the gradient of a potential, $\vec{F} = -q\,\nabla V$. For such a force the work done in carrying a charge q from point A to point B is

$$W = -q\int_A^B \nabla V \cdot d\vec{l} = q\,(V_A - V_B)$$

This result contains only the potentials V_A and V_B at the end points, so it is the same for every route taken from A to B.

Equivalently, the work done by the electrostatic force around any closed loop is zero, $\oint \vec{F}\cdot d\vec{l} = 0$, which is the defining property of a conservative field.

Hence the work done is path independent because the electrostatic force is conservative.

Final answer. The work is path independent because the electrostatic force is a conservative force.

← *Back to Q81*

Q82

A charged particle $(+q)$ moves in a uniform electric field $(\vec{E})$ in the direction opposite to $\vec{E}$. What will be the effect on its electrostatic potential energy during its motion?

[2020 • Set 55-4-3]

SOLUTION

The force on a positive charge in a uniform field is $\vec{F} = q\vec{E}$, which points along $\vec{E}$.

When the charge $(+q)$ moves opposite to $\vec{E}$, it moves against this electric force. An external agent must do positive work on the charge, and this work is stored as electrostatic potential energy.

The same conclusion follows from $\vec{E} = -\nabla V$: moving opposite to $\vec{E}$ takes the charge to a region of higher potential V. Since $U = qV$ with $q > 0$, a larger V means a larger U.

Therefore the electrostatic potential energy of the charge increases during its motion.

Final answer. The electrostatic potential energy of the charge increases.

← *Back to Q82*

Q83

A point charge is placed at the centre of a hollow conducting sphere of internal radius r and outer radius 2r. *The ratio of the surface charge density of the inner surface to that of the outer surface will be* _____ .

[2020 • Set 55-5-1]

SOLUTION

A point charge $+q$ at the centre induces a charge $-q$ on the inner surface of the shell (so that the field inside the conducting wall is zero, by Gauss's law) and a charge $+q$ on the outer surface (by charge conservation, the shell being neutral).

The induced charges spread uniformly over each spherical surface, so the surface charge densities are

$$\sigma_{\text{inner}} = \frac{-q}{4\pi\, r^2}$$

$$\sigma_{\text{outer}} = \frac{+q}{4\pi\,(2r)^2} = \frac{+q}{16\pi\, r^2}$$

Taking the ratio,

$$\frac{\sigma_{\text{inner}}}{\sigma_{\text{outer}}} = \frac{-q/(4\pi r^2)}{+q/(16\pi r^2)} = \frac{-16\pi r^2}{4\pi r^2} = -4$$

Hence the required ratio is −4.

COMMON MISTAKE

Watch the sign: the inner charge is negative, so the ratio is −4, not 4. Many students drop the minus and lose the mark.

Final answer. The ratio of inner to outer surface charge density is **−4**.

← *Back to Q83*

Q84

Draw the equipotential surfaces due to an isolated point charge.

[2019 • Set 55-1-2]

SOLUTION

The potential due to an isolated point charge q is $V(r) = \frac{kq}{r}$, which depends only on the distance r from the charge.

A surface of constant potential therefore corresponds to a constant value of r, so the equipotential surfaces are a family of concentric spheres centred on the charge.

Diagram description: Mark the point charge $+q$ at the centre. Draw three or four concentric circles (representing spheres) around it; each circle is one equipotential surface and is labelled with a decreasing value of potential as the radius grows, for example $V_1 > V_2 > V_3$ outward. The successive circles are drawn farther apart as r increases, since V falls off like $1/r$. The electric field lines are straight radial arrows pointing outward from q (for a positive charge), and each radial line cuts every equipotential circle at right angles.

The equipotential surfaces of an isolated point charge are concentric spheres centred on the charge.

Final answer. The equipotential surfaces are concentric spheres centred on the point charge.

← Back to Q84

Q85

Why is the electrostatic potential inside a charged conducting shell constant throughout the volume of the conductor?

[2019 • Set 55-5-1]

SOLUTION

Inside a conductor in electrostatic equilibrium the electric field is zero, $\vec{E}_{\text{inside}} = \vec{0}$. If the field were not zero, the free electrons would experience a force and keep drifting, which contradicts the equilibrium condition.

The field and potential are related by $\vec{E} = -\nabla V$. Since the field is zero everywhere inside,

$$\nabla V = \vec{0} \Rightarrow V = \text{constant}$$

So the potential cannot change from point to point inside the conductor.

Because the potential is continuous, this constant interior value equals the value on the surface, $V_{\text{inside}} = V_{\text{surface}} = \frac{kQ}{R}$.

Hence the potential is constant throughout the conductor because the field inside it is zero.

Final answer. Since $\vec{E}=0$ inside the conductor, $\nabla V=0$, so the potential stays constant throughout.

← Back to Q85

Q86

Does the charge given to a metallic sphere depend on whether it is hollow or solid? Give reason for your answer.

[2017]

SOLUTION

No, the charge that can be given to a metallic sphere does not depend on whether the sphere is hollow or solid.

Reason: In electrostatic equilibrium the field inside the conducting material is zero. Apply Gauss's law to any closed (Gaussian) surface drawn entirely within the metal,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enclosed}}{\varepsilon_0} = 0 \Rightarrow q_{enclosed} = 0$$

So no excess charge can sit inside the body of the conductor; all the charge given to it resides only on its outer surface. This is true whether the interior is solid metal or hollow, because the interior carries no charge in either case.

The maximum charge the sphere can hold therefore depends only on its outer radius and the dielectric strength of the surrounding medium, not on its internal structure.

Hence the charge does not depend on whether the sphere is hollow or solid.

Final answer. No; the charge resides only on the outer surface, so it is independent of the sphere being hollow or solid.

← *Back to Q86*

Q87

A point charge Q is placed at point 'O' as shown in the figure. Is the potential at point A, i.e. V_A, greater, smaller or equal to potential, V_B, at point B, when Q is (i) positive, and (ii) negative charge?



[2017]

SOLUTION

The potential due to a point charge Q at distance r is $V = \frac{kQ}{r}$.

From the figure, point A is nearer to O than point B, so $\mathbf{r_A < r_B}$. Because V varies as $1/r$, the magnitude of the potential is larger at the nearer point, $\mathbf{|V_A| > |V_B|}$.

(i) When Q is positive: both V_A and V_B are positive, and the nearer point has the larger value, so

$$V_A > V_B$$

(ii) When Q is negative: both V_A and V_B are negative, and the nearer point is more negative, so

$$V_A < V_B$$

Thus $\mathbf{V_A > V_B}$ for positive Q, and $\mathbf{V_A < V_B}$ for negative Q.

Final answer. For positive Q, $\mathbf{V_A > V_B}$; for negative Q, $\mathbf{V_A < V_B}$.

← *Back to Q87*

Q88

What is the amount of work done in moving a point charge Q around a circular arc of radius r at the centre of which another point charge q is located?

[2016]

SOLUTION

Every point on a circular arc of radius r centred on the charge q is at the same distance r from q, so every point on the arc is at the same potential,

$$V = \frac{kq}{r} = \text{constant}$$

The arc therefore lies on an equipotential surface. The work done in moving a charge Q between two points at the same potential is

$$W = Q\,\Delta V = Q\,(V_{\text{final}} - V_{\text{initial}}) = Q \times 0 = 0$$

Hence the work done in moving the charge Q along the arc is zero.

Final answer. The work done is zero, since the arc is an equipotential surface.

← *Back to Q88*

Q89

"For any charge configuration, equipotential surface through a point is normal to the electric field." Justify.

[2014]

SOLUTION

For any small displacement $d\vec{l}$ taken along an equipotential surface, the potential does not change, so **dV = 0**.

The change in potential for a displacement in a field is

$$dV = -\vec{E} \cdot d\vec{l}$$

Setting **dV = 0** gives

$$\vec{E} \cdot d\vec{l} = 0$$

This must hold for every displacement $d\vec{l}$ lying in the surface. A non-zero vector whose dot product with every tangent vector of the surface is zero can only be perpendicular to the surface.

Hence $\vec{E}$ is normal (perpendicular) to the equipotential surface at every point.

CONCEPT

If the field had a component along the surface, work would be done moving a charge over the equipotential, which is impossible since $\Delta V = 0$. That is the quick physical justification.

Final answer. Since $\vec{E} \cdot \vec{dl} = 0$ for every displacement on the surface, $\vec{E}$ must be normal to it.

← *Back to Q89*

Q90

What is the geometrical shape of equipotential surfaces due to a single isolated charge?

[2013]

SOLUTION

For a single isolated point charge Q, the potential is

$$V(r) = \frac{kQ}{r}$$

which depends only on the distance r from the charge.

A surface of constant potential is therefore a surface of constant r, that is, a sphere centred on the charge. Different values of V give different radii, producing a family of concentric spheres.

Hence the equipotential surfaces due to a single isolated charge are concentric spheres centred on the charge.

Final answer. The equipotential surfaces are concentric spheres centred on the charge.

← *Back to Q90*

Q91

Define the term 'dielectric constant' of a medium.

[2013]

SOLUTION

The dielectric constant (also called relative permittivity) of a medium is defined as the ratio of the permittivity ε of the medium to the permittivity ε_0 of free space.

$$K = \frac{\varepsilon}{\varepsilon_0}$$

Equivalently, it is the factor by which the capacitance of a capacitor increases when the space between its plates is completely filled with the dielectric medium instead of vacuum.

$$K = \frac{C_{\text{medium}}}{C_{\text{vacuum}}}$$

It is a **dimensionless** quantity and its value is always **greater than or equal to 1**.

Final answer. Dielectric constant $\boldsymbol{K = \frac{\varepsilon}{\varepsilon_0}}$, a dimensionless number with $\boldsymbol{K \geq 1}$.

← *Back to Q91*

Q92

Name the physical quantity whose S.I. unit is JC^{-1}. *Is it a scalar or a vector quantity?*

[2012]

SOLUTION

The SI unit $J\,C^{-1}$ means joule per coulomb, i.e. energy per unit charge, which is the same as the volt (V).

The physical quantity having this unit is the **electric potential** (or potential difference).

It is a **scalar** quantity, because it is defined as the work done per unit positive charge in bringing the charge from infinity to the given point, and both work and charge are scalars, so their ratio is also a scalar.

Final answer. The quantity is electric potential (potential difference); it is a scalar.

← *Back to Q92*

Q93

Why is the potential inside a hollow spherical charged conductor constant and has the same value as on its surface?

[2012]

SOLUTION

Inside the material of a charged hollow spherical conductor in electrostatic equilibrium, the electric field is zero.

$$\vec{E}_{\text{inside}} = \vec{0}$$

The electric field is related to the potential by $\vec{E} = -\frac{dV}{dr}$. Since the field is zero everywhere inside,

$$\frac{dV}{dr} = 0 \Rightarrow V = \text{constant}$$

So the potential does not change inside the conductor. Because the potential is continuous across the conductor's surface, this constant interior value must equal the value on the surface.

$$V_{\text{inside}} = V_{\text{surface}} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R}$$

Hence the potential inside is constant and equal to the surface potential.

Final answer. Since $\vec{E}=0$ inside, V is constant and equals the surface value $\frac{kQ}{R}$.

← Back to Q93

Q94

A hollow metal sphere of radius 5 cm *is charged such that the potential on its surface is* 10 *V. What is the potential at the centre of the sphere?*

[2011]

SOLUTION

For a charged hollow conducting sphere, the electric field at every point inside the conductor is zero.

$$\vec{E}_{\text{inside}} = \vec{0}$$

Since $\vec{E} = -\dfrac{dV}{dr}$, a zero field means the potential is constant throughout the interior. By continuity of the potential, this constant value equals the surface potential.

$$V_{\text{centre}} = V_{\text{surface}} = 10\ V$$

Therefore the potential at the centre is **10 *V***, independent of the radius.

COMMON MISTAKE

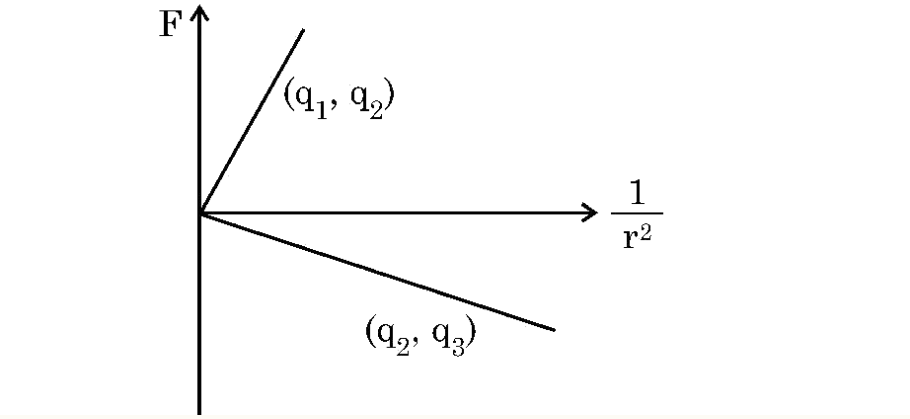
The radius value 5 cm *is a distractor; the inside potential equals the surface potential regardless of radius.*

Final answer. The potential at the centre is **10 *V***.

← Back to Q94

Q95

A point charge Q is placed at point O as shown in the figure. Is the potential difference $V_A - V_B$ *positive, negative or zero, if Q is (i) positive (ii) negative?*



[2011 • Set 55-1-1]

SOLUTION

The potential due to a point charge Q at a distance r from it is given by:

$$V(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$$

From the figure, point A is closer to O than point B, so $\mathbf{r_A < r_B}$.

(i) When Q is positive: both potentials are positive, and the closer point A has the larger value, so $\mathbf{V_A > V_B}$.

$\mathbf{V_A - V_B > 0} \Rightarrow$ the potential difference is **positive**.

(ii) When Q is negative: both potentials are negative, and since A is closer, V_A is more negative than V_B, so $\mathbf{V_A < V_B}$.

$\mathbf{V_A - V_B < 0} \Rightarrow$ the potential difference is **negative**.

Final answer. $\mathbf{V_A - V_B}$ is positive for positive Q and negative for negative Q.

← *Back to Q95*

Q96

What is the electrostatic potential due to an electric dipole at an equatorial point?

[2009]

SOLUTION

The electrostatic potential at a point due to a dipole at distance r, making angle θ with the dipole axis, is:

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\theta}{r^2}$$

For a point on the equatorial line, the angle is $\theta = 90°$, so $\theta = \cos 90° = 0$.

$$V_{\text{equatorial}} = \frac{1}{4\pi\varepsilon_0}\frac{p(0)}{r^2} = 0$$

Physically, every point on the equatorial plane is equidistant from the two equal and opposite charges, so their potential contributions cancel exactly.

Hence the electrostatic potential due to a dipole at any equatorial point is **zero**.

Final answer. The potential at an equatorial point of a dipole is zero.

← *Back to Q96*

Q97

A 500 μ *C charge is at the centre of a square of side* 10 cm. *Find the work done in moving a charge of* 10 μ *C between two diagonally opposite points on the square.*

[2008]

SOLUTION

Two points that are equidistant from a point charge lie on the same equipotential surface, so the potential difference between them is zero.

The central charge is at the centre of the square, and the two diagonally opposite corners are each at the same distance from the centre, namely half the diagonal.

$$r = \frac{\text{diagonal}}{2} = \frac{10\sqrt{2}}{2} = 5\sqrt{2}\ \text{cm}$$

Since both corners are equidistant from the charge, they are at the same potential.

$$\Delta V = V_1 - V_2 = 0$$

The work done in moving a charge between two points is $W = q\,\Delta V$.

$$W = (10\ \mu\text{C}) \times 0 = 0$$

Therefore the work done is **zero**.

Final answer. The work done is zero, since both corners are equidistant from the charge (equipotential).

← *Back to Q97*

Q98

Derive the expression for the electric potential at any point along the axial line of an electric dipole.

[2008]

SOLUTION

Consider a dipole with charge $-q$ at point A and $+q$ at point B, separated by a distance 2a, with the centre O between them. Let P be a point on the axial line at distance r from the centre, on the side of $+q$.

The distances of P from the two charges are:

$$BP = r - a \quad AP = r + a$$

The potential at P is the sum of the potentials due to $+q$ and $-q$.

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r-a} + \frac{1}{4\pi\varepsilon_0}\frac{(-q)}{r+a}$$

$$V = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r-a} - \frac{1}{r+a}\right)$$

Taking the common denominator:

$$V = \frac{q}{4\pi\varepsilon_0}\cdot\frac{(r+a)-(r-a)}{(r-a)(r+a)}$$

$$V = \frac{q}{4\pi\varepsilon_0}\cdot\frac{2a}{r^2-a^2}$$

Using the dipole moment $\boldsymbol{p = q \times 2a = 2qa}$:

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^2-a^2}$$

For a short dipole, $r \gg a$, so a^2 is negligible compared with r^2.

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^2}$$

Thus the axial potential of a dipole falls off as $1/r^2$, faster than the $1/r$ dependence of a point charge.

Final answer. The axial potential is $\boldsymbol{V = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^2-a^2} \approx \frac{1}{4\pi\varepsilon_0}\frac{p}{r^2}}$ for a short dipole.

← *Back to Q98*

Q99

Draw an equipotential surface for a system consisting of two charges Q, −Q separated by a distance 'r' in air.

[2008]

SOLUTION

Draw two charges placed a distance r apart on a horizontal line: $+Q$ on the left and $-Q$ on the right. Mark the midpoint O between them.

Draw a straight vertical line passing through the midpoint O, perpendicular to the line joining the charges. Label this line $\boldsymbol{V = 0}$. This perpendicular bisector plane is the zero-potential equipotential, because every point on it is equidistant from $+Q$ and $-Q$, so the two potentials cancel.

Around $+Q$, draw a set of closed, nearly circular loops (spheres in 3-D) that surround only $+Q$. Label them with positive values, getting larger as they shrink towards $+Q$. These are the $\boldsymbol{V > 0}$ equipotentials.

Around $-Q$, draw a similar set of closed loops surrounding only $-Q$, labelled with negative values, becoming more negative closer to $-Q$. These are the $\boldsymbol{V < 0}$ equipotentials.

The loops are spherical close to each charge but become distorted (oval/dumbbell-shaped) farther out, and they are symmetric on the two sides of the central $\boldsymbol{V = 0}$ plane.

The equipotential surfaces are everywhere **perpendicular** to the electric field lines, which run from $+Q$ to $-Q$.

TIP

Always show the flat $V=0$ surface through the midpoint; examiners look for it specifically in a $+Q$, $-Q$ system.

Final answer. The mid-plane perpendicular bisector is the $V=0$ surface; closed loops surround each charge ($V>0$ near $+Q$, $V<0$ near $-Q$), all perpendicular to the field lines.

← *Back to Q99*

Q100

Define the term 'dielectric constant' of a medium in terms of capacitance of a capacitor.

[2006]

SOLUTION

The dielectric constant of a medium is defined as the ratio of the capacitance C_m of a capacitor when the space between its plates is completely filled with the medium to the capacitance C_0 of the same capacitor with vacuum (or air) between the plates.

$$K = \frac{C_m}{C_0}$$

This is equivalent to $K = \frac{\varepsilon}{\varepsilon_0}$, the ratio of the medium's permittivity to that of free space.

It is a **dimensionless** quantity, always greater than or equal to 1.

Final answer. Dielectric constant $K = \frac{C_m}{C_0}$, the ratio of capacitance with the medium to that with vacuum.

← Back to Q100

Q101

Two isolated metallic spheres S_1 *and* S_2 *of radii* 1 cm *and* 3 cm *respectively are charged such that both have the same charge density* $(\frac{2}{\pi}\times 10^{-9})$ C/m^2. *They are placed far away from each other and connected by a thin wire. Calculate the new charge on sphere* S_2.

[2024 • Set 55-1-1]

SOLUTION

Given: $\mathbf{r_1 = 1\ cm = 1\times 10^{-2}\ m}$ and $\mathbf{r_2 = 3\ cm = 3\times 10^{-2}\ m}$, with the same surface charge density $\sigma = \frac{2}{\pi}\times 10^{-9}\ C/m^2$ on each sphere before they are joined.

The charge on a sphere is $q = \sigma\times$(surface area) $= \sigma\,(4\pi r^2)$.

Initial charge on S_1: $Q_1 = \frac{2}{\pi}\times 10^{-9}\times 4\pi\,(10^{-2})^2 = 8\times 10^{-13}\ C$.

Initial charge on S_2: $Q_2 = \frac{2}{\pi}\times 10^{-9}\times 4\pi\,(3\times 10^{-2})^2 = 8\times 10^{-13}\times 9 = 72\times 10^{-13}\ C$.

Total charge before connection: $Q = Q_1 + Q_2 = (8 + 72)\times 10^{-13} = 8\times 10^{-12}\ C$.

When the two spheres are joined by a thin wire, charge flows until both reach a common potential. So $\frac{kQ_1'}{r_1} = \frac{kQ_2'}{r_2}$.

$\Rightarrow \frac{Q_1'}{Q_2'} = \frac{r_1}{r_2} = \frac{1}{3} \Rightarrow Q_2' = 3\,Q_1'$.

By conservation of charge, $Q_1' + Q_2' = 8\times 10^{-12}\ C$.

Substituting, $Q_1' + 3Q_1' = 8\times 10^{-12} \Rightarrow 4Q_1' = 8\times 10^{-12} \Rightarrow Q_1' = 2\times 10^{-12}\ C$.

Therefore $Q_2' = 3\times 2\times 10^{-12} = 6\times 10^{-12}\ C$.

The **new charge on sphere S_2 is $6\times 10^{-12}\ C = 6$ pC**.

COMMON MISTAKE

After joining, charge spreads so potential is equal, not charge. A common mistake is to assume the charges become equal.

Final answer. The new charge on sphere S_2 is **6 pC** (6×10^{-12} *C*).

← *Back to Q101*

Q102

(a) Obtain an expression for electrostatic potential energy of a system of three charges q, 2q *and* −3q *placed at the vertices of an equilateral triangle of side a.*

OR

(b) Two small conducting balls A and B of radius r_1 *and* r_2 *have charges* q_1 *and* q_2 *respectively. They are connected by a wire. Obtain the expression for charges on A and B, in equilibrium.*

[2023 • Set 55-4-1]

SOLUTION

(a) The total electrostatic potential energy of a system of point charges equals the sum of the potential energies of every distinct pair of charges. For charges q_i and q_j separated by r_{ij}, the pair energy is $\frac{1}{4\pi\varepsilon_0}\frac{q_i q_j}{r_{ij}}$.

Here *q*, 2q and −3q sit at the corners of an equilateral triangle of side *a*, so every pair separation is *a*.

$$U = \frac{1}{4\pi\varepsilon_0 a}[(q)(2q) + (2q)(-3q) + (-3q)(q)].$$

$$U = \frac{1}{4\pi\varepsilon_0 a}[2q^2 - 6q^2 - 3q^2].$$

$$U = \frac{1}{4\pi\varepsilon_0 a}(-7q^2).$$

$$U = -\frac{7q^2}{4\pi\varepsilon_0 a}.$$

The negative sign shows the system is bound (energy is released in assembling it).

OR (b) Two conducting balls A and B (radii r_1, r_2; charges q_1, q_2) are joined by a wire. Charge redistributes until both reach a common potential.

At equilibrium, $\frac{kQ_1'}{r_1} = \frac{kQ_2'}{r_2} \Rightarrow \frac{Q_1'}{Q_2'} = \frac{r_1}{r_2}$.

By conservation of charge, $Q_1' + Q_2' = q_1 + q_2$.

Writing $Q_1' = \frac{r_1}{r_2} Q_2'$ and substituting: $\frac{r_1}{r_2} Q_2' + Q_2' = q_1 + q_2 \Rightarrow Q_2' \frac{r_1 + r_2}{r_2} = q_1 + q_2$.

$Q_2' = \frac{(q_1 + q_2)\, r_2}{r_1 + r_2}$ and similarly $Q_1' = \frac{(q_1 + q_2)\, r_1}{r_1 + r_2}$.

Final answer. (a) $U = -\frac{7q^2}{4\pi\varepsilon_0 a}$. OR (b) $Q_1' = \frac{(q_1+q_2)r_1}{r_1+r_2}$ and $Q_2' = \frac{(q_1+q_2)r_2}{r_1+r_2}$.

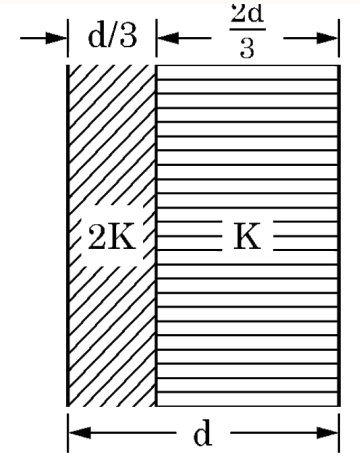
← *Back to Q102*

Q103

(iii) A capacitor A of capacitance C, having charge Q is connected across another uncharged capacitor B of capacitance 2C. Find an expression for (a) the potential difference across the combination and (b) the charge lost by capacitor A.

OR

(iii) Two slabs of dielectric constants 2K and K fill the space between the plates of a parallel plate capacitor of plate area A and plate separation d as shown in figure. Find an expression for capacitance of the system.



[2023 • Set 55-5-1]

SOLUTION

(iii) Capacitor A has capacitance *C* and charge *Q*; capacitor B has capacitance 2C and is uncharged. When connected together, their plates join and they share a common potential difference *V*'.

(a) Charge is conserved, and after connection the two capacitors are effectively in parallel, so the total charge Q is shared at common voltage V': $Q = (C + 2C)\,V'$.

$$Q = 3C\,V' \Rightarrow V' = \frac{Q}{3C}.$$

(b) The charge that stays on A after connection is $Q_A' = C\,V' = C\times\frac{Q}{3C} = \frac{Q}{3}$.

Charge lost by A $= Q - Q_A' = Q - \frac{Q}{3}$.

$\Delta Q_A = \frac{2Q}{3}$ (this charge has flowed onto capacitor B).

OR (iii) Two dielectric slabs of constants 2K and K, each of thickness $d/2$, fill the gap one above the other. The plate area is A. Stacked slabs behave as two capacitors in series.

$$C_1 = \frac{(2K)\varepsilon_0 A}{d/2} = \frac{4K\varepsilon_0 A}{d}.$$

$$C_2 = \frac{K\varepsilon_0 A}{d/2} = \frac{2K\varepsilon_0 A}{d}.$$

For series: $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d}{4K\varepsilon_0 A} + \frac{d}{2K\varepsilon_0 A}$.

$$\frac{1}{C} = \frac{d}{4K\varepsilon_0 A} + \frac{2d}{4K\varepsilon_0 A} = \frac{3d}{4K\varepsilon_0 A}.$$

$$C = \frac{4K\varepsilon_0 A}{3d}.$$

CONCEPT

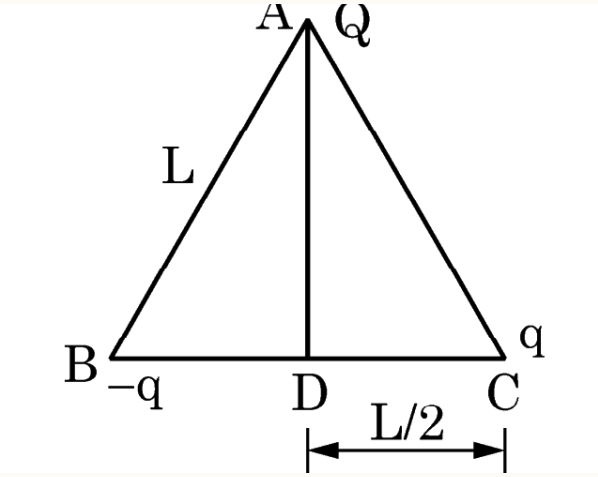
Slabs stacked one above the other (splitting the gap d) act in series; slabs placed side by side (splitting the area A) act in parallel.

Final answer. (iii)(a) $V' = \frac{Q}{3C}$ and (b) charge lost by A $= \frac{2Q}{3}$. OR $C = \frac{4K\varepsilon_0 A}{3d}$.

← *Back to Q103*

Q104

Three point charges Q, q and −q are kept at the vertices of an equilateral triangle of side L as shown in figure. What is
(i) the electrostatic potential energy of the arrangement? and
(ii) the potential at point D?



[2023 • Set 55-5-3]

SOLUTION

The triangle ABC has side L with Q at A, q at B and $-q$ at C. Point D is the midpoint of side BC.

Geometry: $\mathbf{DB = DC = \frac{L}{2}}$, and the altitude $\mathbf{DA = \frac{\sqrt{3}}{2}L}$.

(i) The potential energy of the system is the sum over all three pairs, each separated by L.

$$U = \frac{1}{4\pi\varepsilon_0 L}[(Q)(q) + (q)(-q) + (-q)(Q)].$$

$$U = \frac{1}{4\pi\varepsilon_0 L}[Qq - q^2 - Qq].$$

$$U = \frac{1}{4\pi\varepsilon_0 L}(-q^2).$$

$$U = -\frac{q^2}{4\pi\varepsilon_0 L}$$ (the two Q-terms cancel because q and $-q$ are equidistant from A).

(ii) Potential at D is the algebraic sum of the potentials due to each charge (potential is a scalar).

$$V_D = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q}{DA} + \frac{q}{DB} + \frac{-q}{DC}\right].$$

$$V_D = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q}{\frac{\sqrt{3}}{2}L} + \frac{q}{L/2} - \frac{q}{L/2}\right].$$

The contributions of $+q$ and $-q$ cancel (equal distances $L/2$), leaving

$$V_D = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2Q}{\sqrt{3}L} = \frac{Q}{2\sqrt{3}\,\pi\varepsilon_0 L}.$$

$$V_D = \frac{Q}{2\sqrt{3}\,\pi\varepsilon_0 L} = \frac{\sqrt{3}\,Q}{6\pi\varepsilon_0 L}.$$

CONCEPT

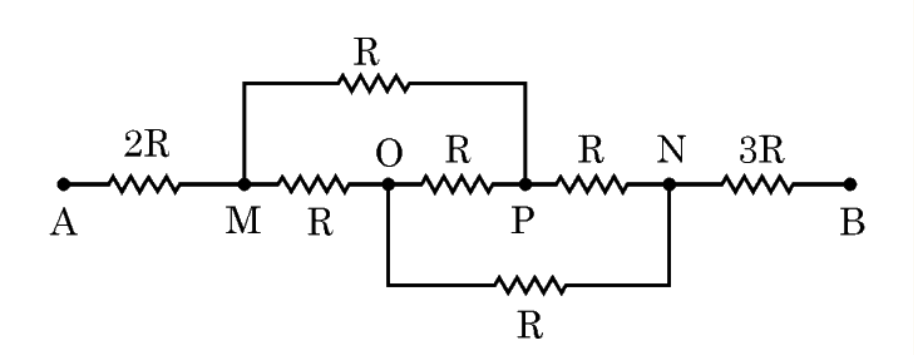
Potential energy uses pair products of charges; potential at a point adds each charge over its own distance. Keep the signs of the charges throughout.

Final answer. (i) $U = -\frac{q^2}{4\pi\varepsilon_0 L}$ and (ii) $V_D = \frac{\sqrt{3}\,Q}{6\pi\varepsilon_0 L}$.

← *Back to Q104*

Q105

The space between the plates of a parallel plate capacitor is completely filled in two ways. In the first case, it is filled with a slab of dielectric constant K. In the second case, it is filled with two slabs of equal thickness and dielectric constants K_1 *and* K_2 *respectively as shown in the figure. The capacitance of the capacitor is same in the two cases. Obtain the relationship between K,* K_1 *and* K_2.



[2020 • Set 55-1-1]

SOLUTION

Let the plate area be A and plate separation be d.

Case 1: the gap is fully filled with a single dielectric of constant K.

$$C_{\text{single}} = \frac{K\varepsilon_0 A}{d}.$$

Case 2: two slabs of equal thickness $\frac{d}{2}$ and constants K_1, K_2 are stacked one above the other. Stacked slabs act as two capacitors in series.

$$C_1 = \frac{K_1\varepsilon_0 A}{d/2} = \frac{2K_1\varepsilon_0 A}{d}.$$

$$C_2 = \frac{K_2\varepsilon_0 A}{d/2} = \frac{2K_2\varepsilon_0 A}{d}.$$

For the series combination: $\frac{1}{C_{\text{stacked}}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d}{2\varepsilon_0 A}\left(\frac{1}{K_1} + \frac{1}{K_2}\right).$

$$\frac{1}{C_{\text{stacked}}} = \frac{d}{2\varepsilon_0 A} \cdot \frac{K_1 + K_2}{K_1 K_2}.$$

$$C_{\text{stacked}} = \frac{2K_1 K_2 \varepsilon_0 A}{(K_1 + K_2)\, d}.$$

The two capacitances are equal: $\frac{K\varepsilon_0 A}{d} = \frac{2K_1 K_2 \varepsilon_0 A}{(K_1 + K_2)\, d}.$

Cancelling $\frac{\varepsilon_0 A}{d}$ from both sides,

$$K = \frac{2K_1 K_2}{K_1 + K_2}.$$

The equivalent constant K is the harmonic mean of K_1 and K_2.

Final answer. $K = \frac{2K_1 K_2}{K_1 + K_2}$.

← *Back to Q105*

Q106

N small conducting liquid droplets, each of radius r, are charged to a potential V each. These droplets coalesce to form a single large drop without any charge leakage. Find

the potential of the large drop.

[2020 • Set 55-1-2]

SOLUTION

Each small droplet has radius r and is charged to surface potential V. For a charged sphere, $V = \frac{kq}{r}$, so the charge on one droplet is $q = \frac{Vr}{k}$.

When the N droplets coalesce into one big drop, the total volume is conserved (the liquid does not change volume).

$$\frac{4}{3}\pi R^3 = N \times \frac{4}{3}\pi r^3.$$

$$R^3 = N r^3 \Rightarrow R = N^{1/3} r.$$

Charge is also conserved (no leakage), so the big drop carries

$$Q = N q = N \times \frac{Vr}{k}.$$

The potential of the big drop is

$$V_{big} = \frac{kQ}{R} = \frac{k \times \frac{NVr}{k}}{N^{1/3} r}.$$

$$V_{big} = \frac{NVr}{N^{1/3} r} = N^{1 - 1/3} V.$$

$$V_{big} = N^{2/3} V.$$

The potential of the large drop is $N^{2/3}$ times the potential of each small droplet, because charge grows by N while radius grows only by $N^{1/3}$.

TIP

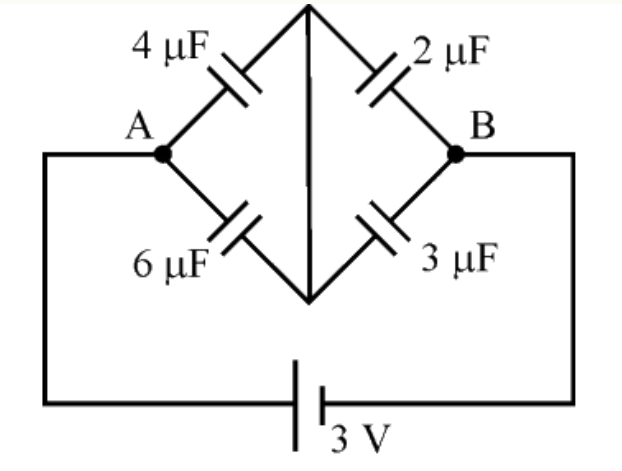
Volume conservation gives the new radius ($R = N^{1/3} r$), and charge conservation gives the new charge. Combine both before finding the potential.

Final answer. The potential of the large drop is $V_{big} = N^{2/3} V$.

← *Back to Q106*

Q107

Find the total charge stored in the network of capacitors connected between A and B as shown in figure:



[2020 • Set 55-4-1]

SOLUTION

The network is reduced to a single equivalent capacitance C_{eq} between terminals *A* and *B*, after which the total stored charge is $\mathbf{Q_{total} = C_{eq}\, V}$.

For the given arrangement, two branches are connected in parallel across the 3 *V* supply. One branch has 4 μF and 2 μF in series; the other has 6 μF and 3 μF in series.

Series combination of the first branch: $\mathbf{C_{4,2} = \frac{4\times 2}{4+2} = \frac{8}{6} = \frac{4}{3}\ \mu F.}$

Series combination of the second branch: $\mathbf{C_{6,3} = \frac{6\times 3}{6+3} = \frac{18}{9} = 2\ \mu F.}$

The two branches are in parallel, so $\mathbf{C_{eq} = C_{4,2} + C_{6,3} = \frac{4}{3} + 2 = \frac{10}{3}\ \mu F.}$

Total charge stored: $\mathbf{Q_{total} = C_{eq}\, V = \frac{10}{3}\times 10^{-6}\times 3.}$

$\mathbf{Q_{total} = 10\times 10^{-6}\ C.}$

$\mathbf{Q_{total} = 10\ \mu C.}$

TIP

Reduce series capacitors with $\frac{1}{C_s}=\Sigma\frac{1}{C_i}$ and parallel ones with $C_p=\Sigma C_i$, then use $Q=C_{eq}V$ for total charge.

Final answer. The total charge stored in the network is $Q_{total} = 10\ \mu C$.

← *Back to Q107*

Q108

You are given three capacitors of 2 μF, 3 μF *and* 4 μF, *respectively.*

(a) Form a combination of all these capacitors of equivalent capacitance $\frac{13}{3}$ μF.

(b) What is the maximum and minimum value of the equivalent capacitance that can be obtained by connecting these capacitors?

[2020 • Set 55-4-3]

SOLUTION

The three capacitors are $C_1 = 2\ \mu F$, $C_2 = 3\ \mu F$ and $C_3 = 4\ \mu F$.

(a) Required equivalent capacitance is $\frac{13}{3}$ μF. Connect the 2 μF and 4 μF in series, and put this combination in parallel with the 3 μF.

Series of 2 μF and 4 μF: $C_{2,4} = \frac{2\times 4}{2+4} = \frac{8}{6} = \frac{4}{3}\ \mu F$.

This in parallel with 3 μF: $C_{eq} = \frac{4}{3} + 3 = \frac{4+9}{3} = \frac{13}{3}\ \mu F$, as required.

Circuit description: the 2 μF and 4 μF form one chain (series) between terminals A and B, and the 3 μF is connected directly across the same terminals A and B.

(b) Maximum capacitance is obtained with all three in parallel.

$C_{max} = 2 + 3 + 4 = 9\ \mu F$.

Minimum capacitance is obtained with all three in series.

$$\frac{1}{C_{min}} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{6+4+3}{12} = \frac{13}{12}.$$

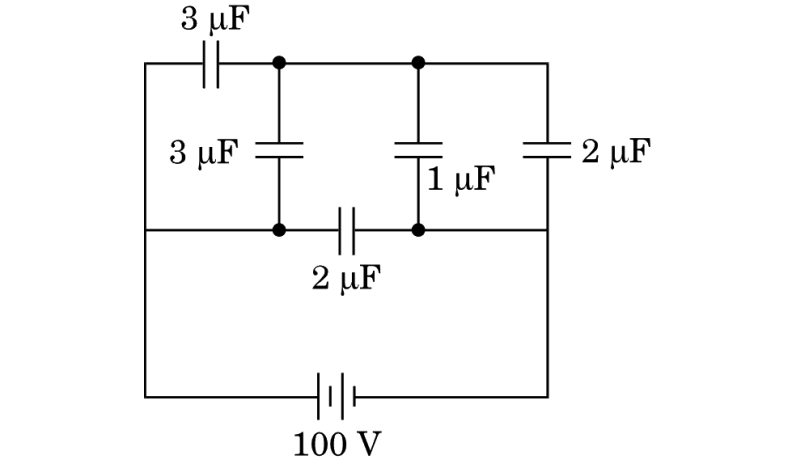
$C_{min} = \frac{12}{13}\ \mu F \approx 0.92\ \mu F$, while $C_{max} = 9\ \mu F$.

Final answer. (a) Series (**2 μF**, **4 μF**) in parallel with **3 μF** gives $\frac{13}{3}\ \mu F$. (b) $C_{max} = 9\ \mu F$, $C_{min} = \frac{12}{13}\ \mu F$.

← *Back to Q108*

Q109

The figure shows a network of five capacitors connected to a 100 *V supply. Calculate the total energy stored in the network.*



[2019 • Set 55-3-1]

SOLUTION

The total energy stored in the network is $U = \frac{1}{2} C_{eq} V^2$, so first reduce the five capacitors to a single equivalent capacitance C_{eq} across the 100 *V* supply.

For the given figure, two branches sit in parallel across the supply. Branch 1 has two 3 μF capacitors in series; branch 2 has a 2 μF capacitor in series with the parallel pair of two 1 μF capacitors.

Branch 1: $C_a = \frac{3 \times 3}{3 + 3} = \frac{9}{6} = 1.5\ \mu F$.

Branch 2, parallel pair: $1 + 1 = 2\ \mu F$.

Branch 2 in series with 2 μF: $C_b = \frac{2 \times 2}{2 + 2} = 1\ \mu F$.

The two branches are in parallel: $C_{eq} = C_a + C_b = 1.5 + 1 = 2.5\ \mu F$.

Total energy: $U = \frac{1}{2} C_{eq} V^2 = \frac{1}{2} \times (2.5 \times 10^{-6}) \times (100)^2$.

$U = \frac{1}{2} \times 2.5 \times 10^{-6} \times 10^4$.

$U = 1.25 \times 10^{-2}\ J$.

$U = 12.5\ mJ$.

Final answer. The total energy stored in the network is $U = 12.5\ mJ$ ($1.25 \times 10^{-2}\ J$).

← *Back to Q109*

Q110

Two identical capacitors of 12 pF each are connected in series across a 50 V battery. Calculate the electrostatic energy stored in the combination. If these were connected in parallel across the same battery, find out the value of the energy stored in this combination.

[2019 • Set 55-5-1]

SOLUTION

Two identical capacitors, each $C = 12\ pF = 12 \times 10^{-12}\ F$, are connected to a $V = 50\ V$ battery.

(i) Series combination. $C_s = \frac{C \times C}{C + C} = \frac{12 \times 12}{12 + 12} = \frac{144}{24} = 6\ pF$.

Energy stored: $U_s = \frac{1}{2} C_s V^2 = \frac{1}{2} \times (6 \times 10^{-12}) \times (50)^2$.

$U_s = \frac{1}{2} \times 6 \times 10^{-12} \times 2500$.

$U_s = 7.5 \times 10^{-9}\ J = 7.5\ nJ$.

(ii) Parallel combination. $C_p = C + C = 12 + 12 = 24\ pF$.

Energy stored: $U_p = \frac{1}{2} C_p V^2 = \frac{1}{2} \times (24 \times 10^{-12}) \times (50)^2$.

$U_p = \frac{1}{2} \times 24 \times 10^{-12} \times 2500.$

$U_p = 3.0 \times 10^{-8}\, J = 30\ \text{nJ}.$

So the **series combination stores 7.5 nJ** and the **parallel combination stores 30 nJ**, the parallel value being four times larger since $C_p/C_s = 24/6 = 4$.

COMMON MISTAKE

At a fixed battery voltage, use $U = \frac{1}{2}CV^2$; do not use $U = \frac{Q^2}{2C}$ blindly, since Q differs between the series and parallel cases.

Final answer. Series energy = **7.5 nJ** and parallel energy = **30 nJ**.

← *Back to Q110*

Q111

Two identical capacitors of 10 pF each are connected in turn (i) in series, and (ii) in parallel across a 20 V battery. Calculate the potential difference across each capacitor in the first case and charge acquired by each capacitor in the second case.

[2019 • Set 55-5-2]

SOLUTION

When identical capacitors are joined in series, the same charge Q resides on each, so they share the supply voltage in inverse proportion to their capacitance. Being identical, each takes an equal share of the supply.

(i) Series across the 20 V battery:

$$V_{\text{each}} = \frac{V_{\text{supply}}}{2} = \frac{20}{2}$$

$$V_{\text{each}} = 10\ V$$

So the potential difference across each capacitor is 10 V.

(ii) Parallel across the 20 V battery: each capacitor is connected directly across the battery, so the full 20 V appears across each.

$$Q_{each} = CV$$

$$Q_{each} = (10\times 10^{-12})(20)$$

$$Q_{each} = 2\times 10^{-10}\ C = 200\ \text{pC}$$

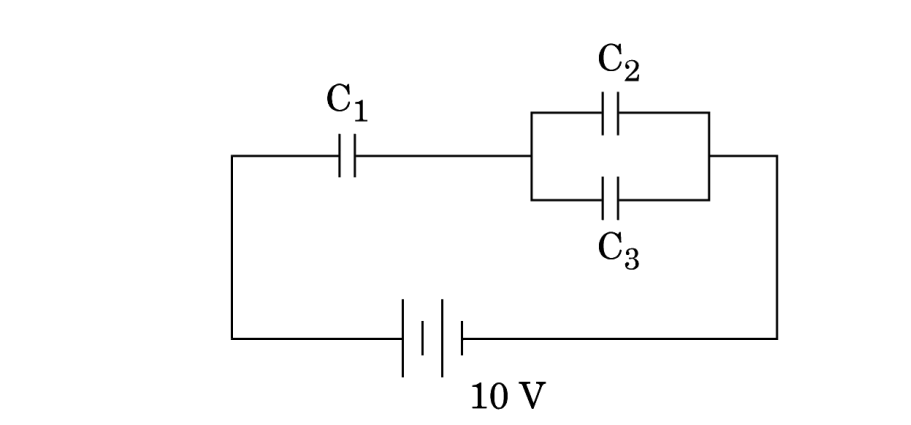
Hence each capacitor in parallel acquires a charge of **200 pC (=0.2 nC)**.

Final answer. Series: 10 V across each; Parallel: 200 pC (0.2 nC) on each capacitor.

← Back to Q111

Q112

The figure shows a network of three capacitors $\mathbf{C_1 = 2\ \mu F}$; $\mathbf{C_2 = 6\ \mu F}$ *and* $\mathbf{C_3 = 3\ \mu F}$ *connected across a battery of 10 V. If a charge of* 6 μ *C is acquired by the capacitor* C_3, *calculate the charge acquired by* C_1.



[2019 • Set 55-5-3]

SOLUTION

From the figure, C_2 and C_3 are connected in parallel with each other, and this parallel block is in series with C_1, the whole set being across the 10 *V* battery.

First find the voltage across C_3, which is the same as the voltage across the parallel block:

$$V_{C_3} = \frac{Q_{C_3}}{C_3} = \frac{6\ \mu\text{C}}{3\ \mu\text{F}}$$

$$V_{C_3} = 2\ V$$

Since C_2 and C_3 are in parallel, $\mathbf{V_{C_2} = V_{C_3} = 2\ V}$. The charge on C_2 is:

$$Q_{C_2} = C_2\, V_{C_2} = (6\ \mu F)(2\ V)$$

$$Q_{C_2} = 12\ \mu C$$

Because C_1 is in series with the parallel block, the charge on C_1 equals the total charge stored in that block:

$$Q_{C_1} = Q_{C_2} + Q_{C_3} = 12 + 6$$

$$Q_{C_1} = 18\ \mu C$$

Hence the charge acquired by C_1 is **18 µC**.

TIP

In a series-parallel network, the charge through the series element always equals the total charge of the block it feeds, not the charge of just one branch.

Final answer. The charge acquired by C_1 is 18 µC.

← *Back to Q112*

Q113

Why does current in a steady state not flow in a capacitor connected across a battery? However momentary current does flow during charging or discharging of the capacitor. Explain.

[2017]

SOLUTION

A capacitor is two conducting plates separated by an insulating dielectric. Free charges cannot cross this insulating gap, so a continuous (conduction) current cannot pass through a capacitor.

In the steady state the capacitor is fully charged. The potential difference across its plates becomes exactly equal to the emf of the battery:

$$V_C = \varepsilon$$

When the two are equal there is no net driving potential left in the circuit, so no charge flows. Therefore the steady-state current is zero.

During charging or discharging the situation is different. While the capacitor is still charging, its voltage V_C is less than the source emf, so a momentary current flows in the external circuit:

$$i(t) = \frac{\varepsilon - V_C}{R}$$

This current carries electrons through the connecting wires from one plate to the other, building up the charge. As V_C rises towards ε, the current falls and finally becomes zero. (During discharging the stored charge flows back, again giving a momentary current that dies away.)

Thus a momentary transient current exists only while the charge on the plates is changing, but **no steady current flows through a fully charged capacitor**.

Final answer. No steady current flows as charges cannot cross the dielectric; transient current flows only while V_C is changing during charging/discharging.

← *Back to Q113*

Q114

What is electrostatic shielding? How is this property used in actual practice? Is the potential in the cavity of a charged conductor zero?

[2016]

SOLUTION

Electrostatic shielding is the phenomenon by which a conducting cavity completely protects the region inside it from any external electric field. Whatever charges or fields exist outside the conductor, the electric field inside the empty cavity is always zero.

This follows because, in electrostatic equilibrium, the field inside the body of a conductor is zero ($\vec{E} = \vec{0}$). Applying Gauss's law to any surface drawn within the conducting wall shows that no field can exist in the enclosed cavity either.

Practical uses of this property:

A Faraday cage (a metal enclosure) shields sensitive electronic instruments from outside electromagnetic disturbance.

Coaxial cables use a surrounding metallic sheath to keep external fields from disturbing the signal.

A car or aircraft acts like a metal box, so passengers inside stay safe during a lightning strike.

Is the potential in the cavity zero? No. Although the electric field in the cavity is zero, the potential there is constant and equals the potential of the conductor. This value is fixed by the conductor's charge and surroundings and need not be zero.

So inside the cavity the **field is zero but the potential is constant (not necessarily zero)**.

Final answer. Electrostatic shielding keeps the cavity field zero (used in Faraday cages, coaxial cables, vehicles); the potential there is constant but not necessarily zero.

← Back to Q114

Q115

A parallel plate capacitor of capacitance C is charged to a potential V. It is then connected to another uncharged capacitor having the same capacitance. Find out the ratio of the energy stored in the combined system to that stored initially in the single capacitor.

[2014]

SOLUTION

Initially the single capacitor C is charged to potential V, so the energy stored in it is:

$$U_0 = \frac{1}{2}CV^2$$

It is then connected to an identical uncharged capacitor of the same capacitance C. The charge redistributes until both plates reach a common potential. Total charge is conserved:

$$Q_{\text{total}} = CV$$

The two capacitors are now in parallel, giving total capacitance $2C$, so the common final voltage is:

$$V_f = \frac{Q_{\text{total}}}{2C} = \frac{CV}{2C} = \frac{V}{2}$$

The energy of the combined system is:

$$U_f = \frac{1}{2}(2C)V_f^2 = \frac{1}{2}(2C)\left(\frac{V}{2}\right)^2$$

$$U_f = \frac{CV^2}{4}$$

Taking the required ratio:

$$\frac{U_f}{U_0} = \frac{CV^2/4}{CV^2/2} = \frac{1}{2}$$

Hence the energy stored finally is half of the initial energy, $\mathbf{U_f : U_0 = 1 : 2}$. The missing half is lost as heat and radiation in the connecting wires during redistribution.

COMMON MISTAKE

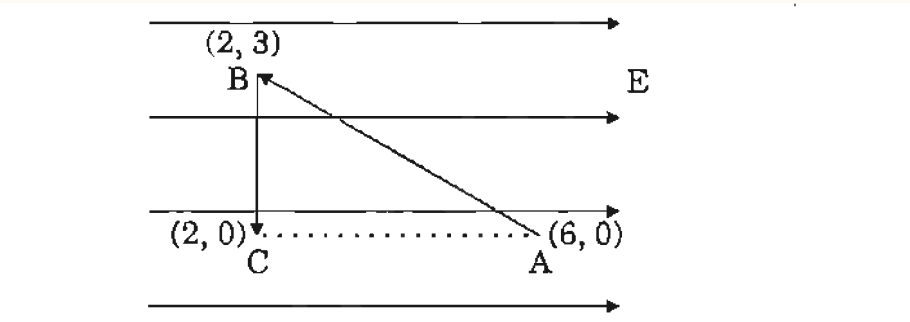
A common error is to say energy is conserved on connecting capacitors. Charge is conserved, but energy is not; half is always dissipated, whatever the wire resistance.

Final answer. The ratio of final to initial energy is 1 : 2 (half the energy is lost in the wires).

← Back to Q115

Q116

A test charge 'q' is moved without acceleration from A to C along the path from A to B and then from B to C in electric field $\vec{E}$ as shown in the figure. (i) Calculate the potential difference between A and C. (ii) At which point (of the two) is the electric potential more and why?



[2012]

SOLUTION

In an electrostatic (conservative) field, the potential difference between two points depends only on their positions, not on the path. So moving the charge from A to B and then B to C gives the same result as going straight from A to C:

$$V_A - V_C = -\int_C^A \vec{E} \cdot d\vec{l}$$

From the figure, $\vec{E}$ is uniform and directed along $+\hat{i}$, with A at the origin and C at $x = 6\ m$. For a uniform field the potential difference is simply E times the displacement along the field:

$$V_A - V_C = -E\,(x_A - x_C) = -E\,(0 - 6)$$

$$V_A - V_C = 6E$$

(For example, if $E = 10\ V/m$, then $V_A - V_C = 60\ V$.)

(ii) Since $V_A - V_C$ is positive, point A is at the higher potential. The reason is that an electric field always points from a region of higher potential to one of lower potential. Here $\vec{E}$ points from A towards C, so A (where the field starts) is at the **higher potential**.

Final answer. V_A − V_C = 6E (= 60 V if E = 10 V/m); point A is at the higher potential since E points from high to low potential.

← Back to Q116

Q117

Draw a plot showing the variation of (i) electric field (E) and (ii) electric potential (V) with distance r due to a point charge Q.

[2012]

SOLUTION

For a point charge Q, the electric field and potential at a distance r are:

$$E(r) = \frac{kQ}{r^2},\ V(r) = \frac{kQ}{r}$$

(i) Plot of E versus r: Since $\boldsymbol{E \propto \frac{1}{r^2}}$, the graph is a steeply falling curve. Mark E on the vertical axis and r on the horizontal axis. The curve starts very high near $\boldsymbol{r \to 0}$ (where $\boldsymbol{E \to \infty}$), drops rapidly, and approaches the r-axis asymptotically (never touching it) as r becomes large. It is a rectangular-hyperbola-like curve that decreases faster than V.

(ii) Plot of V versus r: Since $\boldsymbol{V \propto \frac{1}{r}}$, the graph is also a falling curve but less steep than the E curve. Plot V on the vertical axis against r. It begins very high near $\boldsymbol{r \to 0}$ (where $\boldsymbol{V \to \infty}$ for $\boldsymbol{Q > 0}$) and decreases gradually, approaching the r-axis asymptotically as r increases.

At any given large distance, the V curve lies above the E curve because the potential falls off more slowly ($1/r$) than the field ($1/r^2$). Both curves are positive for $\boldsymbol{Q > 0}$ and would be reflected below the axis (negative) for $\boldsymbol{Q < 0}$.

Final answer. $E \propto 1/r^2$ gives a steeply falling curve; $V \propto 1/r$ gives a less steep falling curve, both tending to zero as $r \to \infty$.

← Back to Q117

Q118

Net capacitance of three identical capacitors in series is 1 μF. *What will be their net capacitance if connected in parallel?*
Find the ratio of energy stored in the two configurations if they are both connected to the same source.

[2011]

SOLUTION

Let each identical capacitor have capacitance C. For three in series, the net capacitance is $C/3$, and this is given as 1 μF:

$$C_s = \frac{C}{3} = 1\ \mu F$$

$$C = 3\ \mu F$$

When the same three capacitors are connected in parallel, the net capacitance adds up:

$$C_p = 3C = 3 \times 3\ \mu F$$

$$C_p = 9\ \mu F$$

For the energy ratio, both combinations are connected to the same source V. The energy stored is $U = \frac{1}{2}CV^2$, which is proportional to capacitance when V is fixed:

$$\frac{U_s}{U_p} = \frac{C_s}{C_p} = \frac{1\ \mu F}{9\ \mu F}$$

$$\frac{U_s}{U_p} = \frac{1}{9}$$

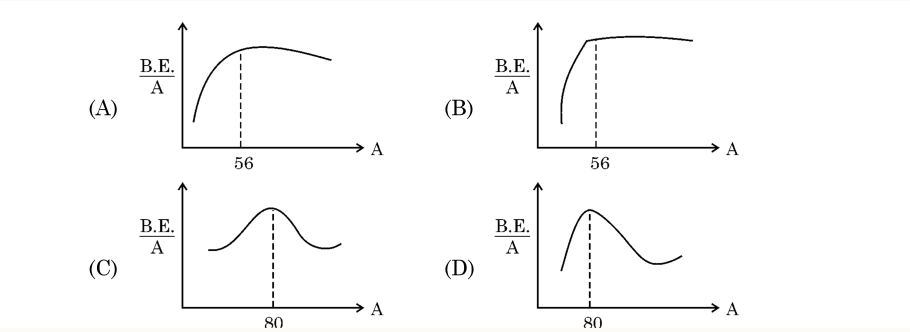
Hence the parallel capacitance is **9 μF** and the energy ratio is $\mathbf{U_s : U_p = 1 : 9}$.

Final answer. Parallel capacitance = 9 μF, and the energy ratio (series : parallel) = 1 : 9.

← *Back to Q118*

Q119

Figure shows two identical capacitors, C_1 and C_2, each of 1 μF capacitance connected to a battery of 6 V. Initially switch S is closed. After sometime S is left open and dielectric slabs of dielectric constant $K = 3$ *are inserted to fill completely the space between the plates of the two capacitors. How will the (i) charge and (ii) potential difference between the plates of the capacitors be affected after the slabs are inserted?*



[2011 • Set 55-1-1]

SOLUTION

With switch *S* closed, each 1 μF capacitor is charged by the 6 *V* battery. The initial charge on each is:

$$Q_0 = CV = (1\ \mu F)(6\ V)$$

$$Q_0 = 6\ \mu C,\ V_0 = 6\ V$$

Now *S* is opened, disconnecting the battery, so each capacitor becomes isolated. Then a dielectric slab of $K = 3$ is inserted to fill each capacitor completely. The capacitance increases to:

$$C' = KC = 3 \times 1\ \mu F = 3\ \mu F$$

(i) Charge: An isolated capacitor (battery removed) cannot gain or lose charge, so the charge is unchanged.

$$Q_{new} = Q_0 = 6\ \mu C$$

(ii) Potential difference: With charge fixed but capacitance tripled,

$$V_{new} = \frac{Q_{new}}{C'} = \frac{6\ \mu C}{3\ \mu F}$$

$$V_{new} = 2\ V$$

So after inserting the slabs, the **charge stays at 6 μC (unchanged)** while the **potential difference falls to 2 *V***, that is, reduced to one-third by the factor K.

TIP

Battery open means charge is fixed (Q constant); battery still connected would mean voltage is fixed (V constant). Always check the switch first.

Final answer. Charge stays 6 μC (unchanged) on each; potential difference drops to 2 V (reduced to one-third).

← Back to Q119

Q120

Two uniformly large parallel thin plates having charge densities +σ and −σ are kept in the X-Z plane at a distance d apart. Sketch an equipotential surface due to electric field between the plates. If a particle of mass m and charge −q remains stationary between the plates, what is the magnitude and direction of this field?

[2011 • Set 55-1-1]

SOLUTION

The two large plates carry charge densities +σ (upper) and −σ (lower) and lie horizontally in the x-z plane, a distance d apart. Between them the field is uniform, perpendicular to the plates, directed from the positive plate to the negative plate (downward), with magnitude:

$$E = \frac{\sigma}{\varepsilon_0}$$

Equipotential surfaces: An equipotential surface is everywhere perpendicular to $\vec{E}$. Since $\vec{E}$ is vertical (along $-y$), the equipotential surfaces are horizontal planes parallel to the two plates, lying in between them. In a sketch, draw the two plates as horizontal lines and a set of evenly spaced horizontal dashed lines between them to represent these equipotential planes.

For the particle of mass m and charge $-q$ to stay stationary, the upward electric force must balance the downward weight. The weight is mg (downward). The electric force on the negative charge is opposite to $\vec{E}$, hence upward, of magnitude qE. For equilibrium:

$$qE = mg$$

$$E = \frac{mg}{q}$$

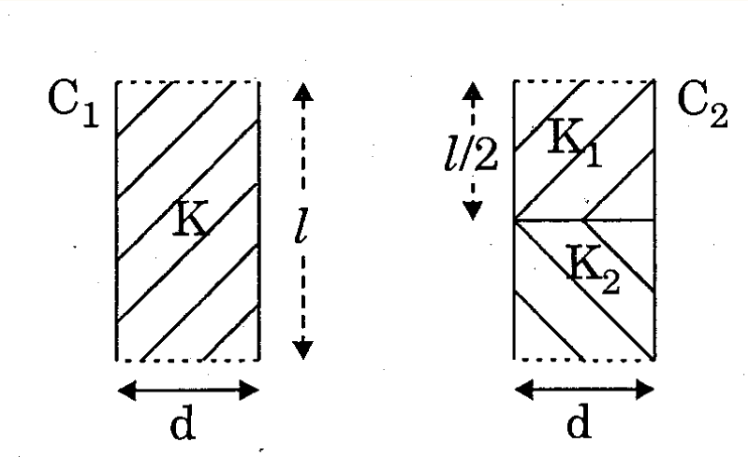
Hence the field has magnitude $\boldsymbol{E = \frac{mg}{q}}$, directed vertically downward (from the $+\sigma$ plate to the $-\sigma$ plate), so that the force on the negative charge is upward and balances gravity.

Final answer. Equipotentials are horizontal planes parallel to the plates; E = mg/q, directed downward (from $+\sigma$ to $-\sigma$ plate).

← *Back to Q120*

Q121

Two identical parallel plate (air) capacitors C_1 *and* C_2 *have capacitances C each. The space between their plates is now filled with dielectrics as shown. If the two capacitors still have equal capacitance, obtain the relation between dielectric constants K,* K_1 *and* K_2.



[2011 • Set 55-2-1]

SOLUTION

Each air capacitor has capacitance $C = \frac{\varepsilon_0 A}{d}$, where A is the plate area and d the separation.

In C_1 the whole gap is filled with a single dielectric of constant K, so its new capacitance is

$$C_1' = \frac{K \varepsilon_0 A}{d}.$$

In C_2 the gap is filled by two slabs, each of thickness $\frac{d}{2}$ and dielectric constants K_1 and K_2. These act as two capacitors in series, each of plate separation $\frac{d}{2}$:

$$C_a = \frac{K_1 \varepsilon_0 A}{d/2} = \frac{2K_1 \varepsilon_0 A}{d}, \; C_b = \frac{2K_2 \varepsilon_0 A}{d}.$$

For the series combination,

$$\frac{1}{C_2'} = \frac{1}{C_a} + \frac{1}{C_b} = \frac{d}{2K_1 \varepsilon_0 A} + \frac{d}{2K_2 \varepsilon_0 A}.$$

$$\frac{1}{C_2'} = \frac{d}{2\varepsilon_0 A}\left(\frac{K_1 + K_2}{K_1 K_2}\right).$$

$$C_2' = \frac{2K_1 K_2}{K_1 + K_2} \cdot \frac{\varepsilon_0 A}{d}.$$

Since the two capacitors are still equal, $\mathbf{C_1' = C_2'}$:

$$\frac{K \varepsilon_0 A}{d} = \frac{2K_1 K_2}{K_1 + K_2} \cdot \frac{\varepsilon_0 A}{d}.$$

Cancelling $\frac{\varepsilon_0 A}{d}$ from both sides gives the required relation:

$$\mathbf{K = \frac{2K_1 K_2}{K_1 + K_2}}.$$

Thus the single dielectric constant K equals the harmonic mean of K_1 and K_2.

COMMON MISTAKE

The two stacked slabs are in SERIES (charge flows through both), not parallel. A very common slip is to add K_1 and K_2 as if they were parallel.

Final answer. The required relation is $K = \dfrac{2K_1 K_2}{K_1 + K_2}$.

← Back to Q121

Q122

(i) Can two equi-potential surfaces intersect each other? Give reasons.
(ii) Two charges $-q$ and $+q$ are located at points $A(0,0,-a)$ and $B(0,0,+a)$ respectively. How much work is done in moving a test charge from point $P(7,0,0)$ to $Q(-3,0,0)$?

[2009]

SOLUTION

(i) No, two equipotential surfaces can never intersect each other.

Electric potential is a single-valued function of position, so at any given point in space the potential has only one definite value. If two equipotential surfaces crossed, the point of intersection would have two different potentials at the same time, which is impossible.

Also, the electric field $\vec{E}$ is always perpendicular to an equipotential surface. At an intersection there would be two normals, implying two directions of $\vec{E}$ at one point, which again cannot happen.

(ii) The charges $-q$ at $A(0,0,-a)$ and $+q$ at $B(0,0,+a)$ form an electric dipole with its axis along the z-axis.

The points $P(7,0,0)$ and $Q(-3,0,0)$ both lie in the plane $\boldsymbol{z = 0}$, which is the equatorial plane of this dipole.

The potential of a dipole at a point making angle θ with the axis, at distance r, is

$$V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{p\theta}{r^2}.$$

On the equatorial plane $\boldsymbol{\theta = 90°}$, so $\boldsymbol{\theta = 0}$ and hence $\boldsymbol{V = 0}$ at every point of this plane.

Therefore $\mathbf{V_P = 0}$ and $\mathbf{V_Q = 0}$.

The work done in moving a test charge q_0 from P to Q is

$$W = q_0 (V_Q - V_P) = q_0 (0 - 0) = 0.$$

So **no work is done** in moving the test charge from P to Q.

Final answer. Equipotential surfaces cannot intersect, and the work done from P to Q is zero.

← *Back to Q122*

Q123

Draw 3 equipotential surfaces corresponding to a field that uniformly increases in magnitude but remains constant along Z-direction. How are these surfaces different from that of a constant electric field along Z-direction?

[2009]

SOLUTION

Let the field point along the z-axis with a magnitude that grows with z, say $\vec{E} = \alpha z \hat{k}$ with $\alpha > 0$.

The potential is obtained from $V = -\int E\, dz$:

$$V = -\int \alpha z\, dz = -\frac{\alpha z^2}{2} + \text{constant}.$$

V depends only on z, so each equipotential is a surface of constant z, that is, a plane perpendicular to the z-axis.

Diagram: draw three horizontal straight lines (a side view of three horizontal planes) lying one above another, perpendicular to a vertical z-axis. Label them V_1, V_2, V_3 at heights $z_1 < z_2 < z_3$, with arrows of $\vec{E}$ pointing upward and getting longer at greater z. The spacing between successive lines decreases as z increases.

Difference from a uniform field $\vec{E} = E_0 \hat{k}$:

For a uniform field, equal potential differences correspond to equal spacings, since $\Delta z = \frac{\Delta V}{E_0}$ is constant. The equipotential planes are therefore equally spaced.

For the increasing field, $\Delta z = \frac{\Delta V}{E}$ becomes smaller where E is larger. So for the same potential interval the planes crowd closer together as z increases.

Thus both fields give parallel planes perpendicular to z, but **uniform-field equipotentials are equally spaced while increasing-field equipotentials get closer together as the field grows stronger**.

Final answer. Both give planes perpendicular to z; for increasing E they crowd closer together, while for uniform E they are equally spaced.

← *Back to Q123*

Q124

Two point charges 4 μC *and* −2 μC *are separated by a distance of 1 m in air. Calculate at what point on the line joining the two charges is the electric potential zero.*

[2007]

SOLUTION

Take $\mathbf{q_1 = +4\ \mu C}$ at $x = 0$ and $\mathbf{q_2 = -2\ \mu C}$ at $x = 1\ m$. The total potential at a point on the line is the algebraic sum of the two potentials, and we set it to zero.

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r_1} + \frac{q_2}{r_2}\right) = 0.$$

Case 1: A point between the charges at distance x from q_1 (so $\mathbf{r_1 = x}$, $\mathbf{r_2 = 1 - x}$).

$$\frac{4}{x} + \frac{(-2)}{1-x} = 0.$$

$$\frac{4}{x} = \frac{2}{1-x}.$$

$$4(1-x) = 2x.$$

$$4 - 4x = 2x \Rightarrow 6x = 4 \Rightarrow x = \frac{2}{3}\ m.$$

So one zero-potential point is at $\frac{2}{3}\ \boldsymbol{m} = \mathbf{0.67\ m}$ from the +4 μC charge.

Case 2: A point beyond the negative charge at distance x from q_1 ($x > 1$), so $r_1 = x$, $r_2 = x - 1$.

$$\frac{4}{x} + \frac{(-2)}{x-1} = 0.$$

$$\frac{4}{x} = \frac{2}{x-1}.$$

$$4(x-1) = 2x.$$

$$4x - 4 = 2x \Rightarrow 2x = 4 \Rightarrow x = 2\ m.$$

So the second zero-potential point is at 2 m from the +4 μC charge (i.e. 1 m beyond the −2 μC charge).

Hence the potential is **zero at $\frac{2}{3}$ m (between the charges) and at 2 m (beyond the −2 μC charge)**, both measured from the +4 μC charge.

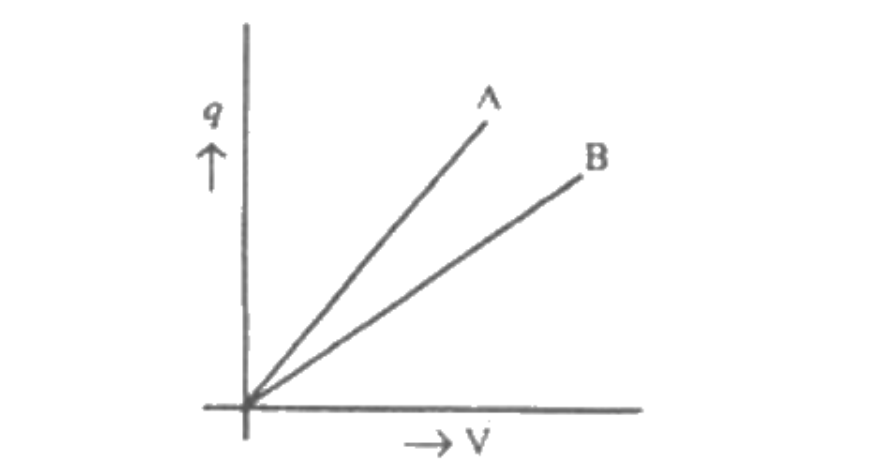
> **TIP**
>
> Potential is a scalar, so add the contributions WITH their signs. There are usually two null points on the line for unequal opposite charges, so check both the in-between region and beyond the smaller charge.

Final answer. V = 0 at 0.67 m between the charges and at 2 m from the +4 muC charge.

← *Back to Q124*

Q125

The given graph shows the variation of charge q versus potential difference V for two capacitors C_1 and C_2. The two capacitors C_1 and C_2 have same plate separation but the plate area of C_2 is double than that of C_1. Which of the lines in the graph correspond to C_1 and C_2 and why?



[2006]

SOLUTION

For any capacitor the charge and voltage are related by $q = CV$.

Rearranging, $C = \frac{q}{V}$, so on a graph of q (vertical) against V (horizontal), the slope of the straight line equals the capacitance C.

For a parallel plate capacitor,

$$C = \frac{\varepsilon_0 A}{d}.$$

Both capacitors have the same plate separation d, but the plate area of C_2 is double that of C_1, i.e. $A_2 = 2A_1$. Therefore

$$C_2 = \frac{\varepsilon_0 A_2}{d} = \frac{\varepsilon_0 (2A_1)}{d} = 2C_1.$$

Since slope equals capacitance, C_2 has twice the slope of C_1.

On the graph, the steeper line (larger slope) represents the capacitor of larger capacitance, which is **C_2 (the larger plate area)**, and the line with the smaller slope represents **C_1**.

Reason: for the same applied voltage V, the larger capacitor stores more charge ($q = CV$), so its q-V line rises more steeply.

Final answer. The steeper line is C2 and the less steep line is C1, since slope = C and C2 = 2C1.

← *Back to Q125*

Q126

(a) *Why does the electric field inside a dielectric decrease when it is placed in an external electric field?*

(b) *A parallel plate capacitor with air between the plates has a capacitance of* 8 pF. *What will be the capacitance if the distance between the plates be reduced by half and the space between them is filled with a substance of dielectric constant* $K = 6$*?*

[2005]

SOLUTION

(a) When a dielectric is placed in an external field $\vec{E}_0$, the field polarises the dielectric: its molecules acquire (or align) dipole moments. As a result, bound (induced) charges appear on the two faces of the slab, negative on the face towards the positive plate and positive on the other face.

These bound surface charges set up their own field $\vec{E}_p$ inside the dielectric directed opposite to the applied field $\vec{E}_0$. The net field inside is therefore the difference:

$$\vec{E} = \vec{E}_0 - \vec{E}_p = \frac{\vec{E}_0}{K}.$$

Because the polarisation field only partly cancels the applied field, the resultant field is reduced (to $\vec{E}_0/K$) but does not become zero. Hence the field inside a dielectric is smaller than the applied field.

(b) Initially, with air between the plates,

$$C_0 = \frac{\varepsilon_0 A}{d} = 8 \text{ pF}.$$

The separation is reduced to $\frac{d}{2}$ and the gap is filled with a dielectric of constant $K = 6$. The new capacitance is

$$C_{new} = \frac{K \varepsilon_0 A}{d/2} = 2K \cdot \frac{\varepsilon_0 A}{d} = 2K\, C_0.$$

$$C_{new} = 2 \times 6 \times 8 \text{ pF}.$$

$$C_{new} = 96 \text{ pF}.$$

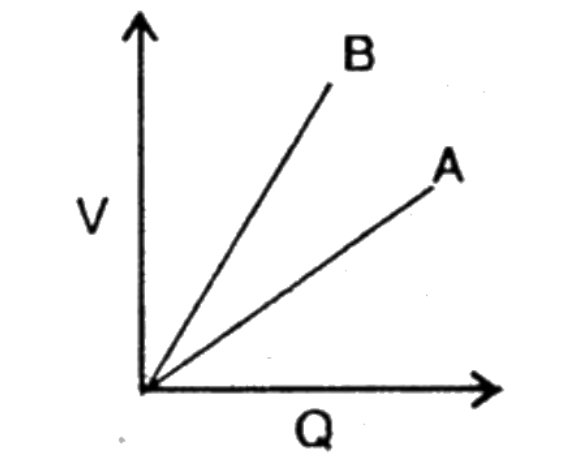
So the new capacitance is **96 pF**, which is 12 times the original air-filled value.

Final answer. The field falls to E0/K because of opposing polarisation charges, and the new capacitance is 96 pF.

← *Back to Q126*

Q127

The graph shows the variation of voltage, 'V' across the plates of two capacitors A and B versus increase of charge, 'Q' stored on them. Which of the two capacitors has higher capacitance? Give reason for your answer.



[2004]

SOLUTION

The charge and voltage of a capacitor are related by $Q = CV$.

Here the graph plots V on the vertical axis against Q on the horizontal axis, so rearranging,

$$V = \frac{1}{C} Q.$$

This is a straight line through the origin whose slope equals $\frac{1}{C}$.

A smaller slope therefore means a larger capacitance, since $\textbf{slope} = \frac{1}{C}$ is small when C is large.

From the figure, capacitor A gives the steeper line and capacitor B the flatter (less steep) line.

The flatter line B has the smaller slope, hence the larger capacitance. So **capacitor B has the higher capacitance**.

Reason: for the same charge Q stored, the capacitor with larger C develops a smaller voltage ($V = Q/C$). On a V-versus-Q graph this appears as a flatter line.

> **COMMON MISTAKE**
>
> *On a V-vs-Q graph the slope is 1/C, not C. The flatter line is the bigger capacitor, the opposite of a q-vs-V graph. Always check which quantity is on which axis before reading the slope.*

Final answer. Capacitor B has the higher capacitance, because its V-Q line is flatter (smaller slope = 1/C).

← *Back to Q127*

Q128

A parallel plate capacitor of capacitance C has a dielectric slab between its plates. It is charged to a potential difference V by connecting it across a battery. The battery is then disconnected. If the dielectric slab is now withdrawn from the capacitor, how will the following be affected?
(a) Capacitance of the capacitor,
(b) Energy stored in the capacitor, and
(c) The potential difference between the plates of the capacitor.
Justify your answer in each case.

[2026 • Set 55-1-1]

SOLUTION

Initially the capacitor with the dielectric has capacitance C and is charged to V, so the stored charge is $Q = CV$.

The battery is then disconnected, so the charge Q on the plates stays constant while the dielectric is withdrawn. Let the dielectric constant be K.

(a) Capacitance: With the dielectric, $C = \frac{K\varepsilon_0 A}{d}$. After it is withdrawn the gap is air, so

$$C_{\text{new}} = \frac{\varepsilon_0 A}{d} = \frac{C}{K}.$$

The capacitance decreases to $\frac{1}{K}$ of its value.

(b) Energy stored: The charge Q is fixed, so we use

$$U = \frac{Q^2}{2C}.$$

$$U_{\text{new}} = \frac{Q^2}{2\,C_{\text{new}}} = \frac{Q^2}{2\,(C/K)} = K \cdot \frac{Q^2}{2C} = K\,U.$$

The energy increases K times. This extra energy equals the work done by the external agent in pulling the dielectric out against the attractive pull of the polarised slab.

(c) Potential difference: With Q fixed,

$$V_{\text{new}} = \frac{Q}{C_{\text{new}}} = \frac{Q}{C/K} = K \cdot \frac{Q}{C} = K\,V.$$

The potential difference increases K times.

Hence on withdrawing the dielectric (battery disconnected): **capacitance becomes $\frac{C}{K}$, energy becomes KU, and potential difference becomes KV**.

CONCEPT

Battery DISCONNECTED means charge Q is the conserved quantity, so reason through U = Q^2/2C and V = Q/C. If the battery had stayed connected, V would be constant instead.

Final answer. Capacitance falls to C/K, while energy rises to KU and potential difference rises to KV.

← *Back to Q128*

Q129

A 12.0 μF capacitor is charged to a potential difference of 150 V. The terminals of the charged capacitor are then connected to those of an uncharged 6.0 μF capacitor. Calculate final potential difference across and charge on, each capacitor.

[2026 • Set 55-1-2]

SOLUTION

The 12.0 μF capacitor is first charged to 150 V. Its initial charge is

$$Q_0 = C_1\, V = 12.0\ \mu F \times 150\ V = 1800\ \mu C.$$

When its terminals are joined to the uncharged 6.0 μF capacitor, the two are in parallel and share charge until they reach a common potential difference V'. No charge is lost, so the total charge is conserved:

$$Q_0 = (C_1 + C_2)\, V'.$$

$$1800\ \mu C = (12.0 + 6.0)\ \mu F \times V'.$$

$$V' = \frac{1800\ \mu C}{18.0\ \mu F} = 100\ V.$$

This common potential difference, 100 V, appears across both capacitors.

Charge on the 12.0 μF capacitor:

$$Q_1 = C_1\, V' = 12.0\ \mu F \times 100\ V = 1200\ \mu C.$$

Charge on the 6.0 μF capacitor:

$$Q_2 = C_2\, V' = 6.0\ \mu F \times 100\ V = 600\ \mu C.$$

Check: $\mathbf{Q_1 + Q_2 = 1200 + 600 = 1800\ \mu C = Q_0}$, so charge is conserved.

Hence the **final potential difference is 100 V across each capacitor, with charges 1200 μC on the 12.0 μF and 600 μC on the 6.0 μF capacitor**.

Final answer. Common p.d. is 100 V; the charges are 1200 muC on the 12 muF and 600 muC on the 6 muF.

← Back to Q129

Q130

A parallel plate capacitor of capacitance C is charged to V volt by a battery. After sometime the battery is disconnected and the distance between the plates is doubled. A slab of dielectric constant $\boldsymbol{k = 1.8}$ *is then introduced to completely fill the space between the plates. How will the following be affected?*
(a) The capacitance of the capacitor.
(b) The electric field between the plates of the capacitor.
(c) The energy stored in the capacitor.
Justify your answer in each case.

[2026 • Set 55-1-3]

SOLUTION

Initially the capacitor has capacitance C at voltage V, so the stored charge is $\boldsymbol{Q = CV}$. The battery is then disconnected, so Q stays constant for all later changes. The separation is doubled ($\boldsymbol{d \to 2d}$) and then a dielectric of constant $\boldsymbol{k = 1.8}$ fills the new gap.

(a) Capacitance: For a parallel plate capacitor $\boldsymbol{C = \frac{\varepsilon_0 A}{d}}$. Doubling d halves the capacitance, and filling with k multiplies it by k:

$$C_{new} = \frac{k\,\varepsilon_0 A}{2d} = \frac{k}{2}\cdot\frac{\varepsilon_0 A}{d} = \frac{k}{2}\,C.$$

$$C_{new} = \frac{1.8}{2}\,C = 0.9\,C.$$

The capacitance decreases to $0.9\,C$ (a drop of about 10%).

(b) Electric field between the plates: The battery is off, so Q and hence the surface charge density $\boldsymbol{\sigma = \frac{Q}{A}}$ stay fixed. In air the field would be $\boldsymbol{E_0 = \frac{\sigma}{\varepsilon_0}}$. With the dielectric present the field is reduced by k:

$$E_{new} = \frac{E_0}{k} = \frac{E_0}{1.8}.$$

The field decreases to $\frac{E_0}{1.8}$ (about 0.56 E_0). Note doubling d alone does not change the field, since $\mathbf{E_0 = \sigma/\varepsilon_0}$ is independent of d at fixed charge.

(c) Energy stored: With Q fixed, use $\boldsymbol{U = \frac{Q^2}{2C}}$:

$$U_{new} = \frac{Q^2}{2\,C_{new}} = \frac{Q^2}{2\,(0.9C)} = \frac{1}{0.9}\cdot\frac{Q^2}{2C} = \frac{10}{9}\,U_0.$$

$$U_{new} \approx 1.11\,U_0.$$

The energy increases to about 1.11 times the original (a rise of about 11%). The increase comes from the work done by the external agent in pulling the plates apart, partly offset by the dielectric being drawn in.

Hence: $\mathbf{C_{new} = 0.9\,C}$, $\mathbf{E_{new} = \frac{E_0}{1.8}}$, **and** $\mathbf{U_{new} = \frac{10}{9}\,U_0 \approx 1.11\,U_0}$.

Final answer. Capacitance becomes 0.9C, field becomes E0/1.8, and energy rises to (10/9)U0 about 1.11 times.

← *Back to Q130*

Q131

(a) An electric field $\vec{E} = \mathbf{E_0}\hat{i}$ *exists in a region of space. Draw three equipotential surfaces in the region.*
(b) Two point charges $-q$ *and* $+q$ *are located at points* $(-a, 0, 0)$ *and* $(a, 0, 0)$ *respectively. Find the electrostatic potential at point* $(x, 0, 0)$ *where* $x \gg a$.

[2026 • Set 55-2-1]

SOLUTION

(a) For a uniform field $\vec{E} = \mathbf{E_0}\hat{i}$, the potential depends only on the x-coordinate, since $\mathbf{dV = -\vec{E}\cdot d\vec{l} = -E_0\,dx}$, giving $\boldsymbol{V(x) = -E_0\,x + \text{constant}}$.

Therefore an equipotential surface (constant V) is a surface of constant x, that is a plane perpendicular to the x-axis. Three equipotential surfaces are drawn as three parallel vertical planes (lying in the y-z plane), all perpendicular to $\vec{E}$.

They are drawn equally spaced for equal potential intervals ΔV, with the potential decreasing in the direction of $\vec{E}$ (the $+\hat{i}$ direction). The field lines are horizontal, pointing along $+\hat{i}$, and cross every plane at right angles.

(b) The charge $+q$ at $(a,0,0)$ is at distance $(x-a)$ from the point $(x,0,0)$, and the charge $-q$ at $(-a,0,0)$ is at distance $(x+a)$ from it (taking $\boldsymbol{x>0}$, $x \gg a$).

By the superposition principle,

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{x-a} + \frac{(-q)}{x+a}\right]$$

$$V = \frac{q}{4\pi\varepsilon_0} \cdot \frac{(x+a)-(x-a)}{(x-a)(x+a)}$$

$$V = \frac{q}{4\pi\varepsilon_0} \cdot \frac{2a}{x^2-a^2}$$

Since $x \gg a$, we have $\mathbf{x^2 - a^2 \approx x^2}$, and writing the dipole moment $\boldsymbol{p = q(2a)}$,

$$V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{p}{x^2}$$

Thus the **potential at $(x,0,0)$ for $x \gg a$ is** $\boldsymbol{V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{2qa}{x^2} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{p}{x^2}}$, falling off as $1/x^2$.

CONCEPT

On the axis of a dipole the potential falls as $1/x^2$, faster than the $1/x$ of a single point charge. Do not forget the squared distance.

Final answer. Equipotentials are parallel planes perpendicular to the x-axis, and $\boldsymbol{V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{p}{x^2}}$ with $\boldsymbol{p = 2qa}$.

← *Back to Q131*

Q132

(i) Draw electric field lines and equipotential surfaces for a system of two equal and opposite point charges separated by some distance.

(ii) Why electric field $\vec{E}$ at a point on an equipotential surface must be perpendicular to the surface at that point?

[2026 • Set 55-2-2]

SOLUTION

(i) Draw the two charges $+q$ (left) and $-q$ (right) separated by a small distance.

Field lines: They originate on the positive charge $+q$ and terminate on the negative charge $-q$. Close to each charge the lines are nearly radial. In the region between the charges they curve smoothly from $+q$ across to $-q$. One field line runs straight along the axis joining the two charges, directed from $+q$ to $-q$. The lines are denser near the charges (stronger field) and sparser far away.

Equipotential surfaces: Drawn as dashed curves cutting every field line at right angles. The perpendicular bisector plane of the line joining the charges is the $V = 0$ equipotential, a flat plane. Near $+q$ the equipotentials are small closed surfaces at positive potential; near $-q$ they are small closed surfaces at negative potential. Everywhere the equipotentials and field lines are mutually perpendicular.

(ii) The work done in moving a test charge through a small displacement $d\vec{l}$ is $\mathbf{dW = q\,\vec{E}\cdot d\vec{l}}$, and the change in potential is

$$dV = -\vec{E}\cdot d\vec{l}$$

On an equipotential surface the potential is the same at every point, so for any displacement $d\vec{l}$ lying in the surface, $\mathbf{dV = 0}$.

$$\therefore \vec{E}\cdot d\vec{l} = 0$$

Since $\vec{E}$ and $d\vec{l}$ are both non-zero, their dot product vanishes only when they are mutually perpendicular. As $d\vec{l}$ can point in any direction tangent to the surface, $\vec{E}$ must be perpendicular to every tangent, hence **perpendicular to the equipotential surface at that point**.

If $\vec{E}$ had a component along the surface, work would be done in moving a charge along the surface, and the potential would change, contradicting the constant-potential condition.

Final answer. Field lines run from $+q$ to $-q$ cutting the equipotentials at right angles; $\vec{E}$ is perpendicular to an equipotential because $\mathbf{dV} = -\vec{E} \cdot \mathbf{d}\vec{l} = 0$ for any displacement on the surface.

← Back to Q132

Q133

(a) Draw electric field lines and equipotential surfaces for a system of two equal positive charges separated by some distance.
(b) Using the drawing in (a) above, explain that electric field need not be a constant over each point of an equipotential surface.

[2026 • Set 55-2-3]

SOLUTION

(a) Draw the two equal positive charges $+q$ and $+q$ separated by a distance.

Field lines: Lines radiate outward from each charge. In the region between the two charges the lines from one charge are repelled by the other, so they bend away from each other and never connect the two charges. At the midpoint of the line joining the charges the two equal and opposite field contributions cancel, giving a neutral point where $\vec{E} = \vec{0}$. Far from both charges the pattern resembles that of a single charge +2q placed at the midpoint.

Equipotential surfaces: Drawn as dashed curves perpendicular to the field lines everywhere. Close to each charge they are small nearly spherical surfaces at high positive potential. At larger distances the two sets merge into a single dumbbell or peanut shaped closed surface enclosing both charges. Since both charges are positive, the potential is positive everywhere and there is no zero-potential surface at finite distance.

(b) An equipotential surface is, by definition, a surface on which the potential V is constant. It is NOT a surface of constant field magnitude.

Consider one such equipotential surface around the two $+q$ charges. The field $\vec{E}$ is everywhere perpendicular to it, but the spacing between successive field lines varies over the surface.

Since the field magnitude is large where the field lines are crowded and small where they are sparse, $|\vec{E}|$ is greater at points of the surface near the charges and smaller at points farther from them.

Quantitatively, $|\vec{E}| = \left|\frac{dV}{dn}\right|$, the rate of change of potential normal to the surface, and this normal gradient is not the same at every point of an irregularly shaped equipotential.

Hence **the electric field need not be constant over an equipotential surface**, even though the potential is constant on it.

Final answer. For two equal positive charges, $\vec{E}=\vec{0}$ at the midpoint, and $|\vec{E}|$ varies across an equipotential because $|\vec{E}| = |dV/dn|$ differs from point to point on an irregular surface.

← *Back to Q133*

Q134

Two parallel plate capacitors X and Y are connected in series to a 6 V battery. They have the same plate area and same plate separation but capacitor X has air between its plates, whereas capacitor Y contains a material of dielectric constant 4.
(a) Calculate the capacitances of X and Y, if the equivalent capacitance of the combination of X and Y is 4 μF.
(b) Calculate the potential difference across the plates of X and Y.

[2026 • Set 55-5-1]

SOLUTION

Let the air capacitor have capacitance $C_X = C_0$. Since Y has the same area and separation but a dielectric of constant $K = 4$,

$$C_Y = K\,C_0 = 4C_0$$

(a) For series combination,

$$\frac{1}{C_{eq}} = \frac{1}{C_X} + \frac{1}{C_Y} = \frac{1}{C_0} + \frac{1}{4C_0}$$

$$\frac{1}{C_{eq}} = \frac{4+1}{4C_0} = \frac{5}{4C_0}$$

$$C_{eq} = \frac{4C_0}{5}$$

Given $\mathbf{C_{eq} = 4\ \mu F}$,

$$\frac{4C_0}{5} = 4 \Rightarrow C_0 = 5\ \mu F$$

$$\therefore C_X = 5\ \mu F,\ C_Y = 4C_0 = 20\ \mu F$$

(b) In a series combination the charge Q on each capacitor is the same and equals the charge on the equivalent capacitor,

$$Q = C_{eq}\, V = (4\ \mu F)(6\ V) = 24\ \mu C$$

Potential difference across X,

$$V_X = \frac{Q}{C_X} = \frac{24\ \mu C}{5\ \mu F} = 4.8\ V$$

Potential difference across Y,

$$V_Y = \frac{Q}{C_Y} = \frac{24\ \mu C}{20\ \mu F} = 1.2\ V$$

Check: $\mathbf{V_X + V_Y = 4.8 + 1.2 = 6.0\ V = V}$, as required.

Hence $\mathbf{C_X = 5\ \mu F,\ C_Y = 20\ \mu F,\ V_X = 4.8\ V}$ **and** $\mathbf{V_Y = 1.2\ V.}$

TIP

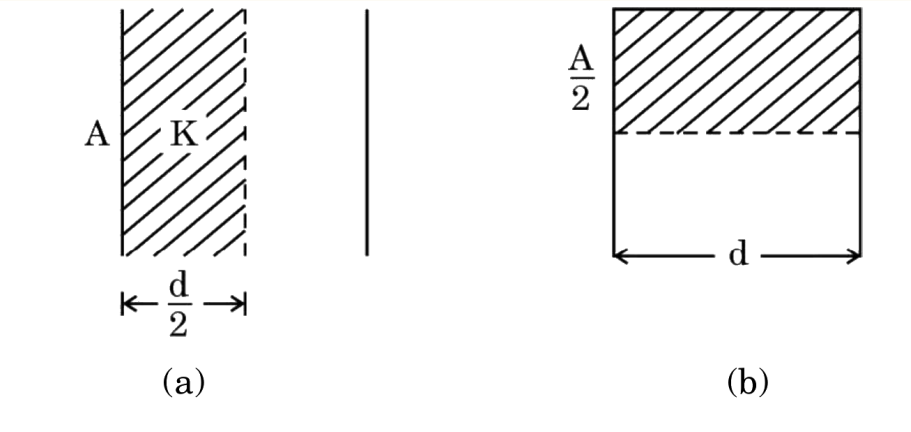
In series the charge is common, so the larger capacitor gets the smaller voltage. The dielectric capacitor Y carries the smaller V here.

Final answer. $\mathbf{C_X = 5\ \mu F}$, $\mathbf{C_Y = 20\ \mu F}$, with $\mathbf{V_X = 4.8\ V}$ and $\mathbf{V_Y = 1.2\ V}$.

← *Back to Q134*

Q135

A parallel plate capacitor has plate area A and plate separation d. Half of the space between the plates is filled with a material of dielectric constant K in two ways as shown in the figure.



Find the values of the capacitance of the capacitors in the two cases.

[2025 • Set 55-5-1]

SOLUTION

Let $C_0 = \frac{\varepsilon_0 A}{d}$ be the capacitance with air filling the whole gap.

Case (a): the dielectric slab of constant K fills the lower half of the gap, occupying the full plate area A but only thickness $d/2$. The arrangement is two capacitors in series, each of plate separation $d/2$.

Air half (thickness $d/2$):

$$C_1 = \frac{\varepsilon_0 A}{d/2} = \frac{2\varepsilon_0 A}{d} = 2C_0$$

Dielectric half (thickness $d/2$):

$$C_2 = \frac{K\varepsilon_0 A}{d/2} = \frac{2K\varepsilon_0 A}{d} = 2KC_0$$

For series,

$$\frac{1}{C_a} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{2C_0} + \frac{1}{2KC_0} = \frac{K+1}{2KC_0}$$

$$C_a = \frac{2K}{K+1} C_0 = \frac{2K\varepsilon_0 A}{(K+1)\, d}$$

Case (b): the dielectric slab fills half the plate area but the full separation d. The arrangement is two capacitors in parallel, each of plate area $A/2$ and separation d.

Air half-area:

$$C_1' = \frac{\varepsilon_0 (A/2)}{d} = \frac{C_0}{2}$$

Dielectric half-area:

$$C_2' = \frac{K\varepsilon_0 (A/2)}{d} = \frac{KC_0}{2}$$

For parallel,

$$C_b = C_1' + C_2' = \frac{C_0}{2} + \frac{KC_0}{2} = \frac{(K+1)C_0}{2} = \frac{(K+1)\varepsilon_0 A}{2d}$$

Hence $\mathbf{C_a = \frac{2K\varepsilon_0 A}{(K+1)d}}$ **(series-type) and** $\mathbf{C_b = \frac{(K+1)\varepsilon_0 A}{2d}}$ **(parallel-type).**

COMMON MISTAKE

Decide series vs parallel by the slab orientation: a slab splitting the thickness gives series, a slab splitting the area gives parallel.

Final answer. $\mathbf{C_a = \frac{2K\varepsilon_0 A}{(K+1)d}}$ and $\mathbf{C_b = \frac{(K+1)\varepsilon_0 A}{2d}}$.

← Back to Q135

Q136

A capacitor of plate area A and plate separation d is charged by a battery to voltage V. The battery is disconnected and plates are slowly pulled apart till the separation becomes 2d. *Find the value of:*
(a) *potential difference between the plates,*
(b) *electric field between the plates,*
(c) *work done in pulling the plates apart.*

[2025 • Set 55-5-2]

SOLUTION

Initially $C = \frac{\varepsilon_0 A}{d}$ and the battery charges the capacitor to voltage V, so the charge is

$$Q = CV = \frac{\varepsilon_0 A V}{d}$$

After the battery is disconnected, this charge Q remains constant while the plates are pulled to separation 2d. The new capacitance is

$$C' = \frac{\varepsilon_0 A}{2d} = \frac{C}{2}$$

(a) New potential difference. With Q fixed,

$$V' = \frac{Q}{C'} = \frac{Q}{C/2} = \frac{2Q}{C} = 2V$$

The potential difference doubles to $\boldsymbol{V' = 2V}$.

(b) New electric field. The field between parallel plates is $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$. Since Q and A are unchanged,

$$E' = \frac{Q}{\varepsilon_0 A} = E = \frac{V}{d}$$

The field stays the same, $\boldsymbol{E' = \frac{V}{d}}$. The voltage rises only because the same field now acts over a larger distance.

(c) Work done. The work done by the external agent equals the increase in the stored energy of the capacitor.

$$U_i = \frac{Q^2}{2C} = \frac{1}{2}CV^2$$

$$U_f = \frac{Q^2}{2C'} = \frac{Q^2}{2(C/2)} = \frac{Q^2}{C} = CV^2$$

$$W = U_f - U_i = CV^2 - \frac{1}{2}CV^2 = \frac{1}{2}CV^2$$

With $\boldsymbol{C = \varepsilon_0 A/d}$,

$$W = \frac{1}{2} \cdot \frac{\varepsilon_0 A}{d} V^2 = \frac{\varepsilon_0 A V^2}{2d}$$

Hence $V' = 2V$, $E' = V/d$ **(unchanged), and** $W = \frac{1}{2}CV^2 = \frac{\varepsilon_0 A V^2}{2d}$.

COMMON MISTAKE

When the battery is disconnected the charge stays fixed (use $U = Q^2/2C$); when it stays connected the voltage stays fixed. Identify which quantity is constant first.

Final answer. $V' = 2V$, $E' = V/d$ (unchanged), and the work done is $W = \frac{1}{2}CV^2 = \frac{\varepsilon_0 A V^2}{2d}$.

← Back to Q136

Q137

(a) *"There is a limit to the amount of charge that can be stored on a given capacitor." Explain.*
(b) *A capacitor is charged by a battery to a potential difference V. It is disconnected from the battery and connected across another identical uncharged capacitor. Calculate the ratio of total energy stored in the combination to the initial energy stored in the capacitor.*

[2025 • Set 55-5-3]

SOLUTION

(a) When more and more charge is placed on a capacitor, the electric field between its plates keeps increasing, since $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$.

The medium between the plates (air or a dielectric) can withstand only a maximum field, called its dielectric strength. If the field exceeds this value, the medium gets ionised and becomes conducting, that is the dielectric breaks down.

Once breakdown occurs, charge leaks across the gap (a spark) and the capacitor cannot hold any more charge. Hence the maximum charge that can be stored is

$$Q_{max} = C\,V_{max} = C\,(E_{breakdown}\,d)$$

So there is a definite limit to the charge a given capacitor can store.

(b) Initial energy stored in the charged capacitor,

$$U_i = \frac{1}{2}CV^2$$

The initial charge is $Q = CV$. When this capacitor is connected across an identical uncharged capacitor (capacitance C), the total charge is shared but conserved,

$$Q_{total} = CV$$

The two identical capacitors in parallel give total capacitance 2C, so the common potential becomes

$$V' = \frac{Q_{total}}{2C} = \frac{CV}{2C} = \frac{V}{2}$$

Final energy of the combination,

$$U_f = \frac{1}{2}(2C)\left(\frac{V}{2}\right)^2 = \frac{1}{2}(2C)\frac{V^2}{4} = \frac{CV^2}{4}$$

Required ratio,

$$\frac{U_f}{U_i} = \frac{CV^2/4}{CV^2/2} = \frac{1}{2}$$

Hence **the ratio of total energy stored in the combination to the initial energy is 1 : 2.** Half the original energy is lost as heat and electromagnetic radiation in the connecting wires during charge sharing.

Final answer. Charge is limited by dielectric breakdown ($Q_{max}=CV_{max}$); the energy ratio is $U_f : U_i = 1 : 2$.

← *Back to Q137*

Q138

Two point charges of −5 μC *and* 2 μC *are located in free space at* (−4 cm, 0) *and* (6 cm, 0) *respectively.*

(a) *Calculate the amount of work done to separate the two charges at infinite distance.*

(b) *If this system of charges was initially kept in an electric field* $\vec{E} = \frac{A}{r^2}\hat{r}$, *where* $A = 8 \times 10^4\ N\,C^{-1}\,m^2$, *calculate the electrostatic potential energy of the system.*

[2025 • Set 55-7-2]

SOLUTION

Here $q_1 = -5\ \mu C = -5 10^{-6}\ C$ at (−4 cm,0) and $q_2 = +2\ \mu C = 2 10^{-6}\ C$ at (6 cm,0). The separation between them is

$$r_{12} = 6 - (-4) = 10\ cm = 0.10\ m$$

(a) The mutual electrostatic potential energy of the pair is

$$U = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_1\, q_2}{r_{12}}$$

$$U = \frac{(9 10^9)(-5 10^{-6})(2 10^{-6})}{0.10}$$

$$U = \frac{-9 10^{-2}}{0.10} = -0.9\ J$$

The work done to separate the charges to infinity equals the energy needed to raise the potential energy from U to zero,

$$W = 0 - U = -(-0.9) = +0.9\ J$$

(b) The external field $\vec{E} = \frac{A}{r^2}\hat{r}$ corresponds to an external potential $V_{ext}(r) = \frac{A}{r}$, with $A = 8 10^4\ N\,C^{-1}m^2$. The charges are at $r_1 = 0.04\ m$ and $r_2 = 0.06\ m$ from the field origin.

$$V_{ext}(r_1) = \frac{8 10^4}{0.04} = 2 10^6\ V$$

$$V_{ext}(r_2) = \frac{8 10^4}{0.06} = 1.33 10^6\ V$$

The potential energy of the charges in the external field is

$$U_{ext} = q_1\, V_{ext}(r_1) + q_2\, V_{ext}(r_2)$$

$$U_{ext} = (-5 10^{-6})(2 10^6) + (2 10^{-6})(1.33 10^6)$$

$$U_{ext} = -10 + 2.67 = -7.33\,J$$

The total electrostatic potential energy of the system is the sum of the mutual energy and the energy in the external field,

$$U_{total} = U + U_{ext} = -0.9 + (-7.33) = -8.23\,J$$

Hence **the work to separate the charges is +0.9 *J*, and the total potential energy of the system in the field is about −8.23 *J*.**

COMMON MISTAKE

The total energy includes both the mutual interaction of the two charges and the energy of each charge in the external field. Add all three contributions.

Final answer. Work to separate the charges = **0.9 *J***; total potential energy of the system ≈ **−8.23 *J***.

← *Back to Q138*

Q139

An electric dipole (dipole moment $\vec{p} = p\hat{i}$), consisting of charges −q and q, separated by distance 2a*, is placed along the x-axis, with its centre at the origin. Show that the potential V, due to this dipole, at a point x, (x≫ a) is equal to* $\frac{1}{4\pi\varepsilon_0}\cdot\frac{\vec{p}\cdot\hat{i}}{x^2}$.

[2024 • Set 55-1-1]

SOLUTION

The dipole has charge +*q* at (+*a*,0) and −*q* at (−*a*,0), so the dipole moment is $\vec{p} = q(2a)\hat{i} = p\hat{i}$. The field point *P* lies on the *x*-axis at (*x*,0) with $x>0$ and $x \gg a$.

Distance of *P* from +*q*:

$$r_+ = x - a$$

Distance of *P* from −*q*:

$$r_- = x + a$$

By the superposition principle the potential at P is

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{x-a} + \frac{(-q)}{x+a}\right]$$

$$V = \frac{q}{4\pi\varepsilon_0}\cdot\frac{(x+a)-(x-a)}{(x-a)(x+a)}$$

$$V = \frac{q}{4\pi\varepsilon_0}\cdot\frac{2a}{x^2-a^2}$$

For a short dipole, $x \gg a$, so $\mathbf{x^2 - a^2 \approx x^2}$. Also $\boldsymbol{q(2a) = p}$, the magnitude of the dipole moment. Therefore

$$V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{2qa}{x^2} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p}{x^2}$$

Since the dipole moment $\vec{p} = p\hat{i}$ points along $\hat{i}$, we can write $p = \vec{p}\cdot\hat{i}$, giving

$$V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\vec{p}\cdot\hat{i}}{x^2}$$

Hence the **potential on the axis of a short dipole is** $\boldsymbol{V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\vec{p}\cdot\hat{i}}{x^2}}$, which decreases as $1/x^2$ with distance.

Final answer. $\boldsymbol{V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{\vec{p}\cdot\hat{i}}{x^2}}$, the axial potential of a short dipole.

← *Back to Q139*

Q140

Three point charges Q_1, Q_2 *and* Q_3 *are located in x–y plane at points* $(-d, 0)$, $(0, 0)$ *and* $(d, 0)$ *respectively.* Q_1 *and* Q_3 *are identical and* Q_3 *is positive. What will be the nature and value of* Q_2 *so that the potential energy of the system is zero?*

[2024 • Set 55-1-2]

SOLUTION

Let $\mathbf{Q_1 = Q_3 = Q}$ (both positive) at $(-d,0)$ and $(+d,0)$, and Q_2 at the origin $(0,0)$. The pairwise separations are

$$r_{12} = d,\ r_{23} = d,\ r_{13} = 2d$$

The total potential energy of the system is the sum of the energies of the three pairs,

$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q_1 Q_2}{r_{12}} + \frac{Q_2 Q_3}{r_{23}} + \frac{Q_1 Q_3}{r_{13}}\right]$$

Substituting the charges and distances,

$$U = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q\,Q_2}{d} + \frac{Q_2\,Q}{d} + \frac{Q\cdot Q}{2d}\right]$$

$$U = \frac{1}{4\pi\varepsilon_0\,d}\left[2Q\,Q_2 + \frac{Q^2}{2}\right]$$

For the potential energy of the system to be zero, $U = 0$, so the bracket must vanish,

$$2Q\,Q_2 + \frac{Q^2}{2} = 0$$

$$2Q\,Q_2 = -\frac{Q^2}{2}$$

$$Q_2 = -\frac{Q}{4}$$

The negative sign shows that Q_2 must be a negative charge, and its magnitude is one-fourth of Q_1 (or Q_3).

Hence **Q_2 is negative with value $Q_2 = -\frac{Q}{4}$**, where Q is the magnitude of each outer charge.

Final answer. Q_2 must be negative, with $Q_2 = -\frac{Q}{4}$ (one-fourth the magnitude of each outer charge).

← *Back to Q140*

Q141

An air-filled parallel plate capacitor with plate separation 1 mm *has a capacitance of* 20 pF. *It is charged to* 4.0 μC. *Calculate the amount of work done to pull its plates to a separation of* 5 mm. *Assume the charge on the plates remains the same.*

[2024 • Set 55-1-3]

SOLUTION

Here the capacitor is isolated (battery disconnected), so the charge $Q = 4.0\ \mu C$ on the plates stays constant while the separation is increased from $d_1 = 1\ mm$ to $d_2 = 5\ mm$.

The capacitance of a parallel plate capacitor is $C = \frac{\varepsilon_0 A}{d}$, so for a fixed area $C \propto \frac{1}{d}$.

$$C_2 = C_1 \cdot \frac{d_1}{d_2} = 20\ pF \times \frac{1\ mm}{5\ mm} = 4\ pF$$

Since the charge is held constant, the energy stored is best written as $U = \frac{Q^2}{2C}$.

Initial energy:

$$U_1 = \frac{Q^2}{2C_1} = \frac{(4.0\times 10^{-6})^2}{2\,(20\times 10^{-12})} = \frac{16\times 10^{-12}}{40\times 10^{-12}} = 0.4\,J$$

Final energy:

$$U_2 = \frac{Q^2}{2C_2} = \frac{(4.0\times 10^{-6})^2}{2\,(4\times 10^{-12})} = \frac{16\times 10^{-12}}{8\times 10^{-12}} = 2.0\,J$$

The plates carry opposite charges and attract each other, so an external agent must do work to pull them apart. Since the battery is disconnected, no energy is supplied by any other source, so the work done by the external agent equals the increase in stored energy.

$$W = U_2 - U_1 = 2.0 - 0.4 = 1.6\,J$$

Hence the **work done to pull the plates apart is 1.6 *J***.

TIP

With the battery disconnected use $U=\frac{Q^2}{2C}$, not $U=12\ CV^2$, because Q is constant while V changes; mixing these up is the commonest slip here.

Final answer. The work done by the external agent is **1.6 J**.

← Back to Q141

Q142

The electric field in a region is given by $\vec{E}=(10x+4)\hat{i}$, *where x is in m and E is in N/C. Calculate the amount of work done in taking a unit charge from*
(i) (5 *m*,0) *to* (10 *m*,0)
(ii) (5 *m*,0) *to* (5 *m*,10 *m*)

[2024 • Set 55-4-1]

SOLUTION

The work done by an external agent in carrying a charge q from point A to point B (with no change in kinetic energy) is

$$W = q\,(V_B - V_A) = -q\int_A^B \vec{E}\cdot d\vec{l}$$

For a unit positive charge $q = 1\ C$, so

$$W = -\int_A^B \vec{E}\cdot d\vec{l}$$

with $\vec{E} = (10x + 4)\,\hat{i}\ N/C$.

(i) From (5 *m*,0) to (10 *m*,0): the motion is along the x-axis, so $d\vec{l} = dx\,\hat{i}$ and $\vec{E}\cdot d\vec{l} = (10x + 4)\,dx$.

$$W = -\int_5^{10}(10x + 4)\,dx$$

$$W = -[\,5x^2 + 4x\,]_5^{10}$$

$$W = -[(5\times 100 + 4\times 10) - (5\times 25 + 4\times 5)]$$

$$W = -[(500 + 40) - (125 + 20)] = -[540 - 145]$$

$$W = -395\ J$$

So the external agent does $-395\ J$ of work; equivalently the field does $+395\ J$ of work as the charge moves along the field.

(ii) From $(5\ m, 0)$ to $(5\ m, 10\ m)$: the motion is along the y-axis, so $\mathbf{d\vec{l}} = \mathbf{dy}\,\hat{j}$.

$\vec{E}\cdot \mathbf{d\vec{l}} = (10x + 4)\,\hat{i}\cdot \mathbf{dy}\,\hat{j} = 0$ since $\hat{i}\cdot\hat{j} = 0$

$$W = -\int_0^{10} 0\ dy = 0$$

The displacement is everywhere perpendicular to $\vec{E}$, so **no work is done in part (ii)**.

Final answer. (i) $W = -395\ J$; (ii) $W = 0$.

← *Back to Q142*

Q143

A thin spherical conducting shell of radius R has a charge q. A point charge Q is placed at the centre of the shell. Find
(i) The charge density on the outer surface of the shell, and
(ii) the potential at a distance of (R/2) from the centre of the shell.

[2024 • Set 55-4-2]

SOLUTION

A point charge $+Q$ sits at the centre of a thin conducting shell of radius R that itself carries charge q.

Inside the conducting material the electric field is zero. Draw a Gaussian sphere lying entirely within the metal of the shell; it must enclose zero net charge. The central charge $+Q$ is enclosed, so the inner surface of the shell must carry an induced charge $-Q$ to make the total enclosed charge zero.

The shell as a whole carries charge q. Since its inner surface holds $-Q$, charge conservation gives the outer surface a charge

$$q_{\text{outer}} = q - (-Q) = q + Q$$

(i) This outer charge spreads uniformly over the sphere of radius R, so the outer surface charge density is

$$\sigma_{\text{outer}} = \frac{q + Q}{4\pi R^2}$$

(ii) The point $r = R/2$ lies inside the cavity. The potential there is the scalar sum of the potentials due to all three charge distributions, with $k = \dfrac{1}{4\pi\varepsilon_0}$.

Central point charge $+Q$ at distance $r = R/2$:

$$V_1 = \frac{kQ}{R/2} = \frac{2kQ}{R}$$

Inner surface charge $-Q$ on a sphere of radius R (point is inside it, where V is constant and equal to its surface value):

$$V_2 = \frac{k(-Q)}{R} = -\frac{kQ}{R}$$

Outer surface charge $(q + Q)$ on a sphere of radius R (point is inside it):

$$V_3 = \frac{k(q + Q)}{R}$$

Adding:

$$V(R/2) = \frac{2kQ}{R} - \frac{kQ}{R} + \frac{k(q + Q)}{R} = \frac{k}{R}[2Q - Q + q + Q] = \frac{k(2Q + q)}{R}$$

$$V(R/2) = \frac{2Q + q}{4\pi\varepsilon_0 R}$$

Hence the **outer charge density is** $\dfrac{q+Q}{4\pi R^2}$ and the **potential at $R/2$ is** $\dfrac{2Q+q}{4\pi\varepsilon_0 R}$.

COMMON MISTAKE

Inside a uniformly charged sphere the potential is constant and equals the surface value kq/R *(not* kq/r*); use* kq/r *only for the central point charge.*

Final answer. $\sigma_{\text{outer}} = \dfrac{q+Q}{4\pi R^2}$ and $V(R/2) = \dfrac{2Q+q}{4\pi\varepsilon_0 R}$.

← Back to Q143

Q144

Two conducting spherical shells A and B of radii R and 2R *are kept far apart and charged to the same charge density* σ. *They are connected by a wire. Obtain an expression for final potential of shell A.*

[2024 • Set 55-4-3]

SOLUTION

Shell A has radius R and shell B has radius 2R. Both start with the same surface charge density σ, and they are far apart so each behaves as an isolated sphere.

Initial charge on each shell (Q = σ× surface area):

$$Q_A = \sigma\,(4\pi\,R^2)$$

$$Q_B = \sigma\,4\pi(2R)^2 = 4\,\sigma\,(4\pi\,R^2) = 4Q_A$$

Total charge:

$$Q_A + Q_B = Q_A + 4Q_A = 5Q_A = 20\pi\sigma\,R^2$$

When the shells are joined by a wire, charge flows until both are at the same potential. For a sphere $V = \frac{kQ}{r}$, so equal potentials give

$$\frac{kQ_A'}{R} = \frac{kQ_B'}{2R} \Rightarrow Q_B' = 2\,Q_A'$$

Charge is conserved on connection:

$$Q_A' + Q_B' = 5Q_A$$

$$Q_A' + 2Q_A' = 5Q_A \Rightarrow 3Q_A' = 5Q_A$$

$$Q_A' = \frac{5Q_A}{3} = \frac{5}{3}\,(4\pi\sigma\,R^2) = \frac{20\pi\sigma\,R^2}{3}$$

The final potential of shell A is

$$V_A = \frac{kQ_A'}{R} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{1}{R}\cdot\frac{20\pi\sigma\,R^2}{3}$$

$$V_A = \frac{20\pi\sigma\,R^2}{4\pi\varepsilon_0 \times 3R} = \frac{5\sigma\,R}{3\varepsilon_0}$$

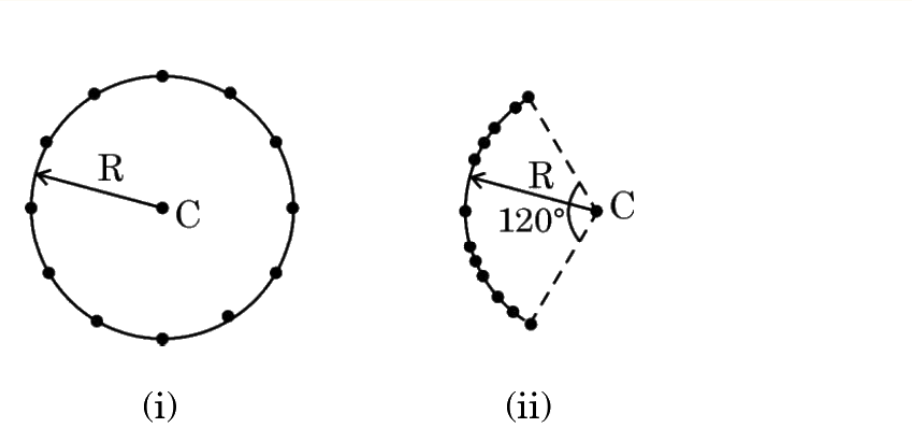
Hence the **final potential of shell A is** $\frac{5\sigma R}{3\varepsilon_0}$ (and shell B is at the same potential, since they are connected).

Final answer. $V_A = \frac{5\sigma R}{3\varepsilon_0}$.

← *Back to Q144*

Q145

(a) Twelve negative charges of same magnitude are equally spaced and fixed on the circumference of a circle of radius R as shown in Fig. (i). Relative to potential being zero at infinity, find the electric potential and electric field at the centre C of the circle. (b) If the charges are unequally spaced and fixed on an arc of 120° *of radius R as shown in Fig. (ii), find electric potential at the centre C.*



[2023 • Set 55-1-1]

SOLUTION

(a) Twelve equal negative charges, each $-q$, lie evenly on a circle of radius R, all at the same distance R from the centre C.

Potential is a scalar, so the contributions simply add. With $k = \frac{1}{4\pi\varepsilon_0}$:

$$V_C = \Sigma \frac{k(-q)}{R} = 12\times\frac{-kq}{R} = -\frac{12kq}{R}$$

$$V_C = -\frac{12q}{4\pi\varepsilon_0 R} = -\frac{3q}{\pi\varepsilon_0 R}$$

For the field, each charge produces a field vector at C pointing along the line joining C to that charge. Because the twelve charges are equally spaced, they form six diametrically opposite pairs; the field of each charge is exactly cancelled by its diametrically opposite partner.

$$\vec{E}_C = \vec{0}$$

(b) Now the same twelve charges are crowded, unequally spaced, onto a 120° arc of the same radius R. Each charge is still at distance R from C, and there are still twelve charges of magnitude q.

The potential at C depends only on the magnitudes of the charges and their distances from C, not on their angular positions:

$$V_C = -\frac{12q}{4\pi\varepsilon_0 R} = -\frac{3q}{\pi\varepsilon_0 R}$$

So the **potential at C is unchanged**, the same as in part (a). (The field at C is, however, no longer zero, since the cancelling symmetry is now broken.)

CONCEPT

Potential is a scalar that ignores direction, so rearranging the charges on the same-radius circle never changes V_C, even though it can change $\vec{E}_C$.

Final answer. (a) $V_C = -\dfrac{12q}{4\pi\varepsilon_0 R}$, $\vec{E}_C = \vec{0}$; (b) $V_C = -\dfrac{12q}{4\pi\varepsilon_0 R}$ (same as a).

← *Back to Q145*

Q146

A 100 μF *capacitor is charged by a* 12 *V battery.*
(a) How much electrostatic energy is stored by the capacitor?
(b) The capacitor is disconnected from the battery and connected in parallel to another uncharged 100 μF *capacitor. What is the electrostatic energy stored by the system?*

[2023 • Set 55-1-2]

SOLUTION

(a) A 100 μF capacitor is charged by a 12 V battery. The energy stored in a capacitor is

$$U_i = \frac{1}{2}CV^2$$

$$U_i = \frac{1}{2}(100\times 10^{-6})(12)^2$$

$$U_i = \frac{1}{2}(100\times 10^{-6})(144) = 7.2\times 10^{-3}\,J$$

$$U_i = 7.2\text{ mJ}$$

The charge stored is $Q = CV = (100\times 10^{-6})(12) = 1.2\times 10^{-3}\,C = 1.2$ **mC**.

(b) The charged capacitor is disconnected from the battery and joined in parallel with an identical uncharged 100 μF capacitor. The total charge is conserved at $Q = 1.2$ **mC**, and the two equal capacitors in parallel share a common potential difference.

Combined capacitance = $C + C = 2C = 200$ **μF**.

Common potential difference:

$$V' = \frac{Q}{2C} = \frac{1.2\times 10^{-3}}{200\times 10^{-6}} = 6\ V$$

Energy now stored by the system:

$$U_f = \frac{1}{2}(2C)\,V'^2 = \frac{1}{2}(200\times 10^{-6})(6)^2$$

$$U_f = \frac{1}{2}(200\times 10^{-6})(36) = 3.6\times 10^{-3}\,J = 3.6\text{ mJ}$$

So the system now stores **3.6 mJ, exactly half of the initial 7.2 mJ**. The missing 3.6 mJ is lost as heat in the connecting wires during the redistribution of charge.

Final answer. (a) $U_i = 7.2$ **mJ**; (b) $U_f = 3.6$ **mJ**.

← *Back to Q146*

Q147

Three point charges Q_1 *(−15 μC),* Q_2 *(10 μC) and* Q_3 *(16 μC) are located at* (0 cm, 0 cm), (0 cm, 3 cm) *and* (4 cm, 3 cm) *respectively. Calculate the electrostatic potential energy of this system of charges.*

[2023 • Set 55-1-3]

SOLUTION

The charges are $\mathbf{Q_1 = -15\ \mu C}$ at (0,0), $\mathbf{Q_2 = 10\ \mu C}$ at (0,3) cm and $\mathbf{Q_3 = 16\ \mu C}$ at (4,3) cm.

The mutual distances are:

$\mathbf{r_{12}}$ = distance **(0,0)→(0,3) = 3 cm** = $3\times 10^{-2}\ m$

$\mathbf{r_{23}}$ = distance **(0,3)→(4,3) = 4 cm** = $4\times 10^{-2}\ m$

$\mathbf{r_{13}}$ = distance **(0,0)→(4,3)** = $\sqrt{4^2+3^2} = \sqrt{25} = 5$ **cm** = $5\times 10^{-2}\ m$

The total electrostatic potential energy is the sum of the energies of the three pairs, with $k = 9\times 10^9\ N\,m^2\,C^{-2}$:

$$U = k\left(\frac{Q_1 Q_2}{r_{12}} + \frac{Q_2 Q_3}{r_{23}} + \frac{Q_1 Q_3}{r_{13}}\right)$$

$$U_{12} = \frac{k\,Q_1 Q_2}{r_{12}} = \frac{(9\times 10^9)(-15\times 10^{-6})(10\times 10^{-6})}{3\times 10^{-2}} = -45\,J$$

$$U_{23} = \frac{k\,Q_2 Q_3}{r_{23}} = \frac{(9\times 10^9)(10\times 10^{-6})(16\times 10^{-6})}{4\times 10^{-2}} = +36\,J$$

$$U_{13} = \frac{k\,Q_1 Q_3}{r_{13}} = \frac{(9\times 10^9)(-15\times 10^{-6})(16\times 10^{-6})}{5\times 10^{-2}} = -43.2\,J$$

Adding:

$$U = U_{12} + U_{23} + U_{13} = -45 + 36 - 43.2$$

$$U = -52.2\,J$$

Hence the **total electrostatic potential energy of the system is −52.2 *J***. The negative value means it is a bound system; 52.2 *J* of external work is needed to separate the charges to infinity.

COMMON MISTAKE

Keep the sign of every charge inside the formula; one pair here is positive and two are negative, and dropping a sign turns −52.2 J into a wrong answer.

Final answer. $U = -52.2\ J$.

← Back to Q147

Q148

Two charged conducting spheres of radii a and b are connected to each other by a wire. Find the ratio of the electric fields at their surfaces.

[2023 • Set 55-2-1]

SOLUTION

When two charged conducting spheres of radii a and b are joined by a thin wire, charge flows between them until they reach the same potential.

For an isolated sphere of charge Q and radius r, the potential is $V = \frac{kQ}{r}$. Equal potentials give

$$\frac{kQ_a}{a} = \frac{kQ_b}{b}$$

$$\frac{Q_a}{Q_b} = \frac{a}{b}$$

The electric field just outside the surface of each sphere is, by Gauss's law,

$$E = \frac{kQ}{r^2}$$

So the ratio of the surface fields is

$$\frac{E_a}{E_b} = \frac{kQ_a/a^2}{kQ_b/b^2} = \frac{Q_a}{Q_b} \cdot \frac{b^2}{a^2}$$

Substituting $\frac{Q_a}{Q_b} = \frac{a}{b}$:

$$\frac{E_a}{E_b} = \frac{a}{b} \cdot \frac{b^2}{a^2} = \frac{b}{a}$$

Hence the **ratio of the surface fields is $E_a : E_b = b : a$**. The smaller sphere has the larger surface field, which is why charge leaks most readily from sharp points (the point-discharge effect).

Final answer. $E_a : E_b = b : a$.

← *Back to Q148*

Q149

A parallel plate capacitor (A) of capacitance C is charged by a battery to voltage V. The battery is disconnected and an uncharged capacitor (B) of capacitance 2C is connected across A. Find the ratio of
(i) final charges on A and B.
(ii) total electrostatic energy stored in A and B finally and that stored in A initially.

[2023 • Set 55-2-1]

SOLUTION

Capacitor A (capacitance C) is charged by the battery to voltage V, so its initial charge is

$$Q_i = CV$$

The battery is then disconnected and A is connected across an uncharged capacitor B (capacitance $2C$). They are now in parallel and share a common potential difference V', with the total charge conserved.

$$Q_i = (C + 2C)\, V'$$

$$CV = 3C\, V' \Rightarrow V' = \frac{V}{3}$$

(i) Final charges on A and B:

$$Q_A = C\, V' = \frac{CV}{3}$$

$$Q_B = 2C\,V' = \frac{2CV}{3}$$

$$\frac{Q_A}{Q_B} = \frac{CV/3}{2CV/3} = \frac{1}{2}$$

(ii) Initial energy stored in A alone:

$$U_i = \frac{1}{2}CV^2$$

Final total energy stored in A and B:

$$U_f = \frac{1}{2}(C + 2C)\,V'^2 = \frac{1}{2}\,(3C)\left(\frac{V}{3}\right)^2 = \frac{1}{2}\cdot 3C\cdot\frac{V^2}{9} = \frac{CV^2}{6}$$

$$\frac{U_f}{U_i} = \frac{CV^2/6}{CV^2/2} = \frac{2}{6} = \frac{1}{3}$$

Hence $\mathbf{Q_A : Q_B = 1 : 2}$ and $\mathbf{U_f : U_i = 1 : 3}$. Two-thirds of the original energy is lost as heat in the connecting wires during charge redistribution.

Final answer. (i) $\mathbf{Q_A : Q_B = 1 : 2}$; (ii) $\mathbf{U_f : U_i = 1 : 3}$.

← *Back to Q149*

Q150

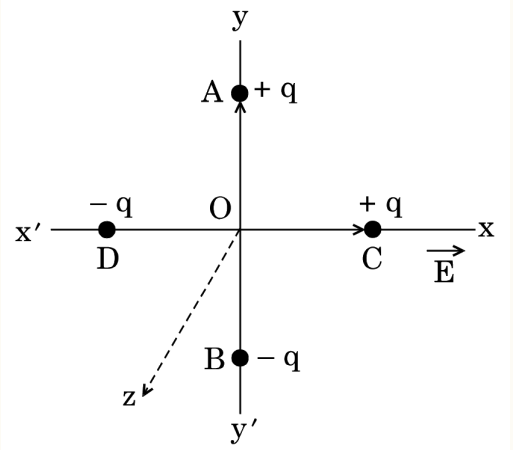
A capacitor of 4 μ*F is charged by a battery of* 12 *V. The battery is disconnected and a dielectric slab of dielectric constant* 8 *is inserted in between the plates of the capacitor to fill the space completely. Find the change in the (a) charge stored in the capacitor, (b) potential difference between the plates of the capacitor, and (c) energy stored in the capacitor.*

OR

Two small identical electric dipoles AB and CD, each of dipole moment $\vec{p}$ *are kept at an angle of* 90° *to each other in an external electric field* $\vec{E}$ *pointing along the x-axis as shown in the figure. Find the*

(a) dipole moment of the arrangement, and

(b) magnitude and direction of the net torque acting on it.



[2020 • Set 55-2-3]

SOLUTION

$\mathbf{C_0 = 4\ \mu F}$ is charged to $\mathbf{V_0 = 12\ V}$, then the battery is disconnected and a dielectric of constant $\boldsymbol{K = 8}$ fills the gap completely. Because the capacitor is isolated, the charge is conserved.

Initial charge:

$$Q_0 = C_0 V_0 = (4\times 10^{-6})(12) = 48\ \mu C$$

Initial energy:

$$U_0 = \frac{1}{2} C_0 V_0^2 = \frac{1}{2}(4\times 10^{-6})(12)^2 = 288\ \mu J$$

After inserting the dielectric, the capacitance becomes

$$C' = K C_0 = 8\times 4 = 32\ \mu F$$

The charge is unchanged, so $\boldsymbol{Q' = Q_0 = 48\ \mu C}$.

New potential difference:

$$V' = \frac{Q_0}{C'} = \frac{48\ \mu C}{32\ \mu F} = 1.5\ V$$

New energy:

$$U' = \frac{Q_0^2}{2C'} = \frac{(48\times 10^{-6})^2}{2(32\times 10^{-6})} = 36\ \mu J$$

(a) Change in charge: $\boldsymbol{\Delta Q = Q' - Q_0 = 0}$ (the battery is disconnected, so no charge can be added or removed).

(b) Change in potential difference: $\Delta V = V' - V_0 = 1.5 - 12 = -10.5\ V$, i.e. the voltage falls by 10.5 V.

(c) Change in energy: $\Delta U = U' - U_0 = 36 - 288 = -252\ \mu J$, i.e. the energy falls by 252 μJ (this energy goes into polarising the dielectric, which is pulled in).

OR

Two identical dipoles AB and CD, each of moment magnitude p, are set at 90° to each other in a uniform field $\vec{E} = E\hat{i}$. By symmetry, place the two moments at ± 45° to the x-axis.

(a) Net dipole moment. Adding the two equal moments like vectors, their y-components (± psin 45°) cancel and their x-components add:

$$p_{net} = 2p\cos 45° = 2p \times \frac{1}{\sqrt{2}} = \sqrt{2}\, p$$

directed along the x-axis (along $\hat{i}$).

(b) Net torque. The torque on a dipole is $\vec{\tau} = \vec{p} \times \vec{E}$, so its magnitude is $\tau = pE\theta$. The resultant moment $\vec{p}_{net} = \sqrt{2}\, p$ is along $\hat{i}$, the same direction as $\vec{E}$, so $\theta = 0$:

$$\tau = p_{net} E \sin 0° = 0$$

So the **net torque on the arrangement is zero**, because the resultant dipole moment is aligned with $\vec{E}$.

CONCEPT

With the battery removed the charge is fixed, so inserting a dielectric lowers V and U; if the battery had stayed connected, V would be fixed instead and Q, U would rise.

Final answer. (a) $\Delta Q = 0$; (b) $\Delta V = -10.5\ V$; (c) $\Delta U = -252\ \mu J$. (OR: $p_{net} = \sqrt{2}\, p$ along $\vec{E}$, net torque = 0.)

← *Back to Q150*

Q151

(a) Two point charges q_1 and q_2 are kept at a distance of r_{12} in air. Deduce the expression for the electrostatic potential energy of this system.
(b) If an external electric field ($\vec{E}$) is applied on the system, write the expression for the total energy of this system.

[2020 • Set 55-5-1]

SOLUTION

(a) To find the electrostatic potential energy of the system, we build the configuration by bringing the charges from infinity one at a time.

First, we bring q_1 from infinity to its location. Since no other charge or field is present yet, no work is done in this step, so $\mathbf{W_1 = 0}$.

Next, we bring q_2 from infinity to its position at a distance r_{12} from q_1. The charge q_1 produces a potential at the location of q_2 given by $\mathbf{V_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_{12}}}$.

The work done in bringing q_2 against this potential is $\mathbf{W_2 = q_2 V_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}}}$.

This work is stored as the electrostatic potential energy of the system. Hence,

$$\boldsymbol{U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}}}.$$

This is the **electrostatic potential energy** of the two-charge system.

(b) When an external electric field $\vec{E}$ (with associated potential V_{ext}) is applied, each charge acquires additional energy by being placed in this external potential, over and above their mutual interaction energy.

The charge q_1 at position $\vec{r}_1$ has energy $q_1\, V_{ext}(\vec{r}_1)$, and the charge q_2 at position $\vec{r}_2$ has energy $q_2\, V_{ext}(\vec{r}_2)$.

Adding the mutual interaction term, the total energy of the system is

$$\mathbf{U_{total} = q_1\, V_{ext}(\vec{r}_1) + q_2\, V_{ext}(\vec{r}_2) + \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}}}.$$

Final answer. The mutual PE is $\boldsymbol{U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}}}$; with an external field the total energy is $\mathbf{U_{total} = q_1\, V_{ext}(\vec{r}_1) + q_2\, V_{ext}(\vec{r}_2) + \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}}}$.

← Back to Q151

Q152

(a) Two point charges $+Q_1$ and $-Q_2$ are placed r distance apart. Obtain the expression for the amount of work done to place a third charge Q_3 at the midpoint of the line joining the two charges.
(b) At what distance from charge $+Q_1$ on the line joining the two charges (in terms of Q_1, Q_2 and r) will this work done be zero.

[2020 • Set 55-5-3]

SOLUTION

(a) The midpoint M of the line joining $+Q_1$ and $-Q_2$ lies at a distance $\frac{r}{2}$ from each charge.

The net electric potential at M due to the two charges is

$$V_M = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q_1}{r/2} + \frac{(-Q_2)}{r/2}\right].$$

$$V_M = \frac{1}{4\pi\varepsilon_0} \cdot \frac{2(Q_1 - Q_2)}{r}.$$

$$V_M = \frac{Q_1 - Q_2}{2\pi\varepsilon_0 r}.$$

The work done by an external agent in bringing the charge Q_3 from infinity to M equals $Q_3 V_M$. Therefore,

$$W = Q_3 V_M = \frac{Q_3(Q_1 - Q_2)}{2\pi\varepsilon_0 r}.$$

This is the **work done** to place Q_3 at the midpoint.

(b) The work done in placing Q_3 at any point is $W = Q_3 V$, where V is the potential at that point due to Q_1 and Q_2. The work will be zero where the net potential is zero.

Let the point lie at a distance x from $+Q_1$ along the line, so its distance from $-Q_2$ is $(r - x)$. Setting the net potential to zero,

$$\frac{1}{4\pi\varepsilon_0}\frac{Q_1}{x} + \frac{1}{4\pi\varepsilon_0}\frac{(-Q_2)}{r-x} = 0.$$

$$\frac{Q_1}{x} = \frac{Q_2}{r - x}.$$

$$Q_1(r - x) = Q_2\, x.$$

$$Q_1\, r = x(Q_1 + Q_2).$$

$$x = \frac{Q_1\, r}{Q_1 + Q_2}.$$

Thus at a distance $x = \dfrac{Q_1\, r}{Q_1 + Q_2}$ from $+Q_1$, the work done is zero.

Final answer. The work to place Q_3 at the midpoint is $W = \dfrac{Q_3(Q_1 - Q_2)}{2\pi\varepsilon_0\, r}$, and it becomes zero at $x = \dfrac{Q_1\, r}{Q_1 + Q_2}$ from $+Q_1$.

← *Back to Q152*

Q153

Draw the equipotential surfaces corresponding to a uniform electric field in the z-direction.

[2019 • Set 55-1-1]

SOLUTION

A uniform electric field directed along the z-axis can be written as $\vec{E} = E_0\, \hat{k}$, where E_0 is constant.

The relation between field and potential along z is $E_z = -\dfrac{dV}{dz}$, so $V = -E_0\, z +$ **constant**.

An equipotential surface is the locus of points having the same potential V. Since V depends only on z, the condition $V =$ **constant** means $z =$ **constant**.

Hence the equipotential surfaces are a family of **parallel planes perpendicular to the z-axis** (that is, planes parallel to the x-y plane).

Diagram description: Draw several equally spaced horizontal parallel lines (representing planes seen edge-on), each labelled with a constant potential value. The field $\vec{E}$ is shown as vertical arrows pointing along $+z$, crossing each plane at right angles. Since the field is uniform, equal increments of potential

give equally spaced planes, with $\Delta z = \frac{\Delta V}{E_0}$. The potential decreases in the direction of $\vec{E}$ (increasing z).

CONCEPT

Equipotential surfaces are always perpendicular to the field lines and are equally spaced only when the field is uniform.

Final answer. The equipotential surfaces are equally spaced parallel planes perpendicular to the z-axis.

← *Back to Q153*

Q154

Draw equipotential surfaces corresponding to the electric field that uniformly increases in magnitude along with the z-direction.

[2019 • Set 55-1-2]

SOLUTION

Let the field point along the z-axis with a magnitude that increases uniformly with z, so $\vec{E}(z) = E(z)\,\hat{k}$ with $E(z) = \alpha z$ where α is a positive constant.

The potential is found from $E_z = -\frac{dV}{dz}$, giving

$$V(z) = -\int E(z)\,dz = -\int \alpha z\,dz = -\frac{\alpha z^2}{2} + \text{constant}.$$

Since V depends only on z, the condition $V = \text{constant}$ again gives $z = \text{constant}$. So the equipotential surfaces are still **parallel planes perpendicular to the z-axis**.

However, the spacing is no longer uniform. For equal potential steps ΔV, the spacing satisfies $|\Delta V| = E(z)\,\Delta z$, so $\Delta z = \frac{\Delta V}{\alpha z}$.

As z increases, $E(z)$ becomes larger, so the spacing Δz between consecutive equipotentials of equal ΔV becomes smaller.

Diagram description: Draw a stack of horizontal parallel lines perpendicular to the z-axis. The lines are drawn far apart near small z (weak field) and become progressively closer together as z increases (strong field). Vertical arrows along $+z$ represent the field, crossing each plane at right angles, with the arrows shown longer at larger z to indicate the growing magnitude.

Final answer. The equipotentials are parallel planes perpendicular to the z-axis, getting progressively closer together as z increases.

← *Back to Q154*

Q155

Two charges $-q$ and $+q$ are located at points $(0, 0, -a)$ and $(0, 0, a)$. What is the electrostatic potential at the points $(0, 0, +z)$ and $(x, y, 0)$?

[2019 • Set 55-1-2]

SOLUTION

The charge $-q$ is at $(0, 0, -a)$ and $+q$ is at $(0, 0, +a)$, forming a dipole of moment $p = q(2a) = 2qa$ directed along $+z$.

(i) Potential at the axial point $(0, 0, +z)$: The distance from $+q$ to this point is $(z - a)$ and from $-q$ is $(z + a)$.

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{z-a} + \frac{-q}{z+a}\right].$$

$$V = \frac{q}{4\pi\varepsilon_0}\left[\frac{(z+a)-(z-a)}{(z-a)(z+a)}\right].$$

$$V = \frac{q}{4\pi\varepsilon_0} \cdot \frac{2a}{z^2 - a^2}.$$

$$V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{p}{z^2 - a^2}, \text{ where } p = 2qa.$$

For $z \gg a$, this reduces to the standard axial dipole result $V \approx \frac{1}{4\pi\varepsilon_0}\frac{p}{z^2}$.

(ii) Potential at the equatorial point $(x, y, 0)$: This point lies in the plane $z = 0$, which is equidistant from both charges. The distance from each of $\pm q$ to $(x, y, 0)$ is the same, namely $\sqrt{x^2 + y^2 + a^2}$.

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{\sqrt{x^2+y^2+a^2}} + \frac{-q}{\sqrt{x^2+y^2+a^2}}\right].$$

$V = 0.$

Hence the potential everywhere on the equatorial plane of the dipole is **zero**.

Final answer. At $(0,0,z)$ the potential is $V = \frac{1}{4\pi\varepsilon_0}\frac{p}{z^2-a^2}$ (with $p=2qa$); at $(x,y,0)$ the potential is zero.

← *Back to Q155*

Q156

Draw the equipotential surfaces due to an electric dipole.

[2019 • Set 55-1-3]

SOLUTION

An electric dipole consists of charges $+q$ and $-q$ separated by a small distance 2a, with dipole moment $\vec{p}$ pointing from $-q$ to $+q$.

The potential at a point distant r from the centre, making an angle θ with the dipole axis, is $V = \frac{1}{4\pi\varepsilon_0}\frac{p\theta}{r^2}$.

Because V depends on both r and θ, the equipotential surfaces are not simple spheres or planes; they have a characteristic shape that surrounds each charge.

Diagram description: Draw the dipole horizontally with $+q$ on the right and $-q$ on the left, joined by the dipole axis. Around the $+q$ end draw a set of closed curves (roughly oval/peanut shaped) labelled with positive potentials ($+V_1$, $+V_2$, ...), the innermost curves hugging $+q$ tightly. Around the $-q$ end draw a mirror-image set of closed curves labelled with negative potentials ($-V_1$, $-V_2$, ...).

The plane passing through the centre and perpendicular to the dipole axis (the equatorial plane, $\theta = 90°$) is the $V = 0$ equipotential; show it as a straight vertical line through the centre.

Key features to mark: the surfaces near each charge are nearly spherical, they become elongated and distorted farther out, they never intersect one another,

and at every point they are drawn perpendicular to the dipole field lines (which start on $+q$ and end on $-q$).

Final answer. The equipotentials are closed peanut-shaped surfaces wrapping each charge (positive around $+q$, negative around $-q$), with the perpendicular bisector plane as the $V=0$ surface.

← *Back to Q156*

Q157

A 200 μF *parallel plate capacitor having plate separation of* 5 mm *is charged by a* 100 *V dc source. It remains connected to the source. Using an insulated handle, the distance between the plates is doubled and a dielectric slab of thickness* 5 mm *and dielectric constant* 10 *is introduced between the plates. Explain with reason, how the (i) capacitance, (ii) electric field between the plates, (iii) energy density of the capacitor will change?*

[2019 • Set 55-2-1]

SOLUTION

Given data: $\mathbf{C_0 = 200\ \mu F}$, original separation $\mathbf{d_0 = 5\ mm}$, $\boldsymbol{V = 100\ V}$, with the source remaining connected so V stays fixed at 100 V. The gap is doubled to $\boldsymbol{d = 10\ mm}$, and a slab of thickness $\boldsymbol{t = 5\ mm}$ and dielectric constant $\boldsymbol{K = 10}$ is inserted, leaving 5 mm of air.

(i) Capacitance: For a slab of thickness t in a gap d, the capacitance is $\boldsymbol{C = \dfrac{\varepsilon_0 A}{d - t + \dfrac{t}{K}}}$.

The denominator is $\boldsymbol{d - t + \dfrac{t}{K} = 10 - 5 + \dfrac{5}{10} = 5.5\ mm}$.

Using $\boldsymbol{\varepsilon_0 A = C_0 d_0 = 200 \times 5 = 1000\ \mu F}$,

$$\boldsymbol{C = \frac{1000}{5.5} \approx 181.8\ \mu F}.$$

So the capacitance **decreases** from 200 μF to about 181.8 μF.

(ii) Electric field: The battery keeps $V = 100\ V$. Let the field in the air gap be E_{air} and in the dielectric $E_{diel} = \frac{E_{air}}{K}$. The potential drop across the gap is

$E_{air}\, t_{air} + E_{diel}\, t = V.$

$E_{air}(5) + \frac{E_{air}}{10}(5) = 100.$

$E_{air}(5.5) = 100 \Rightarrow E_{air} = 18.2\ V/mm \approx 1.82 \times 10^4\ V/m.$

Originally the field was $E_0 = \frac{V}{d_0} = \frac{100}{5} = 20\ V/mm = 2 \times 10^4\ V/m$, so the field in the air region **decreases** (and is even smaller inside the dielectric, $E_{diel} = 1.82\ V/mm$).

(iii) Energy density: In the air region, $u = \frac{1}{2}\varepsilon_0 E_{air}^2$. Since E_{air} (18.2 V/mm) is smaller than the original E_0 (20 V/mm), the energy density in the air region **decreases**. In the dielectric region $u = \frac{1}{2}K\varepsilon_0 E_{diel}^2$, which is also smaller than the original value. Hence the energy density decreases.

> **TIP**
>
> Battery connected means V is held fixed; battery disconnected means Q is held fixed. Always decide which one is constant before judging how E and u change.

Final answer. Capacitance decreases to about **181.8 μF**, the air-gap field decreases from **20** to about **18.2 V/mm**, and the energy density decreases.

← *Back to Q157*

Q158

A 100 μF *parallel plate capacitor having plate separation of* 4 mm *is charged by* 200 *V dc. The source is now disconnected. When the distance between the plates is doubled and a dielectric slab of thickness* 4 mm *and dielectric constant* 5 *is introduced between the plates, how will (i) its capacitance, (ii) the electric field between the plates, and (iii) energy density of the capacitor get affected? Justify your answer in each case.*

[2019 • Set 55-2-2]

SOLUTION

Given data: $C_0 = 100\ \mu F$, original separation $d_0 = 4\ mm$, $V_0 = 200\ V$. The source is then disconnected, so the charge Q_0 stays constant. The gap is doubled to $d = 8\ mm$, and a slab of thickness $t = 4\ mm$ with dielectric constant $K = 5$ is inserted, leaving 4 mm of air.

Initial charge: $Q_0 = C_0 V_0 = 100\ \mu F \times 200\ V = 0.02\ C$.

Original field: $E_0 = \frac{V_0}{d_0} = \frac{200}{4} = 50\ V/mm$.

(i) Capacitance: $C = \frac{\varepsilon_0 A}{d - t + \frac{t}{K}}$.

The denominator is $8 - 4 + \frac{4}{5} = 4.8\ mm$.

Using $\varepsilon_0 A = C_0 d_0 = 100 \times 4 = 400\ \mu F$,

$C = \frac{400}{4.8} \approx 83.3\ \mu F$.

So the capacitance **decreases** from 100 μF to about 83.3 μF.

(ii) Electric field: With the source disconnected, the charge Q_0 and hence the surface charge density $\sigma = \frac{Q_0}{A}$ remain unchanged. In the air region the field is $E_{air} = \frac{\sigma}{\varepsilon_0}$, which is exactly the original value, so $E_{air} = E_0 = 50\ V/mm$ (unchanged). Inside the dielectric the field is reduced: $E_{diel} = \frac{E_{air}}{K} = \frac{50}{5} = 10\ V/mm$.

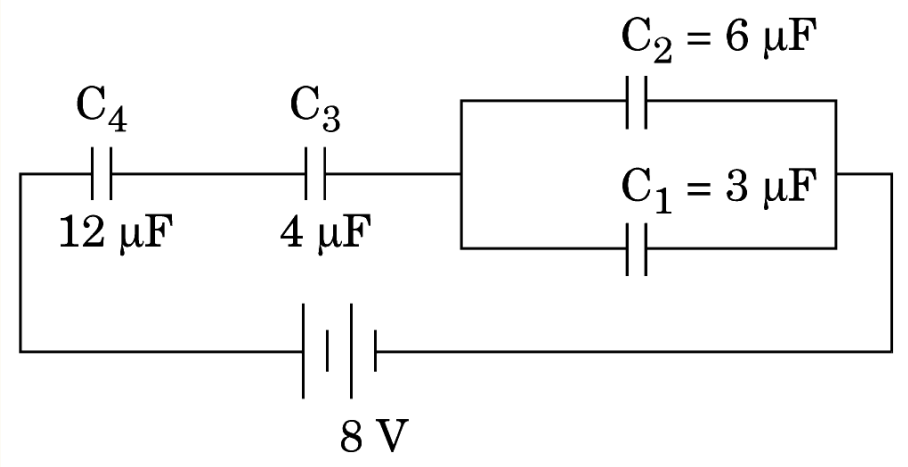
(iii) Energy density: In the air region $u_{air} = \frac{1}{2}\varepsilon_0 E_{air}^2$. Since E_{air} is unchanged, the energy density in the air region is **unchanged**. In the dielectric, $u_{diel} = \frac{1}{2} K\varepsilon_0 E_{diel}^2 = \frac{1}{2} K\varepsilon_0 (\frac{E_0}{K})^2 = \frac{u_0}{K}$, so it is reduced to one-fifth of the original. Thus the energy density stays the same in the air gap and decreases inside the dielectric.

Final answer. Capacitance decreases to about **83.3 μF**; the air-gap field stays at **50 *V*/mm** while the dielectric field is **10 *V*/mm**; energy density is unchanged in air and falls to **1/*K*** inside the dielectric.

← Back to Q158

Q159

In a network, four capacitors C_1, C_2, C_3 and C_4 are connected as shown in the figure.



(a) Calculate the net capacitance in the circuit.
(b) If the charge on the capacitor C_2 is 6 μC, (i) calculate the charge on the capacitors C_3 and C_4, and (ii) net energy stored in the capacitors C_3 and C_4 connected in series.

[2019 • Set 55-2-3]

SOLUTION

From the network, $\mathbf{C_1 = 12\ \mu F}$ is connected in parallel with the series combination of $\mathbf{C_2 = 6\ \mu F}$, $\mathbf{C_3 = 3\ \mu F}$ and $\mathbf{C_4 = 4\ \mu F}$.

(a) First find the series combination of C_2, C_3, C_4:

$$\frac{1}{C_s} = \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4} = \frac{1}{6} + \frac{1}{3} + \frac{1}{4}.$$

$$\frac{1}{C_s} = \frac{2+4+3}{12} = \frac{9}{12} = \frac{3}{4}.$$

$$C_s = \frac{4}{3}\ \mu F.$$

This is in parallel with C_1, so the net capacitance is

$$C_{net} = C_1 + C_s = 12 + \frac{4}{3} = \frac{40}{3} \approx 13.33\ \mu F.$$

(b)(i) Since C_2, C_3 and C_4 are in series, they all carry the same charge. Given the charge on C_2 is 6 μC,

$$\mathbf{Q_{C_3} = Q_{C_4} = 6\ \mu C.}$$

(b)(ii) Energy stored in C_3 using $U = \frac{Q^2}{2C}$:

$$\mathbf{U_3 = \frac{(6 \times 10^{-6})^2}{2 \times (3 \times 10^{-6})} = \frac{36 \times 10^{-12}}{6 \times 10^{-6}} = 6 \times 10^{-6}\,J = 6\ \mu J.}$$

Energy stored in C_4:

$$\mathbf{U_4 = \frac{(6 \times 10^{-6})^2}{2 \times (4 \times 10^{-6})} = \frac{36 \times 10^{-12}}{8 \times 10^{-6}} = 4.5 \times 10^{-6}\,J = 4.5\ \mu J.}$$

Net energy stored in C_3 and C_4 together:

$$\mathbf{U_{3+4} = U_3 + U_4 = 6 + 4.5 = 10.5\ \mu J.}$$

Final answer. Net capacitance is $\frac{40}{3} \approx 13.33\ \mu F$; charges on C_3 and C_4 are each 6 μC; total energy in C_3 and C_4 is 10.5 μJ.

← *Back to Q159*

Q160

(a) Explain briefly, using a proper diagram, the difference in behaviour of a conductor and a dielectric in the presence of external electric field.
(b) Define the term polarization of a dielectric and write the expression for a linear isotropic dielectric in terms of electric field.

[2019 • Set 55-3-1]

SOLUTION

(a) Behaviour of a conductor versus a dielectric in an external field $\vec{E}_0$:

In a conductor there are many free electrons. When placed in $\vec{E}_0$, these free electrons drift opposite to the field and pile up on one face, leaving an equal positive charge on the opposite face. These induced surface charges set up an internal field $\vec{E}_{ind}$ that opposes $\vec{E}_0$. The redistribution continues until $\vec{E}_{ind}$ exactly cancels $\vec{E}_0$, so the net field inside the conductor becomes zero: $\vec{E}_{inside} = 0$.

In a dielectric there are no free charges; the electrons are bound to their molecules. In $\vec{E}_0$ the bound charges shift slightly (or permanent molecular dipoles align), so bound surface charges appear on the two faces. These produce an opposing field that only partially cancels $\vec{E}_0$. The net field inside is reduced but not zero: $\vec{E}_{\text{inside}} = \frac{\vec{E}_0}{K}$, where $K > 1$.

Diagram description: Draw two identical slabs in a uniform field shown by horizontal arrows pointing right. For the conductor, mark − charges densely on the left face and + charges on the right face, and write $\vec{E}_{\text{inside}} = 0$ in the interior with no field lines passing through. For the dielectric, mark fewer (bound) − charges on the left face and + on the right face, and draw a few weaker field lines passing through the interior to show a reduced but non-zero field $\frac{\vec{E}_0}{K}$.

(b) Polarisation: The polarisation $\vec{P}$ of a dielectric is defined as the net electric dipole moment induced per unit volume of the dielectric.

For a linear isotropic dielectric, the polarisation is directly proportional to the applied (net) electric field:

$\vec{P} = \varepsilon_0 \chi_e \vec{E}$,

where χ_e is the electric susceptibility of the dielectric medium.

Final answer. A conductor reduces the internal field to zero ($\vec{E}_{\text{inside}}=0$), while a dielectric only reduces it to $\vec{E}_0/K$; polarisation is dipole moment per unit volume, with $\vec{P}=\varepsilon_0\chi_e\vec{E}$.

← Back to Q160

Q161

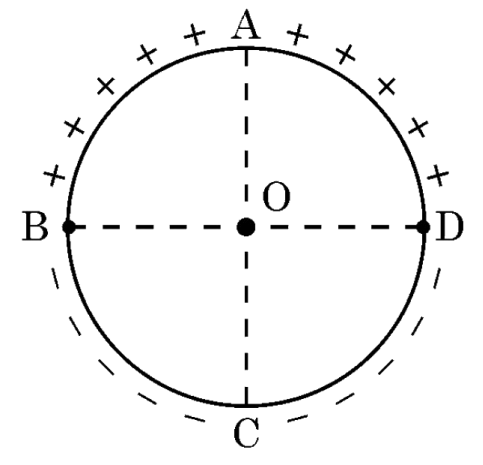
A charge Q is distributed over the surfaces of two concentric hollow spheres of radii r and R **(R >> r)**, *such that their surface charge densities are equal. Derive the expression for the potential at the common centre.*

OR

Three concentric metallic shells A, B and C of radii a, b and c **(a < b < c)** *have surface charge densities +σ, −σ and +σ respectively as shown.*

(a) Obtain the expressions for the potential of three shells A, B and C.

(b) If shells A and C are at the same potential, obtain the relation between a, b and c.



[2019 • Set 55-5-1]

SOLUTION

A point inside a charged spherical shell is at the same potential as its surface, $V = \frac{1}{4\pi\varepsilon_0}\frac{q}{R_{shell}}$, so the centre lies inside both shells and we add the two contributions.

The charge on each sphere follows from its surface charge density:

$$Q_r = \sigma \times 4\pi r^2,\ Q_R = \sigma \times 4\pi R^2$$

Since the two surface densities are equal ($\sigma_r = \sigma_R = \sigma$), the total charge is

$$Q = Q_r + Q_R = 4\pi\sigma (r^2 + R^2)$$

$$\Rightarrow \sigma = \frac{Q}{4\pi(r^2 + R^2)}$$

The potential at the common centre O is the sum of the potentials of the two shells (both treated as constant inside):

$$V_C = \frac{1}{4\pi\varepsilon_0}\frac{Q_r}{r} + \frac{1}{4\pi\varepsilon_0}\frac{Q_R}{R}$$

$$V_C = \frac{1}{4\pi\varepsilon_0}\left(\frac{\sigma\, 4\pi r^2}{r} + \frac{\sigma\, 4\pi R^2}{R}\right)$$

$$V_C = \frac{\sigma}{\varepsilon_0}(r + R)$$

Substituting the value of σ:

$$V_C = \frac{Q\,(r+R)}{4\pi\varepsilon_0\,(r^2+R^2)}$$

OR (three concentric shells A, B, C of radii a, b, c carrying charges +σ, −σ, +σ per unit area, i.e. $\mathbf{Q_A = 4\pi\,a^2\sigma}$, $\mathbf{Q_B = -4\pi\,b^2\sigma}$, $\mathbf{Q_C = 4\pi\,c^2\sigma}$).

(a) The potential of any shell is the sum of contributions from all three; a shell at radius R_i contributes $\frac{kQ}{R_i}$ at points on or inside it and $\frac{kQ}{R_{\text{point}}}$ when the point lies outside it. With $\boldsymbol{k = \frac{1}{4\pi\varepsilon_0}}$:

$$V_A = k\left(\frac{Q_A}{a} + \frac{Q_B}{b} + \frac{Q_C}{c}\right) = \frac{\sigma}{\varepsilon_0}\,(a - b + c)$$

$$V_B = k\left(\frac{Q_A}{b} + \frac{Q_B}{b} + \frac{Q_C}{c}\right) = \frac{\sigma}{\varepsilon_0}\left(\frac{a^2}{b} - b + c\right)$$

$$V_C = k\left(\frac{Q_A}{c} + \frac{Q_B}{c} + \frac{Q_C}{c}\right) = \frac{\sigma}{\varepsilon_0}\left(\frac{a^2}{c} - \frac{b^2}{c} + c\right)$$

(b) Setting $\mathbf{V_A = V_C}$:

$$a - b + c = \frac{a^2}{c} - \frac{b^2}{c} + c$$

$$a - b = \frac{a^2 - b^2}{c} = \frac{(a-b)(a+b)}{c}$$

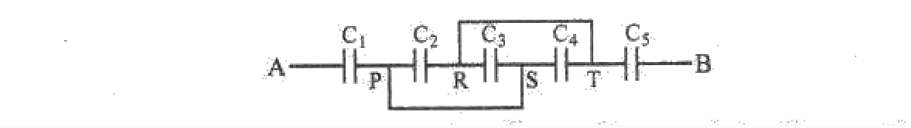
Dividing both sides by $(a - b)$ gives the required relation $\boldsymbol{c = a + b}$.

Final answer. V_C = Q(r+R)/[4\pi\varepsilon_0(r^2+R^2)]; in the OR part, A and C are at equal potential when c = a + b.

← *Back to Q161*

Q162

(i) Find equivalent capacitance between A and B in the combination given below. Each capacitor is of 2 μF *capacitance.*



(ii) If a dc source of 7 V is connected across AB, how much charge is drawn from the source and what is the energy stored in the network?

[2017]

SOLUTION

(i) The circuit is a Wheatstone bridge of five identical 2 μF capacitors: C_1, C_2, C_3, C_4 form the four arms and C_5 is the diagonal bridging the two midpoints.

For a balanced bridge the arm-ratio condition must hold:

$$\frac{C_1}{C_2} = \frac{C_3}{C_4}$$

With all capacitors equal, $\frac{2}{2} = \frac{2}{2}$, so the bridge is balanced. The two midpoints are then at the same potential, no charge resides on C_5, and it can be removed.

The two upper arms C_1, C_2 are now in series, and so are the two lower arms C_3, C_4:

$$C_{12} = \frac{C_1 C_2}{C_1 + C_2} = \frac{2 \times 2}{2 + 2} = 1\ \mu F$$

$$C_{34} = \frac{C_3 C_4}{C_3 + C_4} = \frac{2 \times 2}{2 + 2} = 1\ \mu F$$

These two series branches are in parallel between A and B:

$$C_{AB} = C_{12} + C_{34} = 1 + 1 = \mathbf{2\ \mu F}$$

(ii) Charge drawn from the 7 *V* source:

$$Q = C_{AB}\, V = (2 \times 10^{-6})(7)$$

$$Q = 14 \times 10^{-6}\ C = \mathbf{14\ \mu C}$$

Energy stored in the network:

$$U = \frac{1}{2} C_{AB} V^2 = \frac{1}{2}(2 \times 10^{-6})(7)^2$$

$$U = \frac{1}{2}(2 \times 10^{-6})(49) = 49 \times 10^{-6}\,J = \mathbf{49\ \mu J}$$

TIP

In a balanced Wheatstone bridge of capacitors the diagonal capacitor carries no charge, so drop it before combining the arms; this saves the whole calculation.

Final answer. C_AB = 2 \muF, charge drawn = 14 \muC, energy stored = 49 \muJ.

← *Back to Q162*

Q163

A 12 pF *capacitor is connected to a* 50 *V battery. How much electrostatic energy is stored in the capacitor? If another capacitor of* 6 pF *is connected in series with it with the same battery connected across the combination, find the charge stored and potential difference across each capacitor.*

[2017]

SOLUTION

First, the energy stored in the single 12 pF capacitor charged to 50 *V*:

$$U = \frac{1}{2}CV^2$$

$$U = \frac{1}{2}(12 \times 10^{-12})(50)^2$$

$$U = \frac{1}{2}(12 \times 10^{-12})(2500)$$

$$U = 1.5 \times 10^{-8}\,J = \mathbf{15\ nJ}$$

Now a 6 pF capacitor is joined in series with the 12 pF one across the same 50 *V* battery. The series equivalent is

$$C_s = \frac{C_1\,C_2}{C_1 + C_2} = \frac{12 \times 6}{12 + 6} = \frac{72}{18} = 4\ \text{pF}$$

In series both capacitors carry the same charge, equal to the charge supplied by the battery:

$$Q = C_s V = (4 \times 10^{-12})(50)$$

$$Q = 2 \times 10^{-10}\ C = \mathbf{200\ pC}$$

Potential difference across the 12 pF capacitor:

$$V_{12} = \frac{Q}{C_1} = \frac{200\ \text{pC}}{12\ \text{pF}} \approx \mathbf{16.7\ V}$$

Potential difference across the 6 pF capacitor:

$$V_6 = \frac{Q}{C_2} = \frac{200\ \text{pC}}{6\ \text{pF}} \approx \mathbf{33.3\ V}$$

As a check, $\mathbf{V_{12} + V_6 = 16.7 + 33.3 = 50\ V}$, equal to the battery voltage.

Final answer. Energy in the 12 pF alone = 15 nJ; in series Q = 200 pC on each, with V_12 \approx 16.7 V and V_6 \approx 33.3 V.

← *Back to Q163*

Q164

(i) Derive the expression for the electric potential due to an electric dipole at a point on its axial line.
(ii) Depict the equipotential surfaces due to an electric dipole.

[2017]

SOLUTION

(i) Consider a dipole with charge $-q$ at A and $+q$ at B, separated by 2a, with centre O. Let P be an axial point at distance r from O, on the side of $+q$.

The distances of P from the two charges are

$$\text{BP} = r - a\ (\text{from } +q),\ \text{AP} = r + a\ (\text{from } -q)$$

The net potential at P is the algebraic sum of the potentials due to the two charges:

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r-a} + \frac{1}{4\pi\varepsilon_0}\frac{(-q)}{r+a}$$

$$V = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r-a} - \frac{1}{r+a}\right)$$

$$V = \frac{q}{4\pi\varepsilon_0}\cdot\frac{(r+a)-(r-a)}{(r-a)(r+a)}$$

$$V = \frac{q}{4\pi\varepsilon_0}\cdot\frac{2a}{r^2 - a^2}$$

Writing the dipole moment $\boldsymbol{p = q\,(2a)}$:

$$V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p}{r^2 - a^2}$$

For a short dipole or a far point, $r \gg a$, so a^2 is negligible compared with r^2:

$$V_{\text{axial}} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p}{r^2}$$

(ii) Equipotential surfaces of a dipole: draw the dipole horizontal with $+q$ on the right and $-q$ on the left. Around each charge the equipotentials are small closed loops, nearly spherical close in, with positive values clustered around $+q$ and negative values around $-q$. As we move outward these loops distort into elongated peanut-shaped surfaces wrapping the dipole. The perpendicular bisector plane of the dipole (the equatorial plane) is the single flat $\boldsymbol{V = 0}$ equipotential. The electric field lines, running from $+q$ to $-q$, cross every one of these equipotential surfaces at right angles.

Final answer. On the axis V = p/(4\pi\varepsilon_0 r^2) for a short dipole; equipotentials are closed loops round each charge with the equatorial plane at V = 0.

← *Back to Q164*

Q165

Two identical capacitors of 12 pF *each are connected in series across a battery of* 50 *V. How much electrostatic energy is stored in the combination? If these were connected in parallel across the same battery, how much energy will be stored in the combination*

now?

Also find the charge drawn from the battery in each case.

[2017]

SOLUTION

(i) Series combination across $V = 50\ V$.

$$C_s = \frac{C_1 C_2}{C_1 + C_2} = \frac{12 \times 12}{12 + 12} = \frac{144}{24} = 6\ \text{pF}$$

Energy stored in the series combination:

$$U_s = \frac{1}{2} C_s V^2 = \frac{1}{2}(6 \times 10^{-12})(50)^2$$

$$U_s = \frac{1}{2}(6 \times 10^{-12})(2500) = 7.5 \times 10^{-9}\ J = \mathbf{7.5\ nJ}$$

Charge drawn from the battery:

$$Q_s = C_s\ V = (6 \times 10^{-12})(50) = 3.0 \times 10^{-10}\ C = \mathbf{300\ pC}$$

(ii) Parallel combination across the same $V = 50\ V$.

$$C_p = C_1 + C_2 = 12 + 12 = 24\ \text{pF}$$

Energy stored in the parallel combination:

$$U_p = \frac{1}{2} C_p V^2 = \frac{1}{2}(24 \times 10^{-12})(2500)$$

$$U_p = 3.0 \times 10^{-8}\ J = \mathbf{30\ nJ}$$

Charge drawn from the battery:

$$Q_p = C_p\ V = (24 \times 10^{-12})(50) = 1.2 \times 10^{-9}\ C = \mathbf{1.2\ nC}$$

Since $\mathbf{C_p/C_s = 4}$, both the energy and the charge are four times larger in the parallel case.

Final answer. Series: U_s = 7.5 nJ, Q_s = 300 pC. Parallel: U_p = 30 nJ, Q_p = 1.2 nC.

← Back to Q165

Q166

A charge Q is distributed uniformly over a metallic sphere of radius R. Obtain the expressions for the electric field (E) and electric potential (V) at a point **0 < x < R**. *Show on a plot the variation of E and V with x for* **0 < x < 2R**.

[2017]

SOLUTION

A metallic (conducting) sphere of radius R carries charge Q which, in electrostatic equilibrium, resides entirely on the outer surface; the interior field is zero.

Electric field for $\mathbf{0 \le x < R}$ (inside the sphere): take a spherical Gaussian surface of radius x inside the conductor. It encloses no charge, $\mathbf{q_{enc} = 0}$. By Gauss's law,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\varepsilon_0} = 0$$

$$E(x) = 0 \ (0 \le x < R)$$

Just outside ($x \ge R$) the same law gives $\boldsymbol{E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{x^2}}$, with surface value $\mathbf{E_R = \frac{kQ}{R^2}}$.

Electric potential for $\mathbf{0 \le x < R}$: since $\boldsymbol{E = 0}$ everywhere inside, no work is done moving a charge within the sphere, so the potential is constant and equal to its value at the surface,

$$V_{surface} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R}$$

$$V(x) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R} = \text{constant} \ (0 \le x \le R)$$

For $\boldsymbol{x > R}$ the potential falls as $\boldsymbol{V = \frac{kQ}{x}}$.

Plot description: take x along the horizontal axis. The E curve is flat at zero from $\boldsymbol{x = 0}$ up to $\boldsymbol{x = R}$, then jumps suddenly to the value $\frac{kQ}{R^2}$ at $\boldsymbol{x = R}$, after which it falls off as $\frac{1}{x^2}$. The V curve is a horizontal straight line at the constant value $\frac{kQ}{R}$ from $\boldsymbol{x = 0}$ to $\boldsymbol{x = R}$, and then decreases smoothly as $\frac{1}{x}$ for $\boldsymbol{x > R}$ (for example, halving to $\frac{kQ}{2R}$ at $\boldsymbol{x = 2R}$). Thus V is continuous everywhere while E shows a jump at the surface $\boldsymbol{x = R}$.

Final answer. Inside: E = 0 and V = kQ/R (constant); E jumps to kQ/R^2 at the surface, while V stays continuous and then falls as kQ/x outside.

← *Back to Q166*

Q167

Two identical parallel plate capacitors A and B are connected to a battery of V volts with the switch S closed. The switch is now opened and the free space between the plates of the capacitors is filled with a dielectric of dielectric constant K. Find the ratio of the total electrostatic energy stored in both capacitors before and after the introduction of the dielectric.

[2017]

SOLUTION

Let each identical capacitor have capacitance C. With switch S closed, both A and B are connected in parallel to the V-volt battery, so each is charged to potential difference V.

Energy stored in each capacitor before the dielectric:

$$U_{each} = \frac{1}{2}CV^2$$

Total initial energy of both capacitors:

$$U_i = 2 \times \frac{1}{2}CV^2 = CV^2$$

The charge on each capacitor at this stage is $\boldsymbol{Q = CV}$.

Now the switch S is opened, disconnecting both capacitors from the battery. The charge on each is therefore trapped and stays constant at $Q = CV$ even when the dielectric is inserted.

Inserting the dielectric of constant K raises each capacitance to $C' = KC$. With charge held fixed, the new energy of each capacitor is best written in terms of Q:

$$U'_{\text{each}} = \frac{Q^2}{2C'} = \frac{(CV)^2}{2(KC)} = \frac{CV^2}{2K}$$

Total final energy of both capacitors:

$$U_f = 2 \times \frac{CV^2}{2K} = \frac{CV^2}{K}$$

Taking the ratio of initial to final energy:

$$\frac{U_i}{U_f} = \frac{CV^2}{CV^2/K} = K$$

$$U_i : U_f = K : 1$$

The stored energy falls by the factor K; the lost energy goes into polarising the dielectric.

> **COMMON MISTAKE**
>
> *The switch being opened is the crux: with the battery disconnected charge Q stays fixed, so use U = Q^2/2C, not U = CV^2/2. If S stayed closed, V would be fixed and the ratio would instead be 1 : K.*

Final answer. U_i : U_f = K : 1 (the energy decreases K times because the charge is constant once the switch is opened).

← *Back to Q167*

Q168

Define an equipotential surface. Draw equipotential surfaces:
(i) in the case of a single point charge and

(ii) in a constant electric field in Z-direction.
Why the equipotential surfaces about a single charge are not equidistant?
(iii) Can electric field exist tangential to an equipotential surface? Give reason.

[2016]

SOLUTION

Definition: an equipotential surface is a surface on which every point is at the same electric potential. No work is done by the electric force in moving a charge between any two points on it, since $W = q\,\Delta V = 0$.

(i) Single point charge: as $V = \frac{kQ}{r}$ depends only on the distance r, the equipotentials are a family of concentric spheres centred on the charge. Draw three or four concentric circles around the charge; for $Q > 0$ the potential is positive and decreases outward.

(ii) Uniform field along $+\hat{z}$: since V then depends only on z, the equipotentials are a set of parallel planes lying perpendicular to the Z-axis (drawn as equally spaced horizontal parallel lines), with the field lines along $\hat{z}$ piercing them at right angles. Equal V-steps give equally spaced planes because the field is constant.

Why the equipotentials about a single charge are not equidistant: for fixed ΔV between successive surfaces, $V = \frac{kQ}{r}$ gives

$$\Delta V = \frac{kQ}{r^2}\,\Delta r \Rightarrow \Delta r = \frac{r^2\,\Delta V}{kQ}$$

So the spacing Δr grows as r^2: equipotentials are crowded close to the charge (where E is strong) and spread far apart at large r (where E is weak). Hence they are not equidistant.

(iii) Can the electric field be tangential to an equipotential surface? No. If $\vec{E}$ had a tangential component, moving a test charge along the surface would mean $W = q\int \vec{E}\cdot d\vec{l} \neq 0$, implying a potential difference along the surface. That contradicts the surface being equipotential. Therefore $\vec{E}$ must be everywhere normal (perpendicular) to an equipotential surface.

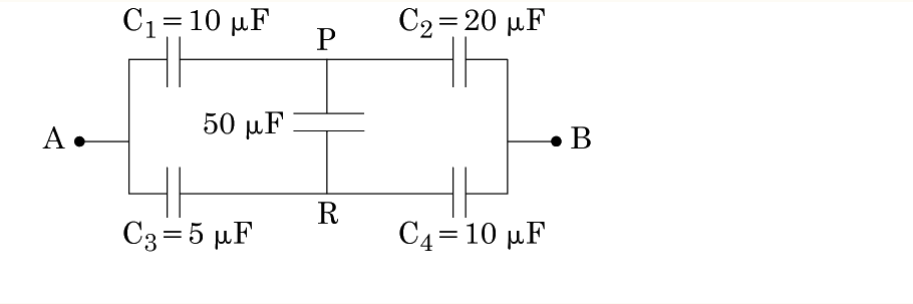
Final answer. An equipotential surface has constant V; spheres for a point charge, parallel planes for a uniform field; spacing grows as r^2 round a charge,

and E is always normal (never tangential) to it.

← *Back to Q168*

Q169

Calculate the equivalent capacitance between points A and B in the circuit below. If a battery of 10 *V is connected across A and B, calculate the charge drawn from the battery by the circuit.*



[2016]

SOLUTION

We reduce the network in stages using the series and parallel rules. (Using the standard CBSE 2016 values: $\mathbf{C_1 = 10\ pF}$ alone in one branch, $\mathbf{C_2 = 20\ pF}$ in parallel with the series pair $\mathbf{C_3 = 5\ pF}$ and $\mathbf{C_4 = 10\ pF}$, all between A and B.)

First combine C_3 and C_4, which are in series:

$$C_{34} = \frac{C_3\,C_4}{C_3 + C_4} = \frac{5 \times 10}{5 + 10} = \frac{50}{15} = \frac{10}{3}\ \text{pF}$$

This series pair is in parallel with C_2:

$$C_{234} = C_2 + C_{34} = 20 + \frac{10}{3} = \frac{70}{3}\ \text{pF}$$

Finally this is in parallel with C_1 across A and B:

$$C_{AB} = C_1 + C_{234} = 10 + \frac{70}{3} = \frac{30 + 70}{3} = \frac{100}{3} \approx 33.3\ \text{pF}$$

With the 10 *V* battery across A and B, the charge drawn from the battery is

$$Q = C_{AB}\,V = \frac{100}{3} \times 10^{-12} \times 10$$

$$Q = \frac{1000}{3} \times 10^{-12}\ C \approx 3.33 \times 10^{-10}\ C \approx \mathbf{333\ pC}$$

(Read the exact capacitor placement from the figure; the working is the same: collapse each smallest series or parallel group, repeat until one equivalent remains, then use $Q = C_{AB}V$.)

Final answer. C_AB \approx 33.3 pF (= 100/3 pF) and the charge drawn at 10 V is Q \approx 333 pC.

← *Back to Q169*

Q170

(a) A parallel plate capacitor (C_1) *having charge Q is connected, to an identical uncharged capacitor* C_2 *in series. What would be the charge accummulated on the capacitor* C_2*?*
(b) Three identical capacitors each of capacitance 3 μF *are connected, in turn, in series and in parallel combination to the common source of V volt. Find out the ratio of the energies stored in two configurations.*

[2016]

SOLUTION

(a) Capacitor C_1 initially holds charge Q. It is then connected in series to an identical uncharged capacitor C_2, with no battery in the loop.

A defining property of a series connection is that the same charge passes through every element, because the inner plates joined by an isolated conductor must carry equal and opposite induced charges. As C_1 shares its charge through the loop, the magnitude of charge that appears on C_2 equals that on C_1.

Since the two capacitors are identical and in series, the charge on C_2 is the full original charge:

$$Q_{C_2} = Q$$

(b) Three identical capacitors of $C = 3\ \mu F$ are connected first all in series, then all in parallel, across the same source of V volts.

Series equivalent:

$$C_s = \frac{C}{3} = \frac{3}{3} = 1\ \mu F$$

Parallel equivalent:

$$C_p = 3C = 3 \times 3 = 9\ \mu F$$

Both combinations are across the same voltage V, and the stored energy is $U = \frac{1}{2} C_{eq} V^2$, so at fixed V the energy is proportional to C_{eq}:

$$\frac{U_s}{U_p} = \frac{\frac{1}{2} C_s V^2}{\frac{1}{2} C_p V^2} = \frac{C_s}{C_p} = \frac{1}{9}$$

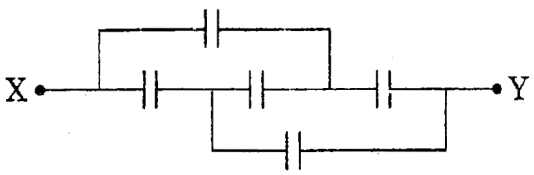
$$U_s : U_p = 1 : 9$$

Final answer. (a) Charge on C_2 equals Q (series carries equal charge); (b) energy ratio U_s : U_p = 1 : 9.

← *Back to Q170*

Q171

Find the equivalent capacitance of the network shown in the figure, when each capacitor is of 1 μ*F. When the ends X and Y are connected to a 6 V battery, find out (i) the charge and (ii) the energy stored in the network.*



[2015]

SOLUTION

The network is the standard CBSE 2015 arrangement of five 1 μF capacitors forming a Wheatstone-type bridge between X and Y: capacitors C_1 and C_2 form

the upper two arms in series, C_3 and C_4 form the lower two arms in series, and the fifth capacitor C_5 bridges the two midpoints.

Since all four arm capacitors are equal, the bridge is balanced, so the potentials at the two midpoints are equal and no charge flows through the bridge capacitor C_5; it can be removed.

Upper arm (two 1 μF in series): $\frac{1}{C_{up}} = \frac{1}{1} + \frac{1}{1} = 2 \Rightarrow C_{up} = 0.5\ \mu F$.

Lower arm (two 1 μF in series): $C_{low} = 0.5\ \mu F$ similarly.

These two arms are in parallel between X and Y: $C_{XY} = C_{up} + C_{low} = 0.5 + 0.5 = 1\ \mu F$.

So the **equivalent capacitance** is $C_{XY} = 1\ \mu F$.

(i) Charge drawn from the battery: $Q = C_{XY}\,V = (1\times 10^{-6})(6) = 6\times 10^{-6}\,C = 6\ \mu C$.

(ii) Energy stored: $U = \frac{1}{2}C_{XY}V^2 = \frac{1}{2}(1\times 10^{-6})(6)^2 = \frac{1}{2}(1\times 10^{-6})(36) = 1.8\times 10^{-5}\,J$.

Hence the **energy stored** is $U = 1.8\times 10^{-5}\,J = 18\ \mu J$.

> **TIP**
>
> Always test for a balanced bridge first: if the two arm ratios are equal, the bridge capacitor carries no charge and the network collapses to simple series-parallel branches.

Final answer. Equivalent capacitance $C_{XY} = 1\ \mu F$; charge $Q = 6\ \mu C$; energy $U = 1.8\times 10^{-5}\,J = 18\ \mu J$.

← *Back to Q171*

Q172

Two capacitors of unknown capacitances C_1 and C_2 are connected first in series and then in parallel across a battery of 100 V. If the energy stored in the two combinations is 0.045 J and 0.25 J respectively, determine the value of C_1 and C_2. Also calculate the charge on each capacitor in parallel combination.

[2015]

SOLUTION

The energy stored in a capacitor (or combination) of capacitance C across a battery of e.m.f. V is $U = \frac{1}{2}CV^2$.

For the series combination, energy $U_s = 0.045\,J$ and $V = 100\,V$, so $C_s = \frac{2U_s}{V^2} = \frac{2(0.045)}{(100)^2} = \frac{0.09}{10000} = 9\times 10^{-6}\,F = 9\ \mu F$.

For the parallel combination, energy $U_p = 0.25\,J$, so $C_p = \frac{2U_p}{V^2} = \frac{2(0.25)}{(100)^2} = \frac{0.5}{10000} = 50\times 10^{-6}\,F = 50\ \mu F$.

In parallel, $C_p = C_1 + C_2 = 50\ \mu F$.

In series, $C_s = \frac{C_1 C_2}{C_1 + C_2}$, so $C_1 C_2 = C_s(C_1 + C_2) = 9\times 50 = 450\ (\mu F)^2$.

Thus C_1 and C_2 are the roots of $x^2 - (C_1+C_2)x + C_1 C_2 = 0$, i.e. $x^2 - 50x + 450 = 0$.

$$x = \frac{50 \pm \sqrt{(50)^2 - 4(450)}}{2} = \frac{50 \pm \sqrt{2500 - 1800}}{2} = \frac{50 \pm \sqrt{700}}{2}.$$

$\sqrt{700} \approx 26.46$, so $x = \frac{50 \pm 26.46}{2}$, giving $x \approx 38.2\ \mu F$ or $x \approx 11.8\ \mu F$.

Hence $C_1 \approx 38.2\ \mu F$ and $C_2 \approx 11.8\ \mu F$ (the two values are interchangeable).

In the parallel combination each capacitor has the full 100 V across it, so the charges are $Q_1 = C_1 V = (38.2\times 10^{-6})(100) = 3.82\times 10^{-3}\,C \approx 3.82\ mC$ and $Q_2 = C_2 V = (11.8\times 10^{-6})(100) = 1.18\times 10^{-3}\,C \approx 1.18\ mC$.

Hence $C_1 \approx 38.2\ \mu F$, $C_2 \approx 11.8\ \mu F$, with $Q_1 \approx 3.82\ mC$ and $Q_2 \approx 1.18\ mC$.

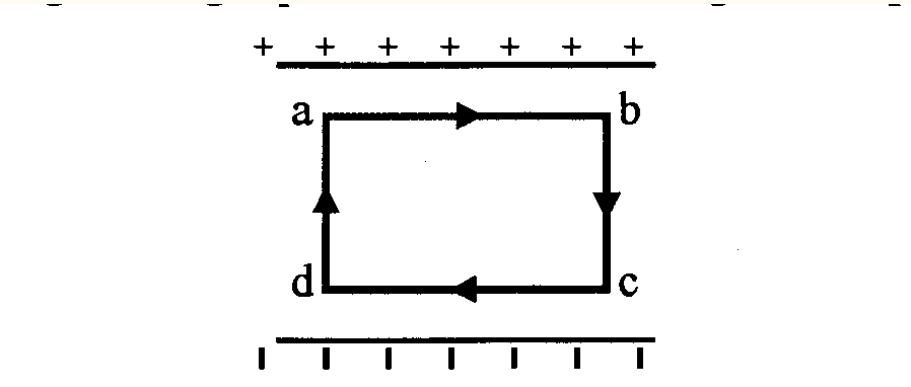
Final answer. $C_1 \approx 38.2\ \mu F$, $C_2 \approx 11.8\ \mu F$; charges in parallel $Q_1 \approx 3.82\ mC$ and $Q_2 \approx 1.18\ mC$.

← *Back to Q172*

Q173

(a) Obtain the expression for the energy stored per unit volume in a charged parallel plate capacitor.

(b) The electric field inside a parallel plate capacitor is E. Find the amount of work done in moving a charge q over a closed rectangular loop abcda.



OR

(a) Derive the expression for the capacitance of a parallel plate capacitor having plate area A and plate separation d.

(b) Two charged spherical conductors of radii R_1 *and* R_2 *when connected by a conducting wire acquire charges* q_1 *and* q_2 *respectively. Find the ratio of their surface charge densities in terms of their radii.*

[2014]

SOLUTION

(a) Energy stored per unit volume in a charged parallel plate capacitor.

The total energy stored is $U = \frac{1}{2}CV^2$, with $C = \frac{\varepsilon_0 A}{d}$ and $V = E\,d$ (uniform field E between plates separated by d).

Substituting, $U = \frac{1}{2}\left(\frac{\varepsilon_0 A}{d}\right)(E\,d)^2 = \frac{1}{2}\varepsilon_0 E^2 (A\,d)$.

The volume occupied by the field is $A\,d$, so the energy per unit volume is $u = \frac{U}{A\,d} = \frac{1}{2}\varepsilon_0 E^2$.

Hence the **energy density** is $u = \frac{1}{2}\varepsilon_0 E^2$.

(b) Work done in moving q around the closed loop abcda.

The electrostatic field is conservative, so the line integral of $\vec{E}$ around any closed path is zero: $\oint \vec{E} \cdot d\vec{l} = 0$.

Therefore the work done by the field on charge q is $W = q\oint \vec{E} \cdot d\vec{l} = 0$.

The **work done over the closed rectangular loop abcda is zero**.

OR

(a) Capacitance of a parallel plate capacitor (plate area A, separation d).

Let each plate carry surface charge density $\sigma = \frac{Q}{A}$. The uniform field between the plates is $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$.

The potential difference between the plates is $V = E\,d = \frac{Q\,d}{\varepsilon_0 A}$.

By definition, $C = \frac{Q}{V} = \frac{Q}{Q\,d/(\varepsilon_0 A)} = \frac{\varepsilon_0 A}{d}$.

Hence $C = \frac{\varepsilon_0 A}{d}$.

OR (b) Ratio of surface charge densities of two connected spheres.

When joined by a wire, the two spheres reach the same potential: $\frac{1}{4\pi\varepsilon_0}\frac{q_1}{R_1} = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{R_2}$, so $\frac{q_1}{q_2} = \frac{R_1}{R_2}$.

Surface charge density $\sigma = \frac{q}{4\pi R^2}$, so $\frac{\sigma_1}{\sigma_2} = \frac{q_1/R_1^{\,2}}{q_2/R_2^{\,2}} = \frac{q_1}{q_2}\cdot\frac{R_2^{\,2}}{R_1^{\,2}} = \frac{R_1}{R_2}\cdot\frac{R_2^{\,2}}{R_1^{\,2}}$.

Hence $\frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$, i.e. the smaller sphere has the larger surface charge density.

CONCEPT

The surface charge density is larger on the sphere of smaller radius; this is why charge (and field) concentrates at sharp points and small curvatures.

Final answer. Energy density $u = \frac{1}{2}\varepsilon_0 E^2$ and the closed-loop work is zero; OR $C = \varepsilon_0 A/d$ and $\sigma_1/\sigma_2 = R_2/R_1$.

← *Back to Q173*

Q174

A capacitor of unknown capacitance is connected across a battery of V volts. The charge stored in it is 360 μC. *When potential across the capacitor is reduced by* 120 *V, the charge stored in it becomes* 120 μC.

Calculate:

(i) The potential V and the unknown capacitance C.

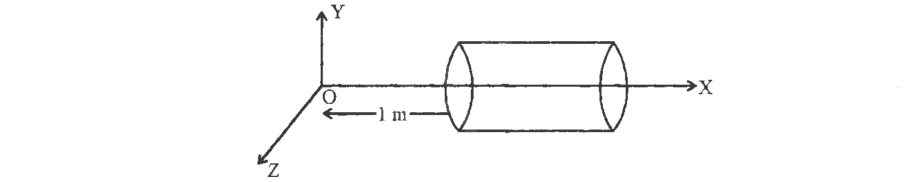
(ii) What will be the charge stored in the capacitor, if the voltage applied had increased by 120 *V?*

OR

A hollow cylindrical box of length 1 *m and area of cross-section* 25 cm^2 *is placed in a three dimensional coordinate system as shown in the figure. The electric field in the region is given by* $\vec{E} = 50\,x\,\hat{i}$*, where E is in N* C^{-1} *and x is in metres. Find*

(i) Net flux through the cylinder.

(ii) Charge enclosed by the cylinder.



[2013]

SOLUTION

(i) For a fixed capacitor, $Q = CV$, so charge is proportional to the applied voltage.

Initially $Q_1 = 360\ \mu C$ at voltage V, and after reducing the voltage by 120 V the charge is $Q_2 = 120\ \mu C$ at voltage $(V - 120)$.

Taking the ratio, $\frac{Q_1}{Q_2} = \frac{V}{V-120}$, i.e. $\frac{360}{120} = \frac{V}{V-120}$.

$3(V - 120) = V \Rightarrow 3V - 360 = V \Rightarrow 2V = 360 \Rightarrow V = 180\ V.$

Capacitance: $C = \frac{Q_1}{V} = \frac{360\ \mu C}{180\ V} = 2\ \mu F.$

So $V = 180\ V$ and $C = 2\ \mu F$.

(ii) If the voltage is increased by 120 V, the new voltage is $V' = 180 + 120 = 300\ V$.

New charge: $Q' = C\,V' = (2\times 10^{-6})(300) = 6.0\times 10^{-4}\ C = 600\ \mu C.$

The **charge stored becomes** 600 μC.

OR Hollow cylindrical box in $\vec{E} = 50\,x\,\hat{i}\ N\,C^{-1}$.

The cylinder lies along the x-axis with area of cross-section $A = 25\ cm^2 = 25\times 10^{-4}\ m^2$, its two flat faces at $x = 1\ m$ and $x = 2\ m$ (length 1 m).

The curved side surface is parallel to $\vec{E}$ (its area vector is radial, perpendicular to $\hat{i}$), so its flux is zero.

Left face at $x = 1\ m$: outward normal is $-\hat{i}$, field $E = 50(1) = 50\ N\,C^{-1}$, so $\Phi_L = -E\,A = -(50)(25\times 10^{-4}) = -0.125\ N\,m^2C^{-1}$.

Right face at $x = 2\ m$: outward normal is $+\hat{i}$, field $E = 50(2) = 100\ N\,C^{-1}$, so $\Phi_R = +E\,A = (100)(25\times 10^{-4}) = +0.25\ N\,m^2C^{-1}$.

(i) Net flux: $\Phi = \Phi_R + \Phi_L = 0.25 - 0.125 = 0.125\ N\,m^2C^{-1}$.

(ii) By Gauss's law, enclosed charge $q = \varepsilon_0\,\Phi = (8.85\times 10^{-12})(0.125) = 1.106\times 10^{-12}\ C \approx 1.11\ pC$.

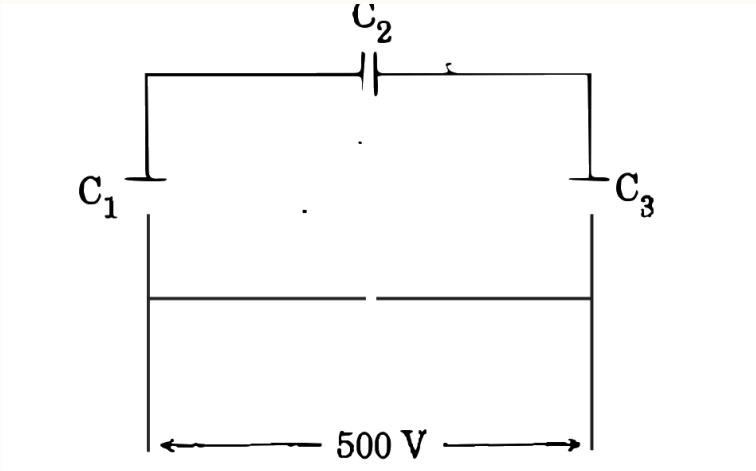
Hence $\Phi = 0.125\ N\,m^2C^{-1}$ and $q \approx 1.11\times 10^{-12}\ C$.

Final answer. $V = 180\ V$, $C = 2\ \mu F$, and at $+120\ V$ the charge is $600\ \mu C$; OR net flux $0.125\ N\,m^2C^{-1}$ with enclosed charge $\approx 1.11\times 10^{-12}\ C$.

← *Back to Q174*

Q175

A network of four capacitors each of 12 μF *capacitance is connected to a* 500 *V supply as shown in the figure. Determine (a) equivalent capacitance of the network and (b) charge on each capacitor.*



[2012]

SOLUTION

In the standard CBSE configuration the network has two parallel branches between the supply terminals, each branch containing two 12 µF capacitors in series.

For each branch (two capacitors in series): $\frac{1}{C_{br}} = \frac{1}{12} + \frac{1}{12} = \frac{2}{12} \Rightarrow C_{br} = 6\ \mu F$.

The two identical branches are in parallel: $C_{eq} = C_{br} + C_{br} = 6 + 6 = 12\ \mu F$.

(a) The **equivalent capacitance** of the network is $C_{eq} = 12\ \mu F$.

(b) Total charge drawn from the supply: $Q = C_{eq}V = (12\times 10^{-6})(500) = 6\times 10^{-3}\ C = 6\ mC$.

This charge divides equally between the two identical parallel branches, so each branch carries $\frac{6\ mC}{2} = 3\ mC$.

In a series branch the same charge passes through both capacitors, so each of the four 12 µF capacitors stores 3 mC.

Hence the **charge on each capacitor** is $Q_{each} = 3\times 10^{-3}\ C = 3\ mC$, and the voltage across each is $V_{each} = \frac{Q}{C} = \frac{3\times 10^{-3}}{12\times 10^{-6}} = 250\ V$, which correctly adds to 500 V across each branch.

> **COMMON MISTAKE**
>
> *Do not assume every capacitor in the network carries the total* 6 mC*; the source charge splits between parallel branches, and within a series branch each capacitor carries only the branch charge.*

Final answer. Equivalent capacitance $C_{eq} = 12\ \mu F$; charge on each of the four capacitors is **3 mC**.

← *Back to Q175*

Q176

Deduce the expression for the electrostatic energy stored in a capacitor of capacitance 'C' and having charge 'Q'.

How will the (i) energy stored and (ii) the electric field inside the capacitor be affected when it is completely filled with a dielectric material of dielectric constant 'K'?

[2012]

SOLUTION

Energy stored in a charged capacitor.

At an intermediate stage let the charge on the plates be q, so the instantaneous voltage is $v = \frac{q}{C}$.

To transfer a further small charge dq from one plate to the other, work done is $dW = v\,dq = \frac{q}{C}\,dq$.

The total work in charging the capacitor from 0 to Q is $W = \int_0^Q \frac{q}{C}\,dq = \frac{1}{C}\left[\frac{q^2}{2}\right]_0^Q = \frac{Q^2}{2C}$.

This work is stored as electrostatic energy, so $U = \frac{Q^2}{2C}$ (equivalently $\frac{1}{2}CV^2$ or $\frac{1}{2}QV$).

Effect of filling the gap completely with a dielectric of constant K (the charging battery is disconnected, so Q stays fixed).

The capacitance increases to $C' = KC$.

(i) Energy stored: $U' = \frac{Q^2}{2C'} = \frac{Q^2}{2KC} = \frac{U}{K}$. The **energy decreases by a factor K**, the lost energy going into polarising the dielectric.

(ii) Electric field inside: the polarised dielectric sets up an opposing field, reducing the net field to $E' = \frac{E}{K}$. The **field decreases by a factor K**.

Final answer. $U = \frac{Q^2}{2C}$; on filling with dielectric K at fixed Q, energy becomes U/K and the field becomes E/K (both decrease by factor K).

← *Back to Q176*

Q177

A capacitor of 200 pF *is charged by a* 300 *V battery. The battery is then disconnected and the charged capacitor is connected to another uncharged capacitor of* 100 pF. *Calculate the difference between the final energy stored in the combined system and the initial energy stored in the single capacitor.*

[2012]

SOLUTION

Initial state of the single capacitor.

$C_1 = 200$ pF charged to $V_0 = 300\ V$, so charge $Q_0 = C_1 V_0 = (200\times 10^{-12})(300) = 6.0\times 10^{-8}\ C$.

Initial energy $U_i = \frac{1}{2} C_1 V_0^2 = \frac{1}{2}(200\times 10^{-12})(300)^2 = \frac{1}{2}(200\times 10^{-12})(9\times 10^4) = 9.0\times 10^{-6}\ J = 9.0\ \mu J$.

After connecting to the uncharged $C_2 = 100$ pF (battery disconnected, so charge is conserved).

The two capacitors are now in parallel and share a common final voltage V_f, with total charge unchanged: $Q_0 = (C_1 + C_2)V_f$.

$$V_f = \frac{Q_0}{C_1 + C_2} = \frac{6.0\times 10^{-8}}{(200 + 100)\times 10^{-12}} = \frac{6.0\times 10^{-8}}{300\times 10^{-12}} = 200\ V.$$

Final energy $U_f = \frac{1}{2}(C_1 + C_2)V_f^2 = \frac{1}{2}(300\times 10^{-12})(200)^2 = \frac{1}{2}(300\times 10^{-12})(4\times 10^4) = 6.0\times 10^{-6}\ J = 6.0\ \mu J$.

Difference: $\Delta U = U_f - U_i = 6.0 - 9.0 = -3.0\ \mu J$.

So the system **loses 3.0 μJ** of energy; this energy is dissipated as heat (and a little radiation) in the connecting wires during the redistribution of charge.

CONCEPT

When two capacitors share charge, the total energy always falls even though charge is conserved; the difference is unavoidably dissipated in the wires.

Final answer. The final energy is less than the initial energy by **3.0 μJ** (a loss of $3.0\times 10^{-6}\ J$).

← Back to Q177

Q178

State the principle of the device that can build up high voltages of the order of a few million volts. Draw its labelled diagram. A stage reaches in this device when the potential at the outer sphere cannot be increased further by piling up more charge on it. Explain why.

[2011 • Set 55-2-1]

SOLUTION

The device is the **Van de Graaff generator**.

Principle. If a small charged conductor is placed inside a larger hollow conductor and the two are connected, all the charge on the inner conductor is transferred to the outer surface of the larger conductor, no matter how much charge the larger conductor already carries. This is because the field inside a conductor cavity is zero (electrostatic shielding) and any added charge resides on the outer surface; this allows charge to be piled up to build very high potentials.

Labelled diagram (described). A large hollow metal sphere (dome) sits on top of a tall insulating support column. An endless insulating belt runs over two pulleys, the lower pulley near the base and the upper pulley inside the dome, driven by a motor. A pointed comb B_1 (spray comb) near the base is connected to a high-voltage source and sprays positive charge onto the belt; a second pointed comb B_2 (collecting comb) inside the dome collects this charge and passes it to the dome. The whole assembly is enclosed in an earthed steel chamber filled with gas at high pressure. Labels: metal dome, insulating belt, motor-driven pulleys, spray comb B_1, collecting comb B_2, insulating column, grounded enclosure.

Why the potential cannot be raised indefinitely.

For a charged dome of radius R at potential V, the surface field is $E = \frac{V}{R}$.

As more charge is piled on, V and hence E keep rising. When E at the surface reaches the dielectric strength (breakdown field) of the surrounding gas, the

gas gets ionised and becomes conducting.

The ionised air then leaks the charge away (corona discharge) as fast as it is supplied, so the dome's potential cannot be increased further. This sets the **maximum attainable potential** of the generator.

Final answer. The device is the Van de Graaff generator; its potential saturates once the surface field reaches the breakdown field of the surrounding gas, which then leaks charge away as fast as it is added.

← *Back to Q178*

Q179

A parallel plate capacitor is charged by a battery. After sometime the battery is disconnected and a dielectric slab with its thickness equal to the plate separation is inserted between the plates. How will (i) the capacitance of the capacitor, (ii) potential difference between the plates and (iii) the energy stored in the capacitor be affected? Justify your answer in each case.

[2010]

SOLUTION

The capacitor is charged by a battery, then the battery is disconnected, so the **charge Q on the plates stays constant** while a dielectric slab of constant K (thickness equal to plate separation d) fully fills the gap.

(i) Capacitance. Inserting a dielectric multiplies the capacitance by K: $\boldsymbol{C' = KC}$.

Justification: the dielectric reduces the effective field between the plates, lowering the voltage for the same charge, which raises $C = Q/V$. So the **capacitance increases** (becomes K times).

(ii) Potential difference. Since $V = \frac{Q}{C}$ with Q fixed and $C \to KC$, we get $V' = \frac{Q}{KC} = \frac{V}{K}$.

The **potential difference decreases** to $\frac{1}{K}$ of its original value.

(iii) Energy stored. Since $U = \frac{Q^2}{2C}$ with Q fixed and $C \to KC$, we get $U' = \frac{Q^2}{2KC} = \frac{U}{K}$.

The **energy stored decreases** to $\frac{1}{K}$ of its original value; the lost energy is spent in polarising the dielectric (and it gets pulled in).

COMMON MISTAKE

Identify the conserved quantity first: here the battery is disconnected so Q is fixed (not V). If the battery had stayed connected, V would be constant and the conclusions would be different.

Final answer. With Q fixed: capacitance increases to KC, potential difference falls to V/K, and energy falls to U/K.

← Back to Q179

Q180

(a) Depict the equipotential surfaces for a system of two identical positive point charges placed a distance 'd' apart.
(b) Deduce the expression for the potential energy of a system of two point charges q_1 *and* q_2 *brought from infinity to the points* $\vec{r}_1$ *and* $\vec{r}_2$ *respectively in the presence of external electric field* $\vec{E}$.

[2010]

SOLUTION

(a) Equipotential surfaces for two identical positive point charges a distance d apart (described).

Close to each charge the equipotentials are small near-spherical surfaces surrounding that individual charge separately, since each charge dominates its own neighbourhood.

At intermediate distances the surfaces become distorted ovals; the perpendicular bisector of the line joining the charges is a plane of symmetry

but is NOT a zero-potential surface, because both charges are positive and contribute positive potential everywhere.

Far from the pair the equipotentials merge into large surfaces that enclose both charges and look almost spherical, as the system then behaves like a single charge +2q at the midpoint.

At the midpoint of the line joining the charges the field is zero (a saddle/neutral point), and the equipotential through that region pinches inward between the two charges.

(b) Potential energy of two charges q_1, q_2 brought from infinity in an external field $\vec{E}$.

Let $V(\vec{r})$ be the external potential. Bringing q_1 from infinity to $\vec{r}_1$ requires work $\mathbf{W_1 = q_1\, V(\vec{r}_1)}$.

Now bring q_2 from infinity to $\vec{r}_2$. It moves through the external field AND through the field of the already-placed q_1.

Work against the external field: $q_2\, V(\vec{r}_2)$.

Work against the field of q_1: $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1\, q_2}{r_{12}}$, where $\mathbf{r_{12} = |\vec{r}_1 - \vec{r}_2|}$.

So $\mathbf{W_2 = q_2\, V(\vec{r}_2) + \dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1\, q_2}{r_{12}}}$.

The total potential energy of the system is the sum $\boldsymbol{U = W_1 + W_2}$:

$$\boldsymbol{U = q_1\, V(\vec{r}_1) + q_2\, V(\vec{r}_2) + \frac{1}{4\pi\varepsilon_0}\frac{q_1\, q_2}{|\vec{r}_1 - \vec{r}_2|}}.$$

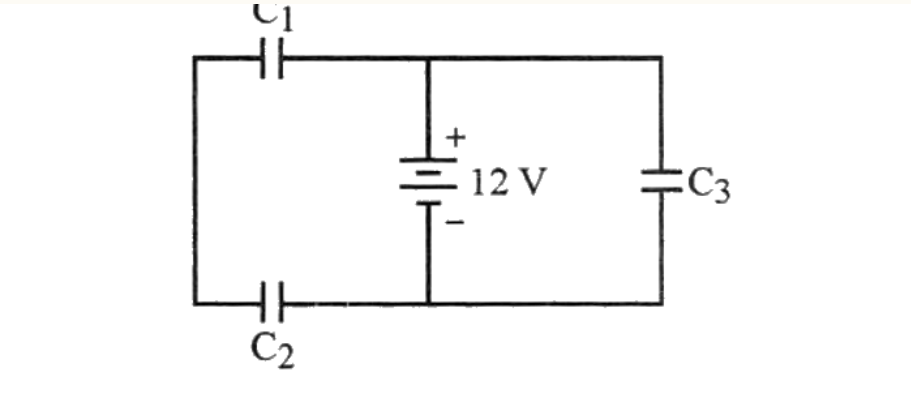
This is the required **expression for the potential energy** of the two-charge system in the external field.

Final answer. Equipotentials are separate near each charge and merge far away (saddle point at the midpoint); the system PE is $\boldsymbol{U = q_1\, V(\vec{r}_1) + q_2\, V(\vec{r}_2) + \dfrac{1}{4\pi\varepsilon_0}\dfrac{q_1\, q_2}{|\vec{r}_1 - \vec{r}_2|}}$.

← *Back to Q180*

Q181

Three identical capacitors C_1, C_2 *and* C_3 *of capacitance* 6 μ *F each are connected to a* 12 *V battery as shown.*



Find

(i) charge on each capacitor

(ii) equivalent capacitance of the network

(iii) energy stored in the network of capacitors

[2009]

SOLUTION

Here C_1 is in series with the parallel combination of C_2 and C_3, and the whole network is across $V = 12\ V$.

First find the equivalent capacitance. The two capacitors C_2 and C_3 are in parallel, so they add directly.

$$C_{23} = C_2 + C_3 = 6 + 6 = 12\ \mu F$$

Now C_1 is in series with C_{23}.

$$C_{eq} = \frac{C_1 \times C_{23}}{C_1 + C_{23}} = \frac{6 \times 12}{6 + 12}$$

$$C_{eq} = \frac{72}{18} = 4\ \mu F$$

The total charge drawn from the battery is

$$Q = C_{eq}\ V = 4 \times 10^{-6} \times 12 = 48 \times 10^{-6}\ C = 48\ \mu C$$

Since C_1 is in series with the parallel block, this full charge passes through C_1.

$$Q_{C_1} = 48\ \mu C$$

The voltage across C_1 is

$$V_{C_1} = \frac{Q_{C_1}}{C_1} = \frac{48}{6} = 8\ V$$

So the voltage across the parallel block is

$$V_{23} = V - V_{C_1} = 12 - 8 = 4\ V$$

The charges on C_2 and C_3 are

$$Q_{C_2} = C_2\, V_{23} = 6 \times 4 = 24\ \mu C$$

$$Q_{C_3} = C_3\, V_{23} = 6 \times 4 = 24\ \mu C$$

Finally, the total energy stored in the network is

$$U = \frac{1}{2}\, C_{eq}\, V^2 = \frac{1}{2} \times 4 \times 10^{-6} \times (12)^2$$

$$U = \frac{1}{2} \times 4 \times 10^{-6} \times 144 = 2.88 \times 10^{-4}\, J = 288\ \mu J$$

Hence the charges are $\mathbf{Q_{C_1} = 48\ \mu C}$, $\mathbf{Q_{C_2} = Q_{C_3} = 24\ \mu C}$, the equivalent capacitance is **4 µF**, and the energy stored is **288 µJ**.

TIP

Charge is always the same on capacitors in series, while voltage is the same on capacitors in parallel. Mixing these up is the most common slip in network problems.

Final answer. Charges: $\mathbf{Q_{C_1}}$**=48 µC,** $\mathbf{Q_{C_2}}$**=**$\mathbf{Q_{C_3}}$**=24 µC;** $\mathbf{C_{eq}}$**=4 µF**; energy ***U*****=288 µJ.**

← *Back to Q181*

Q182

A positive point charge (+q) is kept in the vicinity of an uncharged conducting plate. Sketch electric field lines originating from the point on to the surface of the plate. Derive the expression for the electric field at the surface of a charged conductor.

[2009]

SOLUTION

Field-line sketch (described): Draw the positive point charge $+q$ a short distance in front of a flat conducting plate. The lines of force start out radially from $+q$ in all directions. Those heading towards the plate bend and strike the front face of the plate at right angles, ending on the induced negative charges there. Mark a layer of induced − charges on the near face of the plate (facing $+q$) and a layer of induced + charges on the far face, so the plate stays overall neutral. Inside the metal and behind the plate, draw no lines, because the field there is zero. The whole pattern in front looks the same as that of the real charge $+q$ together with an image charge $-q$ placed at the mirror position behind the plate.

Now derive the field just outside a charged conductor. Let the conductor carry a surface charge density σ at the point considered. Take a small Gaussian pillbox (a tiny cylinder) of cross-sectional area ΔS, placed so that one flat face lies just outside the surface and the other lies just inside the conductor.

Inside a conductor in electrostatic equilibrium the field is zero.

$$\vec{E}_{\text{inside}} = \vec{0}$$

So the inner face contributes no flux. The curved side is parallel to the field (the field just outside a conductor is normal to its surface), so it also contributes no flux. Only the outer face contributes.

$$\Phi = E\,\Delta S$$

The charge enclosed by the pillbox is

$$q_{\text{enc}} = \sigma\,\Delta S$$

By Gauss's law,

$$\Phi = \frac{q_{enc}}{\varepsilon_0}$$

$$E\,\Delta S = \frac{\sigma\,\Delta S}{\varepsilon_0}$$

$$E = \frac{\sigma}{\varepsilon_0}$$

In vector form, along the outward normal $\hat{n}$,

$$\vec{E} = \frac{\sigma}{\varepsilon_0}\,\hat{n}$$

Hence the electric field just outside the surface of a charged conductor is $\vec{\boldsymbol{E}} = \frac{\sigma}{\varepsilon_0}\,\hat{\boldsymbol{n}}$, directed normally outward.

Final answer. The field at the surface of a charged conductor is $\vec{\boldsymbol{E}}=\frac{\sigma}{\varepsilon_0}\,\hat{\boldsymbol{n}}$, directed along the outward normal.

← *Back to Q182*

Q183

A parallel plate capacitor is charged by a battery. After some time the battery is disconnected and a dielectric slab of dielectric constant K is inserted between the plates. How would (i) the capacitance, (ii) the electric field between the plates and (iii) the energy stored in the capacitor, be affected? Justify your answer.

[2009]

SOLUTION

The capacitor is first charged by a battery and then the battery is disconnected before the dielectric is inserted. So the charge Q on the plates stays fixed throughout, because there is no external path for it to leave.

(i) Capacitance. Inserting a dielectric of constant K that fills the gap multiplies the capacitance by K.

$$C' = K\,C$$

So the capacitance increases by the factor K.

(ii) Electric field between the plates. The surface charge density on the plates is unchanged, since Q and area A are both fixed.

$$\sigma = \frac{Q}{A} = \text{constant}$$

Inside the dielectric the field is reduced by the factor K.

$$E' = \frac{\sigma}{K\,\varepsilon_0} = \frac{E}{K}$$

So the electric field decreases by the factor K. This is because the dielectric gets polarised and sets up its own opposing field.

(iii) Energy stored. With charge fixed, write the energy as

$$U = \frac{Q^2}{2C}$$

Since $C \to KC$,

$$U' = \frac{Q^2}{2(KC)} = \frac{U}{K}$$

So the energy stored decreases by the factor K. Physically, with Q fixed and C multiplied by K, the voltage $V = Q/C$ falls by K, and so $U = \frac{1}{2}QV$ falls by K. The lost energy is used up in polarising the dielectric.

Hence: capacitance becomes **KC (increases)**, field becomes ***E*/*K* (decreases)**, and energy becomes ***U*/*K* (decreases)**.

COMMON MISTAKE

The key trick is to spot that the battery is disconnected, so charge Q is constant. If the battery had stayed connected, voltage V would be constant instead and the answers would be different.

Final answer. Capacitance increases to **KC**; field decreases to ***E*/*K***; energy decreases to ***U*/*K*** (charge ***Q*** stays constant).

← *Back to Q183*

Q184

Deduce an expression for the electric potential due to an electric dipole at any point on its axis. Mention one contrasting feature of electric potential of a dipole at a point as compared to that due to a single charge.

[2007]

SOLUTION

Consider a dipole with charge $-q$ at point A and $+q$ at point B, separated by a distance 2a. Let O be the centre. Take a point P on the axial line at distance r from the centre O, on the side of $+q$.

The distances from P to the two charges are

$$|BP| = r - a \; |AP| = r + a$$

The potential at P is the algebraic sum of the potentials due to the two charges.

$$V_P = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{r-a} + \frac{(-q)}{r+a}\right]$$

$$V_P = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{r-a} - \frac{1}{r+a}\right]$$

$$V_P = \frac{q}{4\pi\varepsilon_0}\left[\frac{(r+a)-(r-a)}{(r-a)(r+a)}\right]$$

$$V_P = \frac{q}{4\pi\varepsilon_0} \cdot \frac{2a}{r^2 - a^2}$$

Writing the dipole moment $\boldsymbol{p = q \times 2a}$,

$$V_P = \frac{1}{4\pi\varepsilon_0} \cdot \frac{p}{r^2 - a^2}$$

For a short dipole, $r \gg a$, so a^2 is negligible compared with r^2.

$$V_P = \frac{1}{4\pi\varepsilon_0} \cdot \frac{p}{r^2}$$

One contrasting feature: the potential due to a single point charge varies as $1/r$, but the potential on the axis of a dipole varies as $\mathbf{1/r^2}$, that is, it falls off much faster with distance. Also, the dipole potential depends on the direction of P (it

is non-zero on the axis but zero on the equatorial line), whereas a point charge potential depends only on the distance r.

Final answer. Axial potential $\mathbf{V_P = \frac{1}{4\pi\varepsilon_0}\,\frac{p}{r^2-a^2} \to \frac{1}{4\pi\varepsilon_0}\,\frac{p}{r^2}}$ for a short dipole; it falls as $\mathbf{1/r^2}$ unlike $\mathbf{1/r}$ for a single charge.

← Back to Q184

Q185

A parallel plate capacitor, each with plate area A and separation d, is charged to a potential difference V. The battery used to charge it is then disconnected. A dielectric slab of thickness d and dielectric constant K is now placed between the plates. What change, if any, will take place in
(i) charge on the plates,
(ii) electric field intensity between the plates,
(iii) capacitance of the capacitor.
Justify your answer in each case.

[2007]

SOLUTION

The capacitor of plate area A and separation d is charged to potential difference V, and then the battery is disconnected. A dielectric slab of thickness d (filling the whole gap) and dielectric constant K is now inserted. Because the battery is removed, the charge on the plates cannot change.

(i) Charge on the plates. With no external connection, the charge stays the same.

$$Q' = Q = C\,V = \frac{\varepsilon_0 A}{d}\,V$$

So the charge is unchanged.

(ii) Electric field between the plates. The surface charge density does not change, since Q and A are both fixed.

$$\sigma = \frac{Q}{A} = \text{constant}$$

With air the field was $E = \sigma/\varepsilon_0 = V/d$. With the dielectric in place,

$$E' = \frac{\sigma}{K\varepsilon_0} = \frac{E}{K} = \frac{V}{K\,d}$$

So the field intensity decreases by the factor K, because the polarised dielectric sets up an opposing field.

(iii) Capacitance. Filling the gap with the dielectric multiplies the capacitance by K.

$$C' = \frac{K\varepsilon_0 A}{d} = K\,C$$

So the capacitance increases by the factor K.

Check for consistency: with Q fixed and $C' = KC$, the new voltage is $V' = Q/C' = V/K$, which agrees with $E'd = (V/Kd)\,d = V/K$.

Hence: charge is **unchanged**, field becomes **V/Kd (decreases by K)**, and capacitance becomes **KC (increases by K)**.

Final answer. Charge stays the same; field decreases to $E/K = V/Kd$; capacitance increases to KC (battery disconnected, so Q is constant).

← *Back to Q185*

Q186

Explain the underlying principle of working of a parallel plate capacitor.
If two similar plates, each of area A having surface charge densities +σ and −σ are separated by a distance d in air, write expressions for
(i) the electric field at points between the two plates.
(ii) the potential difference between the plates.
(iii) the capacitance of the capacitor so formed.

[2007]

SOLUTION

Underlying principle. A parallel plate capacitor works on the principle that the capacitance of a conductor can be increased greatly by bringing an earthed (or

oppositely charged) conductor near it. When a charged plate is placed close to another plate, the induced charges reduce the potential of the charged plate, so for the same charge the potential difference is small. A small voltage can therefore hold a large charge, which means a large capacitance. The charge stored is $Q = CV$, where C depends only on the geometry of the plates and the medium between them.

Now take two parallel plates, each of area A, with surface charge densities $+\sigma$ and $-\sigma$, separated by a distance d in air.

(i) Electric field between the plates. Each charged sheet produces a field of magnitude $\sigma/2\varepsilon_0$ on either side. Between the plates the fields of the two sheets point in the same direction and add up.

$$E = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0}$$

$$E = \frac{\sigma}{\varepsilon_0}$$

(Outside the plates the two fields oppose and cancel, so the field there is zero.)

(ii) Potential difference between the plates. The field is uniform, so

$$V = E\,d$$

$$V = \frac{\sigma\,d}{\varepsilon_0}$$

(iii) Capacitance. The total charge on a plate is $Q = \sigma A$, so

$$C = \frac{Q}{V} = \frac{\sigma A}{\sigma\,d/\varepsilon_0}$$

$$C = \frac{\varepsilon_0 A}{d}$$

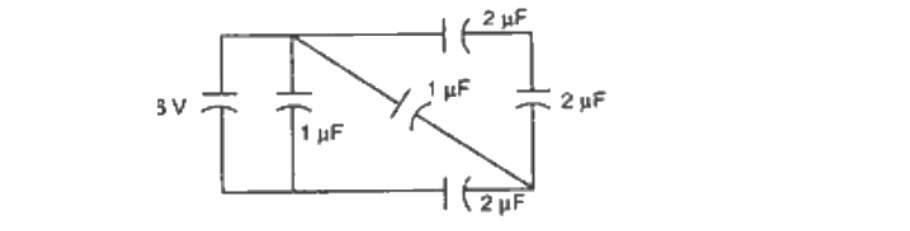
Hence the field is $\boldsymbol{E = \frac{\sigma}{\varepsilon_0}}$, the potential difference is $\boldsymbol{V = \frac{\sigma d}{\varepsilon_0}}$, and the capacitance is $\boldsymbol{C = \frac{\varepsilon_0 A}{d}}$.

Final answer. $\boldsymbol{E = \frac{\sigma}{\varepsilon_0}}$, $\boldsymbol{V = \frac{\sigma d}{\varepsilon_0}}$, and $\boldsymbol{C = \frac{\varepsilon_0 A}{d}}$.

← Back to Q186

Q187

Find the total energy stored in the capacitors in the given network.



[2004]

SOLUTION

To find the total energy stored, first reduce the whole network to a single equivalent capacitance C_{eq} between the two terminals of the source, then use the total energy formula.

Use the two reduction rules. For capacitors in series the reciprocals add,

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots$$

and for capacitors in parallel the capacitances add directly,

$$C_p = C_1 + C_2 + \cdots$$

Apply these step by step: first combine the innermost parallel groups, then fold the resulting series sections, repeating until the whole network collapses to a single value C_{eq} across the supply voltage V.

The total energy stored in the network is then

$$U_{total} = \frac{1}{2} C_{eq} V^2$$

As a cross-check, one may instead find the voltage V_i across each individual capacitor and add the energies,

$$U_{total} = \Sigma \frac{1}{2} C_i V_i^2$$

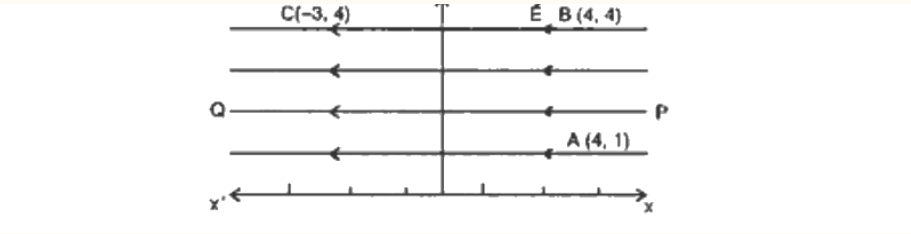
Both routes give the same total. For the given figure, carry out the series/parallel reduction to obtain C_{eq} and multiply by $\frac{1}{2}V^2$ to get the **total energy stored, $U_{total} = \frac{1}{2}C_{eq}V^2$.**

Final answer. Reduce the network to $\mathbf{C_{eq}}$ by series/parallel rules; the total energy stored is $\mathbf{U_{total} = \frac{1}{2}\,C_{eq}\,V^2}$.

← *Back to Q187*

Q188

What is an equipotential surface?
A uniform electric field of $\boldsymbol{E = 300\ NC^{-1}}$ *is directed along PQ. A, B and C are three points in the field having x and y coordinates (in metres) as shown in the figure. Calculate potential difference between the points (i) A and B and (ii) B and C.*



[2003]

SOLUTION

An equipotential surface is a surface on which every point is at the same electric potential. Because the potential is constant over it, no work is done in moving a charge from one point to another on the surface, and the electric field is everywhere perpendicular (normal) to the surface.

For a uniform field $\vec{E}$ the potential difference between two points depends only on the displacement component along $\vec{E}$.

$$V_2 - V_1 = -\vec{E}\Delta\vec{r}$$

Here $\boldsymbol{E = 300\ N\,C^{-1}}$ is directed along PQ, taken along the $+x$ direction. Using the standard CBSE 2003 coordinates $A(2,0)$, $B(2,3)$ and $C(6,3)$ in metres:

(i) Between A and B. The displacement from A to B is along y only; its component along $\vec{E}$ (the x-direction) is zero.

$$V_B - V_A = -E\,(x_B - x_A) = -300\,(2 - 2)$$

$$V_B - V_A = 0\ V$$

So A and B are at the same potential (they lie on the same equipotential surface, since AB is perpendicular to $\vec{E}$).

(ii) Between B and C. The displacement from B to C has an x-component of $(6 - 2) = 4\ m$.

$$V_C - V_B = -E\,(x_C - x_B) = -300\,(6 - 2)$$

$$V_C - V_B = -300 \times 4 = -1200\ V$$

Hence $V_B - V_A = 0\ V$ and $V_C - V_B = -1200\ V$.

CONCEPT

Only the displacement component along $\vec{E}$ matters. Motion perpendicular to the field gives zero potential difference, so always resolve along PQ before substituting.

Final answer. $V_B - V_A = 0\ V$ (perpendicular to $\vec{E}$) and $V_C - V_B = -1200\ V$.

← *Back to Q188*

Q189

Draw a labelled diagram of Van de Graaff generator. State its principle of working.

[2003]

SOLUTION

Labelled diagram (described): Draw a tall insulating supporting column standing on a grounded base. At the top, place a large hollow metallic sphere, the high-voltage dome (terminal), labelled S. Show a wide insulating belt running as a continuous loop over two pulleys, P_1 at the bottom and P_2 inside the dome, driven by a motor at the base. Near the lower pulley draw a pointed

spray comb B_1 connected to a high-voltage source (a few kV), which sprays positive charge onto the moving belt. Near the upper pulley, inside the dome, draw a collecting comb B_2 connected to the inner surface of the dome, which removes the charge from the belt and passes it to the dome. Show an evacuated accelerating tube running down from the dome with an ion source D at the top, where ions are produced and then accelerated downwards to the target T. Label the dome, the belt, both combs, the pulleys, the motor and the accelerating tube.

Principle of working. It is based on two ideas. First, the action of sharp points (corona discharge), by which a pointed conductor sprays or collects charge readily. Second, the fact that charge given to the inner surface of a hollow conductor is transferred entirely to its outer surface, no matter how much charge the conductor already carries. This is because the interior of a charged hollow conductor is field free and all excess charge resides on the outer surface.

$$\text{Charge sprayed on belt} \Rightarrow \text{carried up} \Rightarrow \text{collected inside dome} \Rightarrow \text{moves to outer surface}$$

By continuously feeding charge from the belt to the inner surface of the dome, charge keeps accumulating on the outer surface, building the dome to a very high potential of the order of millions of volts. This high potential is then used to **accelerate charged particles to high energies**.

Final answer. The Van de Graaff generator works on the action of sharp points and on charge transferring to the outer surface of a hollow conductor, building the dome to a very high potential.

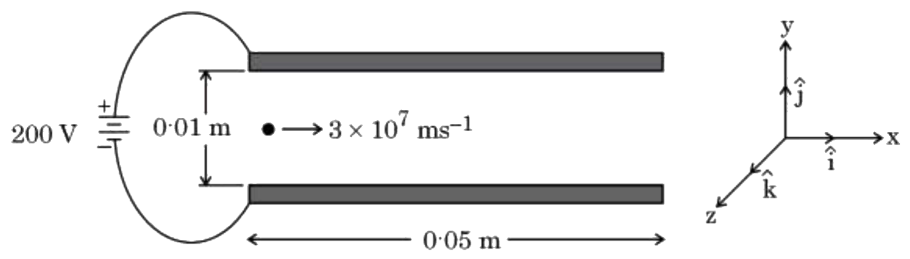
← *Back to Q189*

Q190

The electric potential V and electric field E are closely related concepts in electrostatics. The electric field is a vector quantity that represents the force per unit charge at a given point in space, whereas electric potential is a scalar quantity that represents the potential energy per unit charge at a given point in space. Electric field and electric

potential are related by the equations $\mathbf{E_r = -\frac{dV}{dr}}$ *and* $\vec{E} = -\vec{\nabla} V$*, i.e., electric field is the negative gradient of the electric potential. This means that electric field points in the direction of decreasing potential and its magnitude is the rate of change of potential with distance. The electric field is the force that drives a unit charge to move from higher potential region to lower potential region and electric potential difference between the two points determines the work done in moving a unit charge from one point to the other point.*

A pair of square conducting plates having sides of length 0.05 *m are arranged parallel to each other in x-y plane. They are* 0.01 *m apart along z-axis and are connected to a* 200 *V power supply as shown in the figure. An electron enters with a speed of* 3×10^7 ms^{-1} *horizontally and symmetrically in the space between the two plates. Neglect the effect of gravity on the electron.*



[2026 • Set 55-5-1]

SOLUTION

Set-up (described): Two square conducting plates of side $L = 0.05\ m$ lie parallel in the x-y plane, separated by $d = 0.01\ m$ along the z-axis, connected to a $V = 200\ V$ supply. An electron enters horizontally (along x) with speed $u = 3 \times 10^7\ m\ s^{-1}$, exactly midway between the plates. The field between the plates points along z, so the electron moves like a projectile: uniform velocity along x and uniform acceleration along z.

The uniform field between the plates is the negative gradient of potential; for a uniform gap its magnitude is

$$E = \frac{V}{d} = \frac{200}{0.01}$$

$$E = 2 \times 10^4\ N\ C^{-1}$$

The force on the electron gives an acceleration along z.

$$a = \frac{eE}{m} = \frac{(1.6 \times 10^{-19})(2 \times 10^{4})}{9.1 \times 10^{-31}}$$

$$a = \frac{3.2 \times 10^{-15}}{9.1 \times 10^{-31}} = 3.52 \times 10^{15}\ m\ s^{-2}$$

The horizontal velocity stays constant, so the time spent between the plates (length L) is

$$t = \frac{L}{u} = \frac{0.05}{3 \times 10^{7}}$$

$$t = 1.67 \times 10^{-9}\ s$$

The vertical (along z) deflection during this time, starting from rest in that direction, is

$$y = \frac{1}{2}\, a\, t^{2} = \frac{1}{2}\, (3.52 \times 10^{15})(1.67 \times 10^{-9})^{2}$$

$$y = \frac{1}{2}\, (3.52 \times 10^{15})(2.78 \times 10^{-18})$$

$$y = 4.9 \times 10^{-3}\ m \approx 4.9\ \text{mm}$$

Since it entered at the midplane, the gap on each side is $\boldsymbol{d/2 = 5\ \text{mm}}$; the deflection 4.9 mm is just less than this, so the electron just emerges from the field. Because x grows uniformly while z grows as t^2, the path between the plates is a **parabola**.

Hence: field $\boldsymbol{E = 2 \times 10^{4}\ N\,C^{-1}}$, acceleration $\boldsymbol{a = 3.52 \times 10^{15}\ m\ s^{-2}}$, transit time $\boldsymbol{t = 1.67 \times 10^{-9}\ s}$, vertical deflection $\boldsymbol{y \approx 4.9\ \text{mm}}$, and the trajectory is **parabolic**.

CONCEPT

Inside the plates the path is a parabola, exactly like projectile motion: the velocity along the plates is constant while the field gives a constant sideways acceleration.

Final answer. $\boldsymbol{E = 210^{4}\ N\,C^{-1}}$, $\boldsymbol{a = 3.5210^{15}\ m\ s^{-2}}$, $\boldsymbol{t = 1.6710^{-9}\ s}$, deflection $\boldsymbol{y \approx 4.9\ \text{mm}}$; the path is parabolic.

← *Back to Q190*

Q191

A capacitor is a system of two conductors separated by an insulator. The two conductors have equal and opposite charges with a potential difference between them. The capacitance of a capacitor depends on the geometrical configuration (shape, size and separation) of the system and also on the nature of the insulator separating the two conductors. They are used to store charges. Like resistors, capacitors can be arranged in series or parallel or a combination of both to obtain desired value of capacitance.

[2023 • Set 55-5-1]

SOLUTION

This passage introduces the idea of a capacitor as a charge-storing device. A capacitor is a system of two conductors separated by an insulator (the dielectric), carrying equal and opposite charges $+Q$ and $-Q$ with a potential difference V between them.

The capacitance is defined as the ratio of the charge on either plate to the potential difference between the plates:

$$C = \frac{Q}{V}$$

The SI unit of capacitance is the farad (F), where $\mathbf{1\,F = 1\,C\,V^{-1}}$.

A key point is that the capacitance depends only on the geometry of the system (the shape, size and separation of the conductors) and on the dielectric constant K of the medium between them. It does NOT depend on the actual charge Q or the potential difference V.

For a parallel plate capacitor of plate area A, separation d, with a dielectric of constant K filling the gap:

$$C = \frac{K\varepsilon_0 A}{d}$$

Capacitors can be combined to get a desired value. In **series**, the reciprocals add and the net capacitance falls below the smallest one:

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots$$

In **parallel**, the capacitances simply add and the net value rises:

$$C_p = C_1 + C_2 + \cdots$$

The energy stored in a charged capacitor is:

$$U = \frac{1}{2}CV^2 = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}QV$$

These definitions and combination rules form the basis for the numerical sub-questions of this case study, which apply them to specific networks and to the insertion of a dielectric slab.

Final answer. A capacitor stores charge with $C = Q/V$, depending only on geometry and the dielectric medium; series gives $1/C_s = \sum 1/C_i$ and parallel gives $C_p = \sum C_i$.

← *Back to Q191*

Q192

While travelling back to his residence in the car, Dr. Pathak was caught up in a thunderstorm. It became very dark. He stopped driving the car and waited for thunderstorm to stop. Suddenly he noticed a child walking alone on the road. He asked the boy to come inside the car till the thunderstorm stopped. Dr. Pathak dropped the boy at his residence. The boy insisted that Dr. Pathak should meet his parents. The parents expressed their gratitude to Dr. Pathak for his concern for safety of the child. Answer the following questions based on the above information:
(a) Why is it safer to sit inside a car during a thunderstorm?
(b) Which two values are displayed by Dr. Pathak in his actions?
(c) Which values are reflected in parents' response to Dr. Pathak?
(d) Give an example of a similar action on your part in the past from everyday life.

[2013]

SOLUTION

(a) Why is it safer to sit inside a car during a thunderstorm?

The metal body of a car acts as a Faraday cage. If lightning strikes the car, the charge spreads over the outer metallic surface and the field redistributes so

that the electric field inside the hollow conductor is zero.

Inside any closed conductor the field is zero, a property called **electrostatic shielding**. The redistributed surface charges cancel any external field within the cavity, so the occupants experience no large field and no current passes through them. The charge then flows safely down the metal body to the ground.

(b) Two values displayed by Dr. Pathak:

(i) Compassion and concern for the safety of others. He stopped, invited a stranger child into his car and protected the child until the storm passed.

(ii) Civic and social responsibility. He recognised a vulnerable minor in danger and acted personally to help instead of driving on.

(c) Values reflected in the parents' response:

Gratitude, courtesy and acknowledgement of help received. By thanking Dr. Pathak and insisting that he meet them, the parents show gratefulness and a sense of responsibility and care towards their child.

(d) Example of a similar action from everyday life:

Helping a stranger find shelter during heavy rain, offering a ride or a seat to someone stranded, or guiding an elderly person safely across a busy road. Any honest act of helping a person in distress is acceptable.

> **TIP**
>
> In a value-based question the physics reason (electrostatic shielding) carries the marks, so always include it; do not write only the moral values.

Final answer. A car is safe in a thunderstorm because its metal body acts as a Faraday cage (electrostatic shielding makes the interior field zero); the values shown are compassion, responsibility and gratitude.

← Back to Q192

Q193

(a) Derive an expression for the capacitance of a parallel plate capacitor of plate area

A and plate separation d with air present between the plates.

[2026 • Set 55-2-1]

SOLUTION

Consider two parallel conducting plates, each of area A, separated by air (or vacuum) by a small distance d. The plates are given equal and opposite charges $+Q$ and $-Q$.

The surface charge density on each plate is:

$$\sigma = \frac{Q}{A}$$

Each charged plate behaves like a large (effectively infinite) charged sheet, which produces a uniform field $\frac{\sigma}{2\varepsilon_0}$ on either side of it.

In the region between the two oppositely charged plates, the fields due to the two plates point in the same direction and add up:

$$E = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$$

$$E = \frac{Q}{\varepsilon_0 A}$$

(Outside the plates the two fields are opposite and cancel, so the field there is zero.)

The field between the plates is uniform, so the potential difference between the plates equals field times separation:

$$V = E\,d = \frac{Q\,d}{\varepsilon_0 A}$$

By the definition of capacitance, $C = \frac{Q}{V}$:

$$C = \frac{Q}{\frac{Q\,d}{\varepsilon_0 A}}$$

$$C = \frac{\varepsilon_0 A}{d}$$

This is the capacitance of a parallel plate capacitor with air between the plates. It depends only on the geometry (area A and separation d). If a dielectric of constant K fills the gap, ε_0 is replaced by $K\varepsilon_0$, giving $C = \dfrac{K\varepsilon_0 A}{d}$.

Final answer. The capacitance of an air-filled parallel plate capacitor is $C = \dfrac{\varepsilon_0 A}{d}$.

← Back to Q193

Q194

(b) Two air-filled capacitors of capacitances C_1 and C_2 are connected in parallel with a dc battery. After the capacitors are fully charged, a slab of dielectric constant K is inserted between the plates of each capacitor. How will the (i) charge on each capacitor and (ii) energy stored in the capacitor be affected after the slab is introduced?

[2026 • Set 55-2-1]

SOLUTION

Two air-filled capacitors C_1 and C_2 are connected in parallel across a battery of emf V. The crucial point is that the **battery remains connected** while the slabs are inserted, so the potential difference across each capacitor stays fixed at V.

When a dielectric slab of constant K is inserted, the capacitance of each capacitor increases by the factor K:

$$C_1 \to KC_1,\ C_2 \to KC_2$$

(i) Charge on each capacitor:

With the battery enforcing the voltage V, the charge is $Q = CV$.

Before insertion: $Q_1 = C_1 V$ and $Q_2 = C_2 V$.

After insertion:

$$Q_1' = (KC_1)V = K\,Q_1$$

$$Q_2' = (KC_2)V = K\,Q_2$$

So the charge on each capacitor **increases by the factor K**. The extra charge is supplied by the battery, which pushes additional charge onto the plates.

(ii) Energy stored in each capacitor:

The energy is $U = \frac{1}{2}CV^2$.

Before insertion: $U_1 = \frac{1}{2}C_1 V^2$ and $U_2 = \frac{1}{2}C_2 V^2$.

After insertion (with V unchanged and $C \to KC$):

$$U_1' = \frac{1}{2}(KC_1)V^2 = K\,U_1$$

$$U_2' = \frac{1}{2}(KC_2)V^2 = K\,U_2$$

So the energy stored in each capacitor also **increases by the factor K**. The extra energy is drawn from the battery as it does work in supplying the additional charge $(K-1)Q$ at potential V.

COMMON MISTAKE

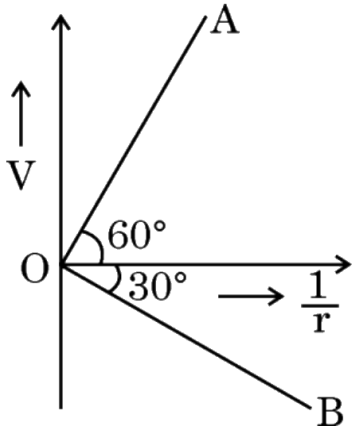
Battery connected means V is constant; battery disconnected means Q is constant. Always check which quantity is fixed before deciding how charge and energy change with K.

Final answer. With the battery connected, inserting the slab multiplies both the charge and the energy of each capacitor by K (each becomes K times its earlier value).

← *Back to Q194*

Q195

(i) In the figure, OA *and* OB *show the variation of electric potential V at a point due to two point charges* Q_1 *and* Q_2 *with* $\frac{1}{r}$ *respectively. Here r represents the distance of the point from the two point charges.*



(I) Identify the nature of the two charges Q_1 *and* Q_2.

(II) What is the value of $(\frac{Q_1}{Q_2})$*? Justify your answer.*

[2026 • Set 55-3-1]

SOLUTION

The potential at a distance r from a point charge Q is:

$$V = \frac{kQ}{r} = kQ(\frac{1}{r})$$

So a graph of V against $\frac{1}{r}$ is a straight line passing through the origin, with slope equal to kQ.

In the figure both lines OA and OB start at the origin and rise (positive slope), going up and to the right.

(I) Nature of the two charges:

Since the slope kQ is positive for each line, and k is positive, both charges must be positive. Therefore **Q_1 and Q_2 are both positive charges**.

(II) Value of $\frac{Q_1}{Q_2}$:

The slope of each line equals kQ, so the ratio of charges equals the ratio of slopes.

Line OA makes 60° with the $1/r$ axis:

$$\text{slope of OA} = \tan 60° = \sqrt{3}$$

Line OB makes 30° with the $1/r$ axis:

$$\text{slope of OB} = \tan 30° = \frac{1}{\sqrt{3}}$$

Therefore:

$$\frac{Q_1}{Q_2} = \frac{\text{slope of OA}}{\text{slope of OB}} = \frac{\sqrt{3}}{\frac{1}{\sqrt{3}}}$$

$$\frac{Q_1}{Q_2} = 3$$

Justification: at the same distance r (same value of $1/r$), $V \propto Q$, so the ratio of the potentials read off the graph equals the ratio of the charges, which is the ratio of the two slopes.

Final answer. Both charges are positive and $\frac{Q_1}{Q_2} = 3$.

← *Back to Q195*

Q196

(ii) Two point charges −2 μC *and* 5 μC *are placed at* (−30 cm, 0) *and* (30 cm, 0) *respectively in an external electric field* $\vec{E} = \frac{A}{x}\hat{i}$, *where* $A = 9 \times 10^5\ N\ m^2\ C^{-1}$. *Find the electrostatic potential energy of this configuration.*

[2026 • Set 55-3-1]

SOLUTION

We have $q_1 = -2\ \mu C$ at $x_1 = -0.30\ m$ and $q_2 = +5\ \mu C$ at $x_2 = +0.30\ m$, in an external field $\vec{E} = \frac{A}{x}\hat{i}$ with $A = 9\times 10^5\ N\ m^2\ C^{-1}$.

The external potential corresponding to this radial field is:

$$V_{ext}(x) = \frac{A}{|x|}$$

At each charge location $|x| = 0.30\ m$, so:

$$V_{ext} = \frac{9\times 10^{5}}{0.30} = 3\times 10^{6}\ V$$

Energy of the charges in the external field:

$$U_{ext} = q_1\, V_{ext}(x_1) + q_2\, V_{ext}(x_2)$$

$$U_{ext} = (-2\times 10^{-6})(3\times 10^{6}) + (5\times 10^{-6})(3\times 10^{6})$$

$$U_{ext} = -6 + 15 = +9\ J$$

Mutual potential energy of the two point charges:

Their separation is $\mathbf{r_{12} = 0.30 + 0.30 = 0.60\ m}$.

$$U_{mutual} = \frac{k\, q_1\, q_2}{r_{12}}$$

$$U_{mutual} = \frac{(9\times 10^{9})(-2\times 10^{-6})(5\times 10^{-6})}{0.60}$$

$$U_{mutual} = \frac{-90\times 10^{-3}}{0.60} = -0.15\ J$$

Total electrostatic potential energy:

$$U = U_{ext} + U_{mutual} = 9 + (-0.15)$$

$$U = 8.85\ J$$

COMMON MISTAKE

The total energy has two parts here: each charge interacting with the external field ($\sum q_i V_{ext}$) PLUS the mutual interaction of the charges (kq_1q_2/r_{12}). Forgetting the mutual term is a common slip.

Final answer. The total electrostatic potential energy of the configuration is $\boldsymbol{U \approx 8.85\ J}$.

← Back to Q196

Q197

(a)

(i) Explain the following statements giving reason:

(I) An equipotential surface through a point is normal to the electric field at that point.

(II) When a dielectric is placed in an external electric field, the electric field inside the dielectric is less than that outside it.

(III) The potential difference between the plates of a charged parallel plate capacitor decreases when its plates are brought closer.

(ii) Obtain an expression for the work done to dissociate the system of three charges q, −4q and 2q placed at the vertices A, B and C respectively of an equilateral triangle of side 'a'.

OR

(b)

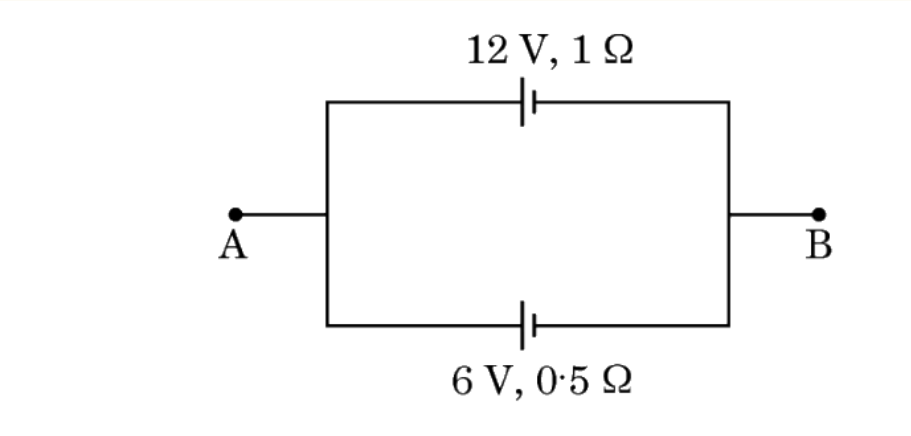
(i) Answer the following giving reason:

(I) The electron drift speed is estimated to be only a few mm/s for currents in the range of a few amperes. How, then, is the current established almost the instant a circuit is closed?

(II) A low voltage supply from which one needs high currents must have very low internal resistance. Why?

(III) The assertion that V = IR is a statement of Ohm's law is not true. Why?

(ii) Two cells of emfs 12 V and 6 V are connected in parallel as shown in the figure. Their internal resistances are 1 Ω and 0·5 Ω respectively. Calculate the emf and internal resistance of the equivalent cell between points A and B.



[2026 • Set 55-4-1]

SOLUTION

(a)(i)(I) An equipotential surface through a point is normal to $\vec{E}$ there.

On an equipotential surface the potential is the same everywhere, so for any small displacement $d\vec{l}$ lying in the surface, $\mathbf{dV = 0}$. Using $\mathbf{dV = -\vec{E}\cdot d\vec{l}}$, this gives $\mathbf{\vec{E}\cdot d\vec{l} = 0}$ for every tangential direction. Hence $\vec{E}$ has no component along the surface; it must point perpendicular (normal) to the surface. If it had a tangential component, work would be needed to move a charge along the surface and the potential would not stay constant.

(a)(i)(II) Field inside a dielectric is less than the field outside.

When a dielectric is placed in an external field $\vec{E}_0$, it gets polarised. Bound charges appear on its surfaces and set up an internal field $\vec{E}_p$ opposite to $\vec{E}_0$. The net field inside is reduced:

$$E = \frac{E_0}{K}$$

Since $K > 1$, the field inside is smaller than the applied field outside.

(a)(i)(III) The potential difference decreases when the plates are brought closer.

With the charge on the plates kept fixed (battery disconnected), $\sigma = Q/A$ is unchanged, so the field $E = \sigma/\varepsilon_0$ stays the same. Since $V = E\,d$, reducing d reduces V. Equivalently $C = \varepsilon_0 A/d$ increases as d falls, and with Q fixed $V = Q/C$ decreases.

(a)(ii) Work done to dissociate the system q, $-4q$, $2q$ on an equilateral triangle of side a.

Charges are at A(q), B($-4q$), C($2q$), each pair separated by a. The total potential energy is the sum over the three pairs:

$$U = \frac{k}{a}[(q)(-4q) + (-4q)(2q) + (2q)(q)]$$

$$U = \frac{k}{a}[-4q^2 - 8q^2 + 2q^2]$$

$$U = -\frac{10\,kq^2}{a}$$

The work done to dissociate the system (to separate the charges to infinity) equals the negative of this energy:

$$W = -U = +\frac{10\,kq^2}{a}$$

$$W = \frac{10\,q^2}{4\pi\varepsilon_0\,a}$$

CONCEPT

Work to dissociate (separate to infinity) equals $-U$ of the system. Compute U first as the sum over all distinct pairs, then change the sign.

Final answer. Field is normal to an equipotential, dielectric reduces the field by $\mathbf{1/K}$, closer plates lower $\boldsymbol{V}$; and the work to dissociate the three charges is $\boldsymbol{W} = \frac{\mathbf{10\,kq^2}}{\boldsymbol{a}} = \frac{\mathbf{10\,q^2}}{\mathbf{4\pi\varepsilon_0\,a}}$.

← Back to Q197

Q198

(a)

(i) Two point charges 5 μC *and* −1 μC *are placed at points* (−3 cm, 0, 0) *and* (3 cm, 0, 0) *respectively. An external electric field* $\vec{\boldsymbol{E}} = \frac{\boldsymbol{A}}{\mathbf{r^2}}\,\hat{\boldsymbol{r}}$, *where* $\boldsymbol{A} = \mathbf{3 \times 10^5\ V\,m}$, *is switched on in the region. Calculate the change in electrostatic energy of the system due to the electric field.*

(ii) A system of two conductors is placed in air and they have net charge of +80 μC *and* −80 μC *which causes a potential difference of* 16 *V between them.*

(1) Find the capacitance of the system.

(2) If the air between the capacitor is replaced by a dielectric medium of dielectric constant 3, *what will be the potential difference between the two conductors?*

(3) If the charges on two conductors are changed to +160 μC *and* −160 μC, *will the capacitance of the system change? Give reason for your answer.*

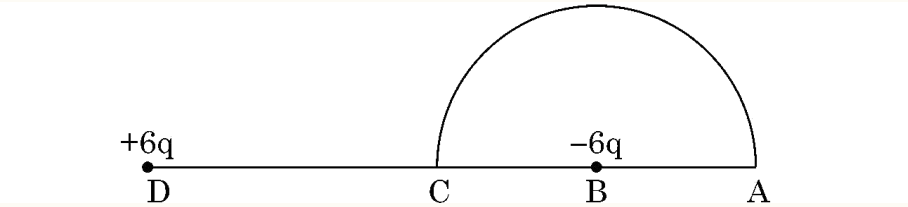
OR

(b)

(i) Consider three metal spherical shells A, B and C, each of radius R. Each shell is

having a concentric metal ball of radius R/10. The spherical shells A, B and C are given charges +6q, −4q *and* 14q *respectively. Their inner metal balls are also given charges* −2q, +8q *and* −10q *respectively. Compare the magnitude of the electric fields due to shells A, B and C at a distance* 3R *from their centres.*

(ii) A charge −6 μC *is placed at the centre B of a semicircle of radius* 5 cm, *as shown in the figure. An equal and opposite charge is placed at point D at a distance of* 10 cm *from B. A charge* +5 μC *is moved from point C to point A along the circumference. Calculate the work done on the charge.*



[2025 • Set 55-1-1]

SOLUTION

(a)(i) Change in electrostatic energy due to the external field.

The field $\vec{E} = \frac{A}{r^2}\hat{r}$ corresponds to the potential $V_{ext}(r) = \frac{A}{r}$, with $A = 3\times 10^5\ V\,m$. Each charge sits at $r = 3\ \text{cm} = 0.03\ m$ from the origin.

$$V_{ext} = \frac{A}{r} = \frac{3\times 10^5}{0.03} = 10^7\ V$$

The change in energy when the field is switched on is the interaction energy of the charges with the field:

$$\Delta U = q_1 V_{ext} + q_2 V_{ext}$$

$$\Delta U = (5\times 10^{-6})(10^7) + (-1\times 10^{-6})(10^7)$$

$$\Delta U = 50 - 10 = 40\ J$$

(a)(ii) Two-conductor capacitor.

(1) Capacitance:

$$C = \frac{Q}{V} = \frac{80\times 10^{-6}}{16} = 5\ \mu F$$

(2) Inserting a dielectric of constant $K = 3$ with the charge fixed multiplies C by K, so the new capacitance is $C' = 3C = 15\ \mu F$. With Q fixed, the new potential

difference is:

$$V' = \frac{Q}{C'} = \frac{V}{K} = \frac{16}{3} \approx 5.33\ V$$

(3) No, the capacitance does not change. Capacitance is a geometric property fixed by the shape, size and separation of the conductors and the medium between them. Doubling the charges to ± 160 μC also doubles the potential difference to 32 *V*, so the ratio $C = Q/V = 5\ \mu F$ stays the same.

OR (b)(i) Comparing the fields of shells A, B, C at distance 3R.

The point 3R lies outside each composite system (inner ball + outer shell). By Gauss's law the field outside depends only on the total enclosed charge:

$$E = \frac{1}{4\pi\varepsilon_0} \frac{Q_{enclosed}}{(3R)^2}$$

Total charge of each system (outer shell + inner ball):

A: **+6q + (−2q) = +4q**

B: **−4q + 8q = +4q**

C: **14q + (−10q) = +4q**

All three enclose the same net charge +4q at the same distance 3R, so:

$$E_A = E_B = E_C$$

The magnitudes of the electric fields are equal (ratio 1:1:1).

(b)(ii) Work done in moving +5 μC from C to A.

Charge −6 μC is at centre B, and +6 μC is at D, 10 cm from B. C and A are the two ends of the diameter of the semicircle of radius 5 cm. By the symmetry of the figure, points A and C are at equal distances from both the −6 μC charge at B and the +6 μC charge at D, so they are at the same potential:

$$V_A = V_C$$

Therefore the work done in moving the charge from C to A is:

$$W = q(V_A - V_C) = 0$$

$$W = 0\ J$$

Final answer. Change in energy due to the field is **40 *J***; ***C* = 5 μF**, ***V'* = 16/3 ≈ 5.33 *V***, and ***C*** is unchanged when charge doubles. (OR: $\mathbf{E_A:E_B:E_C = 1:1:1}$ and the work C to A is **0 *J***.)

← *Back to Q198*

Q199

(i) A small conducting sphere A of radius r charged to a potential V_1 *is enclosed by a spherical conducting shell B of radius R. If A and B are connected by a thin wire, calculate the final potential on sphere A and shell B.*
(ii) Write two characteristics of equipotential surfaces. A uniform electric field of 50 NC^{-1} *is set up in a region along +x axis. If the potential at the origin (0, 0) is 220 V, find the potential at a point (4 m, 3 m).*

[2025 • Set 55-2-1]

SOLUTION

(i) Final potentials of sphere A and shell B after connection.

A small conducting sphere A of radius r is charged to potential V_1 and is enclosed by a conducting shell B of radius R. The charge on A initially is found from $\mathbf{V_1 = \frac{kQ}{r}}$:

$$Q = \frac{V_1 r}{k}$$

When A and B are joined by a wire, the entire charge of the inner sphere flows out to the outer shell, because charge on a conductor resides on its outer surface and the interior of the enclosure is shielded. Sphere A is left with zero net charge, and the whole charge Q now sits on shell B of radius R.

Both conductors are connected, so they are at the same potential, equal to the potential of the shell:

$$V_A = V_B = \frac{kQ}{R}$$

Substituting $\mathbf{Q = \frac{V_1 r}{k}}$:

$$V_A = V_B = \frac{k}{R} \cdot \frac{V_1 r}{k}$$

$$V_A = V_B = \frac{r}{R} V_1$$

(ii) Two characteristics of equipotential surfaces:

(1) The electric field $\vec{E}$ is everywhere perpendicular (normal) to an equipotential surface.

(2) No work is done in moving a charge between two points lying on the same equipotential surface.

Potential at (4 *m*, 3 *m*):

The field is $\vec{E} = 50\,\hat{i}\,N\,C^{-1}$, directed along $+x$. Only the x-displacement matters; the y-displacement contributes nothing because $\vec{E}$ has no y-component.

$$V - V_0 = -E\,\Delta x$$

$$V = V_0 - E\,\Delta x = 220 - (50)(4)$$

$$V = 220 - 200 = 20\ V$$

$$V = 20\ V$$

CONCEPT

Potential drops along the field direction. Moving 4 m along $+x$ in a 50 NC^{-1} field drops V by 200 V; the 3 m along y does nothing since E has no y-part.

Final answer. After connection $V_A = V_B = \frac{r}{R} V_1$; and the potential at **(4 *m*, 3 *m*)** is **20 *V***.

← *Back to Q199*

Q200

(a) (i) The electric field in a region is given by $\vec{E} = 40x\,\hat{i}$ **N/C.** *Find the amount of work done in taking a unit positive charge from a point* (0, 3 *m*) *to the point* (5 *m*, 0).
(ii) A charge Q is distributed over two concentric hollow spheres of radii r and **R (>r)**

such that their surface charge densities are equal. Find:

(I) the electric field, and

(II) the potential

at their common centre.

OR

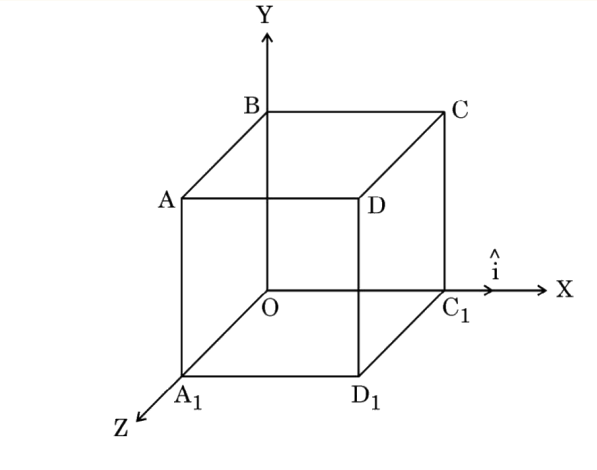
(b) (i) Obtain an expression for the electric field $\vec{E}$ *due to a dipole of dipole moment* $\vec{p}$ *at a point on its equatorial plane and specify its direction. Hence, find the value of electric field:*

(I) at the centre of the dipole $(r = 0)$*, and*

(II) at a point $r \gg a$*,*

where 2a *is the length of the dipole.*

(ii) An electric field $\vec{E} = (10x + 5)\,\hat{i}\ N/C$ *exists in a region in which a cube of side L is kept as shown in the figure. Here x and L are in metres. Calculate the net flux through the cube.*



[2025 • Set 55-6-1]

SOLUTION

(a)(i) Work done in $\vec{E} = 40x\,\hat{i}\ N\,C^{-1}$**.**

The field has only an x-component, so the y-displacement contributes nothing. Going from (0,3) to (5,0), only x varies from 0 to 5 m.

$$V_B - V_A = -\int_A^B \vec{E}\cdot d\vec{l} = -\int_0^5 40x\,dx$$

$$V_B - V_A = -[20x^2]_0^5 = -20(25) = -500\ V$$

Work done by an external agent in carrying a unit positive charge:

$$W_{ext} = q(V_B - V_A) = (1)(-500) = -500\ J$$

So the external agent does $-500\,J$ (the field itself does $+500\,J$).

(a)(ii) Charge Q on two concentric shells of radii r and R (>r) with equal surface charge densities.

Equal densities give charges proportional to area:

$$Q_r = \sigma\, 4\pi\, r^2,\ Q_R = \sigma\, 4\pi\, R^2$$

Total: $Q = \sigma\, 4\pi(r^2 + R^2)$, so $\sigma = \dfrac{Q}{4\pi(r^2 + R^2)}$.

(I) Field at the common centre: the centre lies inside both shells, and the field inside any uniformly charged shell is zero, so

$$\vec{E}_{\text{centre}} = \vec{0}$$

(II) Potential at the centre: each shell contributes $\dfrac{kQ_{\text{shell}}}{R_{\text{shell}}}$ at interior points:

$$V_{\text{centre}} = \frac{kQ_r}{r} + \frac{kQ_R}{R} = 4\pi\, k\sigma\, (r + R)$$

$$V_{\text{centre}} = \frac{Q(r + R)}{4\pi\varepsilon_0(r^2 + R^2)}$$

OR (b)(i) Equatorial field of a dipole.

For a point on the equatorial line at distance r from the centre of a dipole of moment $\vec{p}$ (length 2a), the components along the axis from the two charges add and the perpendicular components cancel:

$$E = \frac{1}{4\pi\varepsilon_0}\ \frac{p}{(r^2 + a^2)^{3/2}}$$

Its direction is **anti-parallel to $\vec{p}$** (from the $+q$ end towards the $-q$ end).

(I) At the centre $r = 0$:

$$E = \frac{1}{4\pi\varepsilon_0}\ \frac{p}{a^3} = \frac{1}{4\pi\varepsilon_0}\ \frac{2qa}{a^3} = \frac{2q}{4\pi\varepsilon_0\, a^2}$$

(II) For $r \gg a$, neglect a^2 beside r^2:

$$E = \frac{1}{4\pi\varepsilon_0}\ \frac{p}{r^3}$$

(b)(ii) Net flux through the cube in $\vec{E} = (10x + 5)\hat{i}\ N/C$.

The field is along x, so only the two faces perpendicular to the x-axis contribute. Let the near face be at $\boldsymbol{x = x_0}$ and the far face at $\boldsymbol{x = x_0 + L}$, each of area L^2.

At $\boldsymbol{x = x_0}$ (inward normal $-\hat{i}$): $\boldsymbol{\Phi_{in} = -(10x_0 + 5)L^2}$

At $\boldsymbol{x = x_0 + L}$ (outward normal $+\hat{i}$): $\boldsymbol{\Phi_{out} = (10(x_0 + L) + 5)L^2}$

Net flux:

$$\Phi = \Phi_{out} + \Phi_{in} = [(10x_0 + 10L + 5) - (10x_0 + 5)]L^2$$

$$\Phi = 10L \cdot L^2$$

$$\Phi = 10L^3\ N\,m^2\,C^{-1}$$

TIP

For flux in a position-dependent field $\boldsymbol{E=(10x+5)i}$, only the constant-gradient part survives: the +5 cancels between the two x-faces, leaving net flux $10L^3$ from the 10x term.

Final answer. Work = $\boldsymbol{-500\ J}$ (agent); $\boldsymbol{E_{centre} = 0}$ and $\boldsymbol{V_{centre} = \frac{Q(r+R)}{4\pi\varepsilon_0(r^2+R^2)}}$. (OR: equatorial $\boldsymbol{E=\frac{kp}{(r^2+a^2)^{3/2}}}$ anti-parallel to $\boldsymbol{p}$; net flux $\boldsymbol{=10L^3\ N\,m^2C^{-1}}$.)

← *Back to Q200*

Q201

(a) (i) *A parallel plate capacitor with plate area A and plate separation d has a capacitance* C_0. *A slab of dielectric constant K having area A and thickness* $(\frac{d}{2})$ *is inserted in the capacitor, parallel to the plates. Find the new value of its capacitance.*
(ii) *You are provided with a large number of* 1 μF *identical capacitors and a power supply of* 1200 *V. The dielectric medium used in each capacitor can withstand up to* 200 *V only. Find the minimum number of capacitors and their arrangement, required to build a capacitor system of equivalent capacitance of* 2 μF *for use with this supply.*
OR
(b) (i) *An electric dipole of dipole moment* $\vec{p}$ *consists of point charges q and −q,*

separated by 2a. *Derive an expression for electric potential in terms of its dipole moment at a point at a distance x (» a) from its centre and lying (I) along its axis, and (II) along its bisector line.*

(ii) *An electric dipole of dipole moment* $\vec{p} = (0.8\hat{i} + 0.6\hat{j}) \times 10^{-29}\ C\ m$ *is placed in an electric field* $\vec{E} = 1.0 \times 10^{7}\ \hat{k}\ V/m$. *Calculate the magnitude of the torque acting on it and the angle it makes with the x-axis, at this instant.*

[2025 • Set 55-7-1]

SOLUTION

(a)(i) When a dielectric slab of thickness $\frac{d}{2}$ fills half the gap, the arrangement behaves as two capacitors in series: an air gap of thickness $\frac{d}{2}$ and a dielectric gap of thickness $\frac{d}{2}$.

The air half has capacitance $C_1 = \frac{\varepsilon_0 A}{d/2} = \frac{2\varepsilon_0 A}{d}$.

The dielectric half has capacitance $C_2 = \frac{K\varepsilon_0 A}{d/2} = \frac{2K\varepsilon_0 A}{d}$.

For series combination, $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d}{2\varepsilon_0 A} + \frac{d}{2K\varepsilon_0 A}$.

$$\frac{1}{C} = \frac{d}{2\varepsilon_0 A}\left(1 + \frac{1}{K}\right) = \frac{d(K+1)}{2K\varepsilon_0 A}.$$

$$C = \frac{2K\varepsilon_0 A}{(K+1)\,d}.$$

Since $C_0 = \frac{\varepsilon_0 A}{d}$, the new capacitance is $C = \frac{2K}{K+1} C_0$.

(a)(ii) Each capacitor can withstand only 200 *V*, but the supply is 1200 *V*. To divide the voltage, capacitors must be joined in series so that each one shares an equal part of the total.

Number of capacitors needed in one series row, $n = \frac{1200\ V}{200\ V} = 6$. Each capacitor then carries 200 *V*, which is safe.

Capacitance of one series row of six 1 μF capacitors, $C_{row} = \frac{1}{6}$ μF.

Let m such rows be joined in parallel to raise the capacitance. For parallel rows,

$\mathbf{C_{eq} = m \times C_{row} = m \times \frac{1}{6}\ \mu F.}$

Setting $\mathbf{C_{eq} = 2\ \mu F}$: $\mathbf{\frac{m}{6} = 2 \Rightarrow m = 12.}$

Total number of capacitors = $\mathbf{n \times m = 6 \times 12 = 72.}$

Arrangement: 12 parallel rows, each row having 6 capacitors in series; total 72 capacitors.

(b)(i) The dipole has charge $+q$ at distance a and $-q$ at distance a from the centre O, giving dipole moment $\boldsymbol{p = q(2a).}$

(I) Along the axis: take point P on the axis at distance x from the centre, on the side of $+q$.

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{x-a} - \frac{q}{x+a}\right].$$

$$V = \frac{q}{4\pi\varepsilon_0}\left[\frac{(x+a)-(x-a)}{x^2-a^2}\right] = \frac{q}{4\pi\varepsilon_0}\cdot\frac{2a}{x^2-a^2}.$$

Since $x \gg a$, we take $\mathbf{x^2 - a^2 \approx x^2}$, and $\boldsymbol{q(2a) = p}$:

$$V_{axial} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{p}{x^2}.$$

(II) Along the bisector (equatorial line): a point P on the perpendicular bisector at distance x is equidistant from $+q$ and $-q$, each at distance $\sqrt{x^2+a^2}$.

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{\sqrt{x^2+a^2}} - \frac{q}{\sqrt{x^2+a^2}}\right] = 0.$$

So $\mathbf{V_{equatorial} = 0}$ at every point on the bisector, because the positive and negative potentials cancel exactly.

(b)(ii) Given $\vec{p} = (0.8\hat{i} + 0.6\hat{j}) \times 10^{-29}\ C\,m$ and $\vec{E} = 1.0 \times 10^7\ \hat{k}\ V/m$.

Torque $\vec{\tau} = \vec{p} \times \vec{E}$.

$$\vec{\tau} = [(0.8\hat{i} + 0.6\hat{j}) \times 10^{-29}] \times (1.0 \times 10^7\ \hat{k}).$$

Using $\hat{i} \times \hat{k} = -\hat{j}$ and $\hat{j} \times \hat{k} = \hat{i}$:

$$\vec{\tau} = (0.8)(10^{-22})(-\hat{j}) + (0.6)(10^{-22})(\hat{i}) = (0.6\hat{i} - 0.8\hat{j}) \times 10^{-22}\ N\,m.$$

Magnitude $|\vec{\tau}| = \sqrt{(0.6)^2 + (0.8)^2} \times 10^{-22} = \sqrt{1.0} \times 10^{-22} = 1.0 \times 10^{-22}\ N\,m$.

Angle with the x-axis: $\phi = \frac{-0.8}{0.6}$, so $\phi = \tan^{-1}(\frac{0.8}{0.6}) \approx 53°$ below the x-axis (i.e. $\phi \approx -53°$).

> **TIP**
>
> In voltage-sharing problems, first fix the number in series from the safe voltage limit, then add parallel rows to reach the required capacitance. Doing it in the other order is a common slip.

Final answer. (a)(i) $C = \frac{2K}{K+1} C_0$; (ii) 72 capacitors (12 parallel rows of 6 in series); (b)(i) $V_{axial} = \frac{p}{4\pi\varepsilon_0 x^2}$, $V_{eq} = 0$; (ii) $\tau = 1.010^{-22}\ N\,m$ at about $53°$ to the x-axis.

← *Back to Q201*

Q202

(i) Obtain the expression for the capacitance of a parallel plate capacitor with a dielectric medium between its plates.
(ii) A charge of 6 μC *is given to a hollow metallic sphere of radius* 0.2 *m. Find the potential at (i) the surface and (ii) the centre of the sphere.*

[2024 • Set 55-3-1]

SOLUTION

(i) Consider a parallel plate capacitor with two plates each of area A, separated by a small distance d, with the gap completely filled by a dielectric of dielectric constant K. Let the charge on the plates be $+Q$ and $-Q$, so the surface charge density is $\sigma = \frac{Q}{A}$.

In the absence of the dielectric, the uniform field between the plates would be $E_0 = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$.

When the dielectric is inserted, it gets polarised and bound charges appear on its faces. These set up an opposing field, reducing the net field by a factor K:

$$E = \frac{E_0}{K} = \frac{\sigma}{K\varepsilon_0} = \frac{Q}{K\varepsilon_0 A}.$$

The potential difference between the plates is $V = E\,d = \dfrac{Q\,d}{K\varepsilon_0 A}$.

By definition, capacitance $C = \dfrac{Q}{V}$.

$$C = \frac{Q}{\dfrac{Q\,d}{K\varepsilon_0 A}} = \frac{K\varepsilon_0 A}{d}.$$

Hence $C = \dfrac{K\varepsilon_0 A}{d}$. Without the dielectric ($K = 1$), this reduces to $C_0 = \dfrac{\varepsilon_0 A}{d}$, so the dielectric increases the capacitance K times.

(ii) Given charge on the hollow metallic sphere $q = 6\ \mu C = 6\times 10^{-6}\ C$ and radius $R = 0.2\ m$, with $k = 9\times 10^9\ N\,m^2C^{-2}$.

A charged conductor has all its charge on the outer surface, so for points on and inside the sphere the potential equals that of a point charge at distance R.

Potential at the surface: $V_{surface} = \dfrac{kq}{R} = \dfrac{(9\times 10^9)(6\times 10^{-6})}{0.2}$.

$$V_{surface} = \frac{54\times 10^3}{0.2} = 2.7\times 10^5\ V.$$

Inside a hollow charged conductor the field is zero, so the potential stays constant and equals the surface value everywhere inside.

Potential at the centre: $V_{centre} = V_{surface} = 2.7\times 10^5\ V$.

Thus the potential at both the surface and the centre is $2.7\times 10^5\ V$.

Final answer. $C = \dfrac{K\varepsilon_0 A}{d}$; potential at the surface and at the centre are both $2.710^5\ V$.

← *Back to Q202*

Q203

(i) A dielectric slab of dielectric constant K and thickness t is inserted between plates of a parallel plate capacitor of plate separation d and plate area A. Obtain an expression for its capacitance.
(ii) Two capacitors of different capacitances are connected first (1) in series and then (2) in parallel across a dc source of 100 V. *If the total energy stored in the combination in the two cases are* 40 mJ *and* 250 mJ *respectively, find the capacitance of the capacitors.*

[2024 • Set 55-4-1]

SOLUTION

(i) Consider a parallel plate capacitor with plate area A, plate separation d, carrying charge $\pm Q$ with surface charge density $\sigma = \frac{Q}{A}$. A dielectric slab of constant K and thickness t ($t < d$) is placed between the plates.

In the air gap (total thickness $d - t$) the field is $E_0 = \frac{\sigma}{\varepsilon_0}$.

Inside the slab the field is reduced to $E = \frac{E_0}{K} = \frac{\sigma}{K\varepsilon_0}$.

The potential difference across the plates is the sum across the air gap and the slab:

$$V = E_0(d - t) + \frac{E_0}{K}\, t.$$

$$V = \frac{\sigma}{\varepsilon_0}\left[(d - t) + \frac{t}{K}\right] = \frac{\sigma}{\varepsilon_0}\left[d - t + \frac{t}{K}\right].$$

Capacitance $C = \frac{Q}{V} = \frac{\sigma A}{\frac{\sigma}{\varepsilon_0}\left[d - t + \frac{t}{K}\right]}$.

$$C = \frac{\varepsilon_0 A}{d - t + \frac{t}{K}} = \frac{\varepsilon_0 A}{d - t\left(1 - \frac{1}{K}\right)}.$$

Hence $C = \frac{\varepsilon_0 A}{d - t\left(1 - \frac{1}{K}\right)}$. For $t \to 0$ this gives $\frac{\varepsilon_0 A}{d}$ (air), and for $t = d$ it gives $\frac{K\varepsilon_0 A}{d}$ (fully filled), as expected.

(ii) Energy stored is $U = \frac{1}{2} C_{eq} V^2$, with $V = 100\ V$, so $V^2 = 10^4\ V^2$.

For the series case, $U_s = 40\ mJ = 40 \times 10^{-3}\ J$:

$$C_s = \frac{2U_s}{V^2} = \frac{2(40 \times 10^{-3})}{10^4} = 8 \times 10^{-6}\ F = 8\ \mu F.$$

For the parallel case, $U_p = 250\ mJ = 250 \times 10^{-3}\ J$:

$$C_p = \frac{2U_p}{V^2} = \frac{2(250 \times 10^{-3})}{10^4} = 50 \times 10^{-6}\ F = 50\ \mu F.$$

For parallel: $C_1 + C_2 = 50\ \mu F$.

For series: $\frac{C_1 C_2}{C_1 + C_2} = 8\ \mu F$, so $C_1 C_2 = 8 \times 50 = 400\ (\mu F)^2$.

The two values satisfy the quadratic $C^2 - 50C + 400 = 0$.

$$C = \frac{50 \pm \sqrt{2500 - 1600}}{2} = \frac{50 \pm \sqrt{900}}{2} = \frac{50 \pm 30}{2}.$$

$C = 40\ \mu F$ or $C = 10\ \mu F$.

Hence the capacitances are $C_1 = 40\ \mu F$ and $C_2 = 10\ \mu F$.

Final answer. $C = \frac{\varepsilon_0 A}{d - t(1 - 1/K)}$; the two capacitors are **40 μF** and **10 μF.**

← *Back to Q203*

Q204

(i) Draw equipotential surfaces for an electric dipole.
(ii) Two point charges q_1 *and* q_2 *are located at* $\vec{r}_1$ *and* $\vec{r}_2$ *respectively in an external electric field* $\vec{E}$*. Obtain an expression for the potential energy of the system.*
(iii) The dipole moment of a molecule is 10^{-30} Cm*. It is placed in an electric field* $\vec{E}$ *of* 10^5 *V/m such that its axis is along the electric field. The direction of* $\vec{E}$ *is suddenly changed by* 60° *at an instant. Find the change in the potential energy of the dipole, at that instant.*

[2024 • Set 55-5-1]

SOLUTION

(i) Equipotential surfaces of an electric dipole: take the dipole moment $\vec{p}$ pointing from $-q$ to $+q$ along the vertical axis. Near $+q$ the equipotentials are small closed surfaces enclosing $+q$ with $\boldsymbol{V > 0}$; near $-q$ they are small closed surfaces enclosing $-q$ with $\boldsymbol{V < 0}$.

The perpendicular bisector plane through the centre of the dipole is the $\boldsymbol{V = 0}$ equipotential, a flat plane separating the positive and negative regions. Far away, the surfaces become distorted, closed peanut-shaped curves, symmetric about the dipole axis but not symmetric across the bisector plane. Every equipotential surface is drawn so that field lines (which run from $+q$ to $-q$) cross it at right angles.

(ii) Two point charges q_1 and q_2 at positions $\vec{r}_1$ and $\vec{r}_2$ in an external field $\vec{E}$.

Bringing q_1 from infinity to $\vec{r}_1$ against the external field requires work $\mathbf{W_1 = q_1\, V(\vec{r}_1)}$, where V is the potential of the external field.

Bringing q_2 to $\vec{r}_2$ requires work against the external field and against the field of q_1 already present:

$$\mathbf{W_2 = q_2\, V(\vec{r}_2) + \frac{1}{4\pi\varepsilon_0}\frac{q_1\, q_2}{r_{12}}}, \text{ where } \mathbf{r_{12} = |\vec{r}_1 - \vec{r}_2|}.$$

The total potential energy is the sum of the work done:

$$U = q_1\, V(\vec{r}_1) + q_2\, V(\vec{r}_2) + \frac{1}{4\pi\varepsilon_0}\frac{q_1\, q_2}{|\vec{r}_1 - \vec{r}_2|}.$$

(iii) For a single dipole in a field, $\boldsymbol{U = -pE\theta}$, where θ is the angle between $\vec{p}$ and $\vec{E}$.

Given $\boldsymbol{p = 10^{-30}\ C\ m}$ and $\boldsymbol{E = 10^5\ V/m}$. Initially the axis is along $\vec{E}$, so $\boldsymbol{\theta_i = 0°}$:

$$U_i = -pE\cos 0° = -pE.$$

When $\vec{E}$ turns by 60°, the angle between $\vec{p}$ and the new field is $\boldsymbol{\theta_f = 60°}$:

$$U_f = -pE\cos 60° = -pE\left(\frac{1}{2}\right) = -\frac{pE}{2}.$$

Change in potential energy $\boldsymbol{\Delta U = U_f - U_i = -\frac{pE}{2} - (-pE) = \frac{pE}{2}}$.

$$\Delta U = \frac{1}{2}(10^{-30})(10^5) = \frac{1}{2}\times 10^{-25} = 5\times 10^{-26}\ J.$$

Hence the change in potential energy is $\Delta U = +5\times 10^{-26}\,J$.

Final answer. $U = q_1\,V(\vec{r}_1) + q_2\,V(\vec{r}_2) + \dfrac{q_1\,q_2}{4\pi\varepsilon_0\,r_{12}}$; change in PE = $+510^{-26}\,J$.

← *Back to Q204*

Q205

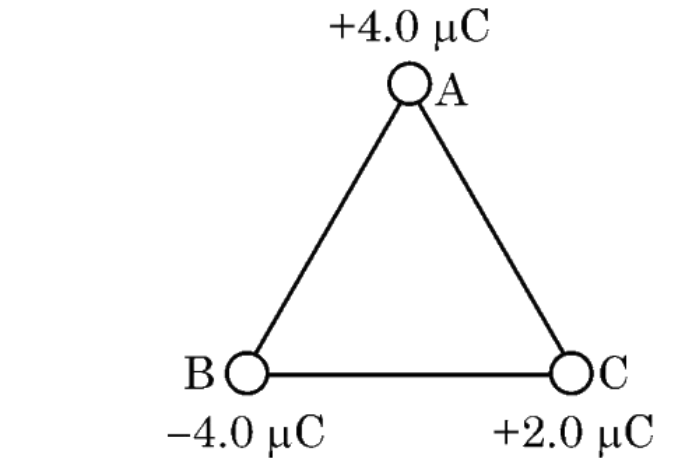
(i) Consider two identical point charges located at points (0,0) *and* (*a*,0).

(1) Is there a point on the line joining them at which the electric field is zero?

(2) Is there a point on the line joining them at which the electric potential is zero? Justify your answers for each case.

(ii) State the significance of negative value of electrostatic potential energy of a system of charges.

Three charges are placed at the corners of an equilateral triangle ABC of side 2.0 *m as shown in figure. Calculate the electric potential energy of the system of three charges.*



[2023 • Set 55-2-1]

SOLUTION

(i)(1) Two identical positive charges +*q* sit at (0,0) and (*a*,0). Yes, the electric field is zero at the midpoint $(\frac{a}{2}, 0)$.

At the midpoint each charge is at the same distance $\frac{a}{2}$, so each produces a field of equal magnitude. The two fields point away from their own charge, i.e. in

opposite directions along the line joining them, and therefore cancel, giving $\vec{E} = \mathbf{0}$.

(i)(2) No, there is no point on the line joining them where the potential is zero. Potential is a scalar and each positive charge contributes $V = \dfrac{kq}{r} > 0$. The sum of two positive quantities can never be zero, so $V \neq 0$ anywhere on the line (it is zero only at infinity).

(ii) Significance of negative potential energy: a negative value of the total electrostatic potential energy means the system is in a bound (stable) state. External work must be done to pull the charges apart to infinity; the magnitude of this work equals $|U|$.

Triangle calculation: charges $q_A = +4\ \mu C$, $q_B = -4\ \mu C$, $q_C = +2\ \mu C$ at the corners of an equilateral triangle of side $r = 2.0\ m$, with $k = 9\times 10^9\ N\,m^2C^{-2}$.

The total energy is the sum over the three pairs: $U = \dfrac{k}{r}(q_A q_B + q_B q_C + q_C q_A)$.

$$q_A q_B = (+4)(-4) = -16\ (\mu C)^2.$$

$$q_B q_C = (-4)(+2) = -8\ (\mu C)^2.$$

$$q_C q_A = (+2)(+4) = +8\ (\mu C)^2.$$

Sum of products $= (-16 - 8 + 8)\times 10^{-12} = -16\times 10^{-12}\ C^2$.

$$U = \frac{9\times 10^9}{2.0}\times(-16\times 10^{-12}).$$

$$U = (4.5\times 10^9)(-16\times 10^{-12}) = -72\times 10^{-3}\ J.$$

Hence the potential energy of the system is $U = -7.2\times 10^{-2}\ J = -72$ mJ.

> **COMMON MISTAKE**
>
> *For three charges, always add exactly three pair terms (AB, BC, CA). Forgetting one pair or double-counting is the most common error in these triangle questions.*

Final answer. Field is zero at the midpoint; potential is never zero on the line; system PE $= -7.210^{-2}\ J$.

← Back to Q205

Q206

(i) (A) Why does the electric field inside a dielectric slab decrease when kept in an external electric field?
(B) Derive an expression for the capacitance of a parallel plate capacitor filled with a medium of dielectric constant K.
(ii) A charge $q = 2\ \mu C$ *is placed at the centre of a sphere of radius* 20 cm. *What is the amount of work done in moving* 4 μC *from one point to another point on its surface?*
(iii) Write a relation for polarisation $\vec{P}$ *of a dielectric material in the presence of an external electric field.*

[2021]

SOLUTION

(i)(A) When a dielectric slab is placed in an external field $\vec{E}_0$, its molecules become polarised: in non-polar dielectrics the positive and negative charge centres are slightly displaced, and in polar dielectrics the permanent dipoles align with the field. As a result, bound (induced) charges appear on the faces of the slab, negative on the face toward the positive plate and positive on the face toward the negative plate.

These bound charges produce their own field $\vec{E}_p$ directed opposite to $\vec{E}_0$. The net field inside the dielectric is therefore reduced to $\vec{E} = \vec{E}_0 - \vec{E}_p = \frac{\vec{E}_0}{K}$, which is smaller than the applied field. This is why the field inside a dielectric decreases.

(i)(B) Consider a parallel plate capacitor of plate area A, separation d, completely filled with a dielectric of constant K, carrying charge $\pm Q$ with $\sigma = \frac{Q}{A}$.

Field inside the dielectric: $E = \frac{\sigma}{K\varepsilon_0} = \frac{Q}{K\varepsilon_0 A}$.

Potential difference: $V = E\,d = \frac{Q\,d}{K\varepsilon_0 A}$.

Capacitance: $C = \frac{Q}{V} = \frac{Q}{\frac{Q\,d}{K\varepsilon_0 A}} = \frac{K\varepsilon_0 A}{d}$.

Hence $C = \frac{K\varepsilon_0 A}{d}$.

(ii) A charge $q = 2\ \mu C$ sits at the centre of a sphere of radius 20 cm. The field of this central charge is spherically symmetric, so every point on the surface of the sphere is at the same potential, i.e. the surface is an equipotential.

Work done in moving a charge between two points at the same potential is $W = q' \Delta V = q'(V_2 - V_1) = q' \times 0$.

$$W = (4\ \mu C)(0) = 0.$$

Hence the work done in moving the 4 μC charge between two points on the surface is zero.

(iii) For a linear isotropic dielectric in an external field $\vec{E}$, the polarisation is proportional to the field:
$\vec{P} = \varepsilon_0 \chi_e \vec{E}$, where χ_e is the electric susceptibility, related to the dielectric constant by $\chi_e = K - 1$.

Final answer. E falls because induced bound charges oppose the field; $C = \frac{K\varepsilon_0 A}{d}$; work done = 0; and $\vec{P} = \varepsilon_0 \chi_e \vec{E}$.

← *Back to Q206*

Q207

(i) Obtain an expression for the potential energy of an electric dipole placed in a uniform electric field.
(ii) Three capacitors of capacitance C_1, C_2 *and* C_3 *are connected in series to a source of V volt. Show that the total energy stored in the combination of capacitors is equal to sum of the energy stored in individual capacitors.*
(iii) A capacitor of capacitance C is connected across a battery. After charging, the battery is disconnected and the separation between the plates is doubled. How will (i) the capacitance of the capacitor, and (ii) the electric field between the plates be affected? Justify your answer.

[2021]

SOLUTION

(i) Consider a dipole of moment $\vec{p}$ placed in a uniform field $\vec{E}$, making an angle θ with the field. The torque on it is $\boldsymbol{\tau = pE\theta}$.

To rotate the dipole through a small angle dθ against this torque, the external work is $\boldsymbol{dW = \tau\, d\theta = pE\theta\, d\theta}$.

Taking the reference position at $\boldsymbol{\theta = 90°}$ (where $\boldsymbol{U = 0}$), the work done in rotating from 90° to θ is:

$$W = \int_{90°}^{\theta} pE\theta'\, d\theta' = pE[-\theta']_{90°}^{\theta}.$$

$$W = pE(-\theta + \cos 90°) = -pE\theta.$$

This work is stored as potential energy: $\boldsymbol{U = -pE\theta = -\vec{p}\cdot\vec{E}}$.

U is minimum (−pE, stable) at $\boldsymbol{\theta = 0°}$ and maximum (+pE, unstable) at $\boldsymbol{\theta = 180°}$.

(ii) Three capacitors C_1, C_2, C_3 in series across V all carry the same charge Q. The voltage across each is $\boldsymbol{V_i = \frac{Q}{C_i}}$.

The supply voltage is $\boldsymbol{V = V_1 + V_2 + V_3 = Q\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right) = \frac{Q}{C_{eq}}}$, where $\boldsymbol{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$.

Total energy stored: $\boldsymbol{U = \frac{1}{2}\frac{Q^2}{C_{eq}} = \frac{1}{2}Q^2\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)}$.

$$U = \frac{1}{2}\frac{Q^2}{C_1} + \frac{1}{2}\frac{Q^2}{C_2} + \frac{1}{2}\frac{Q^2}{C_3} = U_1 + U_2 + U_3.$$

Hence the total energy stored equals the sum of the energies stored in the individual capacitors.

(iii) After charging, the battery is disconnected, so the charge Q on each plate stays fixed while the separation is doubled from d to 2d.

(i) Capacitance: $\boldsymbol{C = \frac{\varepsilon_0 A}{d}}$. Doubling d gives $\boldsymbol{C' = \frac{\varepsilon_0 A}{2d} = \frac{C}{2}}$. The capacitance is halved.

(ii) Electric field: $\boldsymbol{E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}}$. Since Q, A, ε_0 are unchanged, E does not depend on d.

Therefore the field between the plates remains unchanged, while the capacitance is halved.

CONCEPT

When the battery is disconnected, Q is constant (not V). The field $\boldsymbol{E = Q/\varepsilon_0 A}$ then stays the same on changing d; this is the point examiners want you to justify.

Final answer. $\boldsymbol{U = -\vec{p}\cdot\vec{E}}$; series energy equals the sum of individual energies; doubling $\boldsymbol{d}$ halves $\boldsymbol{C}$ but leaves $\boldsymbol{E}$ unchanged.

← *Back to Q207*

Q208

Find the expression for the potential energy of a system of two point charges q_1 *and* q_2 *located at* $\vec{r}_1$ *and* $\vec{r}_2$ *respectively in an external electric field* $\vec{E}$.

[2020 • Set 55-1-1]

SOLUTION

We build the system one charge at a time, in an external field $\vec{E}$ whose potential is $V(\vec{r})$.

First, bring q_1 from infinity to its position $\vec{r}_1$. Since no other charge is present yet, the only work done is against the external field:

$$W_1 = q_1\, V(\vec{r}_1).$$

Next, bring q_2 from infinity to $\vec{r}_2$. Now work is done against the external field and also against the field of the already placed charge q_1:

$$W_2 = q_2\, V(\vec{r}_2) + \frac{1}{4\pi\varepsilon_0}\frac{q_1\, q_2}{|\vec{r}_1 - \vec{r}_2|}.$$

The total potential energy of the system equals the total work done in assembling it:

$$U = W_1 + W_2.$$

$$U = q_1\,V(\vec{r}_1) + q_2\,V(\vec{r}_2) + \frac{1}{4\pi\varepsilon_0}\frac{q_1\,q_2}{|\vec{r}_1 - \vec{r}_2|}.$$

Hence $\mathbf{U = q_1\,V(\vec{r}_1) + q_2\,V(\vec{r}_2) + \frac{1}{4\pi\varepsilon_0}\frac{q_1\,q_2}{|\vec{r}_1 - \vec{r}_2|}}$.

The first two terms are the energies of the individual charges in the external field, while the last term is the mutual interaction energy of the two charges, which would be present even without the external field.

Final answer. $\mathbf{U = q_1\,V(\vec{r}_1) + q_2\,V(\vec{r}_2) + \frac{1}{4\pi\varepsilon_0}\frac{q_1\,q_2}{|\vec{r}_1 - \vec{r}_2|}}$.

← *Back to Q208*

Q209

Draw equipotential surfaces due to an isolated point charge $(-q)$ *and depict the electric field lines.*

[2020 • Set 55-1-1]

SOLUTION

For an isolated point charge $-q$, the potential at distance r is $\mathbf{V(r) = \frac{1}{4\pi\varepsilon_0}\frac{(-q)}{r} = -\frac{kq}{r}}$.

Since V depends only on r, every surface of constant potential is a sphere centred on the charge. The equipotential surfaces are therefore a set of concentric spheres around $-q$.

Diagram description: draw the charge $-q$ as a small dot at the centre. Around it draw three or more concentric circles (representing spherical equipotential surfaces) of increasing radius. Their potentials are all negative, with the potential becoming less negative (closer to zero) as r increases, so the innermost sphere is at the most negative potential.

The spacing between successive equipotentials (taken at equal steps of V) increases with distance, because the field weakens as $\frac{1}{r^2}$ farther from the charge.

Field lines: for $-q$ the field is $\vec{E} = -\frac{kq}{r^2}\hat{r}$, directed radially inward. So draw straight arrows pointing inward from all directions toward $-q$ (lines start at infinity and end on the charge).

The radial field lines cross every spherical equipotential surface at right angles, since field lines are always perpendicular to equipotential surfaces.

Hence the equipotentials are concentric spheres around $-q$, and the field lines are radial arrows pointing inward, meeting the spheres perpendicularly.

Final answer. Equipotentials are concentric spheres centred on $-q$; field lines are radial arrows pointing inward, crossing the spheres at right angles.

← *Back to Q209*

Q210

Three point charges +1 μC, −1 μC *and* +2 μC *are initially infinite distance apart. Calculate the work done in assembling these charges at the vertices of an equilateral triangle of side* 10 cm.

[2020 • Set 55-1-1]

SOLUTION

Side of the triangle $l = 10\ \text{cm} = 0.10\ m$, with $k = 9\times 10^9\ N\,m^2C^{-2}$. Charges: $q_1 = +1\ \mu C$, $q_2 = -1\ \mu C$, $q_3 = +2\ \mu C$.

The work done in assembling the charges equals the total potential energy, which is the sum of the energies of the three pairs.

$$U_{12} = \frac{k\,q_1\,q_2}{l} = \frac{(9\times 10^9)(+1\times 10^{-6})(-1\times 10^{-6})}{0.10}.$$

$$U_{12} = \frac{-9\times 10^{-3}}{0.10} = -9\times 10^{-2}\,J = -0.09\,J.$$

$$U_{23} = \frac{k\,q_2\,q_3}{l} = \frac{(9\times 10^9)(-1\times 10^{-6})(+2\times 10^{-6})}{0.10}.$$

$$U_{23} = \frac{-18\times 10^{-3}}{0.10} = -18\times 10^{-2}\,J = -0.18\,J.$$

$$U_{31} = \frac{k\, q_3\, q_1}{l} = \frac{(9\times 10^{9})(+2\times 10^{-6})(+1\times 10^{-6})}{0.10}.$$

$$U_{31} = \frac{+18\times 10^{-3}}{0.10} = +18\times 10^{-2}\, J = +0.18\, J.$$

Total work done $W = U = U_{12} + U_{23} + U_{31}$.

$$W = -0.09 - 0.18 + 0.18 = -0.09\, J.$$

Hence the work done in assembling the charges is $W = -0.09\, J = -90$ mJ. The negative sign shows the configuration is bound, so the field does positive work as the charges come together.

Final answer. Work done in assembling the charges $W = -0.09\, J = -90$ mJ.

← *Back to Q210*

Q211

Describe briefly the process of transferring the charge between the two plates of a parallel plate capacitor when connected to a battery. Derive an expression for the energy stored in a capacitor.

[2019 • Set 55-1-1]

SOLUTION

Charging process. The two plates of a parallel plate capacitor are connected to the terminals of a battery of emf V. The battery acts as a charge pump: it does work on the free electrons in the connecting wires.

Electrons are pulled away from the plate connected to the positive terminal, leaving that plate with a net positive charge $+Q$. These electrons travel through the external circuit and accumulate on the plate connected to the negative terminal, giving it an equal negative charge $-Q$.

This flow of charge continues, and the potential difference across the plates keeps rising, until it becomes exactly equal to the battery emf V. At that instant the current stops and the capacitor is fully charged with $Q = CV$.

At every stage equal and opposite charges $\pm Q$ appear on the two plates, and no charge actually crosses the gap between the plates.

Energy stored in a capacitor. Let the capacitor at some instant during charging carry a charge q, so the potential difference across it is $v = \frac{q}{C}$.

The small work done by the battery to push an additional charge dq onto the plate against this potential difference is

$$dW = v\, dq = \frac{q}{C}\, dq$$

The total work done in charging the capacitor from 0 to the final charge Q is found by integration:

$$W = {}_0{}^{Q}\frac{q}{C}\, dq = \frac{1}{C}\left[\frac{q^2}{2}\right]_0^{Q} = \frac{Q^2}{2C}$$

This work is stored as electrostatic potential energy U in the capacitor:

$$U = \frac{Q^2}{2C}$$

Using $Q = CV$, this can also be written as

$$\mathbf{U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV}$$

This energy resides in the **electric field** filling the space between the plates, with energy density $u = \frac{1}{2}\varepsilon_0 E^2$.

Final answer. The energy stored is $U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV$.

← *Back to Q211*

Q212

A parallel plate capacitor is charged by a battery to a potential difference V. It is disconnected from battery and then connected to another uncharged capacitor of the

same capacitance. Calculate the ratio of the energy stored in the combination to the initial energy on the single capacitor.

[2019 • Set 55-1-1]

SOLUTION

A capacitor of capacitance C is charged by a battery to potential difference V, so its charge is $Q = CV$ and its initial stored energy is

$$U_i = \frac{1}{2}CV^2$$

The battery is now disconnected, so the total charge Q is fixed and cannot change. This charged capacitor is then connected to an identical uncharged capacitor of capacitance C.

When the two are joined, they are in **parallel**, so they share charge until both reach a common potential V_f. By conservation of charge, the total charge stays the same:

$$Q = (C + C)\,V_f = 2C\,V_f$$

$$CV = 2C\,V_f$$

$$V_f = \frac{V}{2}$$

The total energy stored in the combination (two capacitors of total capacitance 2C at potential V_f) is

$$U_f = \frac{1}{2}(2C)V_f^2 = \frac{1}{2}(2C)\left(\frac{V}{2}\right)^2 = \frac{1}{2}(2C)\frac{V^2}{4} = \frac{CV^2}{4}$$

The required ratio of final to initial energy is

$$\frac{U_f}{U_i} = \frac{CV^2/4}{CV^2/2} = \frac{1}{2}$$

$$\mathbf{U_f : U_i = 1 : 2}$$

Thus only half of the original energy remains; the other half is lost as **Joule heat** in the connecting wires during the transient redistribution of charge.

COMMON MISTAKE

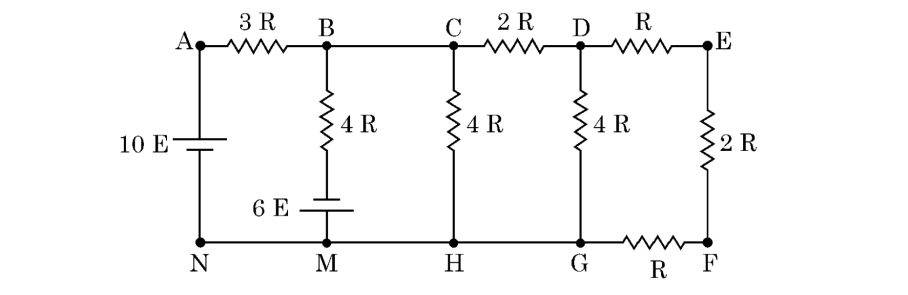
A common mistake is to think energy is conserved here. Charge is conserved, but half the energy is always lost as heat, no matter how small the wire resistance is.

Final answer. The ratio of the energy stored in the combination to the initial energy is **1 : 2**.

← Back to Q212

Q213

(a) When a parallel plate capacitor is connected across a dc battery, explain briefly how the capacitor gets charged.
(b) A parallel plate capacitor of capacitance 'C' is charged to 'V' volt by a battery. After some time the battery is disconnected and the distance between the plates is doubled. Now a slab of dielectric constant $\mathbf{1 < k < 2}$ *is introduced to fill the space between the plates. How will the following be affected?*
(i) The electric field between the plates of the capacitor.
(ii) The energy stored in the capacitor.
Justify your answer in each case.
(c) The electric potential as a function of distance 'x' is shown in the figure. Draw a graph of the electric field E as a function of x.



OR

(a) Derive an expression for the potential energy of an electric dipole in a uniform electric field. Explain conditions for stable and unstable equilibrium.

(b) Is the electrostatic potential necessarily zero at a point where the electric field is zero? Give an example to support your answer.

[2019 • Set 55-4-1]

SOLUTION

(a) How a capacitor gets charged from a DC battery. When the two plates are connected across a DC battery of emf V, the battery does work on the free electrons in the external wires. It removes electrons from the plate joined to the positive terminal (making it $+Q$) and pushes them onto the plate joined to the negative terminal (making it $-Q$).

The potential difference between the plates rises as charge builds up. Charging continues until this potential difference equals the battery emf V, after which the current stops and the capacitor is fully charged with $\boldsymbol{Q = CV}$.

(b) Battery disconnected, separation doubled, dielectric inserted. The capacitor of capacitance C is charged to V, so its charge is $\boldsymbol{Q = CV}$. Once the battery is removed, the charge Q stays fixed for all the following changes.

(i) Electric field between the plates. The field due to the plates is

$$E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{A\varepsilon_0}$$

Since Q and A are unchanged, doubling the separation d does not change E (the field of a uniform sheet is independent of distance). When a dielectric of constant K is then inserted, the field inside it is reduced by the factor K:

$$E_{new} = \frac{E}{K}$$

So the **electric field decreases to** $\boldsymbol{\frac{E}{K}}$ (since $\boldsymbol{K > 1}$).

(ii) Energy stored. With charge fixed, $\boldsymbol{U = \frac{Q^2}{2C}}$. After doubling the gap, $\boldsymbol{C \to \frac{C}{2}}$; after inserting the dielectric, $\boldsymbol{C \to K \times \frac{C}{2} = \frac{KC}{2}}$. So

$$U_{new} = \frac{Q^2}{2C_{new}} = \frac{Q^2}{2\,(KC/2)} = \frac{Q^2}{KC}$$

Compared with the initial $\boldsymbol{U_i = \frac{Q^2}{2C}}$,

$$\frac{U_{new}}{U_i} = \frac{Q^2/(KC)}{Q^2/(2C)} = \frac{2}{K}$$

Since $K < 2$ for the given range $1 < K < 2$, the factor $\frac{2}{K} > 1$, so the **energy increases**. (Doubling the gap doubles the energy by pulling the plates apart; the dielectric then partly reduces it, but for $K<2$ the net energy is still larger than the original.)

(c) E versus x graph. The electric field is $E = -\frac{dV}{dx}$, the negative slope of the V-x graph. Wherever V versus x is a straight line with constant slope, E is constant (a horizontal line). A region where V rises with x gives a negative constant E; a region where V is flat gives $E = 0$; a region where V falls with x gives a positive constant E. The E-x graph is therefore a series of horizontal steps, each equal to the negative slope of the corresponding straight segment of the V-x graph.

OR

(a) Potential energy of a dipole in a uniform field. A dipole of moment $\vec{p}$ in a uniform field $\vec{E}$ experiences a torque $\tau = pE\theta$, where θ is the angle between $\vec{p}$ and $\vec{E}$. To rotate the dipole through dθ, the external work done against the field is

$$dW = \tau\, d\theta = pE\theta\, d\theta$$

Taking $\theta = 90°$ as the reference of zero potential energy, the potential energy at angle θ is

$$U = \int_{90°}^{\theta} pE\theta'\, d\theta' = pE[-\theta']_{90°}^{\theta}$$

$$U = -pE\theta$$

$$\mathbf{U = -\vec{p}\cdot\vec{E}}$$

For **stable equilibrium**, $\theta = 0°$, so $U = -pE$ is minimum (dipole aligned with the field). For **unstable equilibrium**, $\theta = 180°$, so $U = +pE$ is maximum (dipole anti-parallel to the field).

(b) Is potential necessarily zero where field is zero? No. The electric field $E = -\frac{dV}{dx}$ depends only on how V changes with position, not on its value. So E can be zero even where V is not zero.

Example: at the midpoint of the line joining two equal positive charges, the two fields cancel so $\boldsymbol{E = 0}$, but each charge contributes a positive potential, so the net V is non-zero (positive) there.

Final answer. Field decreases to $\boldsymbol{E/K}$; energy increases by the factor $\boldsymbol{2/K}$ (greater than **1** since **K<2**); the E-x graph is constant steps equal to the negative slope of the V-x graph.

← *Back to Q213*

Q214

(a) Distinguish, with the help of a suitable diagram, the difference in the behaviour of a conductor and a dielectric placed in an external electric field. How does polarised dielectric modify the original external field?
(b) A capacitor of capacitance C is charged fully by connecting it to a battery of emf E. It is then disconnected from the battery. If the separation between the plates of the capacitor is now doubled, how will the following change?
(i) charge stored by the capacitor.
(ii) field strength between the plates.
(iii) energy stored by the capacitor.
Justify your answer in each case.

[2016]

SOLUTION

(a) Conductor versus dielectric in an external field. For a conductor placed in an external field $\vec{E}_0$: the free electrons drift opposite to $\vec{E}_0$ and pile up on one face, leaving the other face positive. These induced surface charges set up an internal field that exactly opposes $\vec{E}_0$, so in electrostatic equilibrium the net field inside the conductor is zero.

$$\vec{E}_{\text{inside}} = \vec{0}$$

Diagram for the conductor: draw a metal slab between parallel vertical field lines; show − charges on the left face and + charges on the right face, and show no field lines inside the slab.

For a dielectric placed in $\vec{E}_0$: it has no free electrons. The field instead polarises the molecules, producing bound charges on the two faces. These bound charges create an opposing internal field, but the cancellation is only partial, so a reduced field survives inside.

$$\vec{E}_{\text{inside}} = \frac{\vec{E}_0}{K}$$

Diagram for the dielectric: draw a slab with rows of small aligned dipoles (− on the left edge, + on the right edge), with field lines passing through but weakened inside.

How the polarised dielectric modifies the field: the bound surface charges produce a field opposite to $\vec{E}_0$, so the net field is reduced from E_0 to $\frac{E_0}{K}$, where K is the dielectric constant ($\boldsymbol{K > 1}$).

(b) Capacitor charged to emf E, battery disconnected, separation doubled. The charge stored is $\boldsymbol{Q = CE}$. Once the battery is removed, Q is fixed.

(i) Charge stored. With the battery disconnected, no path exists for charge to flow, so Q is **unchanged**.

(ii) Field strength between the plates. The field is

$$E_{\text{plate}} = \frac{\sigma}{\varepsilon_0} = \frac{Q}{A\varepsilon_0}$$

Both Q and A are unchanged, and the expression has no d in it, so the field is **unchanged** when the separation is doubled.

(iii) Energy stored. The capacitance is $\boldsymbol{C = \frac{\varepsilon_0 A}{d}}$, so doubling d halves it to $\frac{C}{2}$. With Q fixed,

$$U = \frac{Q^2}{2C}$$

$$U_{\text{new}} = \frac{Q^2}{2(C/2)} = \frac{Q^2}{C} = 2U$$

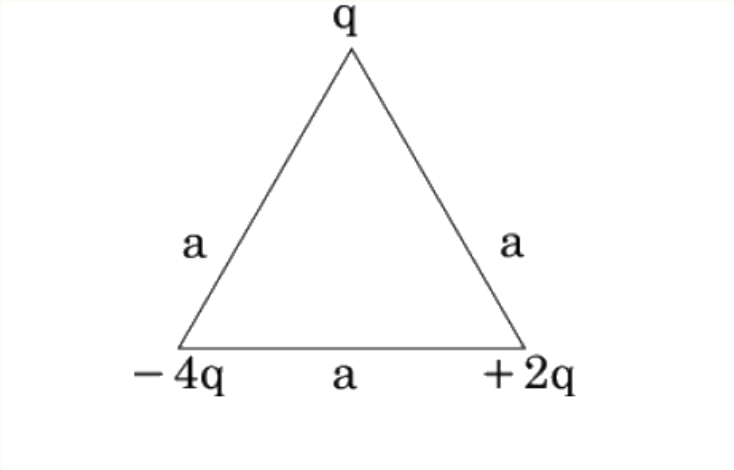
So the **energy stored doubles**. The extra energy equals the external work done in pulling the oppositely charged plates apart.

Final answer. Charge is unchanged; field strength is unchanged; energy stored doubles when the separation is doubled.

← *Back to Q214*

Q215

(a) Explain why, for any charge configuration, the equipotential surface through a point is normal to the electric field at that point.
Draw a sketch of equipotential surfaces due to a single charge (−q), depicting the electric field lines due to the charge.
(b) Obtain an expression for the work done to dissociate the system of three charges placed at the vertices of an equilateral triangle of side a as shown below.



[2016]

SOLUTION

(a) Why an equipotential surface is normal to the field. Consider a small displacement $d\vec{l}$ made along an equipotential surface. By definition every point on this surface is at the same potential, so the change in potential for this displacement is zero:

$$dV = 0$$

The relation between potential change and field is

$$dV = -\vec{E} \cdot d\vec{l}$$

Therefore

$$\vec{E}\cdot d\vec{l} = 0$$

Since $d\vec{l}$ is any non-zero displacement lying in the surface and the dot product is zero, $\vec{E}$ must be perpendicular to every such tangent vector. Hence $\vec{E}$ is **normal to the equipotential surface** at every point. (If $\vec{E}$ had a component along the surface, work would be done in moving a charge along it and the potential would change, contradicting the definition.)

Sketch for a single charge $-q$: draw the charge $-q$ at the centre. Around it draw concentric circles (the cross-sections of spherical equipotential surfaces). Draw straight field lines as radial arrows pointing inward toward $-q$, each arrow crossing the circles at right angles.

(b) Work to dissociate three charges on an equilateral triangle of side a. Let the charges be q, −4q and +2q at the three vertices. The total potential energy of the configuration is the sum of the energies of all three pairs, each pair separated by a:

$$U = \frac{k}{a}[(q)(-4q) + (-4q)(2q) + (2q)(q)]$$

$$U = \frac{k}{a}[-4q^2 - 8q^2 + 2q^2]$$

$$U = \frac{k}{a}(-10q^2) = -\frac{10\,kq^2}{a}$$

To dissociate the system means to separate all charges to infinity, where $\boldsymbol{U = 0}$. The work done by the external agent equals the change in potential energy:

$$W_{ext} = U_{final} - U_{initial} = 0 - \left(-\frac{10\,kq^2}{a}\right)$$

$$\boldsymbol{W_{ext} = \frac{10\,kq^2}{a} = \frac{10\,q^2}{4\pi\varepsilon_0\,a}}$$

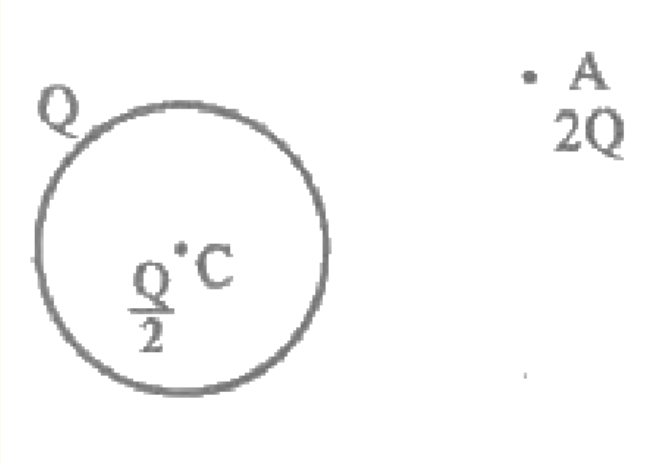
The positive sign shows the system is bound, so external work must be supplied to pull the charges apart.

Final answer. $\vec{E}$ is normal to the equipotential since **dV=0** along it; work to dissociate = $\boldsymbol{\frac{10\,kq^2}{a} = \frac{10\,q^2}{4\pi\varepsilon_0\,a}}$.

← Back to Q215

Q216

(a) Explain, using suitable diagrams, the difference in the behaviour of a (i) conductor and (ii) dielectric in the presence of external electric field. Define the terms polarization of a dielectric and write its relation with susceptibility.
(b) A thin metallic spherical shell of radius R carries a charge Q on its surface. A point charge $\frac{Q}{2}$ *is placed at its centre C and an other charge* +2Q *is placed outside the shell at a distance x from the centre as shown in the figure. Find (i) the force on the charge at the centre of shell and at the point A, (ii) the electric flux through the shell.*



[2015]

SOLUTION

(a) Conductor versus dielectric. A conductor placed in an external field $\vec{E}_0$ has free electrons that redistribute on its surfaces until the field inside the conductor is exactly cancelled.

$$\vec{E}_{\text{inside}} = \vec{0}$$

Diagram: a metal slab between parallel field lines, with induced − charge on one face and + on the other, and no field lines inside.

A dielectric placed in $\vec{E}_0$ has no free electrons. The field polarises its molecules, producing bound surface charges that set up an opposing field. The cancellation is only partial, leaving a reduced field inside.

$$\vec{E}_{\text{inside}} = \frac{\acute{E}_0}{K}$$

Diagram: a slab of aligned dipoles, with bound − charge on one face and + on the other, and weakened field lines inside.

Polarisation $\vec{P}$ is the dipole moment induced per unit volume of the dielectric. For a linear dielectric it is proportional to the field:

$$\vec{P} = \varepsilon_0 \chi_e \vec{E}$$

where χ_e is the **electric susceptibility**, related to the dielectric constant by $\chi_e = K - 1$.

(b) Spherical shell of radius R carrying charge Q, with $\frac{Q}{2}$ at the centre C and +2Q outside at distance x from the centre.

(i) Force on the charge at the centre. By symmetry, the charge Q spread uniformly on the shell produces zero field at the centre. So only the external charge +2Q exerts a force on $\frac{Q}{2}$:

$$F_C = \frac{k(\frac{Q}{2})(2Q)}{x^2} = \frac{kQ^2}{x^2} = \frac{Q^2}{4\pi\varepsilon_0 x^2}$$

directed along the line CA (away from the centre, since both are positive).

Force on the external charge +2Q at point A. From outside, the shell charge Q together with the central charge $\frac{Q}{2}$ behave as a single point charge $(Q + \frac{Q}{2}) = \frac{3Q}{2}$ located at the centre. Hence the force on +2Q is

$$F_A = \frac{k(\frac{3Q}{2})(2Q)}{x^2} = \frac{3kQ^2}{x^2} = \frac{3Q^2}{4\pi\varepsilon_0 x^2}$$

directed outward along CA (repulsive).

(ii) Electric flux through the shell. By Gauss's law, only the charge enclosed by the shell matters. The charge enclosed is the central charge $\frac{Q}{2}$ (the external +2Q is outside and contributes zero net flux; the shell's own charge Q lies on the surface, not enclosed). Therefore

$$\Phi = \frac{Q_{\text{enclosed}}}{\varepsilon_0} = \frac{Q/2}{\varepsilon_0} = \frac{Q}{2\varepsilon_0}$$

Final answer. $F_C = \dfrac{Q^2}{4\pi\varepsilon_0 x^2}$, $F_A = \dfrac{3Q^2}{4\pi\varepsilon_0 x^2}$, and the flux through the shell is $\dfrac{Q}{2\varepsilon_0}$.

← *Back to Q216*

Q217

Define the term 'capacitance' of a capacitor. Derive an expression for the total capacitance of 'n' capacitors arranged in series.

[2010]

SOLUTION

Definition of capacitance. The capacitance C of a capacitor is the ratio of the magnitude of the charge Q on either plate to the potential difference V between the plates:

$$C = \frac{Q}{V}$$

It measures how much charge a capacitor can store per unit potential difference. Its SI unit is the **farad (F)**, where $\mathbf{1\ F = 1\ C\ V^{-1}}$.

n capacitors in series. Connect n capacitors $C_1, C_2, , C_n$ in series across a source of potential difference V.

In a series combination, the same magnitude of charge Q appears on every capacitor. This is because the inner plates are isolated, so the charge induced on each capacitor must be equal.

The potential difference across the i-th capacitor is

$$V_i = \frac{Q}{C_i}$$

The total potential difference is the sum of the individual potential differences:

$$V = V_1 + V_2 + + V_n$$

$$V = \frac{Q}{C_1} + \frac{Q}{C_2} + + \frac{Q}{C_n}$$

$$V = Q(\frac{1}{C_1} + \frac{1}{C_2} + + \frac{1}{C_n})$$

If C_{eq} is the equivalent capacitance, then by definition $V = \frac{Q}{C_{eq}}$. Substituting,

$$\frac{Q}{C_{eq}} = Q(\frac{1}{C_1} + \frac{1}{C_2} + + \frac{1}{C_n})$$

Cancelling Q,

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + + \frac{1}{C_n} = {}_{i=1}{}^{n}\frac{1}{C_i}$$

Thus the equivalent capacitance of a series combination is always smaller than the smallest individual capacitance.

Final answer. $C = \frac{Q}{V}$ (SI unit farad), and for n capacitors in series $\frac{1}{C_{eq}} = {}_{i=1}{}^{n}\frac{1}{C_i}$.

← *Back to Q217*

Q218

Define electric potential at a point due to a point charge. Write its SI unit. Derive an expression for the electric potential at a point outside a charged sphere.

[2010]

SOLUTION

Definition of electric potential. The electric potential at a point due to a point charge is the work done per unit positive charge in bringing a small positive test charge from infinity (the reference of zero potential) to that point, without acceleration, against the electric force:

$$V = \frac{W}{q_0} = -{}_{\infty}{}^{P}\vec{E}\cdot \mathrm{d}\vec{l}$$

It is a scalar quantity. Its SI unit is the **volt (V)**, where $1\ V = 1\ J\ C^{-1}$.

Field of a charged sphere outside it. Consider a sphere of radius R carrying total charge Q. For a point at distance $r > R$ from the centre, the field is radial by symmetry. Apply Gauss's law over a concentric Gaussian sphere of radius r:

$$E(r) \times 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$E(r) = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{kQ}{r^2}$$

Potential at an outside point. Taking infinity as zero potential, the potential at distance r is the negative line integral of the field from infinity to r:

$$V(r) = -\int_{\infty}^{r} E(r')\, dr' = -\int_{\infty}^{r} \frac{kQ}{r'^2}\, dr'$$

$$V(r) = -kQ\left[-\frac{1}{r'}\right]_{\infty}^{r} = kQ\left(\frac{1}{r} - 0\right)$$

$$\mathbf{V(r) = \frac{kQ}{r} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}} \quad (r > R)$$

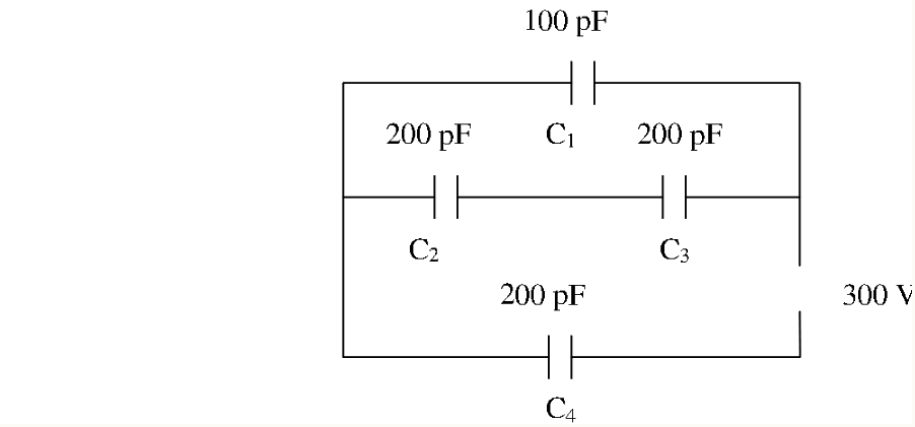
So outside the sphere, the potential is the same as that of a point charge Q placed at the centre.

Final answer. Potential is $V = \dfrac{W}{q_0}$ (SI unit volt), and outside a charged sphere $V(r) = \dfrac{kQ}{r} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{r}$ for $r > R$.

← Back to Q218

Q219

(a) Derive an expression for the energy stored in a parallel plate capacitor C, charged to a potential difference V.
(b) Obtain the equivalent capacitance of the network given below. For a supply of 300 *V, determine the charge and voltage across* C_4.



[2008]

SOLUTION

(a) Energy stored in a parallel plate capacitor. Let the capacitor at some instant during charging carry charge q, so the instantaneous potential difference is $v = \frac{q}{C}$. The small work done to bring an additional charge dq from the negative to the positive plate is

$$dW = v\, dq = \frac{q}{C}\, dq$$

The total work done in charging from 0 to final charge Q is

$$U = {}_0{}^{Q}\frac{q}{C}\, dq = \frac{1}{C}\left[\frac{q^2}{2}\right]_0{}^{Q} = \frac{Q^2}{2C}$$

Using $Q = CV$,

$$\mathbf{U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV}$$

(b) Equivalent capacitance and charge and voltage across C_4. In the standard NCERT network, $\mathbf{C_1 = 100\ pF}$, $\mathbf{C_2 = 200\ pF}$, $\mathbf{C_3 = 200\ pF}$, $\mathbf{C_4 = 100\ pF}$, with C_2 and C_3 in series, that combination in parallel with C_1, and C_4 in series with the whole, across $V = 300\ V$.

Series of C_2 and C_3:

$$\frac{1}{C_{23}} = \frac{1}{200} + \frac{1}{200} = \frac{2}{200} = \frac{1}{100}$$

$$C_{23} = 100\ pF$$

This is in parallel with C_1:

$$C_{123} = C_1 + C_{23} = 100 + 100 = 200 \text{ pF}$$

This is in series with C_4:

$$\frac{1}{C_{eq}} = \frac{1}{C_{123}} + \frac{1}{C_4} = \frac{1}{200} + \frac{1}{100} = \frac{1}{200} + \frac{2}{200} = \frac{3}{200}$$

$$C_{eq} = \frac{200}{3} \approx 66.7 \text{ pF}$$

Total charge supplied (this is the charge on C_4, since C_4 is in the main series line):

$$Q_{total} = C_{eq}\, V = \frac{200}{3} \times 10^{-12} \times 300$$

$$Q_{total} = 2 \times 10^{-8}\, C = 0.02\ \mu C$$

So the charge on C_4 is

$$\mathbf{Q_4 = 2 \times 10^{-8}\ C}$$

The voltage across C_4 is

$$V_4 = \frac{Q_4}{C_4} = \frac{2 \times 10^{-8}}{100 \times 10^{-12}}$$

$$\mathbf{V_4 = 200\ V}$$

TIP

The charge on a capacitor in the main series line equals the total charge drawn from the supply, since the same charge flows through every series element.

Final answer. $U = \frac{Q^2}{2C} = \frac{1}{2}CV^2$; for the network $C_{eq} = \frac{200}{3}$ pF, with $Q_4 = 210^{-8}$ C and $V_4 = 200$ V.

← *Back to Q219*

Q220

Explain the principle on which Van de Graaff generator operates. Draw a labeled schematic sketch and write briefly its working.
A Van de Graaff type generator is capable of building up potential difference of 15×10^6 *V. The dielectric strength of the gas surrounding the electrode is* 5×10^7 *V* m^{-1}. *What is the minimum radius of the spherical shell required?*

[2008]

SOLUTION

Principle. The Van de Graaff generator works on two facts: any excess charge given to a conductor resides entirely on its outer surface, and the inside of a hollow charged conductor is electrostatically shielded. So a small charged sphere placed inside a large hollow dome and connected to it transfers all its charge to the outer surface of the dome, however high the dome's own charge already is. By repeating this, the dome can be charged to a very high potential.

Labelled schematic sketch. Draw a large hollow metal sphere (dome) mounted on an insulating column. Inside and below it draw two pulleys carrying a continuous insulating belt. Label a sharp comb B_1 (spray comb) at the lower pulley connected to a high-voltage source, and a comb B_2 (collector comb) at the upper pulley touching the inside of the dome. Mark the motor driving the lower pulley and the grounded base.

Working. The lower comb B_1, at a high positive potential, sprays positive charge onto the moving belt by corona discharge. The belt carries this charge upward into the dome. There the collector comb B_2 removes the charge and passes it to the dome, where it spreads over the outer surface. As the belt runs continuously, charge keeps accumulating and the dome's potential rises to several million volts, until limited by leakage or breakdown of the surrounding gas.

Minimum radius. The dome's potential is limited because the field at its surface must not exceed the dielectric strength (breakdown field) of the surrounding gas. For a charged sphere, the surface field and surface potential are related by

$$E_{surface} = \frac{V}{R}$$

The condition for no breakdown is $E_{surface} \le E_{breakdown}$, so

$$\frac{V_{max}}{R} \le E_{breakdown}$$

The minimum radius corresponds to the equality:

$$R_{min} = \frac{V_{max}}{E_{breakdown}}$$

Substituting $\mathbf{V_{max} = 15\times 10^{6}\ V}$ and $\mathbf{E_{breakdown} = 5\times 10^{7}\ V\,m^{-1}}$:

$$R_{min} = \frac{15\times 10^{6}}{5\times 10^{7}}\ m$$

$$R_{min} = 0.30\ m$$

$$\mathbf{R_{min} = 0.30\ m = 30\ cm}$$

Final answer. The minimum radius of the spherical shell is $\mathbf{R_{min} = 0.30\ m = 30\ cm}$.

← *Back to Q220*